

SpaceCDF — A Course Textbook

Companion textbook for the SpaceCDF 40-hour Concurrent Design Facility programme

SpaceCDF

2026

How to use this book

This textbook is the SpaceCDF course volume. It accompanies the 40-hour Concurrent Design Facility programme and is organised in four parts:

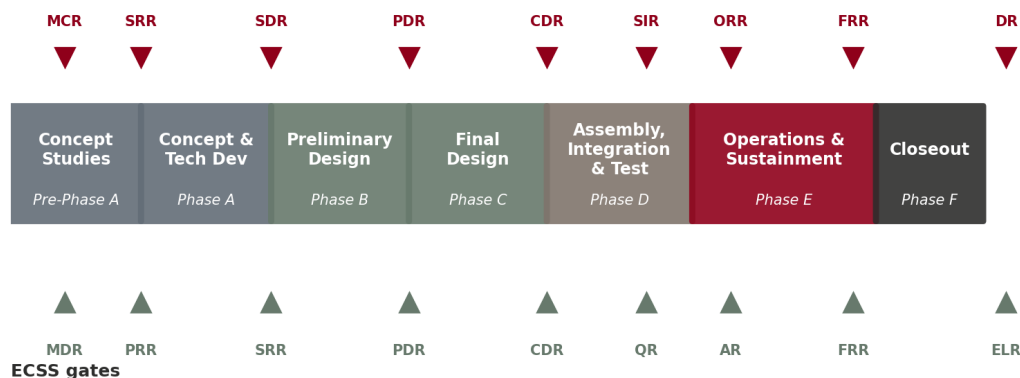
1. **Per-session course chapters** — one chapter per course session (Sessions 1.1 – 5.4). Each contains learning objectives, course chapters, exercises, expected reading, and supporting figures.
2. **Per-role background briefings** — a single chapter giving every CDF position the decisions, formulas, charts, pitfalls, and references they need from Day 1.
3. **Position appendix** — short-form per-position cards kept for quick reference at a console.
4. **Verification appendix** — a first-principles derivation and plot for every formula students uses, with cross-method sanity checks. Suitable as the worked verification trail at PDR.

The book is paired with the *Learner's Workbook*, which carries the worksheets students fills in.

Expected reading before any session. NASA SEH §1 – §3 — <https://www.nasa.gov/reference/systems-engineering-handbook/>. NPR 7123.1D §3 — <https://nodis3.gsfc.nasa.gov/>. NASA CubeSat 101 — https://www.nasa.gov/wp-content/uploads/2017/03/nasa_csli_cubesat_101_508.pdf.

Project Lifecycle Phases & Review Gates — NASA vs ECSS

NASA gates



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The lifecycle students lives.

NASA SEH — 17 Common Technical Processes

SYSTEM DESIGN

- 1. Stakeholder Expectations Definition
- 2. Technical Requirements Definition
- 3. Logical Decomposition
- 4. Design Solution Definition

PRODUCT REALISATION

5. Product Implementation	6. Product Integration
7. Product Verification	8. Product Validation
9. Product Transition	

TECHNICAL MANAGEMENT

10. Technical Planning	11. Requirements Management
12. Interface Management	13. Technical Risk Management
14. Configuration Management	15. Technical Data Management
16. Technical Assessment	17. Decision Analysis

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The 17 NASA SEH processes; each session names which process(es) it sits inside.

Acknowledgement — Generative AI (AIG)

This textbook was produced with the assistance of generative AI as part of the SpaceCDF Concurrent Design Facility workflow. The design-loop convergence, agent rationales, embedded figures and document rendering are generated by the SpaceCDF backend (Python · matplotlib · python-docx · WeasyPrint) guided by ECSS, NASA SEH and SMAD-4 references. Editorial framing, worked examples, and pedagogical commentary remain owned by the SpaceCDF teaching team. The chapter *How This Textbook Was Generated* describes the pipeline in detail.

Attribution follows the AIG (Assisted by Generative AI) framework — Peters (2023), Logos IA-EN, CC BY-NC-SA 4.0 — <https://mpeters.uqo.ca/en/logos-ia-en-peters-2023/>

Any course deliverable that incorporates content from this textbook — or from any document exported by SpaceCDF — must carry the AIG badge and a short note describing how generative AI was used. Refusing to use AI where it is permitted is fine; what is not acceptable is using it without saying so.

How This Textbook Was Generated

This textbook is itself an artefact of the same Concurrent Design Facility methodology that students will exercise during the 40-hour intensive. Its preparation pipeline mirrors a CDF design loop in three respects: the *source* is a structured set of Markdown chapters that describe each session as a self-contained module; the *transformation layer* is a Python toolchain — reorientation scripts, matplotlib figure generators, python-docx and WeasyPrint renderers — shared with the design-review exporters used inside the tool; and the *review layer* consists of the SpaceCDF teaching team, who own the prose voice, the worked examples, and the editorial judgement on what to include or omit.

Source — per-session Markdown chapters

The textbook is assembled from forty-odd Markdown chapter files under `docs/course/facilitator/`. Each chapter is self-contained — learning objectives, references, key equations, worked examples, exercises — so that an instructor can lift any one of them out and use it on its own. The single-source build script (`docs/build_course_pdfs.sh`) concatenates these chapter files, appends the appendices, and feeds them through the PDF renderer.

Voice reorientation

The book began life as a *Course Textbook* filled with stage directions ("Begin by asking the group...", "Students should draw the V-model on the whiteboard..."). A scripted reorientation pass — driven by `scripts/reorient_books.py` — converted those directions into textbook prose, stripped timing cues from headings, removed embedded "Teaching Notes" subsection markers, and made the volume self-paced and self-contained. The same pass keeps the volume aligned with the current tool: it updates the agent count, restores the four-step workflow nomenclature, and inserts the SDR gate into all review-gate tables.

The reorientation rules are documented in the script itself. Each rule is a regular-expression substitution applied to the per-chapter Markdown sources, with the *expanded* book and the workbook kept in lock-step. If a pedagogical decision changes — say, the team chooses to keep discussion prompts as live prompts rather than reflective questions — the rule is edited in one place and the book is regenerated.

Procedural figure generation

Every diagram in this volume (V-model, lifecycle gates, beta-angle eclipse geometry, Tsiolkovsky curves, free-space path loss, antenna patterns, ground-track sketches, link-budget waterfalls, thermal node bars, risk index maps) is generated procedurally by a matplotlib script under `docs/assets/figures/` using the uOttawa Horizon palette (`uottawa_brand.py`). The procedural figures are sources of truth in the same sense as a code listing: they can be re-rendered by re-running the script when the underlying numerics change. There are no hand-drawn diagrams — every line on the page is derived from a small Python program with a reproducible numerical input.

The same matplotlib styling is used by the SpaceCDF design-review exporters: when a study generates a Word SRR, its donut charts and link waterfalls are drawn by the same helpers, with the same colours, that produce the figures in this textbook. The intent is that a student looking at a *figure* on the page of an SRR has already encountered the same kind of figure in the textbook chapter that introduced the underlying physics.

Numerical worked examples — generated by running the tool

Every numerical worked example in this textbook — UniSat-1, the 3U EO CubeSat (the running "EOSAT-1" example), ground-track timings, link-budget closures, thermal radiator sizings — was computed by running the SpaceCDF tool itself, so the textbook and the tool stay numerically consistent. The relevant YAML mission descriptors live under `configs/examples/`. If a Tier 1 agent is updated and a worked-example number changes, the textbook reference can be regenerated by running `python scripts/run_design.py configs/examples/<mission>.yaml` and copying the new values into the chapter source.

This convention applies to the *Course Workbook* too: every answer key hint in the workbook is derived from running the tool on the same YAML mission descriptor used in the textbook.

Document rendering

The PDF you are reading is built by `docs/build_course_pdfs.sh`: Markdown chapters are concatenated, parsed by `python-markdown` (with tables, fenced code, and table-of-contents extensions enabled), wrapped in a uOttawa-styled HTML scaffold (cover banner, bilingual EN/FR "Page X of / de Y" footer, crimson page-header rule), and rendered to A4 PDF by WeasyPrint. The cover banner, the running header, and the SCDF document code are the same elements you will find on every SRR / PDR / CDR design-review bundle that the tool exports during your design study. A reviewer holding both the textbook and a freshly exported SRR side by side should immediately recognise that they belong to the same documentary family.

Validation and citation

Although AI assistance is used at every step of this pipeline, no claim in this textbook is left ungoverned by a human author. ECSS standards, NASA references and SMAD-4 are cited at the point of use; numerical results are reproducible by running the tool against the published YAML; pedagogical framing is owned by the teaching team. The AIG badge (see the *Acknowledgement* section) is attached to every exported document so that downstream readers can trace which contributions are AI-assisted.

What this means for learners

The textbook is generated by the same kind of human-plus-AI loop that the SpaceCDF tool exposes to students. Learning to *read* AI-assisted documents critically — checking the citations, re-deriving the numbers, asking which voice authored which passage — is itself part of the curriculum. The first time a student in students generates an SRR from a converged design state, they should recognise the layout, the voice and the figure conventions of this volume in the document the tool has just produced for them.

In other words: this is not a book *about* a tool, it is a *companion volume* to the tool. The same hands that wrote the chapters wrote the tool, and the same loop that closes a CubeSat design loop closes the loop on the next edition of this book.

Agents, Positions, and the SpaceCDF Architecture

Two distinct concepts run through this textbook and they are sometimes conflated in industry literature. Disentangling them is essential before the first design loop.

A **position** is a *human role* assigned to a person sitting at a console in the Concurrent Design Facility — for example *Power Engineer*, *AOCS Lead*, or *Mission Analyst*. SpaceCDF defines fifteen positions, each with scoped editing rights on a subset of the design parameters.

A **design agent** is a *computational solver* that runs in the background and converges a particular subsystem or analysis. The current release of SpaceCDF ships twenty agents grouped in two tiers:

Tier	Purpose	Agents
Tier 1 — Parametric sizing	Compute the design point of one subsystem from inputs supplied by the others	orbit · link · data · power · thermal · AOCS · propulsion · mass · structure
Tier 2 — Cross-system analysis	Score, check or audit the converged design	cost · risk · reliability · debris · radiation · sustainability · volume · systems · conflicts · community · TRL

Each agent publishes three things to the design state when it runs: its **parameters** (numbers with units, margin and confidence), its **rationale** (a paragraph of textbook-grade reasoning), and a set of **assumptions** that downstream agents and reviewers can interrogate. Several agents also emit *structured intermediates* — the orbit agent exposes a ΔV breakdown, the link agent exposes a full waterfall, the thermal agent exposes hot- and cold-case node temperatures, the reliability agent exposes a FMECA, and so on. Most exports surface these intermediates as tables and figures alongside the bare numbers.

When a human position assignment changes a value, that parameter becomes *sticky* in the design state and the agents do not overwrite it during subsequent design loops. This is the formal handover between the human operator and the computational solver, and it is the reason a CDF session converges so quickly: the agents do the iterative arithmetic while the positions do the engineering judgement.

Design Reviews and the SpaceCDF Exports

A review gate is not merely a meeting. It is a *document set* whose existence and contents are agreed in advance, whose generation is traceable to a converged design state, and whose acceptance moves the project from one ECSS phase to the next. SpaceCDF treats each review gate as the natural anchor of an export bundle.

Review	ECSS phase	Exported bundle	Principal contents
MCR	Pre-Phase A	Mission Concept Report	Stakeholders, MoP/TPM, alternatives, ConOps draft
SRR	Phase A	SRR Design-Review Pack (Word + Excel + Markdown)	Mass / power / data / cost budgets with margin, orbit and architecture figures, compliance matrix, equipment list, TRL roll-up, risk matrix, cost P-curve, WBS bar
SDR	Phase A/B	System Design Review Pack	SRR contents plus block diagrams, derived requirements, preliminary trade studies
PDR	Phase B	PDR Design-Review Pack	SRR + SDR contents plus full trade-study record, verification matrix, interface control documents
CDR	Phase C	CDR Design-Review Pack	PDR contents plus test plan, AIT procedures, qualification status

In addition to the design-review packs, SpaceCDF generates a family of *data item descriptions* (DIDs) in editable Word format: the Mission Requirements Document (MRD), Concept of Operations (ConOps), Verification Plan (VP), Systems Engineering Management Plan (SEMP), Risk Management Plan (RMP), Interface Requirements Document (IRD), Test Plan, and the Bill of Materials. Specialist reports cover the thermal analysis, the end-of-life deorbit plan, the Remote Sensing Space Systems Act (RSSSA) licence application, and the verification test programme.

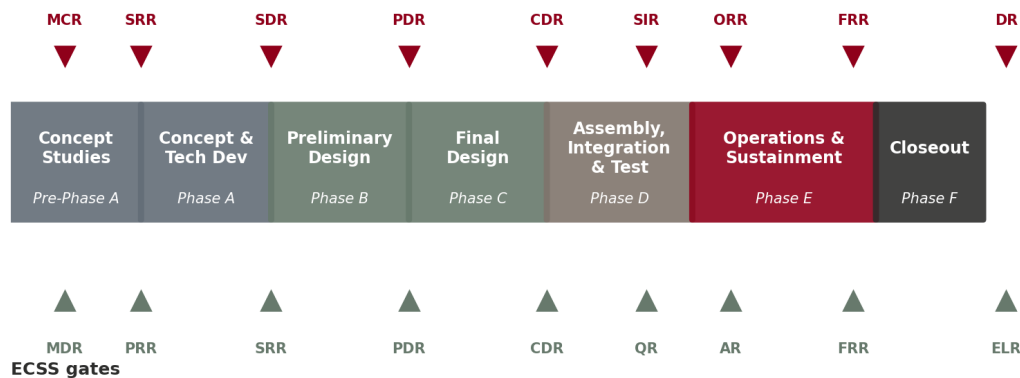
Every exported document uses the same uOttawa SpaceCDF course identity: garnet cover banner with the bilingual "u Ottawa · UNIVERSITÉ D'OTTAWA" mark, slab-bold title block, crimson section headings, the SCDF-{code} document code, and a bilingual page footer reading "Page X of / de Y". The intent is that a converged design state, exported the day before a review, arrives in the reviewers' hands looking and reading the same as the supporting course material.

Part 1 — Course Lectures

Session 1.1: Introduction to Space Mission Design

Project Lifecycle Phases & Review Gates — NASA vs ECSS

NASA gates



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Lifecycle phases and review gates.

NASA SEH — 17 Common Technical Processes

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NASA SEH 17 processes pyramid.

Expected reading before this session. NASA SEH §1 – §3 (~ 60 min) — <https://www.nasa.gov/reference/systems-engineering-handbook/>. NPR 7123.1D §3 (~ 30 min).

Prerequisites: None (engineering background assumed) **References:** - NASA, Systems Engineering Handbook Rev 2 (SP-2016-6105), 2016 - NASA, NPR 7123.1D -- Systems Engineering Processes and Requirements, 2020 - ECSS, ECSS-M-ST-10C Rev.1 -- Space Project Management, 2009 - ESA, "20 Years of Concurrent Design at ESA", 2018

Learning Objectives

By the end of this session, participants will be able to: 1. Describe the purpose and structure of a Concurrent Design Facility (CDF) and contrast it with sequential design 2. Explain the System-V model and its two sides (decomposition + integration) 3. List the 17 common technical processes defined in NPR 7123.1D and explain their recursive application 4. Map NASA lifecycle phases (Pre-A through F) to ECSS phases and identify key review gates 5. Identify the role of each CDF engineering position and their parameter ownership

1. What is a Concurrent Design Facility?

1.1 The Problem with Sequential Design

Consider the question: How is spacecraft design traditionally done?

Traditional spacecraft design follows a **sequential (waterfall) approach**: one discipline completes its work, documents it, and passes the design to the next discipline. This method dominated the space industry from the 1960s through the 1990s. Its limitations are well documented:

- **Interface mismatches** discovered late in integration, requiring costly redesign
- **Budget overruns** in mass, power, and cost not caught until formal reviews
- **Long iteration cycles** -- months between design reviews with limited inter-discipline feedback
- **Knowledge silos** -- critical design rationale locked in individual engineers' notes
- **Rework costs** -- the cost of correcting an error increases by roughly an order of magnitude per lifecycle phase (the "1-10-100 rule")

The aerospace industry's response was **concurrent engineering (CE)**, also called Integrated Product Development (IPD) or Integrated Concurrent Engineering (ICE). The core idea: bring all disciplines together to work simultaneously on a shared parametric model, resolving conflicts in real-time rather than discovering them in review.

Industry Practice: Boeing's Phantom Works implemented concurrent engineering for the X-32 Joint Strike Fighter demonstrator in the late 1990s, reducing design iteration time from weeks to hours. JPL's Team X has been performing rapid mission assessments since 1995, completing Pre-Phase A studies in as little as one week. ESA's CDF, established in November 1998 at ESTEC (Noordwijk, Netherlands), became the most widely replicated model for space-specific concurrent design.

1.2 ESA's CDF -- The Reference Model

ESA's Concurrent Design Facility pioneered the approach for space missions and has been widely replicated at space agencies worldwide (DLR, CNES, JAXA, CSA, and over 60 universities).

Parameter	Value
Established	November 1998, ESTEC, Noordwijk, NL
Team size	15--25 domain specialists per study
Session format	3--4 hour focused design sessions, typically 2 per week
Study duration	3--6 weeks for a complete Phase 0/A mission assessment
Tooling	Originally Excel-based IDM; now Open Concurrent Design Tool (OCDT)
Output	Complete mission feasibility assessment: mass, power, cost, risk, schedule
Track record	200+ studies completed by 20th anniversary (2018)
Cost savings	Studies completed in ~1/5 the time and ~1/3 the cost of traditional approach

[Source: ESA CDF official documentation; Bandecchi et al., "The ESA/ESTEC Concurrent Design Facility", ESA Bulletin No. 107, 2001]

1.3 Key Principles of Concurrent Design

Shared parametric model. All disciplines read from and write to a common data model. When the power engineer changes the solar array area, the structures engineer immediately sees the mass impact, and the thermal engineer sees the changed radiator view factor. This eliminates the "frozen interface document" problem.

Scoped parameter ownership. Each engineering position "owns" a defined set of parameters. Only the power engineer can modify EPS parameters; only the AOCS engineer can modify attitude control parameters. This prevents conflicting edits while maintaining concurrent access.

Real-time conflict detection. When parameter changes create conflicts (e.g., mass budget exceeded, link margin negative), the system flags them immediately. The systems engineer arbitrates.

Design convergence through iteration. A CDF study typically converges through 3--5 major iterations, with each session refining the design toward closure. The parametric model propagates changes automatically, so the team sees the system-level impact of every decision.

1.4 SpaceCDF -- A Web-Based CDF

SpaceCDF implements the CDF concept as a web-based tool accessible from any browser:

Feature	Implementation
Shared model	Real-time synchronisation via WebSocket; all participants see live updates
Parameter ownership	15 engineering positions with scoped editing rights
Design convergence	20 background design agents (9 Tier 1 sizing, 11 Tier 2 analysis) that propagate parametric relationships
Conflict detection	Automated constraint engine with 187 inter-parameter connections
Document generation	ECSS-compliant exports (Word, PDF) with full traceability
Review gates	Automated gate criteria evaluation for MCR, SRR, SDR, PDR, CDR

Discussion point. What advantages does concurrent design offer over sequential? What risks does it introduce (e.g., groupthink, premature convergence)?

2. The System-V Model

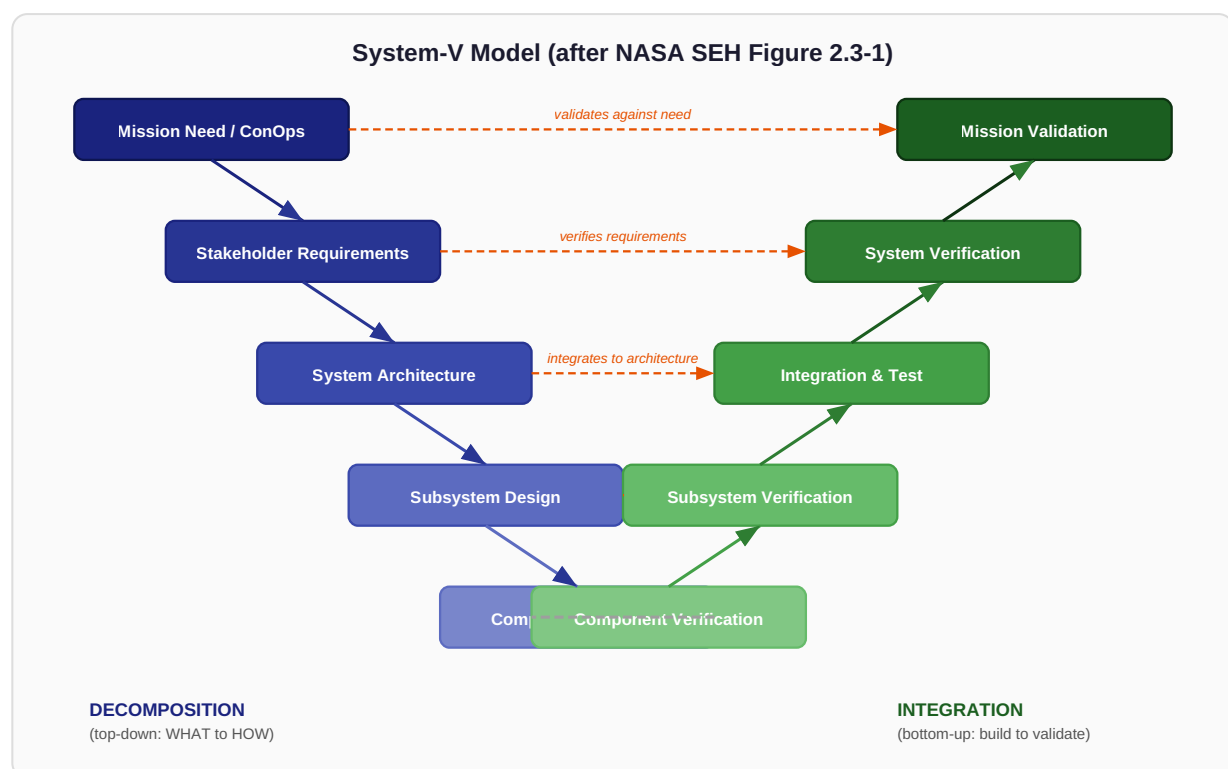
2.1 Origin and Purpose

The "Vee" (V) model is the foundational framework for systems engineering. It appears in NASA SEH Section 2.3 (Figure 2.3-1), in ECSS-E-ST-10C, and in ISO/IEC 15288. It is not a project schedule -- it is a **logical model** showing the relationship between decomposition (breaking the problem down) and integration (building and verifying the solution).

The V-model answers two fundamental questions: 1. **Left side (top-down):** How do we decompose the mission need into buildable components? 2. **Right side (bottom-up):** How do we verify that what we built satisfies the original need?

The key insight is **horizontal traceability**: every element on the left side has a corresponding verification or validation activity on the right side, connected at the same level of abstraction.

2.2 V-Model Diagram



2.3 Left Side: Top-Down Decomposition

The left side of the V answers "how do we break the problem down into solvable pieces?"

Level	Activity	Key Question	Output
1. Mission Need	Define the problem and concept of operations	What problem are we solving? For whom?	ConOps, problem statement, MoEs

Level	Activity	Key Question	Output
2. Stakeholder Requirements	Capture needs of all stakeholders	What must the system achieve? (WHAT, not HOW)	Stakeholder requirements baseline
3. System Architecture	Define segments, elements, interfaces	How is the system structured?	System architecture, ICDs
4. Subsystem Design	Design each subsystem to meet allocated requirements	How does each subsystem work?	Subsystem specifications
5. Component Selection	Choose or design lowest-level components	Which hardware/software fulfils each need?	Component specifications, BoM

2.4 Right Side: Bottom-Up Integration

The right side answers "how do we verify that what we built satisfies the original need?"

Level	Activity	Key Question	Verification Method
5. Component Verification	Test each component against its spec	Does each part work?	Inspection, test
4. Subsystem Verification	Integrate and test each subsystem	Do subsystems meet allocated requirements?	Test, analysis
3. Integration & Test	Assemble the system and test interfaces	Do the subsystems work together?	Test, demonstration
2. System Verification	Verify system against requirements	Does the system meet all requirements?	Test, analysis, inspection, demonstration
1. Mission Validation	Validate against the original need	Does it solve the original problem?	Operations, user acceptance

2.5 Horizontal Traceability

The orange dashed arrows in the diagram represent **horizontal traceability** -- the principle that every element on the left must have a corresponding verification/validation activity on the right:

- Each **requirement** (left) has a **verification method** (right): test, analysis, inspection, or demonstration (TAID)
- Each **design decision** (left) has a **test or analysis** (right) to confirm it works
- This traceability is captured in the **Requirements Verification Matrix (RVM)**

Key Equation: The traceability completeness metric is:

$$T_c = \frac{N_{\text{verified}}}{N_{\text{total_requirements}}} \times 100\%$$

A target of $T_c = 100\%$ is required at CDR. At PDR, $T_c \geq 90\%$ is typical.

2.6 Iteration and the SE Engine

The V is not a single pass. Real design iterates -- requirements change as design constraints are discovered, and design parameters feed back to refine requirements. This iterative loop is captured in the concept of

the **"SE Engine"** (NASA SEH Section 2.1): the 17 common technical processes applied recursively at every level of the system hierarchy.

[Source: NASA SEH Section 2.3, Figure 2.3-1; INCOSE Systems Engineering Handbook, 4th Edition, Section 3.2]

Exercise. Sketch the V-model from memory and label each level. Add the horizontal traceability arrows and name one verification method for each level.

3. The 17 Common Technical Processes

3.1 Overview

NASA's NPR 7123.1D defines 17 processes that apply recursively at every level of the system hierarchy. They are grouped into three categories, collectively called the **"SE Engine"**. These processes are not sequential -- they execute concurrently and iteratively, which is precisely what a CDF facilitates.

[Source: NPR 7123.1D Chapter 3; NASA SEH Section 2.1, Figure 2.1-1]

3.2 System Design Processes (1--4)

These processes decompose the problem into a solution. They correspond to the left side of the V-model.

#	Process	Key Activity	Key Output	NASA SEH Reference
1	Stakeholder Expectations Definition	Elicit needs, define ConOps, establish MoEs	Stakeholder requirements baseline	Section 4.1
2	Technical Requirements Definition	Write "shall" statements, define MoPs and TPMs	Technical requirements baseline	Section 4.2
3	Logical Decomposition	Functional analysis, behavioural modelling, derived requirements	Functional architecture	Section 4.3
4	Design Solution Definition	Trade studies, select among alternatives, produce baseline design	Design solution baseline	Section 4.4

3.3 Product Realisation Processes (5--9)

These processes build, verify, and deliver the solution. They correspond to the right side of the V-model.

#	Process	Key Activity	Key Output	NASA SEH Reference
5	Product Implementation	Make, buy, code, or reuse lowest-level products	Hardware/software products	Section 5.1
6	Product Integration	Assemble per integration plan, verify interfaces	Integrated system	Section 5.2
7	Product Verification	Confirm product meets technical requirements (TAID)	Verification evidence	Section 5.3
8	Product Validation		Validation evidence	Section 5.4

#	Process	Key Activity	Key Output	NASA SEH Reference
		Confirm product meets stakeholder expectations		
9	Product Transition	Deliver, deploy, hand over to operations	Operational system	Section 5.5

3.4 Technical Management Processes (10--17)

These processes manage the engineering work and apply throughout the lifecycle.

#	Process	Key Activity	Governing Document
10	Technical Planning	SEMP, subsidiary plans, WBS	NPR 7123.1D Section 3.10
11	Requirements Management	Baselining, traceability, change control	NPR 7123.1D Section 3.11
12	Interface Management	ICDs, IRDs, internal + external interfaces	NPR 7123.1D Section 3.12
13	Technical Risk Management	Identification, assessment, mitigation per NPR 8000.4	NPR 8000.4B
14	Configuration Management	Baselines, CM plan, CCBs	NPR 7120.5F Chapter 4
15	Technical Data Management	Data rights, retention, dissemination	NPR 7123.1D Section 3.15
16	Technical Assessment	TPMs, reviews, EVM, health checks	NPR 7123.1D Section 3.16
17	Decision Analysis	Structured alternative selection (trade studies)	NASA SEH Section 6.8

3.5 Recursion Across the System Hierarchy

A critical property of the 17 processes is that they apply **recursively** at every level of the system hierarchy:

Mission Level: 17 processes applied (mission-level requirements, ConOps, validation)
 System Level: 17 processes applied (system requirements, architecture, verification)
 Subsystem Level: 17 processes applied (subsystem requirements, design, testing)
 Component Level: 17 processes applied (component specs, procurement, acceptance)

At each level, the same processes execute but with different scope, detail, and duration. In a CDF environment, multiple levels are often worked concurrently -- the systems engineer manages the mission and system levels while subsystem engineers work their respective domains.

Industry Practice: On the James Webb Space Telescope (JWST), requirements management (Process 11) was applied at five levels of the system hierarchy simultaneously: Observatory, Optical Telescope Element, Integrated Science Instrument Module, individual instruments (NIRSpec, MIRI, NIRCам,

FGS), and components (detectors, mechanisms). Each level had its own requirements baseline, traceability matrix, and verification plan. The total requirements count exceeded 10,000.

Exercise. Map each of the 17 processes to a feature in SpaceCDF. Which processes does the tool support directly? Which require human judgment? Complete Part A of Worksheet 1.1.

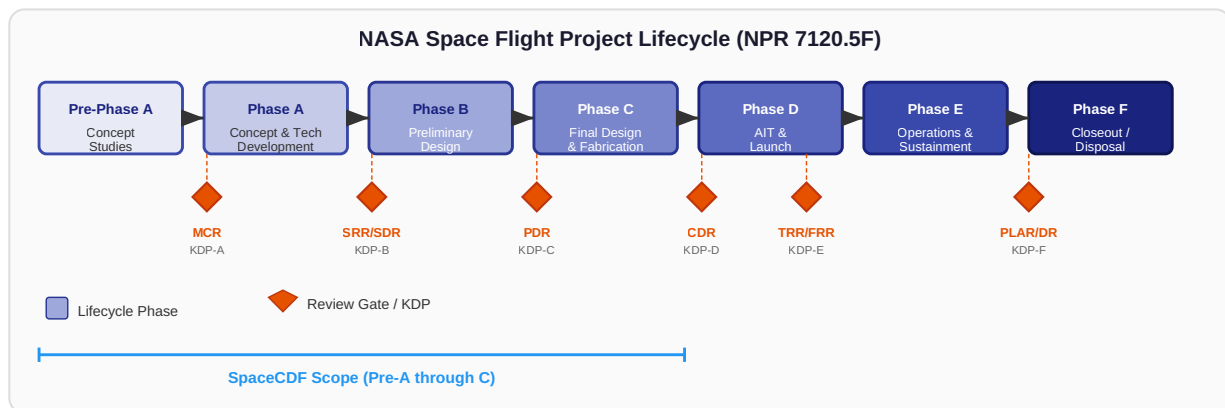
4. Lifecycle Phases and Review Gates

4.1 NASA Lifecycle Phases

NASA defines seven lifecycle phases for space flight projects, governed by NPR 7120.5F. Each phase has specific objectives, activities, and exit criteria evaluated at formal review gates called Key Decision Points (KDPs).

[Source: NPR 7120.5F Chapter 2; NASA SEH Chapter 3]

4.2 Lifecycle Phase Diagram



4.3 Phase Details

Phase	Name	Primary Activity	Exit Review	Key Outputs
Pre-A	Concept Studies	Identify need, explore feasibility, develop ConOps	MCR	Mission concept, preliminary ConOps, feasibility assessment
A	Concept & Technology Development	Develop requirements, mature technology to TRL 4+	SRR/SDR	Requirements baseline, technology development plan
B	Preliminary Design & Tech Completion	Preliminary design, close preliminary budgets, TRL 6+	PDR	Preliminary design, budget allocations, test plans
C	Final Design & Fabrication	Detailed design, build hardware, TRL 8	CDR	Detailed drawings, as-built documentation, flight hardware
D	Assembly, Integration & Test, Launch	System AIT, environmental testing, launch	TRR, ORR, FRR	Tested flight system, launch readiness
E			PLAR	

Phase	Name	Primary Activity	Exit Review	Key Outputs
	Operations & Sustainment	Operate the mission, produce data products		Science data, operational lessons
F	Closeout	Decommission, passivate, dispose, lessons learned	DR	Disposal confirmation, final report

Key Decision Points (KDPs): These are go/no-go authority decisions between phases, lettered A through F. They are distinct from technical reviews -- KDPs are management decisions informed by technical review outcomes. For projects exceeding \$250M (USD), KDP-C requires a Joint Confidence Level (JCL) analysis per NPR 7120.5F.

4.4 ECSS Lifecycle Phases

ECSS defines a similar but not identical lifecycle. The phase letters conveniently align, but the entry/exit criteria and review content differ.

[Source: ECSS-M-ST-10C Rev.1, Section 5.3]

ECSS Phase	Name	NASA Equivalent	Key Difference
0	Mission Analysis / Needs Identification	Pre-A	ECSS Phase 0 includes formal mission analysis
A	Feasibility	A	ECSS-A focuses on demonstrating feasibility
B	Preliminary Definition (B1/B2)	B	ECSS splits into B1 (system) and B2 (detailed)
C	Detailed Definition	C	Similar scope
D	Qualification & Production	D	ECSS-D emphasises qualification models
E	Utilisation	E	Similar scope
F	Disposal	F	ECSS-F explicitly addresses passivation

4.5 Review Gates -- What Each Checks

Review	Full Name	Key Question	Required Evidence
MCR	Mission Concept Review	Is the mission need justified? Is space the right answer?	Problem statement, stakeholder analysis, alternatives analysis, ConOps draft
SRR	System Requirements Review	Are requirements complete, consistent, traceable, and verifiable?	Requirements baseline, traceability matrix, risk register, preliminary ConOps
SDR	System Design Review	Does the system architecture satisfy requirements?	System architecture, functional decomposition, interface definitions
PDR	Preliminary Design Review	Does the preliminary design meet all requirements?	Design description, budget status (mass/power/cost), risk mitigation plans

Review	Full Name	Key Question	Required Evidence
CDR	Critical Design Review	Is the design complete and ready for fabrication?	Detailed drawings, analysis results, test plans, all budgets closed with positive margins
TRR	Test Readiness Review	Is the system ready for formal testing?	Test procedures, facility readiness, acceptance criteria
FRR	Flight Readiness Review	Is everything ready for launch?	All tests passed, waivers documented, launch procedures verified

Industry Practice: RADARSAT-2 (MDA/CSA, launched 2007) went through all NASA-equivalent review gates over a 6-year development. The CDR alone involved over 400 review items and 50 reviewers across 3 weeks. The mission exceeded its 7-year design life, operating for over 15 years -- a testament to thorough verification at each gate.

Exercise. In SpaceCDF, go to the Gate Review tab and examine the MCR exit criteria. Which criteria are auto-evaluated by the tool? Which require manual review by an engineer?

5. CDF Engineering Positions

5.1 Position Overview

In a CDF study, each engineering position owns a set of parameters and is responsible for their domain's design decisions. SpaceCDF supports 15 positions:

Position	Responsibility	Key Owned Parameters	Primary Interfaces
Systems Engineer	Overall architecture, budgets, margins, conflict resolution	Mass margin, power margin, system-level budgets	All positions
Mission Analyst	Orbit design, coverage analysis, ground station access	Altitude, inclination, RAAN, eclipse fraction, contact time	Comms, Payload, Ground
Payload Lead	Instrument performance, data generation rates	GSD, spectral bands, data rate, FOV, SNR	Mission Analyst, Power, AOCS, Comms
Power Engineer	Solar arrays, batteries, EPS architecture, duty cycling	SA area, battery capacity, bus voltage, DoD	All (power is a universal constraint)
AOCS Engineer	Attitude sensors, actuators, pointing modes	Pointing accuracy, slew rate, wheel momentum, sensor suite	Payload, Structures, Propulsion
Thermal Engineer	Temperature control, radiators, heaters, coatings	Max/min temps, radiator area, heater power, thermal margin	Structures, Power, Payload
Comms Engineer	Link budget, transponder, antenna, spectrum licensing	Link margin, data rate, frequency band, EIRP, G/T	Mission Analyst, Payload, Ground
Propulsion Engineer	Delta-V budget, thruster selection, propellant mass	Isp, propellant mass, total impulse, thrust vector	Mission Analyst, AOCS, Structures

Position	Responsibility	Key Owned Parameters	Primary Interfaces
Structures Engineer	Primary structure, mechanisms, launch loads	Structure mass, natural frequency, margin of safety	All (mass is a universal constraint)
Cost Engineer	WBS, cost estimating relationships, schedule	Total cost, per-subsystem cost, launch cost, risk-adjusted cost	Systems Engineer, all subsystems
Compliance Engineer	Standards, frequency licensing, export control	Standard applicability, ITAR/EAR status, filing status	All
User Representative	End-user needs, data product requirements	Data format, latency, accessibility, coverage	Payload, Comms, Ground
Mission Operations	Ground ops concept, staffing, automation	Contact schedule, automation level, anomaly procedures	Comms, Ground, Systems
Ground Segment	Ground stations, data processing pipeline	Station network, processing latency, archive capacity	Comms, Mission Ops, User Rep
Software Engineer	Flight software, FDIR, TC/TM interfaces	FSW architecture, command dictionary, FDIR rules	OBC, AOCS, Comms, Payload

5.2 Interface Conflicts

The most productive CDF sessions are those where **interface conflicts** are identified and resolved. Common conflict patterns:

Conflict	Positions Involved	Resolution Approach
Mass budget exceeded	Systems, Structures, all subsystems	Re-allocate margins, descope payload, select lighter components
Power budget exceeded	Systems, Power, Payload, Comms	Adjust duty cycles, increase SA area, reduce payload power
Pointing requirement vs actuator capability	Payload, AOCS, Structures	Relax pointing requirement, add fine steering mirror, improve isolation
Downlink capacity vs data generation	Comms, Payload, Ground	Add ground stations, increase TX power, reduce data rate, add compression
Volume constraint vs thermal rejection	Structures, Thermal, Payload	Redesign radiator location, add deployable radiator, reduce heat dissipation

Discussion point. Which positions would interact most frequently in your mission? Where do you expect the most difficult trade-offs?

6. Session Summary

Topic	Key Takeaway
CDF	Concurrent multi-discipline design resolves conflicts in real-time; SpaceCDF implements this as a web-based tool

Topic	Key Takeaway
System-V	Left side decomposes (need to components); right side integrates (build to validate); horizontal traceability links them
17 Processes	Recursive at every system level; grouped as Design (1--4), Realisation (5--9), Management (10--17)
Lifecycle Phases	Pre-A through F with KDP gates; ECSS phases approximately align; SpaceCDF covers Pre-A through C
Review Gates	Each gate answers a specific question with required evidence; auto-evaluated where possible in SpaceCDF
Positions	Each owns a parameter domain; conflicts arise at interfaces; the systems engineer arbitrates

1U Worked Example: UniSat-1

Throughout this course, we use a second running example alongside the 3U EO CubeSat: **UniSat-1**, a 1U CubeSat technology demonstrator designed by a university team. This is the simplest realistic spacecraft design.

Mission: Demonstrate a novel MEMS-based magnetometer for space weather monitoring from LEO.

Why 1U? The 1U form factor (100 x 100 x 113.5 mm, up to 1.33 kg) is the smallest standard CubeSat and the entry point for many university and educational missions. It forces extreme design discipline -- every gram, every milliwatt, and every cubic centimetre matters.

Parameter	Value
Form factor	1U (100 x 100 x 113.5 mm)
Mass limit	1.33 kg (CDS Rev 14)
Target mass	1.0 kg
Orbit	400 km circular, 51.6 deg (ISS rideshare)
Design lifetime	6 months
Payload	MEMS magnetometer (50 g, 0.2 W, < 1 kbps)
Comms	UHF 437 MHz, 9600 bps
Power	~2 W orbit average (body-mounted solar cells)
AOCS	Passive magnetic (permanent magnet + hysteresis rods)
Propulsion	None (natural deorbit in ~1 year)
Estimated cost	50--200 kEUR
Development time	6--12 months

UniSat-1 illustrates that a meaningful space mission can be accomplished with just five subsystems (EPS, OBC, Comms, Structure, Payload), no active attitude control, no propulsion, and no thermal hardware beyond surface coatings. As the course progresses, each session will show how the same design processes apply to UniSat-1, but with radically simpler solutions at every step.

Discussion point. *How does the CDF process differ when the team has only 5 subsystems instead of 8--9? Which engineering positions are still needed, and which can be combined?*

7. Tool Exercise

1. Open SpaceCDF and navigate through the workflow steps (Step 1 Mission Need · Step 2 Concept · Step 3 Requirements · Step 4 Design)
2. On the Design Dashboard, identify which KPI cards correspond to which engineering positions
3. Go to the Positions tab and review the key questions for your assigned position
4. Go to the Gate Review tab and examine the MCR exit criteria

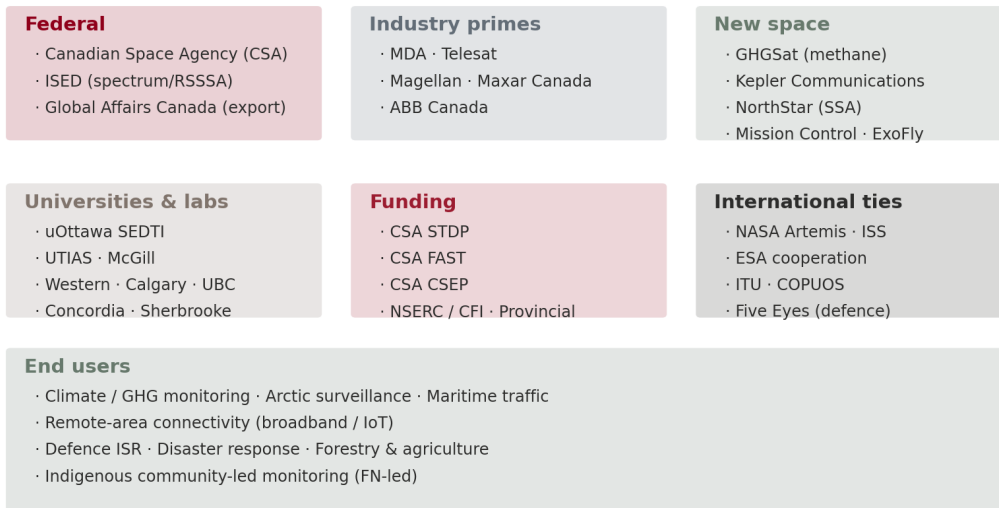
Complete Worksheet 1.1: Map the 17 processes to SpaceCDF features and identify lifecycle phases for given activities.

References

1. NASA, Systems Engineering Handbook (SP-2016-6105 Rev 2), 2016
2. NASA, NPR 7123.1D -- Systems Engineering Processes and Requirements, 2020
3. NASA, NPR 7120.5F -- NASA Space Flight Program and Project Management Requirements, 2021
4. ECSS, ECSS-M-ST-10C Rev.1 -- Space Project Management, 2009
5. INCOSE, Systems Engineering Handbook, 4th Edition, 2015
6. Bandecchi, M. et al., "The ESA/ESTEC Concurrent Design Facility", ESA Bulletin No. 107, 2001
7. Wertz, J.R. et al., Space Mission Engineering: The New SMAD (SMAD4), Microcosm Press, 2011

Session 1.2: The Canadian Space Ecosystem & Regulatory Environment

The Canadian Space Sector — illustrative map of actors



SpaceCDF · uOttawa SEDTI

Representative stakeholder map for Canadian missions.

Expected reading before this session. NASA SEH §4.1 (~ 30 min); SMAD4 Ch. 1 (mission objectives).

Prerequisites: Session 1.1 **References:** - Government of Canada, Canadian Space Agency Act (S.C. 1990, c. 13) - Government of Canada, Remote Sensing Space Systems Act (RSSSA, S.C. 2005, c. 45) - Government of Canada, Radiocommunication Act (R.S.C., 1985, c. R-2) - ISED Canada, Spectrum Management and Telecommunications - Government of Canada, Export and Import Permits Act (EIPA) - NASA, Systems Engineering Handbook (SP-2016-6105 Rev 2), 2016 - ITU, Radio Regulations, Edition 2020

Learning Objectives

By the end of this session, participants will be able to: 1. Describe the Canadian Space Agency's mandate and organisational structure 2. Map the Canadian space industry ecosystem -- prime contractors, SMEs, universities, and funding programs 3. Identify the regulatory bodies that govern space activities in Canada (CSA, ISED, GAC, DND) 4. Explain the RSSSA licensing process for remote sensing satellites 5. Describe the spectrum licensing process through ISED and ITU coordination 6. Identify Canadian export control obligations under EIPA, ITAR, and EAR

Part 1: The Canadian Space Ecosystem

1. The Canadian Space Agency

1.1 Mandate and History

The Canadian Space Agency (CSA) was established by the *Canadian Space Agency Act* (S.C. 1990, c. 13) with the mandate to "promote the peaceful use and development of space, to advance the knowledge of space through science and to ensure that space science and technology provide social and economic benefits for Canadians."

Milestone	Year	Significance
Alouette 1	1962	Canada becomes the 3rd nation to build and design a satellite
Hermes (CTS)	1976	World's first high-power direct broadcast satellite (CSA/NASA)
Canadarm	1981	Robotic arm for Space Shuttle; established Canada's space robotics dominance
RADARSAT-1	1995	First Canadian Earth observation satellite; C-band SAR
CSA established	1990	Canadian Space Agency Act creates the agency
Canadarm2	2001	ISS robotic arm; continues robotics legacy
RADARSAT-2	2007	Commercial SAR satellite (MDA); RSSSA applies
RADARSAT Constellation Mission (RCM)	2019	Three C-band SAR satellites; maritime surveillance, disaster management
Canadarm3	In dev.	For the Lunar Gateway; next-generation autonomous robotics
Lunar Gateway participation	2019+	Canada committed as partner; astronaut flights planned

[Source: CSA, "Canada's Space Milestones", www.asc-csa.gc.ca]

1.2 CSA Organisational Focus Areas

The CSA is headquartered in Saint-Hubert, Quebec, and organises its activities around three strategic priorities:

Priority	Activities	Key Programs
Space Exploration	ISS operations, Lunar Gateway, Canadarm3, astronaut program	Lunar Exploration Accelerator Program (LEAP)
Space Utilisation	Earth observation, satellite communications, GNSS augmentation	RADARSAT, SmartEarth, eHealth
Space Science & Technology	Astronomy, planetary science, technology development	CASTOR, JWST/NIRISS, space health

1.3 Canadian Space Strategy (2019)

The *Exploration, Imagination, Innovation: A New Space Strategy for Canada* (2019) committed \$2.05B over 24 years and established five priorities: 1. Ensure astronaut health and well-being in deep space 2. Develop and use advanced AI and robotics in space 3. Connect rural and remote Canadians through next-generation satellite communications 4. Use space data and technology to manage natural resources sustainably 5. Position the Canadian space sector for the future

Industry Practice: The \$150M Lunar Exploration Accelerator Program (LEAP) specifically targets Canadian SMEs and universities to develop technologies for lunar surface science, health, AI, and robotics. This "pull-through" model -- where a strategic government investment creates opportunities for industry -- is a hallmark of the Canadian space ecosystem.

2. Canadian Space Industry Map

2.1 Industry Structure

The Canadian space industry generated approximately \$2.7B in revenue (2022) and employs ~12,000 people. It is structured in tiers:

Tier	Role	Key Players	Revenue Share
Tier 1: Primes	System integration, large platforms, constellation operations	MDA Space, Telesat, SFL (U of T)	~60%
Tier 2: Subsystem/Component	Subsystem design, specialised components, ground systems	Honeywell Canada, ABB, Magellan Aerospace, NGC Aerospace	~25%
Tier 3: SMEs & Startups	Niche technologies, applications, data analytics	GHGSat, NorthStar, Kepler Communications, Wyvern	~10%
Tier 4: Academia	R&D, training, technology incubation	U of T (SFL), York U, UBC, U of Calgary, Western	~5%

2.2 Key Canadian Space Capabilities

Canada has world-leading capabilities in several domains:

Capability	Key Player(s)	Notable Missions	Global Standing
Space robotics	MDA Space	Canadarm, Canadarm2, Dextre, Canadarm3	World leader
SAR Earth observation	MDA Space, CSA	RADARSAT-1/2/RCM	Top 3 globally
Satellite communications	Telesat	Anik fleet, Telesat Lightspeed (LEO constellation)	Pioneer in GEO DBS
Small satellite systems	SFL (U of T)	MOST, NEOSat, GHGSat-D, BRITE	Top university lab globally
GHG monitoring	GHGSat	GHGSat constellation (methane)	World's only commercial GHG monitoring constellation
Optical instruments	ABB, Honeywell	JWST/FGS-NIRISS, OSIRIS-REx laser altimeter	Tier 1 instrument builder
Star trackers/AOCS	NGC Aerospace/Honeywell	Numerous missions	Widely used sensors

2.3 Funding Programs

Canadian space activities are funded through several mechanisms:

Program	Administered By	Scope	Typical Value
Space Technology Development Program (STDP)	CSA	TRL advancement for flight heritage	\$100K--\$3M per project
Class Grants and Contributions	CSA	Space science, awareness, STEM	Variable
LEAP	CSA	Lunar technology development	Up to \$3M per project
IRAP	NRC	SME innovation support	\$50K--\$1M
SIF	ISED	Strategic innovation projects	\$10M--\$100M+
DRDC contracts	DND	Dual-use technology	Classified--\$10M
NSERC / CFI	Federal	Academic research infrastructure	\$50K--\$5M

Industry Practice: GHGSat (Montreal) progressed from a CSA-funded technology demonstrator (GHGSat-D "Claire", launched 2016) to a commercial constellation of 12+ satellites monitoring methane emissions from individual industrial facilities. This trajectory -- government seed funding enabling commercial scale-up -- is the canonical Canadian space industry growth model.

Discussion point. For your mission concept, which industry tier would you engage? What CSA funding programs might apply? What capabilities would need to be sourced internationally?

3. Comparison with Other National Models

Understanding Canada's space ecosystem requires comparing it with partner agencies:

Parameter	Canada (CSA)	USA (NASA)	Europe (ESA)	Japan (JAXA)
Annual budget (approx.)	~\$300M CAD	~\$25B USD	~\$7.5B EUR	~\$1.5B USD
Staff	~670	~18,000	~2,300	~1,500
Launch capability	None (uses partners)	SLS, commercial (SpaceX, ULA)	Ariane 6, Vega-C	H3, Epsilon
Key strength	Robotics, SAR, instruments	Full spectrum	Science, EO, navigation	Rockets, science, ISS (Kibo)
Industry model	Government seed + commercial scale	Government prime + commercial	Juste retour (geographic return)	Government-led + commercial

Canada's lack of indigenous launch capability means all Canadian satellites launch on foreign vehicles, which has implications for export control (Section 5 below).

4. Canadian Space in the International Context

4.1 International Partnerships

Canada participates in space through multilateral and bilateral agreements:

Partnership	Canadian Contribution	Canadian Benefit
ISS	Canadarm2, Dextre, SPDM	Astronaut access, research time, technology demonstrations
Lunar Gateway	Canadarm3	Astronaut flights, lunar science access
Copernicus (ESA)	Data access agreement	Free access to Sentinel data for Canadian users
COSPAS-SARSAT	Canadian ground stations, MEOSAR payloads	Search and rescue capability for Canadian territory
Five Eyes	Intelligence sharing, dual-use technology	Access to allied space intelligence capabilities

4.2 The Canadian Niche Strategy

Canada has historically pursued a "niche excellence" strategy: rather than attempting full-spectrum space capability, Canada develops world-class expertise in specific domains (robotics, SAR, instruments) and trades access to these capabilities for partnership in larger programs.

Key Equation: The "partnership leverage ratio" -- informal but useful -- describes how Canada trades capability for access:

$$L = \frac{V_{\text{access}}}{C_{\text{contribution}}}$$

Where V_{access} is the value of program access obtained and $C_{\text{contribution}}$ is the cost of Canada's contribution. For the ISS, Canada's Canadarm2/Dextre contribution (~\$2B CAD over 30 years) provides access to a 150B+ program, yielding $L \approx 75$. This is the highest leverage ratio of any ISS partner.

Part 2: Regulatory Environment

5. Regulatory Framework Overview

Space activities in Canada are governed by multiple federal bodies. There is no single "Space Act" -- instead, regulation is distributed across several statutes:

Regulatory Body	Statute	What It Regulates	Relevance to Mission Design
CSA	Canadian Space Agency Act	Space program coordination, policy	Mission approval, funding
ISED	Radiocommunication Act		

Regulatory Body	Statute	What It Regulates	Relevance to Mission Design
		Radio spectrum allocation, licensing	Frequency licensing for all RF transmissions
ISED	Remote Sensing Space Systems Act (RSSSA)	Operation of remote sensing satellites	Licensing for any satellite that images Earth
GAC	Export and Import Permits Act (EIPA)	Export control of strategic goods	Export permits for satellite technology
GAC	Controlled Goods Program	Access to controlled (ITAR/CG) goods	Personnel security clearances
DND/DRDC	National Defence Act	Dual-use technology, military space	Classification, security requirements
Transport Canada	(No current statute)	Launch operations (future)	Canada has no domestic launch regulation yet

6. Remote Sensing Space Systems Act (RSSSA)

6.1 Purpose and Scope

The RSSSA (S.C. 2005, c. 45) regulates the operation of remote sensing satellites from Canada. It was enacted primarily in response to the commercialisation of RADARSAT-2 (the first Canadian commercial EO satellite) and addresses national security and foreign policy concerns related to Earth imagery.

[Source: Government of Canada, RSSSA, <https://laws-lois.justice.gc.ca/eng/acts/r-5.4/>]

6.2 Who Needs a Licence?

An RSSSA licence is required for any person or entity that: - **Operates** a remote sensing space system from Canada, or - **Controls** the collection or distribution of remote sensing data from a Canadian system

This applies regardless of where the satellite is manufactured or launched. If it is operated from Canadian soil, RSSSA applies.

6.3 Licence Categories and Conditions

Licence Type	Scope	Typical Conditions
System Operating Licence	Operate a remote sensing satellite system	Data handling plan, priority access for Canadian government, shutter control provisions
Interoperability Arrangements	Share data with foreign entities	Approval for each foreign government or commercial partner

Key conditions typically imposed:

Condition	Description	Rationale
Priority access	Government of Canada can request priority imaging of any area	National security
Shutter control	Government can restrict imaging of specified areas	

Condition	Description	Rationale
		National security, diplomatic relations
Data disposition	Approved plan for data storage, distribution, and destruction	Prevent unauthorised access
System disposal	Plan for end-of-life decommissioning	Space debris mitigation
Raw data protection	Restrictions on distribution of unprocessed data	Prevent circumvention of controls

6.4 Implications for CubeSat and University Missions

Even small CubeSat missions with cameras must comply with RSSSA if they produce images of Earth. Key considerations:

Parameter	Threshold	Implication
GSD	No formal threshold in the Act	All resolutions require licensing
Orbit type	Any orbit imaging Earth	LEO, MEO, GEO all in scope
Data distribution	Any distribution of imagery	Must have approved data handling plan
Foreign participation	Any foreign access to data	Requires interoperability arrangement

Industry Practice: MDA's RADARSAT-2 operates under an RSSSA licence that includes shutter control provisions. During the 2010 Haiti earthquake response, the Government of Canada exercised priority access to direct RADARSAT-2 to image the disaster zone within hours. This demonstrates both the operational utility and the regulatory mechanism of RSSSA.

Discussion point. Does your mission concept involve any remote sensing of Earth? If so, what RSSSA conditions might apply?

7. Spectrum Management and Licensing

7.1 Why Spectrum Licensing Matters

Every satellite that transmits or receives radio signals requires spectrum authorisation. This is not optional -- unauthorised RF transmission is illegal in all ITU member states and can result in harmful interference to other systems.

The process involves two levels: 1. **National:** ISED (Innovation, Science and Economic Development Canada) issues the radio licence 2. **International:** ITU (International Telecommunication Union) coordinates the frequency filing to prevent interference with other national administrations

7.2 The ITU Framework

The International Telecommunication Union allocates spectrum globally through the Radio Regulations (RR), updated at World Radiocommunication Conferences (WRC) every 3–4 years.

ITU Concept	Description	Relevance
Frequency allocation	Spectrum divided among radio services (Fixed Satellite, Mobile, Earth Exploration Satellite, etc.)	Determines which bands are available for your mission type
ITU Regions	Region 1 (Europe/Africa), Region 2 (Americas), Region 3 (Asia/Pacific)	Canada is in Region 2; allocations may differ by region
Coordination	Bilateral/multilateral agreements to avoid interference	Required before a new satellite system can operate
Notification	Formal filing with the ITU Radiocommunication Bureau (BR)	Records the satellite system in the Master International Frequency Register (MIFR)

7.3 Common Satellite Frequency Bands

Band	Frequency Range	Typical Use	Key Characteristics
UHF	400--450 MHz	CubeSat TT&C, AIS, ADS-B	Simple antennas, low data rate (1--9.6 kbps), crowded
S-band	2.0--2.3 GHz	TT&C (uplink + downlink)	Standard for housekeeping; moderate antenna gain required
X-band	8.025--8.4 GHz	EO payload data downlink	Higher data rates (10--150 Mbps); requires tracking antenna on ground
Ka-band	26.5--40 GHz	High-throughput data downlink, inter-satellite links	Very high data rates (100+ Mbps); rain attenuation significant
L-band	1.5--1.6 GHz	GNSS, mobile satellite service	Navigation, SAR beacons
C-band	3.7--4.2 / 5.9--6.4 GHz	Legacy GEO comms (DBS), SAR (RADARSAT)	Being repurposed for 5G in some countries

7.4 The Licensing Process (Canada)

The ISED spectrum licensing process for a satellite system follows these steps:

Step	Activity	Duration	Documents
1	Pre-application consultation with ISED	2--4 weeks	Preliminary technical description
2	Formal application submission	--	ISED application form, technical annex
3	ISED technical review	3--6 months	Interference analysis, coordination requirements
4	ITU filing (Advance Publication Information -- API)	--	ITU API filing (submitted by ISED on Canada's behalf)
5	ITU coordination with affected administrations	2--7 years	Coordination agreements
6	ITU notification and recording in MIFR	--	MIFR entry
7	ISED licence issuance	--	Radio licence

Key Equation: The link budget equation governs whether a given frequency allocation provides sufficient performance. The received signal power is:

$$P_r = P_t + G_t + G_r - L_{fs} - L_{atm} - L_{misc}$$

Where all values are in dB: - P_t = transmit power (dBW) - G_t = transmit antenna gain (dBi) - G_r = receive antenna gain (dBi) - $L_{fs} = 20\log_{10}(4\pi d/\lambda)$ = free-space path loss (dB) - L_{atm} = atmospheric loss (dB) - L_{misc} = miscellaneous losses (pointing, polarisation, etc.)

The link margin is then:

$$M = P_r - P_{r, min} = (E_b/N_0)_{received} - (E_b/N_0)_{required}$$

A positive margin of at least 3 dB is typically required; 6 dB for critical links.

7.5 CubeSat-Specific Considerations

Many CubeSat operators use amateur radio frequencies (VHF/UHF) during development but must transition to commercial bands for operational missions, especially if there is any commercial data distribution.

Approach	Licensing Path	Limitations
Amateur radio (VHF/UHF)	IARU coordination; national amateur licence	No commercial use; data must be freely available; limited power/bandwidth
Commercial UHF	ISED commercial licence; ITU coordination	Low data rate; crowded band; longer licensing timeline
S-band TT&C	ISED licence; SFCG coordination	Standard path; moderate timeline (6--12 months)
X-band payload	ISED licence; ITU coordination	Higher data rate; requires tracking ground station

Industry Practice: Kepler Communications (Toronto) initially operated its first pathfinder satellites on Ku-band for store-and-forward IoT data relay. Obtaining Ku-band spectrum rights required coordination with incumbent GEO operators, a process that took over 2 years. The lesson: begin spectrum licensing as early as Phase A.

8. Export Control

8.1 The Export Control Landscape

Satellite technology is controlled goods in most countries due to dual-use (civil/military) applications. Canadian space companies and universities must navigate three overlapping regimes:

Regime	Administered By	Scope	Key List
EIPA (Canada)	Global Affairs Canada (GAC)	Canadian-origin goods and technology	Export Control List (ECL), Groups 1--7
ITAR (USA)	US State Department (DDTC)	US-origin defence articles and services	US Munitions List (USML), Category XV (spacecraft)
EAR (USA)	US Commerce Department (BIS)	US-origin dual-use goods and technology	Commerce Control List (CCL), ECCN 9Axx

8.2 Canadian Export Controls (EIPA)

The Export and Import Permits Act requires an export permit for goods on Canada's Export Control List (ECL). Space-relevant ECL groups:

ECL Group	Description	Examples
Group 1	Dual-use goods (Wassenaar Arrangement)	Radiation-hardened electronics, space-qualified components
Group 2	Munitions (defence goods)	Military satellite systems, encrypted comms
Group 4	Nuclear-related (NSG)	RTGs, nuclear propulsion components
Group 6	Missile technology (MTCR)	Complete rocket systems, guidance systems, propulsion
Group 7	Chemical/biological weapons	Not typically space-relevant

8.3 US Export Controls (ITAR and EAR)

Any mission using US-origin components (which includes nearly all commercial space electronics) must comply with US export controls:

ITAR (International Traffic in Arms Regulations): - USML Category XV covers spacecraft and related articles - Requires a Technical Assistance Agreement (TAA) or Manufacturing License Agreement (MLA) - Applies to US-origin defence articles regardless of where they are located - Violation penalties: up to \$1M per violation and 20 years imprisonment

EAR (Export Administration Regulations): - Controls dual-use items (commercial space components, software, technology) - Uses Export Control Classification Numbers (ECCNs) - Requires a licence for export to most countries (except Canada for many items under bilateral agreements) - The US-Canada Defence Production Sharing Agreement provides some exemptions

8.4 Practical Implications for Mission Design

Design Decision	Export Control Implication
Using US-origin reaction wheels	EAR licence may be required; ITAR if military-grade
Collaborating with a non-Five Eyes university	TAA required for ITAR-controlled technical data
Launching on a non-US rocket	ITAR Technology Transfer Agreement for US components on foreign launcher
Open-sourcing satellite software	Must verify no controlled algorithms (e.g., encryption above EAR thresholds)
Hosting a foreign payload	May require export permit under EIPA and/or ITAR approval
Publishing detailed design in a paper	Technical data review required if ITAR-controlled components discussed

Industry Practice: MDA's export of RADARSAT-2 ground segment technology to international partners required extensive EIPA permitting and ITAR compliance review (since US-origin components were embedded in the system). The process took 18+ months and required dedicated compliance staff.

Even university CubeSat projects using US-origin COTS components must track their ECCN classifications.

Discussion point. *Does your mission concept involve any US-origin components? Any international partners? What export control obligations might apply?*

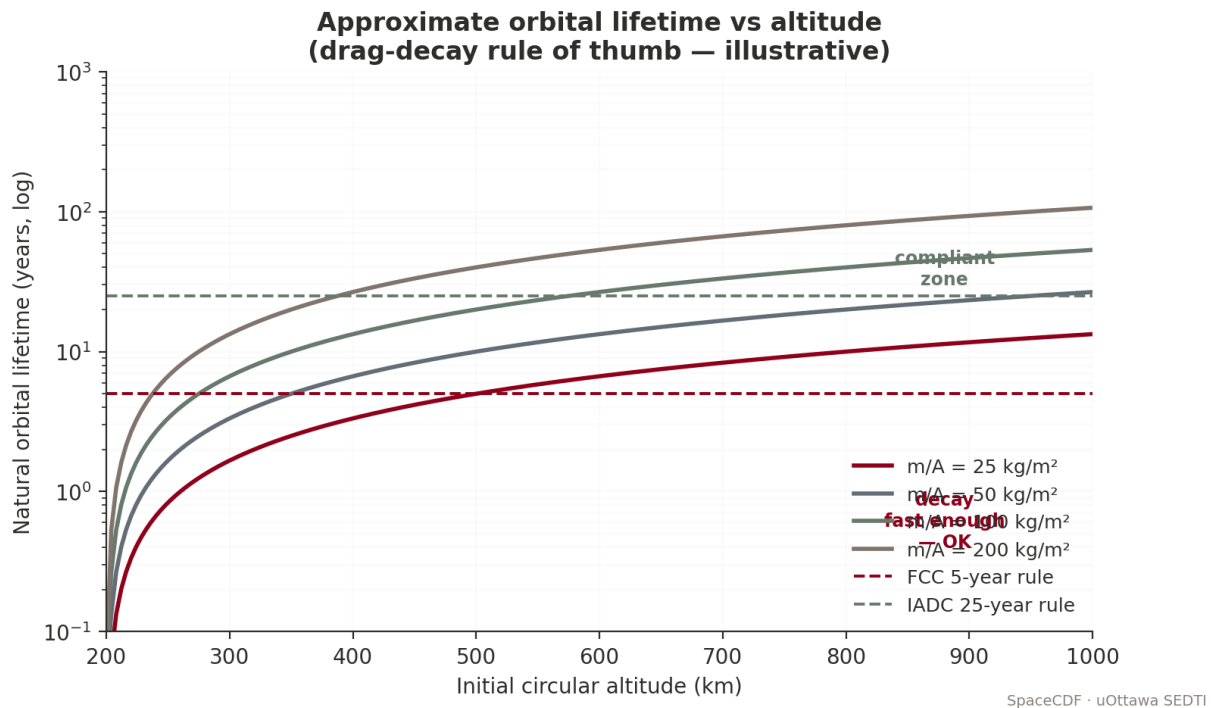
Session Summary

Topic	Key Takeaway
CSA mandate	Promote peaceful use of space; niche excellence strategy (robotics, SAR, instruments)
Industry structure	4-tier ecosystem; government seed funding enables commercial scale-up
RSSSA	Any remote sensing satellite operated from Canada requires a licence; includes shutter control
Spectrum licensing	Two-level process (ISED national + ITU international); start in Phase A; 2+ year timeline
Export control	Three overlapping regimes (EIPA, ITAR, EAR); nearly all missions encounter US controls
Key lesson	Regulatory compliance is not optional -- it shapes design decisions from the earliest phases

References

1. Government of Canada, Canadian Space Agency Act (S.C. 1990, c. 13)
2. Government of Canada, Remote Sensing Space Systems Act (S.C. 2005, c. 45)
3. Government of Canada, Radiocommunication Act (R.S.C., 1985, c. R-2)
4. Government of Canada, Export and Import Permits Act (R.S.C., 1985, c. E-19)
5. ISED, Spectrum Management and Telecommunications
6. ITU, Radio Regulations, Edition 2020
7. CSA, Exploration, Imagination, Innovation: A New Space Strategy for Canada, 2019
8. CSA, State of the Canadian Space Sector Report, 2022
9. US Department of State, ITAR (22 CFR 120-130)
10. US Department of Commerce, EAR (15 CFR 730-774)

Session 1.3: International Standards for Space Systems



Orbital lifetime vs altitude with FCC/IADC limits.

Expected reading before this session. NASA SEH §6.5 — Decision Analysis (≈ 30 min); SMAD4 Ch. 6.

Prerequisites: Sessions 1.1--1.2 **References:** - ECSS, ECSS System -- Description, Implementation and General Requirements (ECSS-S-ST-00C), 2020 - NASA, Systems Engineering Handbook (SP-2016-6105 Rev 2), 2016 - CDS, Interface Definition Document for Launch Vehicle Adapters, Rev 14, 2023 - IADC, Space Debris Mitigation Guidelines (IADC-02-01 Rev 3), 2021 - ISO, ISO 24113:2023 -- Space Debris Mitigation Requirements, 2023 - UNOOSA, COPUOS Guidelines for the Long-term Sustainability of Outer Space Activities, 2019 - ESA, Space Debris Mitigation Compliance Verification Guidelines (ESSB-HB-U-002), 2023

Learning Objectives

By the end of this session, participants will be able to: 1. Describe the ECSS standard framework and its hierarchical structure (Management, Engineering, Product Assurance, Sustainability) 2. Compare ECSS and NASA standards and identify equivalences 3. Explain the CDS Rev 14 interface standard and its role in launch vehicle integration 4. Apply space debris mitigation guidelines (IADC, ISO 24113, FCC 5-year rule) to mission design 5. Describe the role of COPUOS and the Outer Space Treaty framework 6. Use SpaceCDF's compliance tracking features

Part 1: The ECSS Framework

1. Why Standards Matter in Space Engineering

1.1 The Cost of Non-Compliance

Space missions operate in an environment where failures are catastrophic, repair is impossible, and the consequences of interference or debris affect all space users. Standards exist to:

1. **Ensure mission safety and reliability** -- by codifying lessons learned from decades of space operations
2. **Enable interoperability** -- by defining common interfaces, data formats, and protocols
3. **Reduce cost** -- by providing proven design approaches rather than re-inventing solutions
4. **Satisfy regulatory requirements** -- by demonstrating compliance with national and international obligations
5. **Facilitate technology transfer** -- by establishing a common engineering language

Industry Practice: The loss of the Mars Climate Orbiter (1999) is the canonical example of what happens without rigorous interface standards. Lockheed Martin's ground software produced thruster force data in pound-force seconds, while NASA's navigation software expected Newton-seconds. The spacecraft entered the Martian atmosphere at 57 km altitude instead of the planned 226 km and was destroyed. Total loss: \$327.6M. This failure directly led to NASA's strengthening of Process 12 (Interface Management) in NPR 7123.1D. The lesson: standards and interface control are not bureaucratic overhead -- they are mission-critical.

1.2 The Major Standard Frameworks

Three major standard frameworks govern space activities globally:

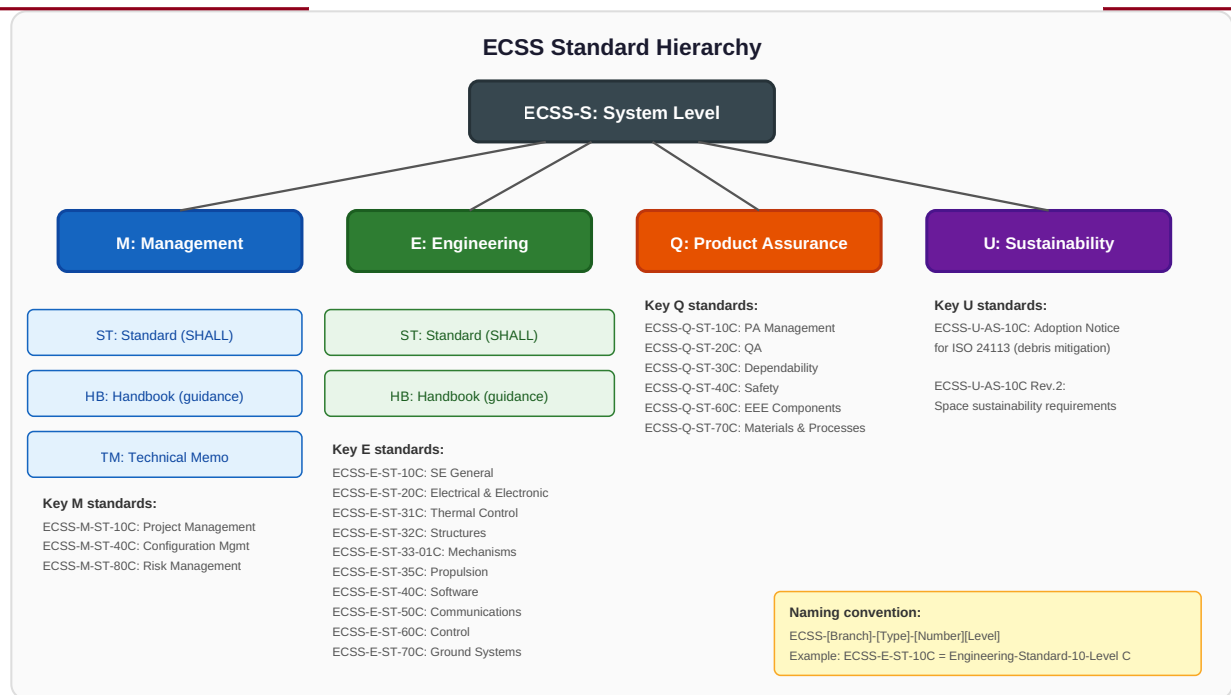
Framework	Scope	Typical Adopter	Document Count
ECSS (European Cooperation for Space Standardization)	Full lifecycle, all disciplines	ESA, European industry, CSA (partial)	144 standards + 56 handbooks
NASA Technical Standards	NASA programs and projects	NASA centres, US contractors	NPR/NPD/NASA-STD series
ISO TC 20/SC 14	Space systems and operations	International (reference standard)	~40 standards

These frameworks are not mutually exclusive. Many missions (including Canadian ones) adopt a tailored combination: ECSS for systems engineering and product assurance, NASA standards for specific technical domains, and ISO for debris mitigation.

2. ECSS Standard Hierarchy

2.1 Structure

The ECSS system organises standards into four branches, each with three levels:



2.2 The Four Branches

Branch	Code	Scope	Example Standard
Management	M	Project management, planning, reviews, configuration management, risk management	ECSS-M-ST-10C Rev.1 (Space Project Management)
Engineering	E	Technical disciplines: systems, structures, thermal, EPS, comms, AOCS, software, ground	ECSS-E-ST-10C (Systems Engineering General Requirements)
Product Assurance	Q	Quality, reliability, safety, EEE components, materials, contamination	ECSS-Q-ST-10C (Product Assurance Management)
Sustainability	U	Space debris mitigation, end-of-life disposal, space environment protection	ECSS-U-AS-10C Rev.2 (adoption of ISO 24113)

2.3 Document Types

Type	Code	Meaning	Obligation
Standard	ST	Contains "shall" requirements -- mandatory when invoked	Must comply or formally request a waiver
Handbook	HB	Contains guidance, best practices, worked examples	Advisory -- not mandatory but strongly recommended
Technical Memorandum	TM	Contains background information, state of the art	Informative only

2.4 Tailoring

A critical concept in ECSS (and all standard frameworks) is **tailoring** -- the process of selecting which standards and which requirements within those standards apply to a given project.

ECSS-S-ST-00C defines three tailoring levels:

Tailoring Level	Description	Typical Application
Full application	All requirements of the invoked standard apply	Large ESA missions (Sentinel, JUICE)
Partial application	Selected clauses apply; others are waived with rationale	Medium missions, CSA projects
Not applicable	Standard is not invoked for this project	CubeSat missions, technology demonstrators

The tailoring rationale is documented in the **Product Assurance and Safety Plan (PASP)** and the **Systems Engineering Management Plan (SEMP)**.

Key Equation: The cost of standards compliance scales non-linearly with mission class. A rough relationship observed in ESA studies:

$$C_{\text{compliance}} \approx k \cdot M_{\text{SC}}^{0.7} \cdot N_{\text{standards}}^{0.3}$$

Where M_{SC} is the spacecraft dry mass (kg), $N_{\text{standards}}$ is the number of invoked standards, and k is a constant dependent on the organisation's maturity. For a 10 kg CubeSat invoking 5 standards, compliance costs are roughly 5--10% of mission cost. For a 1000 kg satellite invoking 40 standards, compliance costs can reach 15--20%.

3. Key ECSS Engineering Standards

3.1 Systems Engineering (ECSS-E-ST-10C)

This is the master engineering standard, equivalent to NASA's NPR 7123.1D. It defines: - System engineering processes and activities - Requirements engineering (writing, verification, traceability) - Functional analysis and decomposition - Interface management - Configuration management (technical aspects)

ECSS-E-ST-10C Concept	NASA Equivalent	Key Difference
System requirements specification	Technical requirements baseline	ECSS requires a formal Requirements Specification Document (RSD)
Functional analysis	Logical decomposition (Process 3)	Similar scope
Verification matrix	Requirements verification matrix	ECSS uses a formal DRD (Document Requirements Definition)
Design justification file	Design solution baseline	ECSS requires a DJF that traces every design decision

3.2 Key Subsystem Standards

Standard	Discipline	Key Requirements	NASA Equivalent
ECSS-E-ST-20C	Electrical & Electronic	EPS design, grounding, EMC, harness	NASA-STD-4003A (EEE), NASA-HDBK-4001

Standard	Discipline	Key Requirements	NASA Equivalent
ECSS-E-ST-31C	Thermal Control	Thermal design, analysis, testing	NASA-STD-5001B (Structural Design)
ECSS-E-ST-32C	Structures	Structural design, factors of safety, testing	NASA-STD-5001B, GEVS (GSFC-STD-7000A)
ECSS-E-ST-33-01C	Mechanisms	Mechanism design, testing, lubrication	NASA-STD-5017
ECSS-E-ST-35C	Propulsion	Propulsion system design, testing	--
ECSS-E-ST-40C	Software	SW development, verification, FDIR	NASA-STD-8739.8
ECSS-E-ST-50C	Communications	Comms system design, link budget, protocols	CCSDS standards
ECSS-E-ST-60C	Control	AOCS design, pointing, navigation	--
ECSS-E-ST-70C	Ground Systems	Ground segment design, operations	CCSDS standards

3.3 Structural Design Requirements (Example)

To illustrate the depth of ECSS standards, consider the structural design requirements from ECSS-E-ST-32C:

Requirement	Value	Rationale
Factor of Safety (FoS) -- Yield	≥ 1.25	Prevent permanent deformation under limit loads
Factor of Safety -- Ultimate	≥ 1.5	Prevent structural failure under ultimate loads
Qualification loads	$1.25 \times \text{limit loads}$	Demonstrate margin beyond expected environment
First natural frequency (axial)	$> 25 \text{ Hz typical (launcher-dependent)}$	Avoid coupling with launcher modes
First natural frequency (lateral)	$> 10 \text{ Hz typical (launcher-dependent)}$	Avoid coupling with launcher modes

Key Equation: The margin of safety (MoS) is defined as:

$$\text{MoS} = \frac{\sigma_{\text{allowable}}}{\text{FoS} \times \sigma_{\text{applied}}} - 1$$

A positive MoS ($\text{MoS} > 0$) indicates the structure meets the requirement. A MoS of 0.0 means the structure exactly meets the requirement with no margin to spare.

Example: If the allowable stress is 280 MPa, the applied stress is 150 MPa, and the FoS is 1.5:

$$\text{MoS} = \frac{280}{1.5 \times 150} - 1 = \frac{280}{225} - 1 = 0.244$$

4. NASA Standards Comparison

4.1 NASA Technical Standards Architecture

NASA's standards are organised differently from ECSS. Key document types:

Type	Code	Example	Obligation
NASA Policy Directive	NPD	NPD 8700.1 (Safety and Mission Assurance)	Mandatory for all NASA programs
NASA Procedural Requirement	NPR	NPR 7123.1D (SE Processes)	Mandatory; defines processes and requirements
NASA Technical Standard	NASA-STD	NASA-STD-5001B (Structural Design)	Mandatory when invoked
NASA Handbook	NASA-HDBK	NASA-HDBK-4001 (EEE Parts)	Advisory guidance
Special Publication	SP	SP-2016-6105 (SEH)	Reference text

4.2 Cross-Reference Table

Domain	ECSS Standard	NASA Standard	ISO Standard
Systems Engineering	ECSS-E-ST-10C	NPR 7123.1D, SP-2016-6105	ISO 15288
Project Management	ECSS-M-ST-10C	NPR 7120.5F	ISO 21500
Configuration Management	ECSS-M-ST-40C	NPR 7120.5F Ch. 4	ISO 10007
Risk Management	ECSS-M-ST-80C	NPR 8000.4B	ISO 31000
Structural Design	ECSS-E-ST-32C	NASA-STD-5001B	--
Debris Mitigation	ECSS-U-AS-10C	NASA-STD-8719.14A	ISO 24113
Cleanliness	ECSS-Q-ST-70-01C	NASA-SN-C-0005	ISO 14644
Software	ECSS-E-ST-40C	NASA-STD-8739.8	ISO 12207

Part 2: Launch Interfaces, Debris Mitigation & International Law

5. CDS Rev 14 -- Launch Vehicle Interface Standard

5.1 What is the CDS?

The **Cubesat Design Specification (CDS)**, maintained by the California Polytechnic State University (Cal Poly) and updated by the CubeSat community, defines the mechanical, electrical, and operational interfaces between CubeSats and their deployment systems (P-PODs, ISIPOD, etc.).

[Source: CDS Rev 14, 2022, available via Cal Poly CubeSat Program]

5.2 Key CDS Requirements

Parameter	CDS Rev 14 Requirement	Rationale
Unit dimensions	100 × 100 × 113.5 mm per U	Standard deployer rail spacing
Mass per U	≤ 2.0 kg (with waiver up to 2.66 kg for some deployers)	Deployer spring mechanism limits
Rail material	Hard-anodised aluminium (7075 or 6061-T6)	Deployer contact surface compatibility
Centre of gravity	Within 2 cm of geometric centre	Deployment dynamics, tumble rate control
Deployables	Must be constrained during launch; no protrusion beyond CubeSat envelope	Protect adjacent payloads in deployer
Separation springs	Prohibited on CubeSat (deployer provides)	Standardised deployment mechanism
RF silence	No RF transmission until 30 minutes after deployment	Avoid interference with launch vehicle
Deployment switches	Minimum 1 per deployable; 2 for redundancy	Prevent premature deployment
Battery charge state	Fully charged (recommended); charging from deployer not available	No power interface with deployer
Propulsion	If present: must be inhibited by 3 independent inhibits; no toxic propellants	Safety of primary payload and deployer

5.3 Form Factors

Form Factor	Dimensions (mm)	Mass Limit	Typical Applications
1U	100 x 100 x 113.5	2.0 kg	Technology demonstrators, IoT nodes
1.5U	100 x 100 x 170.2	3.0 kg	Enhanced technology demonstrators
2U	100 x 100 x 227.0	4.0 kg	Simple instruments, store-and-forward
3U	100 x 100 x 340.5	6.0 kg	Standard EO/science missions
6U	100 x 226.3 x 340.5	12.0 kg	Advanced EO, communications
12U	226.3 x 226.3 x 340.5	24.0 kg	High-performance missions
16U	226.3 x 226.3 x 454.0	32.0 kg	Near-microsatellite capability

5.4 Rideshare Launch Interfaces

For non-CubeSat smallsats (microsatellites 10--200 kg), the interface standard depends on the launch provider:

Launch Provider	Interface Standard	Adapter Type	Key Document
SpaceX (Rideshare)	ESPA-class	15" or 24" ESPA port	SpaceX Rideshare User's Guide

Launch Provider	Interface Standard	Adapter Type	Key Document
Rocket Lab (Electron)	Custom separation system	Rocket Lab-provided	Electron Payload User's Guide
Arianespace (Vega-C)	ASAP-S	Multi-payload adapter	Vega-C User Manual
ISRO (PSLV)	Custom adapter	ISRO-provided	PSLV User Manual

Industry Practice: Planet Labs' SuperDove constellation (150+ satellites) uses a standardised 3U-plus form factor that is CDS-compliant but extends the standard with a custom deployer arrangement for batch deployment. Each SuperDove satellite (mass ~5.8 kg) carries a multispectral imager with 8 bands and ~3m GSD. The standardisation of the bus design enabled a manufacturing rate of 2+ satellites per week -- only possible because the CDS provides a stable interface baseline.

6. Space Debris Mitigation

6.1 The Debris Problem

As of 2025, there are approximately: - **36,500+** tracked objects larger than 10 cm in Earth orbit - **1,000,000+** estimated objects 1--10 cm (untracked, lethal to spacecraft) - **130,000,000+** estimated objects 1 mm -- 1 cm (can damage components)

The debris population is growing due to collisions (the Kessler Syndrome), anti-satellite tests (e.g., Chinese ASAT 2007: 3,500+ tracked fragments), and the rapid growth of mega-constellations.

[Source: ESA Space Debris Office, Annual Report 2024; NASA ODPO Orbital Debris Quarterly News]

6.2 Debris Mitigation Guidelines Hierarchy

Level	Document	Scope	Status
International voluntary	IADC Space Debris Mitigation Guidelines (Rev 3, 2021)	Global best practice	Advisory; adopted by COPUOS
International standard	ISO 24113:2023	Requirements for debris mitigation	Standard; invoked by many agencies
European standard	ECSS-U-AS-10C Rev.2	ECSS adoption notice for ISO 24113	Mandatory for ESA missions
US regulation	FCC 47 CFR 25.114(d)(14)	Post-mission disposal for US-licensed satellites	Legally binding for FCC licensees
NASA standard	NASA-STD-8719.14A	NASA process for limiting orbital debris	Mandatory for NASA missions
ESA requirement	ESA/ADMIN/IPOL(2023)2	Clean Space requirements	Mandatory for ESA missions (2023+)

6.3 Key Debris Mitigation Requirements

Requirement	Source	Value	Notes
Post-mission disposal	IADC, ISO 24113	≤ 25 years	

Requirement	Source	Value	Notes
			Voluntary guideline; being tightened
Post-mission disposal	FCC (2024+)	≤ 5 years	Legally binding for FCC-licensed sats
Probability of successful disposal	ISO 24113	≥ 0.9	Must demonstrate reliability of deorbit mechanism
Casualty risk on re-entry	NASA-STD-8719.14A	$\leq 1 : 10,000$ per event	Drives material selection and design-for-demise
Collision avoidance probability	IADC	$< 10^{-4}$ per year (cumulative)	Drives orbit selection and manoeuvre capability
Passivation	ISO 24113, ECSS-U-AS-10C	All stored energy sources depleted at EOL	Batteries, pressure vessels, wheels, RF

6.4 Post-Mission Disposal Options

Method	Applicable Orbit	Mechanism	Time to Re-entry
Atmospheric drag (natural)	LEO < 600 km	Natural orbital decay	Months to years
Atmospheric drag (augmented)	LEO < 700 km	Drag sail, drag tether	Weeks to years
Propulsive deorbit	LEO	Thrusters lower perigee to ~200 km	Days
Graveyard orbit	GEO	Raise orbit ~300 km above GEO	Indefinite (not re-entry)
Heliocentric disposal	Beyond GEO	Escape Earth orbit	N/A

Key Equation: The orbital lifetime of a satellite in LEO due to atmospheric drag is approximately:

$$\tau \approx \frac{C_D \cdot A \cdot \rho \cdot a^2}{2m} \cdot (\text{complex integral})$$

A simpler rule-of-thumb for circular orbits:

$$\tau_{\text{years}} \approx \frac{h - 200}{30} \cdot \frac{m/A}{50}$$

Where h is altitude in km, m is mass in kg, and A is the cross-sectional area in m^2 . This is very approximate; actual lifetime depends on solar activity (which modulates atmospheric density at high altitudes), drag coefficient, and orbit eccentricity. Use NASA's DAS (Debris Assessment Software) or ESA's DRAMA tool for accurate predictions.

More precisely, the ballistic coefficient is:

$$B = \frac{m}{C_D \cdot A}$$

Lower B (lighter, larger area) means faster re-entry. A 3U CubeSat ($m \approx 4 \text{ kg}$, $A \approx 0.03 \text{ m}^2$, $C_D \approx 2.2$) at 400 km has $B \approx 61 \text{ kg/m}^2$ and will re-enter within ~1--3 years depending on solar cycle.

6.5 Practical Implications for Mission Design

Design Decision	Debris Mitigation Impact
Orbit altitude selection	Altitudes > 600 km require active deorbit; > 700 km strongly discouraged for non-maneuvrable spacecraft
Propulsion system	Required for altitudes > 500--600 km to meet 25-year (or 5-year FCC) rule
Passivation design	Must design battery disconnect, pressure relief, wheel spin-down circuits
Material selection	Aluminium structures preferred for design-for-demise; titanium and carbon fibre survive re-entry
Collision avoidance	Manoeuvre capability or conjunction assessment service (e.g., 18th SDS, ESA SSA) needed

Industry Practice: OneWeb (648 satellites at 1200 km) carries propulsion on every satellite specifically for end-of-life deorbit, since natural decay from 1200 km would take centuries. Each satellite carries sufficient propellant for multiple collision avoidance manoeuvres plus a complete deorbit burn to lower perigee below 300 km. The deorbit operation takes approximately 3 months per satellite.

7. International Space Law and COPUOS

7.1 The UN Space Treaties

The legal framework for space activities is established by five UN treaties negotiated under the Committee on the Peaceful Uses of Outer Space (COPUOS):

Treaty	Year	Key Provisions	Ratification
Outer Space Treaty (OST)	1967	Space is free for exploration; no national appropriation; states responsible for national activities; liability for damage	114 parties (incl. Canada)
Rescue Agreement	1968	Return astronauts and space objects	99 parties
Liability Convention	1972	Launching state liable for damage on Earth (absolute) and in space (fault-based)	98 parties
Registration Convention	1976	States must register space objects with UN	72 parties
Moon Agreement	1979	Moon and celestial bodies are "common heritage of mankind"	18 parties (NOT US, Russia, China)

7.2 Key Legal Principles for Mission Design

Principle	Source	Implication
State responsibility	OST Art. VI	The Government of Canada is internationally responsible for all Canadian space activities, including private/university missions
Authorisation and supervision	OST Art. VI	Canada must authorise and continuously supervise all non-governmental space activities (this is why RSSSA exists)
Liability	Liability Convention	Canada (as launching state) is liable for damage caused by Canadian satellites; absolute liability on Earth, fault-based in space
Registration	Registration Convention	Canada must register all space objects with the UN; CSA maintains the Canadian registry
Non-contamination	OST Art. IX	Must avoid harmful contamination of space and celestial bodies (planetary protection)
Due regard	OST Art. IX	Must conduct activities with "due regard" for other states' interests (debris mitigation)

7.3 COPUOS Long-Term Sustainability Guidelines (2019)

In 2019, COPUOS adopted 21 guidelines for the long-term sustainability of outer space activities. These are voluntary but politically significant:

Guideline Category	Count	Key Points
Policy and regulatory	7	Adopt national regulatory frameworks, register space objects, share SSA data
Safety of operations	4	Conjunction assessment, collision avoidance, re-entry risk assessment
International cooperation	4	Share debris mitigation best practices, coordinate spectrum use
Scientific and technical	6	Improve debris models, develop removal technology, research space weather effects

[Source: A/74/20, Report of COPUOS, 2019, Annex II]

8. SpaceCDF Compliance Features

SpaceCDF tracks compliance through the **Compliance Engineer** position:

Feature	What It Tracks	Automation Level
Standard applicability matrix	Which ECSS/NASA/ISO standards apply to this mission	Semi-automated (suggests based on mission type)
Debris mitigation compliance	Post-mission lifetime, passivation plan, casualty risk	Automated (calculates from orbit and mass)
RSSSA checklist	Licence requirements for Canadian remote sensing missions	Manual (checklist with guidance)

Feature	What It Tracks	Automation Level
Spectrum filing status	ISED/ITU filing progress	Manual (status tracking)
Export control classification	ECCN/USML classification of key components	Manual (per-component)
CDS compliance	Mechanical/electrical interface compliance with CDS Rev 14	Semi-automated (checks mass, dimensions, CG)

Exercise. In SpaceCDF, navigate to the Compliance panel. Review the debris mitigation compliance status for your mission. What orbit altitude would be needed to comply with the FCC 5-year rule without propulsion? Use the tool's orbital lifetime calculator.

Session Summary

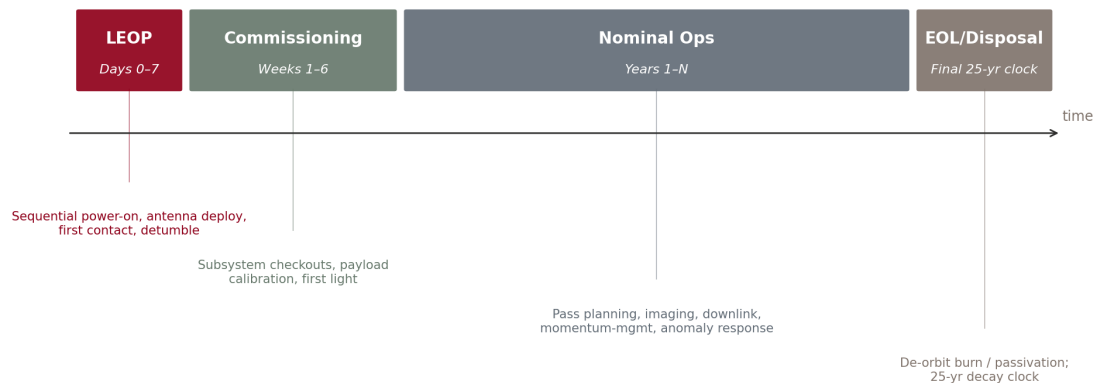
Topic	Key Takeaway
ECSS framework	Four branches (M, E, Q, U); three document types (ST, HB, TM); tailoring is essential
NASA standards	Organised as NPD/NPR/NASA-STD/NASA-HDBK; broadly equivalent to ECSS
CDS Rev 14	Defines CubeSat mechanical/electrical/operational interfaces; compliance is required for rideshare launch
Debris mitigation	IADC 25-year rule; FCC 5-year rule (2024+); passivation required; drives orbit and propulsion design
International law	OST establishes state responsibility; Canada liable for all Canadian space activities
COPUOS	21 sustainability guidelines (2019); voluntary but increasingly influential
Compliance in SpaceCDF	Tracked through the Compliance Engineer position; automated where possible

References

1. ECSS, ECSS-S-ST-00C -- Space Standardization, 2020
2. ECSS, ECSS-E-ST-10C -- System Engineering General Requirements, 2009
3. ECSS, ECSS-E-ST-32C Rev.1 -- Structural General Requirements, 2008
4. ECSS, ECSS-U-AS-10C Rev.2 -- Adoption Notice of ISO 24113, 2023
5. ISO, ISO 24113:2023 -- Space Systems: Space Debris Mitigation Requirements, 2023
6. IADC, IADC-02-01 Rev 3 -- Space Debris Mitigation Guidelines, 2021
7. Cal Poly, CubeSat Design Specification (CDS) Rev 14, 2022
8. NASA, Systems Engineering Handbook (SP-2016-6105 Rev 2), 2016
9. NASA, NASA-STD-8719.14A -- Process for Limiting Orbital Debris, 2019
10. UNOOSA, A/74/20 -- Report of COPUOS, 2019
11. UNOOSA, Outer Space Treaty, 1967
12. FCC, 47 CFR 25.114 -- Satellite Applications

Session 1.4: Mission Needs, Stakeholder Analysis & Trade Studies

Mission Operations Timeline (representative LEO mission)



SpaceCDF · uOttawa SEDTI

Mission operations timeline (LEOP through disposal).

Expected reading before this session. NASA SEH Appendix S — ConOps (≈ 45 min); SMAD4 Ch. 14.

Prerequisites: Sessions 1.1--1.3 **References:** - NASA, Systems Engineering Handbook (SP-2016-6105 Rev 2), 2016 -- Sections 4.1, 4.4, 6.8 - NASA, NPR 7123.1D -- Sections 3.2.1 (Process 1), 3.2.4 (Process 4), 3.5.8 (Process 17) - ECSS, ECSS-E-ST-10C -- Section 5.1 (Requirements Engineering) - ECSS, ECSS-M-ST-10C Rev.1 -- Section 5.3 (Review Process) - Wertz, J.R. et al., Space Mission Engineering: The New SMAD (SMAD4), Microcosm Press, 2011 -- Chapters 1--3 - NASA SEH Appendix C -- How to Write a Good Requirement - NASA SEH Appendix S -- ConOps Outline

Learning Objectives

By the end of this session, participants will be able to: 1. Write a clear problem statement that defines the mission need without prescribing a solution 2. Identify and categorise mission stakeholders using a structured analysis matrix 3. Define mission objectives with measurable success criteria and trace them through the MoE/MoP/TPM hierarchy 4. Structure and execute a trade study using weighted scoring methods (Decision Analysis -- Process 17) 5. Evaluate space vs non-space alternatives objectively and document the decision rationale 6. Design a Concept of Operations (ConOps) including mission phases, operational modes, and data flow 7. Use SpaceCDF Steps 1--2 and the Trade Studies tab to capture and evaluate mission concepts

Part 1: Mission Need & Stakeholder Analysis

1. The Problem Statement

1.1 Why Start with the Problem?

NASA SEH Section 4.1.1 states: *"Clearly define the problem before solving it."*

The single most common failure mode in mission design is not technical -- it is **solving the wrong problem**. Teams jump to solutions ("we need a 6U CubeSat at 500 km") before articulating the actual need ("agricultural monitoring in sub-Saharan Africa requires 10 m resolution multispectral imagery every 5 days"). This premature commitment to a solution:

- Closes off potentially superior alternatives (commercial data purchase, aerial surveys, ground sensors)
- Introduces unnecessary constraints that drive up cost and risk
- Makes it impossible to evaluate mission success (success against what?)
- Violates the fundamental SE principle of **WHAT before HOW**

[Source: NASA SEH Section 4.1; NASA SEH Appendix C -- "How to Write a Good Requirement"]

1.2 Structure of a Good Problem Statement

A rigorous problem statement answers four questions:

Question	Content	Example
What is the problem?	The capability gap -- what is missing or inadequate	"Lack of timely, affordable multispectral imagery at sufficient resolution and revisit"
Who is affected?	Stakeholders and end users who suffer from the gap	"Agricultural monitoring agencies in sub-Saharan Africa"
What is the impact?	Consequence of not solving it -- quantified if possible	"Delayed or inaccurate crop assessments affecting food aid allocation for 300M people"
What constraints exist?	Budget, schedule, political, regulatory, or technical limitations	"Budget: <\$10M total mission cost; timeline: 3 years to first data"

1.3 Good vs Bad Problem Statements

Example -- GOOD:

"Agricultural monitoring agencies in sub-Saharan Africa lack timely, affordable access to multispectral imagery at sufficient resolution (≤ 10 m) and revisit rate (≤ 5 days) to support crop yield prediction and food security planning. Current Sentinel-2 data provides 10 m resolution but only 5-day revisit at the equator, and cloud cover reduces usable observations to approximately 60%. The consequence is delayed or inaccurate crop assessments affecting food aid allocation for 300 million people. Budget constraint: total mission cost $< \$10$ M; first data delivery within 3 years."

Example -- BAD:

"We need a 6U CubeSat with a multispectral imager at 500 km SSO."

The bad example prescribes a solution (6U CubeSat, SSO), specifies a design parameter (500 km), and says nothing about the actual problem. It is a **solution statement**, not a problem statement.

1.4 The WHAT vs HOW Principle

At the mission need level, everything should describe **WHAT** is needed, not **HOW** to achieve it:

WHAT (correct at this stage)	HOW (premature at this stage)
"10 m resolution imagery"	"Use a 15 cm aperture telescope"
"Daily revisit at equator"	"Deploy a Walker delta constellation of 12 satellites"
"Data within 6 hours of acquisition"	"Use X-band downlink at 150 Mbps to 3 ground stations"
"Total cost under \$10M"	"Use COTS components exclusively"
"5-year operational lifetime"	"Select radiation-hardened components"

The HOW column is not wrong -- these are all reasonable design decisions. But they belong in later phases (Phase A/B), not in the problem statement. Fixing the HOW prematurely eliminates design freedom and may lead to suboptimal solutions.

Discussion point. Why is it harmful to specify HOW at this stage? What design options does it close off?

2. Stakeholder Identification & Analysis

2.1 Who is a Stakeholder?

NASA SEH Section 4.1.2 defines stakeholders as "all parties who have a legitimate interest in the system throughout its lifecycle." This is broader than just the end user -- it includes everyone who affects or is affected by the mission.

2.2 Stakeholder Categories for Space Missions

Category	Examples	Typical Needs	Typical Constraints
End Users	Scientists, farmers, shipping companies, emergency responders	Data quality, resolution, latency, format, accessibility	Data rights, training, infrastructure
Operators	Mission control centre, ground station operators	Operability, automation level, staffing, training	Budget for operations, personnel availability
Sponsors / Funders	Space agency (CSA), university, commercial investor, DND	Return on investment, schedule adherence, risk profile	Budget cap, political timelines, reporting requirements
Regulatory	ISED, ITU, FCC, export control (GAC)	Compliance with spectrum, debris, RSSA, export regulations	Licensing timelines, filing requirements
Launch Provider	SpaceX, Rocket Lab, ISRO, Arianespace	CDS compliance, mass/volume limits, schedule, payment	Interface requirements, launch window, manifesting
Ground Segment	KSAT, SSC, SatNOGS, own stations	Frequency compatibility, data volume, contact time	Station availability, geographic coverage

Category	Examples	Typical Needs	Typical Constraints
Data Consumers	Archives (e.g., EODMS), APIs, partner agencies	Data format (NetCDF, GeoTIFF), metadata standards, timeliness	Format standards, distribution agreements
General Public	Taxpayers (for government-funded missions)	Value for money, transparency, societal benefit	Political expectations, public engagement
Orbital Environment	Other satellite operators, debris community	Debris mitigation, spectrum cleanliness, collision avoidance	IADC guidelines, ISO 24113, FCC 5-year rule

2.3 Stakeholder Analysis Matrix

For each stakeholder, capture five attributes:

Attribute	Description	Scale
Name / Role	Who are they?	Text
Needs	What do they require from the mission?	Text (specific, measurable where possible)
Constraints	What limitations do they impose?	Text (mandatory vs desirable)
Priority	How critical is satisfying this stakeholder?	Primary (must satisfy) / Secondary (should satisfy) / Tertiary (nice to have)
Influence	How much power do they have over the mission?	High / Medium / Low

Industry Practice: For the RADARSAT Constellation Mission (RCM), the stakeholder analysis identified over 20 distinct stakeholder groups, including DND (maritime surveillance), Environment Canada (sea ice monitoring), Agriculture and Agri-Food Canada (crop monitoring), Natural Resources Canada (forestry), Public Safety Canada (disaster response), and international partners. Each stakeholder had different priority imaging modes, coverage requirements, and data latency needs. The systems engineering challenge was to design a 3-satellite constellation that satisfied all these needs within a single system architecture -- a classic example of multi-stakeholder optimisation.

Exercise. Complete Part A of Worksheet 1.4 -- identify at least 4 stakeholders for your team's mission and fill in the analysis matrix.

3. Mission Objectives and the MoE/MoP/TPM Hierarchy

3.1 From Need to Objectives

Objectives translate the problem statement into specific, testable goals. Each objective must have:

Attribute	Description	Example
Text	Clear statement of what the mission will achieve	"Provide 10 m GSD multispectral imagery over the target region"

Attribute	Description	Example
Priority	Primary (mission fails without) or secondary (desirable)	Primary
Measurable criterion	How you know the objective is met -- a number with a unit	"GSD $\leq$ 10 m at nadir, 4+ spectral bands, revisit $\leq$ 5 days"
Type	Category of objective	Observation, communication, navigation, science, technology demonstration

3.2 Writing Good Objectives

Good Objective	Why It Is Good
"Provide 10 m GSD multispectral imagery with 4+ bands for the target region (30S--30N), with $\leq$ 5-day revisit and $\leq$ 24 h data latency"	Specific (10 m, 4 bands, geographic scope), measurable (all criteria have numbers + units), relevant to agriculture, achievable with CubeSat technology
"Achieve 99.5% AIS ship detection probability in the North Atlantic within 30 minutes of ship transmission"	Specific (AIS, North Atlantic), measurable (99.5%, 30 min), relevant to maritime safety

Bad Objective	Why It Is Bad
"Take pictures from space"	Not specific, not measurable, no criterion for success
"Build a 3U CubeSat"	This is a solution, not an objective -- it says nothing about what the mission should accomplish
"Demonstrate new technology"	What technology? What does "demonstrate" mean? How do you know it succeeded?

3.3 The MoE / MoP / TPM Hierarchy

These three measures form a chain from operational need to tracked design parameter:

[Source: NASA SEH Section 4.1.4, Section 4.2.4, Section 6.7.3]

Measure	What It Measures	Who Sets It	Example	When Evaluated
MoE (Measure of Effectiveness)	How well the system satisfies the operational need	Users / stakeholders	"% of crop assessments delivered within 5 days of acquisition"	Phase E (operations)
MoP (Measure of Performance)	Technical performance of the system	Systems engineer	"GSD $\leq$ 10 m at nadir"	Phase C/D (verification)
TPM (Technical Performance Measure)	Design parameter tracked over time	Design team	"Current best estimate of imager mass vs allocation (1.5 kg target)"	All phases (continuous)

The hierarchy flows downward:

Stakeholder Need

- > MoE (operational effectiveness)
- > Mission Objective
 - > MoP (technical performance)
 - > Technical Requirement ("shall" statement)
 - > TPM (tracked design parameter)

Key Equation: Measures of Effectiveness often involve probability and time, connecting to system-level performance:

$$\text{MoE} = P_{\text{detection}} \times P_{\text{data_delivery}} \times f(T_{\text{latency}})$$

For an Earth observation mission:

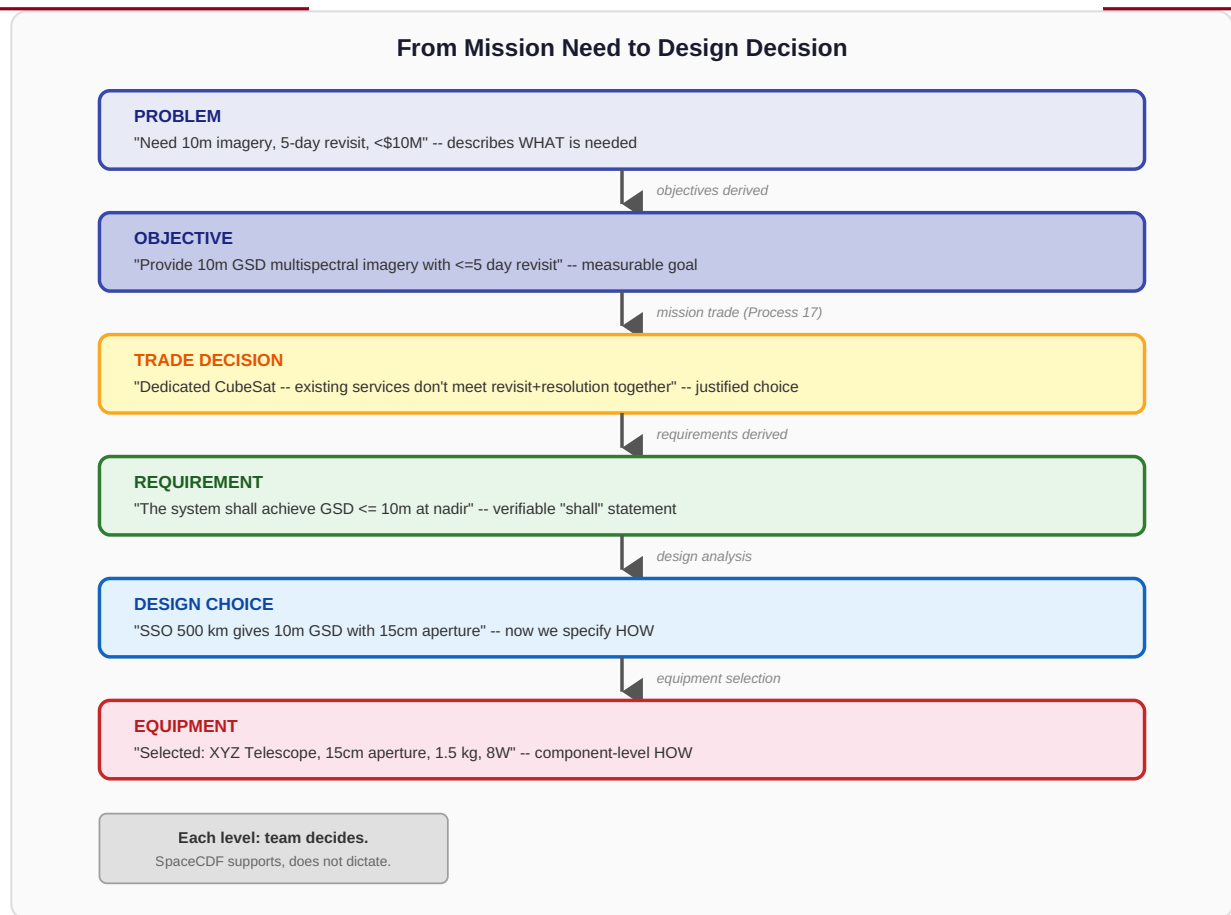
$$\text{MoE}_{\text{coverage}} = \frac{A_{\text{imaged_per_revisit}}}{A_{\text{total_target}}} \times (1 - P_{\text{cloud}})$$

Where $A_{\text{imaged_per_revisit}}$ depends on swath width and orbit ground track spacing, $A_{\text{total_target}}$ is the target area, and P_{cloud} is the cloud cover probability.

Exercise. For one of your objectives, trace the full hierarchy from stakeholder need through MoE, MoP, and TPM. Complete Part B of Worksheet 1.4.

4. From Need to Design: The Flow

Show how the mission need flows through to design decisions without the need itself prescribing the design:



At each level in this hierarchy, the design team makes decisions informed by analysis and trade studies. The tool supports this process with parametric calculations and automated constraint checking, but the team retains decision authority.

Part 2: Trade Study Methodology

5. Decision Analysis Framework

5.1 Process 17: Decision Analysis

NASA SEH Process 17 (Decision Analysis) provides the framework for structured alternative evaluation. Every major design decision should follow this process to ensure decisions are auditable, traceable, and defensible.

[Source: NASA SEH Section 6.8, Process 17: Decision Analysis]

5.2 The Five Elements of a Rigorous Trade Study

Element	Description	Pitfall to Avoid
1. Decision statement	What are we deciding? Framed as a question.	Too broad ("What should we build?") or too narrow ("Which reaction wheel?") for the current phase
2. Alternatives	What are the options? Minimum 3, including "do nothing"	Straw-man alternatives designed to lose; failure to include obvious options

Element	Description	Pitfall to Avoid
3. Criteria	What matters? Performance, cost, schedule, risk, compliance	Missing a critical criterion; criteria that overlap (double-counting)
4. Weightings	How important is each criterion? Normalised to sum to 1.0	All criteria weighted equally (means nothing matters more than anything else); weights set after seeing scores
5. Scoring	How well does each alternative perform?	Inconsistent scoring scale; anchoring bias; group conformity

5.3 Weighting Methods

Method	Procedure	Best When	Limitations
Pairwise comparison	Compare criteria two at a time; count wins. Weight = wins / total comparisons	Small number of criteria (< 8)	Circular preferences possible; does not capture magnitude
Swing weighting	For each criterion, assess the value of moving from worst to best case. Normalise.	Criteria have different units and scales	Requires clear understanding of worst/ best cases
Direct assignment	Team discusses and agrees on weights	Quick preliminary assessment	Subjective; dominated by vocal team members
AHP (Analytic Hierarchy Process)	Pairwise comparison on a 1--9 ratio scale; compute eigenvector	Complex decisions with many stakeholders	Mathematically complex; consistency check needed

Key Equation (Pairwise Comparison): For n criteria, there are $\frac{n(n-1)}{2}$ pairwise comparisons. The raw weight of criterion i is:

$$w_i^{\text{raw}} = \sum_{j \neq i} p_{ij}$$

Where $p_{ij} = 1$ if criterion i is preferred over j , 0.5 for a tie, and 0 if j is preferred. The normalised weight is:

$$w_i = \frac{w_i^{\text{raw}}}{\sum_{k=1}^n w_k^{\text{raw}}}$$

5.4 Scoring Methods

Type	Description	Example	When to Use
Quantitative	Numeric value normalised to 0--1	Mass: 5 kg scores 0.8, 10 kg scores 0.4	When hard numbers exist
Qualitative	Verbal rating mapped to number	"Excellent" = 1.0, "Good" = 0.75, "Fair" = 0.5, "Poor" = 0.25	When only expert judgment is available
Threshold (go/no-go)	Pass/fail gate applied before scoring	"TRL ≥ 6 required" -- below threshold is eliminated	Non-negotiable requirements

5.5 Weighted Score Calculation

For each alternative a across n criteria:

Key Equation:

$$S(a) = \sum_{c=1}^n w_c \cdot s(a, c)$$

Where w_c is the normalised weight of criterion c and $s(a, c)$ is the normalised score (0--1) of alternative a on criterion c .

For "higher is better" criteria:

$$s(a, c) = \frac{v(a, c) - v_{\min}(c)}{v_{\max}(c) - v_{\min}(c)}$$

For "lower is better" criteria:

$$s(a, c) = 1 - \frac{v(a, c) - v_{\min}(c)}{v_{\max}(c) - v_{\min}(c)}$$

5.6 Sensitivity Analysis

A trade study result is only meaningful if it is **robust** -- i.e., the ranking does not change with small perturbations in weights or scores. Sensitivity analysis tests this:

Method	Procedure	What It Reveals
Weight perturbation	Vary each weight by $\pm 20\%$, re-normalise, re-score	Which criteria are "swing" criteria that could flip the outcome
Score perturbation	Vary each score by ± 1 level, re-calculate	Which scores are the most uncertain and impactful
Threshold analysis	For each criterion, find the weight at which the 2nd-place alternative overtakes the 1st	How much the weights would need to change to reverse the decision

Industry Practice: For the Iridium NEXT constellation (66 operational + 6 on-orbit spares), the initial trade study for the constellation architecture compared Walker Delta, Walker Star, and hybrid configurations across 12 criteria including global coverage, revisit time, inter-satellite link geometry, launch cost, and orbital debris risk. Sensitivity analysis revealed that the ranking was robust for all weight perturbations up to $\pm 30\%$, giving high confidence in the selected Walker Star configuration at 780 km altitude.

Exercise. Practice the pairwise comparison method by weighting 4 criteria for a lunch restaurant choice: taste, cost, healthiness, speed. Then score 3 options. This builds the skill on a low-stakes example before applying it to mission design.

6. Space vs Non-Space Alternatives

6.1 The Most Important Trade Study

Before committing to building a satellite, teams must honestly evaluate whether existing solutions meet the need. This is the most important trade study in the mission lifecycle because it determines whether the entire project should proceed.

6.2 Alternative Categories

Category	Examples	Strengths	Weaknesses
Existing free data	Copernicus Sentinel-2, Landsat 8/9	Zero acquisition cost, proven quality, long archive	Fixed resolution/revisit, no tasking control
Commercial data purchase	Planet (SuperDove), Maxar (WorldView), ICEYE (SAR)	High resolution, fast tasking, no development risk	Ongoing cost, data rights limitations, vendor dependency
Aerial (drones/ aircraft)	Survey drones, P-3 Orion, Twin Otter	Very high resolution (< 1 m), flexible scheduling	Limited area coverage, weather dependent, regulatory constraints
Ground sensors	IoT networks, weather stations, tide gauges	Continuous monitoring, low per-unit cost	Point measurements only, no spatial coverage
Dedicated satellite	New CubeSat, SmallSat, or microsatellite	Full control, custom instrument, IP ownership	High cost (\$2--50M), development risk, 2--5 year schedule
Constellation	Multiple dedicated satellites	Global coverage, short revisit (< 1 day)	Much higher cost, operational complexity
Hosted payload	Payload on another operator's bus	Lower cost, shared bus risk	Compromised orbit, limited control, schedule dependency

6.3 Decision Criteria: When is Space Justified?

Space is likely NOT the right answer when: - Existing free data (Sentinel-2, Landsat) meets resolution and revisit needs - Coverage requirement is local (drones or aircraft are cheaper and higher resolution) - Data rate is very low (ground sensors with cellular/satellite IoT backhaul suffice) - Budget does not support minimum viable satellite cost (~\$2M for a basic 3U CubeSat) - Technology readiness is too low for the available schedule

Space is likely justified when: - Global or wide-area coverage is needed simultaneously - Persistent monitoring is required (24/7 or very frequent revisit < 5 days) - User needs data ownership and control over acquisition scheduling - No existing service provides the required measurement type (e.g., specific wavelength, polarisation) - Regulatory or sovereignty requirements demand national control over the sensor

6.4 Constellation Sizing

For missions requiring short revisit, the number of satellites drives cost:

Key Equation: For a Walker Delta constellation at altitude h with half-swath angle η , the number of orbital planes P and satellites per plane S needed for revisit time T_{rev} is approximately:

$$P \times S \geq \frac{2\pi R_E}{v_{\text{ground}} \times T_{\text{rev}} \times \tan(\eta)}$$

Where $v_{\text{ground}} \approx 7.1$ km/s (ground track velocity at 500 km) and η is related to the instrument swath width W by $\eta = W/(2R_E)$ for small angles.

More practically, constellation cost scales sub-linearly due to learning curves:

$$C_{\text{total}} = C_1 \times \sum_{i=1}^N i^{\log_2(L)}$$

Where C_1 is the first unit cost, N is the number of satellites, and L is the learning curve factor (typically 0.90--0.95 for small batches, 0.85 for large batches like Planet's SuperDove).

7. Documenting the Trade Decision

Every trade study must produce an **auditable decision record**. NASA SEH Process 17 requires documentation of:

Element	Content	Purpose
Decision statement	What was decided	Clarity
Alternatives considered	All options evaluated, including rejected ones	Completeness
Criteria and weights	What mattered and how much	Transparency
Scoring rationale	Why each alternative received its score	Auditability
Result and recommendation	The selected alternative and its total score	Decision record
Sensitivity analysis	Robustness of the result	Confidence
Risks of selected option	What could go wrong with the chosen approach	Risk awareness
Responsible person and date	Who decided and when	Accountability

Industry Practice: The Canadian Hydrographic Service evaluated commercial SAR data (ICEYE, Capella) versus a dedicated satellite for Arctic maritime surveillance. The trade study documented 8 alternatives across 11 criteria, with weights set by a stakeholder panel including DND, Transport Canada, and CCG. The decision to use RCM data supplemented by commercial SAR tasking -- rather than building a new satellite -- saved an estimated \$100M while meeting 90% of the maritime domain awareness requirements.

Part 3: Concept of Operations

8. Mission Architecture

8.1 Three Segments of a Space Mission

Every space mission comprises three segments that must be designed together:

Segment	Components	Key Design Drivers
Space Segment	Platform (bus): EPS, AOCS, OBC, thermal, structure, propulsion. Payload: instrument(s). Communications: TT&C + payload data link.	Mass, power, volume, orbit, lifetime
Ground Segment	Ground Operations: commanding, telemetry, orbit determination. Payload Data Centre: data reception, processing (L0-L1-L2-L3), archive.	Contact time, data volume, processing capacity
User Segment	Data products and services. APIs, portals, archives. Training and documentation.	Latency, format, accessibility, user capacity

8.2 Data Interfaces Between Segments

Interface	Direction	Band/Protocol	Content
TM/TC	Space <-> Ground Ops	S-band (typical)	Housekeeping telemetry, telecommands
Payload data	Space -> Data Centre	X-band or Ka-band	Science/imagery data (high volume)
Orbit/TLE	Ground Ops -> Data Centre	Network	Geolocation metadata for data products
Data products	Data Centre -> Users	Internet/API	Processed imagery (L2/L3), analytics

8.3 Interactive Architecture Diagram

SpaceCDF provides a **drag-and-drop architecture diagram editor** in the ConOps tab:

Symbol	Type	Represents
Satellite (blue)	<code>satellite</code>	Space segment (spacecraft + payload)
Ground Station (green)	<code>groundStation</code>	Ground receiving station with antenna
Processing (cyan)	<code>processing</code>	Data processing, MCC, archive
User (amber)	<code>user</code>	End user / data consumer
Sensor (orange)	<code>sensor</code>	Ground sensor, IoT device, in-situ instrument
GNSS/External (purple)	<code>gnss</code>	External system (GNSS, relay sat, other constellation)

Exercise. In the ConOps tab, build your mission architecture diagram. Add all segments, label all connections with data type and frequency band.

9. Mission Phases and Operational Modes

9.1 Operational Mission Phases

Phase	Duration (typical CubeSat)	Activities	Key Risks
LEOP	1--3 days	Deployment, antenna deploy, first contact, initial health check	Deployment failure, tumbling, no contact

Phase	Duration (typical CubeSat)	Activities	Key Risks
Commissioning	2--4 weeks	Subsystem checkout, calibration, first light, orbit determination	Anomalies, calibration issues
Nominal Ops	Months to years	Routine acquisition, downlink, orbit maintenance	Component degradation, anomalies
Extended Ops	Beyond design life	Continued operations with degraded performance	Solar array degradation, propellant depletion
Disposal	Days to months	Passivation, deorbit manoeuvre or natural decay	Failure of deorbit system

9.2 Operational Modes and Power Budgets

Each mode defines which subsystems are active, the pointing configuration, power demand, and data flow:

Mode	Subsystems Active	Pointing	Power (3U typical)	Data Flow
Safe	EPS, OBC, TTC (beacon), AOCS (coarse)	Sun-pointing	~1--2 W	Beacon only
Idle	EPS, OBC, AOCS (standby), TTC (beacon)	Inertial hold	~2 W	Health TM periodic
Science/Imaging	+ Payload, AOCS (fine pointing)	Nadir or target	~6 W	Instrument -> OBC storage
Downlink	+ TTC (full TX power)	Ground station track	~8 W	OBC -> TX -> GS
Eclipse	EPS (battery), OBC, TCS (heaters), AOCS	Inertial hold	~3 W (battery)	None
Orbit Maintenance	+ Propulsion	Thrust direction	~7 W	Manoeuvre TM

9.3 Duty Cycling and Power Analysis

CubeSats have limited power generation (7--25 W for a 3U with deployable panels). Not all modes can run simultaneously. The orbit timeline determines what can happen when:

Typical 95-minute orbit at 500 km SSO: - 60 min sunlight, 35 min eclipse - ~10 min imaging per orbit (~10% duty cycle) - ~8 min downlink per pass (1--2 passes/day over a single ground station) - ~42 min idle - 35 min eclipse (battery-powered)

Key Equations -- Power Budget:

Orbit-average power:

$$P_{\text{avg}} = \sum_i P_{\text{mode},i} \times DC_i$$

Where DC_i is the duty cycle (fraction of orbit) for each mode.

Solar array sizing:

$$P_{SA} = P_{\text{sunlight}} + \frac{P_{\text{eclipse}} \times t_{\text{eclipse}}}{t_{\text{sunlight}} \times \eta_{\text{charge}}}$$

Where $\eta_{\text{charge}} \approx 0.9$ (battery charge efficiency).

Battery sizing:

$$E_{\text{battery}} = \frac{P_{\text{eclipse}} \times t_{\text{eclipse}}}{\text{DoD} \times \eta_{\text{discharge}}}$$

Where DoD is the maximum depth of discharge (typically 0.2--0.3 for long life, up to 0.5 for short missions) and $\eta_{\text{discharge}} \approx 0.95$.

[Source: Wertz et al., SMAD4, Chapter 11; ECSS-E-ST-20C power budget methodology]

10. Data Flow Pipeline

10.1 End-to-End Data Flow

Instrument -> Onboard Storage -> Downlink -> Ground Reception
-> Processing (L0 -> L1 -> L2) -> Archive -> User Delivery

Stage	Key Parameter	Sized By
Data generation	GB/day	Payload data rate $\times$ imaging duty cycle
Onboard storage	GB	Must hold ≥ 1 day of data ($2 \times$ for margin)
Downlink per pass	GB/pass	Link data rate $\times$ contact time per pass
Processing	hours	Algorithm complexity, compute infrastructure
User delivery	hours	Archive API, network bandwidth

10.2 Data Budget Balance

Key Equation: For the system to be sustainable, daily downlink capacity must equal or exceed daily data generation:

$$C_{\text{downlink}} = R_{\text{data}} \times N_{\text{passes}} \times t_{\text{contact}} \times \eta_{\text{protocol}} \geq G_{\text{daily}}$$

Where: - R_{data} = data link rate (Mbps) - N_{passes} = number of ground station passes per day - t_{contact} = average contact time per pass (seconds) - η_{protocol} = protocol overhead factor (~0.8 for CCSDS framing) - G_{daily} = daily data generation (Mb)

If $C_{\text{downlink}} < G_{\text{daily}}$, data accumulates on board and storage eventually fills. Solutions: - Increase R_{data} (higher TX power, higher frequency band, better antenna) - Increase N_{passes} (more ground stations, polar ground station for SSO) - Increase t_{contact} (higher altitude increases contact time but worsens resolution) - Decrease G_{daily} (lower duty cycle, on-board compression, selective downlink)

SpaceCDF exercise: Check the Data Budget on the Dashboard. Does your design balance? If not, which parameter would you change first?

11. ConOps Tool Exercise

1. Navigate to the **ConOps** tab in SpaceCDF
2. Review and edit the **mission architecture diagram** -- add all segments and connections
3. Edit the **mission phases**: adjust durations for your mission type
4. Review the **operational modes**: are the right modes defined? Adjust power values.
5. Check the **data flow pipeline**: does downlink capacity balance data generation?

Complete Worksheet 1.4, Parts E and F: Document your ConOps outline and calculate the orbit-average power budget.

1U Worked Example: UniSat-1

Trade Study: Why 1U instead of 2U or 3U for UniSat-1?

The UniSat-1 team must justify the choice of a 1U form factor for their MEMS magnetometer technology demonstration. This is a classic Process 17 (Decision Analysis) exercise.

Decision statement: "What CubeSat form factor best supports a MEMS magnetometer technology demonstration within the university's budget and schedule constraints?"

Alternatives:

Alternative	Mass Limit	Internal Volume	Typical Cost	Dev Time
1U	1.33 kg	~1000 cm ³	50--200 kEUR	6--12 months
2U	2.66 kg	~2000 cm ³	100--400 kEUR	12--18 months
3U	4.0 kg	~3000 cm ³	200--800 kEUR	18--24 months

Criteria and scoring:

Criterion	Weight	1U	2U	3U	Rationale
Cost	0.35	1.0	0.5	0.2	University budget is 150 kEUR total
Schedule	0.25	1.0	0.6	0.3	Must launch within 12 months
Payload fits	0.20	0.8	1.0	1.0	MEMS sensor is 50 g, 0.2 W -- fits easily in 1U
Design simplicity	0.10	1.0	0.7	0.5	Smaller team, fewer subsystems
Data return	0.10	0.5	0.7	1.0	More volume allows better comms, but 9600 bps is sufficient for < 1 kbps payload
Weighted Total		0.90	0.63	0.41	

Result: 1U wins decisively. The MEMS magnetometer payload (50 g, 0.2 W, < 1 kbps) has no need for the extra volume, mass, or power that 2U/3U would provide. The additional cost and schedule of a larger bus are unjustified.

Sensitivity check: Even if cost weight drops from 0.35 to 0.15 (and schedule from 0.25 to 0.15, redistributing to data return), 1U still wins (0.82 vs 0.68 vs 0.51). The result is robust.

Key lesson: Do not over-design the bus for a simple payload. The 1U form factor imposes healthy constraints that force the team to focus on the mission objective rather than adding unnecessary capability.

Session Summary

Topic	Key Takeaway
Problem statement	Defines WHAT is needed, not HOW to solve it; answers what, who, impact, constraints
Stakeholder analysis	Identify all parties with legitimate interest; capture needs, constraints, priority, influence
Objectives	Must be specific, measurable, and traceable to stakeholder needs via MoE/MoP/TPM chain
Trade study structure	5 elements: decision, alternatives, criteria, weights, scores; must include sensitivity analysis
Space vs non-space	Existing services may already meet the need -- evaluate honestly before committing to build
Decision documentation	Every trade decision must have auditable rationale per Process 17
ConOps	Three segments (space, ground, user); operational modes drive power budget via duty cycling
Data pipeline	Daily downlink capacity must equal or exceed daily data generation

References

1. NASA, Systems Engineering Handbook (SP-2016-6105 Rev 2), 2016
2. NASA, NPR 7123.1D -- Systems Engineering Processes and Requirements, 2020
3. ECSS, ECSS-E-ST-10C -- System Engineering General Requirements, 2009
4. Wertz, J.R. et al., Space Mission Engineering: The New SMAD (SMAD4), Microcosm Press, 2011
5. NASA SEH Appendix C -- How to Write a Good Requirement
6. NASA SEH Appendix S -- ConOps Outline
7. Saaty, T.L., "The Analytic Hierarchy Process", McGraw-Hill, 1980

Session 2.1: The System-V and Requirements Engineering

Expected reading before this session. NASA SEH Appendix C — How to Write a Good Requirement (≈ 30 min); ECSS-E-ST-10C §5.5.

Prerequisites: Week 1 complete (mission need, stakeholder analysis, ConOps defined) **SpaceCDF Tab:** Requirements

References

- NASA, *Systems Engineering Handbook* (NASA/SP-2016-6105 Rev 2), 2016, Ch. 4.2 & Appendix C
- ECSS, *ECSS-E-ST-10-06C: Technical Requirements Specification*, 2009, Sec. 5
- ECSS, *ECSS-E-ST-10C Rev.1: Space Engineering -- System Engineering General Requirements*, 2017, Sec. 5.2
- Wertz, Everett & Puschell, *Space Mission Engineering: The New SMAD*, 2011, Ch. 3--4
- INCOSE, *Systems Engineering Handbook*, 5th ed., 2023, Ch. 2.3.5

Learning Objectives

By the end of this session, participants will be able to:

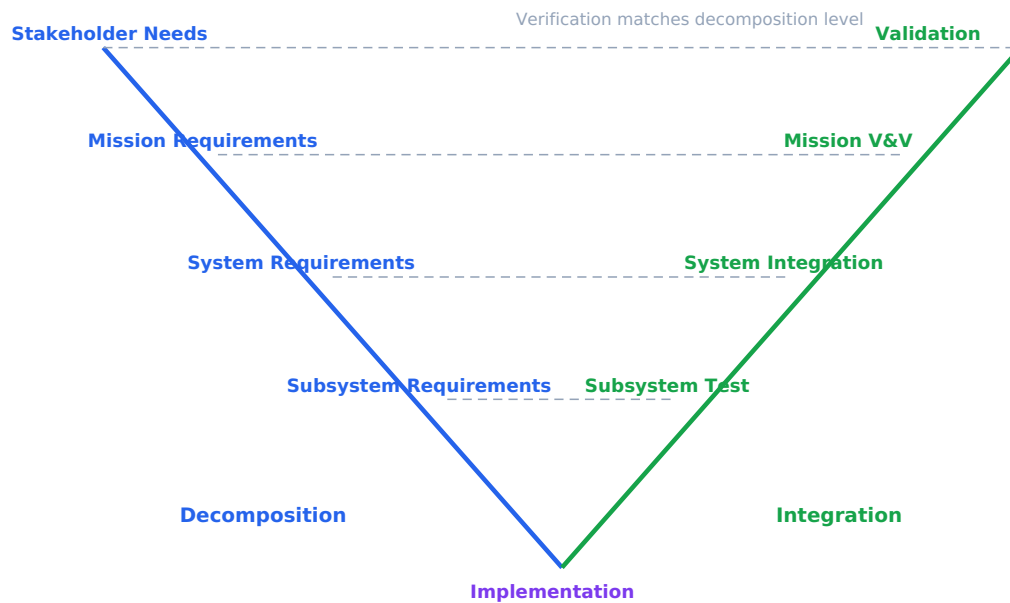
1. Explain the System-V lifecycle model and how requirements flow down the left side while verification flows up the right side
2. Write requirements that satisfy the SMART criteria and express WHAT, not HOW
3. Classify requirements by type: functional, performance, interface, constraint, and environmental
4. Construct a three-level requirement hierarchy (mission, system, subsystem) with bidirectional traceability
5. Assign verification methods (ATID) to requirements and justify each choice
6. Use SpaceCDF's Requirements tab to generate, validate, and manage requirements

1. The System-V Model

The **System-V** (or "Vee model") is the canonical systems engineering lifecycle. It connects the decomposition of requirements on the left branch to the integration and verification on the right branch. Every horizontal level on the V represents a matched pair: requirements at that level are verified by the corresponding test or analysis campaign at the same level on the right side.

[Source: NASA SEH Fig. 3-1; ECSS-E-ST-10C Rev.1 Fig. 5-1; INCOSE SEH 5th ed. Fig. 2.7]

The V-Model Structure



Key insight: The V is not a timeline -- it is a *correspondence map*. The left branch asks "what must be done?" at increasing levels of detail. The right branch asks "did we do it?" at corresponding levels. Each horizontal row is a matched pair.

Phase Mapping to the V

V-Level	Left Branch (Decomposition)	Right Branch (Verification)	NASA Phase
Top	Stakeholder needs & objectives	Operational validation (does the mission deliver value?)	Pre-A / E
Mission	Mission-level requirements	Mission-level V&V (commissioning, in-orbit checkout)	A / E
System	System-level requirements	System integration & test (end-to-end)	B / D
Subsystem	Subsystem requirements	Subsystem qualification & acceptance test	C / D
Bottom	Implementation (detailed design, build)	Component acceptance	C-D

Real Mission Example: Planet SuperDove

Planet's SuperDove constellation (2020--present) illustrates the V-model in commercial practice:

Level	Requirement (Left)	Verification (Right)
Stakeholder	"Daily global coverage at <5 m for agricultural analytics"	In-orbit validation: coverage maps, user feedback

Level	Requirement (Left)	Verification (Right)
Mission	"Revisit $\leq$ 1 day at $\pm 60^\circ$ latitude"	Constellation coverage simulation confirmed
System	"GSD $\leq$ 3.7 m at 475 km SSO"	End-to-end imaging test from orbit
Subsystem	"Telescope aperture $\geq$ 90 mm, 8 spectral bands"	Instrument calibration on ground + in-orbit

[Source: Planet Labs, "Planet Imagery Product Specifications," 2023]

2. What is a Requirement?

[Source: NASA SEH Sec. 4.2; ECSS-E-ST-10-06C Sec. 5.2]

A requirement is a **formal, verifiable statement of what the system must do or how well it must perform**. It is expressed as a "shall" statement with a measurable threshold.

The Shall Convention

Requirements use contractually precise language:

Keyword	Meaning	Contractual Force
"Shall"	Mandatory requirement -- must be met	Binding
"Should"	Goal or guideline -- desired but not mandatory	Advisory
"Will"	Statement of fact or intent -- not a requirement	Informational

Example:

"The system **shall** achieve a ground sample distance of 10 m or better at nadir from the operational orbit."

This is testable: GSD can be computed from optical geometry and verified from calibration imagery.

Requirement Types

Type	Definition	Example
Functional	What the system must <i>do</i> (a capability)	"The system shall acquire multispectral imagery of the target area"
Performance	How <i>well</i> it must do it (quantified)	"The system shall achieve GSD $\leq$ 10 m at nadir"
Interface	Boundary agreements between elements	"The EPS shall provide 5.0 V $\pm$ 0.25 V regulated bus to all subsystems"
Constraint	Non-negotiable external limits	"The system total mass shall not exceed 6.0 kg"
Environmental	Survival conditions	"The system shall survive launch loads of 9 g axial and 4 g lateral"

[Source: NASA SEH Appendix C, Table C-1; ECSS-E-ST-10-06C Sec. 5.2.2]

Requirements vs. Design Choices

Requirement (WHAT)	Design Choice (HOW)
"The system shall achieve GSD ≤ 10 m"	"Use a 150 mm aperture telescope"
"The system shall survive launch loads"	"Use aluminium 7075-T6 structure"
"The system shall provide ≥ 3 dB link margin"	"Use S-band with 2 W transmitter"
"The system shall deorbit within 5 years of EOL"	"Include cold gas propulsion system"
"The system shall operate for ≥ 3 years"	"Use rad-tolerant components"

Key principle: Requirements constrain the solution space without specifying the solution. This preserves design freedom for trade studies.

3. The SMART Framework

While NASA Appendix C provides a detailed quality checklist, the SMART acronym serves as a rapid quality screen:

Letter	Meaning	Test Question
S	Specific	Does it address exactly one concern? Is it unambiguous?
M	Measurable	Does it have a numeric threshold with units? Can it be verified?
A	Achievable	Is it technically feasible with current or near-term technology?
R	Relevant	Does it trace to a stakeholder need or higher-level objective?
T	Traceable	Can you identify its parent (where it came from) and its children (how it will be verified)?

NASA SEH Appendix C: Full Quality Checklist

[Source: NASA SEH Appendix C -- "How to Write a Good Requirement"]

The full NASA checklist adds criteria beyond SMART:

1. **Single requirement per statement** -- no compound "and" requirements
2. **Positive form** -- state what the system SHALL DO, not what it shall NOT do
3. **Active voice** -- "The system shall..." not "It is required that..."
4. **No implementation** -- avoid naming specific hardware, software, or methods
5. **Verifiable** -- must be provable by analysis, test, inspection, or demonstration
6. **No TBDs** -- every threshold must have a value (even if it changes later)
7. **Consistent** -- no contradictions with other requirements in the set
8. **Bounded** -- tolerance or range specified where appropriate

Common Anti-Patterns

Anti-Pattern	Problem	Better Version
"The spacecraft shall operate at 500 km altitude"	Prescribes orbit (design choice)	"The system shall provide global coverage with revisit ≤ 5 days at ± 60 deg latitude"
"The spacecraft shall use triple-junction GaAs solar cells"	Prescribes technology	"The EPS shall generate ≥ 15 W EOL with ≤ 0.5 m ² array area"
"The system shall use AX.25 protocol"	Prescribes implementation	"The TTC system shall provide reliable command reception with BER $\leq 10^{-6}$ "
"The spacecraft shall be a 3U CubeSat"	Prescribes form factor	"The system shall have total mass ≤ 6 kg and fit within a standard 3U deployer envelope"

Exception: Interface requirements ARE specific because they define contractual boundaries:

"The EPS shall provide 28 V $\pm$ 2 V regulated bus voltage to all subsystems."

This is acceptable because bus voltage is a negotiated interface agreement, not a unilateral design choice.

4. Requirement Hierarchy and Traceability

Requirements exist at multiple levels, each decomposed from the level above. Every requirement must trace bidirectionally.

Hierarchy Structure

```

Stakeholder Need: "Timely agricultural monitoring for food security"
|
v derives
Mission Requirement (MR):
  MR-001: "The system shall provide multispectral imagery with
           GSD  $\leq 10$  m and revisit  $\leq 5$  days over target region"
|
v decomposes into
System Requirements (SR):
  SR-PL-001: "The payload shall achieve GSD  $\leq 10$  m at nadir
              from the operational orbit"
  SR-AOCS-001: "The AOCS shall provide pointing accuracy  $\leq 0.1$  deg
                (3-sigma) during imaging"
  SR-LINK-001: "The comms system shall downlink  $\geq 5$  GB/day"
|
v derives
Subsystem Requirements (SSR):
  SSR-PL-001a: "The telescope aperture shall be  $\geq 80$  mm"
  SSR-AOCS-001a: "The star tracker shall provide  $\leq 5$  arcsec
                  attitude knowledge (3-sigma)"
  SSR-LINK-001a: "The X-band transmitter shall provide  $\geq 2$  W RF power"

```

Level Definitions

Level	Scope	Written By	Verified By
Mission	What the mission achieves (MoE-derived)	Systems engineer + stakeholders	Operational validation
System	What the system must do (MoP-derived)	Systems engineer	System-level integration & test
Subsystem	What each subsystem must provide	Subsystem engineers	Subsystem qualification & acceptance
Component	What each component must meet	Component engineers	Component acceptance testing

Traceability Rules

Every requirement must trace in three directions:

1. **Upward:** to its parent requirement or objective (WHY does this exist?)
2. **Downward:** to child requirements or design parameters (HOW is it decomposed?)
3. **Horizontally:** to a verification method (HOW will it be confirmed?)

This three-axis traceability is captured in the **Requirements Traceability Matrix (RTM)**.

[Source: NASA SEH Sec. 6.2 (Process 11: Requirements Management); ECSS-E-ST-10C Sec. 5.2]

Splitting Compound Requirements

A compound requirement that addresses multiple concerns must be split for independent verification:

Compound (Poor)	Split (Correct)
"The system shall provide 10 m GSD with 5-day revisit and 24 h latency"	MR-001: "GSD <= 10 m" + MR-002: "Revisit <= 5 days" + MR-003: "Latency <= 24 h"
"The EPS shall provide positive power margin in all modes including eclipse"	SR-PWR-001: "Positive margin in sunlit modes" + SR-PWR-002: "Positive margin in eclipse" + SR-PWR-003: "Battery DOD <= 30% in worst-case eclipse"

Rationale: Each split requirement can be verified independently. If GSD passes but revisit fails, the team knows exactly what to fix without ambiguity.

5. Verification Methods -- ATID

[Source: ECSS-E-ST-10-02C Rev.1 Sec. 5.3; NASA SEH Sec. 5.3]

Every requirement must have an assigned verification method. The four standard methods (sometimes called ATRI or ATID) are:

Method	Code	What It Proves	When Used
Analysis	A	Requirement satisfied by mathematical model, simulation, or similarity	Phase B--C; when testing is impractical or cost-prohibitive

Method	Code	What It Proves	When Used
Test	T	Requirement satisfied by physical measurement	Phase C--D; when proof requires hardware in a controlled environment
Inspection	I	Requirement confirmed by visual or physical examination	Physical characteristics (dimensions, labels, markings, mass)
Demonstration	D	Requirement confirmed by operational exercise	Functional requirements proven by executing procedures

Verification Method Selection Guide

Requirement Type	Typical Method	Rationale
Mass $\leq$ X kg	I (weigh it)	Direct physical measurement
Pointing $\leq$ Y deg	A (simulation) + T (TVAC/HITL)	Analysis first, confirmed by test
Link margin $\geq$ 3 dB	A (link budget analysis)	Full test requires satellite in orbit
Survival at launch loads	T (vibration test)	Must physically prove structural integrity
Data latency $\leq$ 24 h	D (end-to-end ops exercise)	Pipeline is procedural -- demonstrate it works
Operating temperature range	T (thermal vacuum test)	Thermal environment must be simulated
Software FDIR logic	D (fault injection test)	Demonstrate correct autonomous response

Verification by Project Phase

Phase	Verification Activities
B	Analysis verification (models, simulations, budgets, trade studies)
C	Design verification (detailed analysis, breadboard testing, qualification)
D	Acceptance testing (flight hardware), system integration, environmental test
E	In-orbit validation (commissioning, calibration, operational checkout)

6. Worked Example: 3U Earth Observation CubeSat

Worked Example -- Deriving Requirements for a 3U EO CubeSat

Stakeholder need: "Monitor crop health across the Canadian prairies with weekly updates."

Step 1 -- Mission requirement: MR-001: "The system shall provide NDVI-capable imagery with GSD $\leq$ 10 m and revisit $\leq$ 7 days over 49--54 deg N, 100--115 deg W."

Step 2 -- System requirements (decomposed): - SR-PL-001: "The payload shall acquire imagery in at least RED (630--690 nm) and NIR (760--900 nm) bands." - SR-AOCS-001: "The AOCS shall provide pointing accuracy $\leq$ 0.5 deg (3-sigma) during imaging." - SR-LINK-001: "The comms system shall downlink $\geq$ 2 GB per day to a Canadian ground station." - SR-PWR-001: "The EPS shall provide positive power margin in all operational modes."

Step 3 -- Subsystem requirements (derived): - SSR-PL-001a: "The telescope aperture shall be ≥ 80 mm." [Derived from GSD + altitude] - SSR-AOCS-001a: "The star tracker shall provide ≤ 30 arcsec attitude knowledge." [Derived from pointing budget] - SSR-LINK-001a: "The X-band transmitter shall provide ≥ 2 W RF output power." [Derived from link budget]

Step 4 -- Verification assignment:

Requirement	Method	Phase	Rationale
MR-001 (GSD + revisit)	A + D	B + E	Analysis during design; demonstrated from orbit
SR-PL-001 (spectral bands)	T	D	Spectral calibration in lab
SR-AOCS-001 (pointing)	A + T	B + D	Simulation then hardware-in-loop test
SR-LINK-001 (downlink)	A	B	Link budget analysis
SSR-PL-001a (aperture)	I	D	Physical measurement

1U Worked Example: UniSat-1

Worked Example -- Deriving Requirements for a 1U Technology Demonstrator

Stakeholder need: "Demonstrate the feasibility of a MEMS-based magnetometer for space weather monitoring in LEO."

Step 1 -- Mission requirements:

ID	Requirement	Type	Rationale
MR-001	"The system shall measure the local magnetic field vector with resolution ≤ 10 nT at 1 Hz sampling rate for a minimum of 6 months."	Performance	Minimum science return for technology validation
MR-002	"The system total mass shall not exceed 1.33 kg."	Constraint	CDS Rev 14, 1U mass limit
MR-003	"The system shall downlink at least 1 MB of magnetometer data per day."	Performance	Sufficient for statistical analysis of sensor performance
MR-004	"The system shall operate for a minimum of 6 months in LEO."	Performance	Minimum mission duration for seasonal variation coverage

Step 2 -- System requirements (decomposed): - SR-PWR-001: "The EPS shall provide orbit-average power ≥ 2 W to all subsystems." - SR-PWR-002: "The battery shall provide ≥ 10 Wh capacity." - SR-LINK-001: "The comms system shall provide a downlink data rate ≥ 9600 bps." - SR-LINK-002: "The comms system shall achieve link margin ≥ 3 dB at 10 deg minimum elevation." - SR-OBC-001: "The OBC shall consume ≤ 0.5 W average power." - SR-STR-001: "The structure shall comply with CDS Rev 14, 1U envelope (100 x 100 x 113.5 mm)."

Step 3 -- Subsystem requirements (derived): - SSR-PWR-001a: "Body-mounted solar cells shall generate ≥ 2 W orbit-average power accounting for eclipse and geometry." [Derived from power

budget] - SSR-LINK-001a: "The UHF transmitter shall provide ≥ 0.5 W RF output at 437 MHz." [Derived from link budget]

Step 4 -- Verification assignment:

Requirement	Method	Phase	Rationale
MR-001 (magnetometer performance)	T + D	D + E	Ground calibration then in-orbit demonstration
MR-002 (mass ≤ 1.33 kg)	I	D	Weigh the flight unit
SR-PWR-001 (orbit-avg power)	A	B	Power budget analysis
SR-LINK-001 (9600 bps downlink)	A + D	B + E	Link budget analysis; demonstrated from orbit
SR-STR-001 (CDS compliance)	I	D	Physical measurement against CDS template

Key difference from 3U: The 1U requirement set is much smaller (~15--20 requirements vs ~40--60 for a 3U EO mission). There are no pointing requirements, no imaging requirements, and no propulsion requirements. This makes the verification campaign significantly simpler and cheaper.

7. SpaceCDF Exercise

Instructions

1. Navigate to the **Requirements** tab in SpaceCDF
2. Click "**Generate from Objectives**" -- the AI agent generates SMART requirements from your mission objectives and ConOps
3. For each generated requirement: - Review the SMART validation badges (green = pass, amber = warning, red = fail) - Check: does it say WHAT not HOW? Is it a single concern? - **Accept**, **Edit**, or **Reject** each one
4. Use the **Level filter** (Mission / System / Subsystem) to view by hierarchy level
5. Use the **Type filter** (Functional / Performance / Interface / Constraint) to classify
6. Navigate to the **V&V Matrix** sub-panel to assign verification methods (A/T/I/D) to each accepted requirement

Exercise Tasks

1. Generate requirements for your team's mission
2. Identify at least one requirement that specifies HOW (implementation) -- rewrite it as WHAT
3. Split any compound requirements into individual testable statements
4. Classify 5 requirements by type (functional, performance, interface, constraint, environmental)
5. For 5 key requirements, assign verification method (A/T/I/D) and target phase (B/C/D/E)
6. Complete Worksheet 2.1

Key Equations Reference

There are no equations in requirements engineering per se, but the following relationship governs traceability completeness:

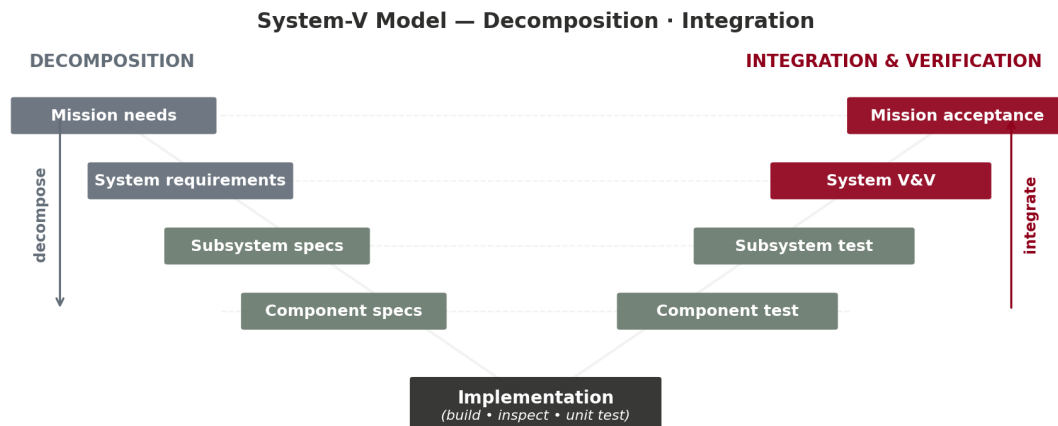
$$\text{Coverage} = \frac{N_{\text{verified requirements}}}{N_{\text{total requirements}}} \times 100\%$$

Target: 100% coverage -- every requirement must have at least one assigned verification method and one parent traceability link.

Session Summary

Topic	Key Takeaway
System-V model	Left branch decomposes requirements; right branch integrates and verifies at matching levels
Shall statements	Formal, verifiable, single-concern statements
WHAT not HOW	Requirements preserve design freedom; design choices come later in trade studies
SMART	Specific, Measurable, Achievable, Relevant, Traceable
Requirement types	Functional, performance, interface, constraint, environmental
Hierarchy	Mission -> System -> Subsystem with bidirectional traceability (up, down, horizontal)
Splitting	One concern per requirement for independent verification
ATID	Analysis, Test, Inspection, Demonstration -- assigned per requirement per phase

Session 2.2: Functional Decomposition and Allocation



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System-V model with course touchpoints.

Expected reading before this session. NASA SEH §4.3 (≈ 30 min); ECSS-E-ST-10C §5.6.

Prerequisites: Session 2.1 (requirements defined and validated) **SpaceCDF Tab:** Functions

References

- NASA, *Systems Engineering Handbook* (NASA/SP-2016-6105 Rev 2), 2016, Sec. 4.3 (Process 3: Logical Decomposition)
- ECSS, *ECSS-E-ST-10C Rev.1: System Engineering General Requirements*, 2017, Sec. 5.3
- INCOSE, *Systems Engineering Handbook*, 5th ed., 2023, Ch. 2.3.5.3
- Wertz, Everett & Puschell, *Space Mission Engineering: The New SMAD*, 2011, Ch. 2
- Blanchard & Fabrycky, *Systems Engineering and Analysis*, 5th ed., 2010, Ch. 3

Learning Objectives

By the end of this session, participants will be able to:

1. Explain the purpose of functional decomposition in bridging requirements to design
2. Decompose mission objectives into a hierarchical function tree (3+ levels)
3. Categorise functions by type (observe, communicate, navigate, point, power, protect, store, process, support)
4. Allocate functions to subsystem domains, including multi-allocation for shared responsibilities
5. Derive requirements from functional analysis

6. Identify and resolve coverage gaps (leaf functions without linked requirements)
7. Use SpaceCDF's function tree editor to build and validate a function architecture

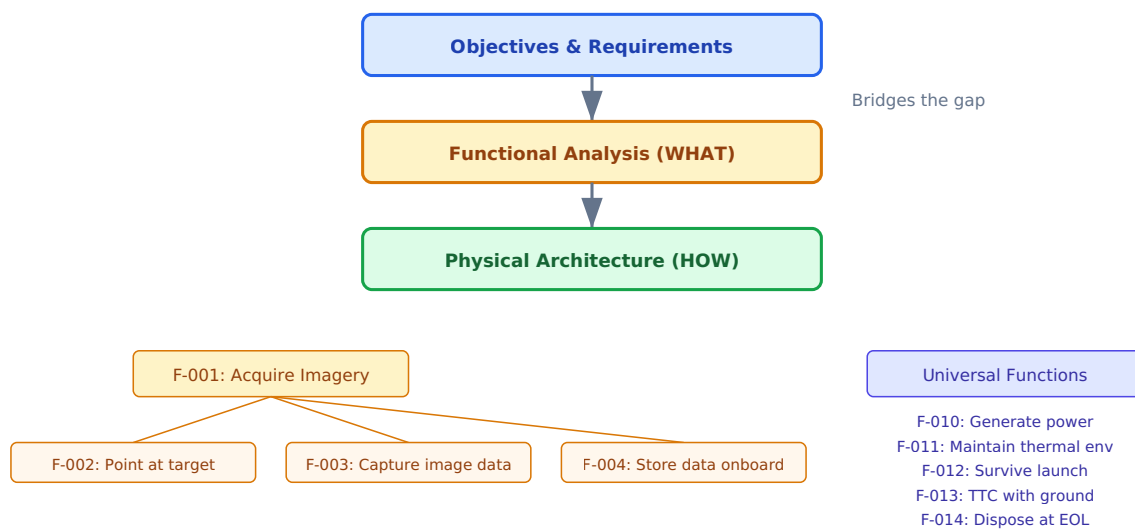
1. What is Functional Decomposition?

[Source: NASA SEH Sec. 4.3 -- Process 3: Logical Decomposition]

Functional decomposition answers: **"What must the system DO to satisfy the requirements?"**

It creates a bridge between requirements (WHAT the system must achieve) and physical design (HOW it will be built). Functions describe *actions* the system must perform, without specifying the physical hardware or software that will perform them.

The Decomposition Flow



Function Types

Every spacecraft function falls into one of these categories:

Type	Verb	Description	Example
Observe	Sense/measure	Acquire data from the environment	Capture multispectral imagery, receive AIS signals
Communicate	Transfer	Move information between elements	Downlink science data, uplink telecommands
Navigate	Determine/control position	Know or change the orbit	Perform orbit maintenance manoeuvre
Point	Orient	Control spacecraft attitude	Nadir pointing during imaging, target tracking
Power	Generate/store/distribute	Provide electrical energy	Solar power generation, battery charge management
Protect	Maintain environment	Keep components within limits	Thermal control, radiation shielding

Type	Verb	Description	Example
Store	Record	Buffer data for later use	Onboard solid-state data recording
Process	Transform	Convert data between forms	On-board image compression, LO framing
Support	Provide structure	Mechanical integrity	Survive launch loads, deploy mechanisms

Universal Functions

Regardless of mission type, **every spacecraft** must perform these five functions:

1. **Generate electrical power** (solar, RTG, battery)
2. **Maintain thermal environment** (passive/active thermal control)
3. **Survive the launch environment** (structural loads, vibration, shock)
4. **Communicate with ground for TTC** (telemetry, tracking, command)
5. **Dispose of the spacecraft at end of life** (deorbit, graveyard, passivation)

SpaceCDF auto-generates these universal functions for every mission.

2. Mission-Type-Specific Function Trees

Different mission types produce fundamentally different primary function trees. The universal functions remain the same; it is the mission-specific branch that differentiates.

Earth Observation (Optical/SAR)

```
F-001: Acquire imagery of target area
+-- F-002: Point instrument at target
+-- F-003: Capture image data (expose detector / SAR pulse)
+-- F-004: Store acquired data onboard
+-- F-005: Downlink data to ground station
```

Real example -- Planet SuperDove: F-001 decomposes into 8-band pushbroom acquisition, on-board radiometric correction, lossless compression, and X-band downlink. Each sub-function traces to specific SuperDove requirements (3.7 m GSD, 8 spectral bands, 200 Mbps downlink).

Communications Relay (Store-and-Forward or Bent-Pipe)

```
F-001: Relay communications between users
+-- F-002: Receive uplink signal from user terminal
+-- F-003: Process and route data (store-and-forward or bent-pipe)
+-- F-004: Transmit downlink signal to destination terminal or gateway
```

Real example -- Astrocast (IoT): F-001 decomposes into L-band receive (from IoT devices), on-board message deduplication and store-and-forward, and UHF/S-band downlink to gateway stations. Each message is ≤ 160 bytes.

[Source: Astrocast, "Astrocast Network Overview," astrocast.com]

AIS/IoT Receiver (Passive)

```
F-001: Receive and process signals of interest
+-- F-002: Receive AIS/IoT signals (passive -- no uplink transmission)
+-- F-003: Decode and validate messages
+-- F-004: Store processed data onboard
+-- F-005: Downlink to ground for distribution
```

Ground Segment Functions

Missions that include ground-side processing need ground-domain functions:

```
F-G01: Receive satellite data at ground station
F-G02: Process data pipeline (L0 -> L1 -> L2 products)
F-G03: Archive and distribute data products to users
F-G04: Operate mission control centre (planning, commanding, monitoring)
```

SpaceCDF supports allocation to ground domains: `ground_station`, `ground_processing`, `ground_sensor`.

3. Allocation to Subsystems

Each function must be **allocated** to one or more responsible subsystem domains. This allocation defines system boundaries and, critically, identifies **interfaces** wherever a function is shared.

Allocation Rules

Function	Typical Allocation	Multi-allocation?
Acquire imagery	Payload	No (single owner)
Point at target	AOCS	No
Store data	OBC/Data Handling	No
Downlink data	Comms + AOCS	Yes -- comms for RF chain, AOCS for antenna pointing
Generate power	EPS	No
Maintain thermal	Thermal	No
Relay communications	Payload + Comms	Yes -- boundary depends on architecture
Survive launch	Structure	No
Dispose at EOL	Propulsion (or Ops)	No (or multi if drag-sail)

Multi-Allocation and Interface Identification

When a function is allocated to more than one subsystem, it creates an **interface** that must be explicitly managed.

Example -- "Downlink data to ground station":

Responsible Subsystem	Contribution	Interface Created
Comms (Link)	Transponder, antenna, modulation, RF chain	Comms <-> AOCS (antenna pointing)
AOCS	Antenna pointing towards ground station during pass	Comms <-> Data Handling (packet routing)
Data Handling	Data packaging, prioritisation, CCSDS framing	Data Handling <-> AOCS (pass scheduling)

SpaceCDF supports multi-allocation by entering comma-separated domains.

Discussion point. For each multi-allocated function in your design, where should the system boundary be drawn? Who "owns" the function, and who is a "contributor"?

Derived Requirements from Functions

Each allocated function generates **derived requirements** -- requirements not explicitly stated by stakeholders but necessary for the function to work:

Function	Derived Requirement	Derivation Logic
"Point instrument at target"	"AOCS shall provide ≤ 0.1 deg pointing accuracy during imaging"	From GSD requirement + optical geometry
"Store data onboard"	"OBDH shall provide ≥ 32 GB solid-state storage"	From data rate x orbit period x missed-pass margin
"Downlink within daily contact window"	"TX shall provide ≥ 50 Mbps effective data rate"	From daily data volume / total contact time
"Generate power in all modes"	"SA shall produce ≥ 15 W EOL"	From worst-case sunlit power demand + recharge

[Source: NASA SEH Sec. 4.3.3 -- derived requirements from functional allocation]

4. Performance Criteria and Coverage Analysis

Each leaf function (a function with no sub-functions) must have **performance criteria** -- quantitative thresholds that define "how well" the function must be performed:

Function	Performance Criteria
Acquire imagery	GSD ≤ 10 m; SNR $\geq 100:1$; ≥ 4 spectral bands
Point at target	Accuracy ≤ 0.1 deg; stability ≤ 0.01 deg/s; slew rate ≥ 1 deg/s
Downlink data	Link margin ≥ 3 dB; daily throughput ≥ 5 GB
Generate power	Positive margin in all modes; battery DOD $\leq 30\%$
Maintain thermal env	All components within operating range with ≥ 5 degC margin

Performance criteria form the quantitative basis for subsystem-level requirements.

Coverage Analysis

Every leaf function must trace to at least one requirement. If it does not, there is a **coverage gap** -- the function is defined but has no verification path.

SpaceCDF shows coverage status with badges:

- **Green badge:** Function has linked requirements (covered)
- **Amber badge:** "No requirements" -- coverage gap detected
- **Red badge:** Function conflicts with existing requirements

Real Mission Example: CAPSTONE Coverage Gap

NASA's CAPSTONE mission (2022, Advanced Space) experienced a coverage gap during development: the "maintain attitude during trajectory correction manoeuvre" function initially had no formal pointing requirement during burns. This gap was identified during functional review and led to the derivation of SSR-AOCS-007: "The AOCS shall maintain pointing accuracy ≤ 2 deg during all propulsive manoeuvres."

[Source: Advanced Space, "CAPSTONE Mission Overview," 2022]

5. SpaceCDF Function Tree Exercise

Instructions

1. Navigate to the **Functions** tab in SpaceCDF
2. The tool auto-generates functions based on your selected mission type
3. **Review the generated tree:** Are the functions appropriate for YOUR mission? - For comms missions: should show relay/receive/transmit (not multispectral imagery) - For EO missions: should show acquire/point/store/downlink - For technology demos: should show demonstrate/characterise/report
4. **Edit functions:** - Click **edit** to change name, domain allocation, or performance criteria - Click **+sub** to add sub-functions (decompose further) - Click **x** to remove inappropriate functions
5. **Add ground segment functions** if your mission includes ground processing
6. **Check multi-allocation:** For any function allocated to multiple domains, verify the interface is captured
7. **Coverage check:** Are there any leaf functions (amber badge) without linked requirements? Derive requirements for them.

Exercise Tasks

1. Build a function tree with at least 3 levels of decomposition
2. For each leaf function, write one derived requirement with a measurable threshold
3. Create a function-to-requirement traceability table (minimum 8 entries)
4. Identify at least one multi-allocated function and define the interface boundary
5. Complete Worksheet 2.2

Worked Example: 3U EO CubeSat Function Tree

Function Tree for Agricultural Monitoring CubeSat

``` F-001: Acquire multispectral imagery +-- F-002: Point telescope at target area (AOCS) +-- F-003: Expose detector and capture image frames (Payload) +-- F-004: Compress and store image data (OBC) +-- F-005: Downlink stored data to ground station (Comms + AOCS)

F-010: Generate electrical power (EPS) +-- F-011: Convert solar energy to electrical (SA) +-- F-012: Store energy for eclipse (Battery) +-- F-013: Regulate and distribute power (EPS board)

F-020: Maintain thermal environment (Thermal) +-- F-021: Reject waste heat from electronics (Radiator) +-- F-022: Maintain battery temperature during eclipse (Heater)

F-030: Survive launch environment (Structure) F-040: Communicate with ground for TTC (Comms) F-050: Dispose at end of life (Propulsion/Ops) ```

**Derived requirements from F-002 (Point telescope at target):** - SSR-AOCS-001: "The AOCS shall achieve  $\leq 0.5$  deg pointing accuracy during imaging mode" - SSR-AOCS-002: "The AOCS shall achieve pointing stability  $\leq 0.01$  deg/s during imaging"

**Multi-allocation for F-005 (Downlink data):** - Comms subsystem: owns RF chain (transponder, antenna, modulation) - AOCS subsystem: provides antenna pointing towards ground station - Interface: AOCS must receive pass-schedule triggers from OBC to initiate slew to ground station pointing

## Session Summary

| Topic                    | Key Takeaway                                                                               |
|--------------------------|--------------------------------------------------------------------------------------------|
| Functional decomposition | Bridges requirements (WHAT) to design (HOW) by identifying system actions                  |
| Function types           | Observe, Communicate, Navigate, Point, Power, Protect, Store, Process, Support             |
| Universal functions      | Power, thermal, launch survival, TTC, disposal -- every spacecraft needs all five          |
| Mission-specific trees   | EO: acquire/point/store/downlink; Comms: receive/route/transmit; AIS: receive/decode/store |
| Allocation               | Each function assigned to subsystem domain(s); multi-allocation creates interfaces         |
| Derived requirements     | Functions generate new requirements not stated by stakeholders                             |
| Coverage analysis        | Every leaf function must trace to at least one requirement -- no gaps allowed              |

## Session 2.3: Orbit Selection and Mission Architecture

**N<sup>2</sup> Interface Matrix — 6U CubeSat (illustrative)**

|       |      |         |       |           |            |          |         |
|-------|------|---------|-------|-----------|------------|----------|---------|
| D     |      |         |       |           |            | D        | Payload |
| D     | D    |         | D     |           |            | Data/OBC | D       |
|       |      | T       |       |           | Propulsion |          |         |
|       | M    |         |       | Structure | M          |          | M       |
| D     | O    |         | Comms |           |            | D        | R       |
| T     |      | Thermal |       |           |            |          | T       |
| D     | AOCS |         |       | M         |            | D        | M       |
| Power | E    | T       | E     |           | E          | E        | E       |

■ Data (D)

■ Electrical (E)

■ Mechanical (M)

■ Thermal (T)

■ RF (R)

■ Optical (O)

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*N<sup>2</sup> interface matrix for a 6U CubeSat.*

**Expected reading before this session.** ECSS-E-ST-10-24C — Interface management (≈ 60 min) — <https://ecss.nl/>.

**Prerequisites:** Sessions 2.1--2.2 (requirements and functions defined) **SpaceCDF Tabs:** Mission Architecture, Orbit Trade Advisor

### References

- Vallado, *Fundamentals of Astrodynamics and Applications*, 4th ed., 2013, Ch. 2--6
- Wertz, Everett & Puschell, *Space Mission Engineering: The New SMAD*, 2011, Ch. 5--7
- Curtis, *Orbital Mechanics for Engineering Students*, 4th ed., 2020, Ch. 2--4
- NASA, *Systems Engineering Handbook*, 2016, Sec. 4.4 (Process 4: Design Solution Definition)

- ECSS, *ECSS-U-AS-10C Rev.2: Space Debris Mitigation Requirements*, 2023
- IADC, *Space Debris Mitigation Guidelines*, IADC-02-01 Rev 3, 2021
- FCC, *Report and Order FCC 22-74: Space Innovation*, 2022

## Learning Objectives

By the end of this session, participants will be able to:

1. Define and compute the six Keplerian orbital elements
2. Calculate orbital period, velocity, and eclipse fraction for circular orbits
3. Compute the Sun-synchronous inclination using J2 precession
4. Evaluate orbit trade-offs across altitude, coverage, lifetime, radiation, and debris compliance
5. Apply the Hohmann transfer  $\Delta V$  equations for orbit raising and deorbit
6. Use SpaceCDF's orbit trade advisor with mission-appropriate scoring weights

## 1. Keplerian Orbital Elements

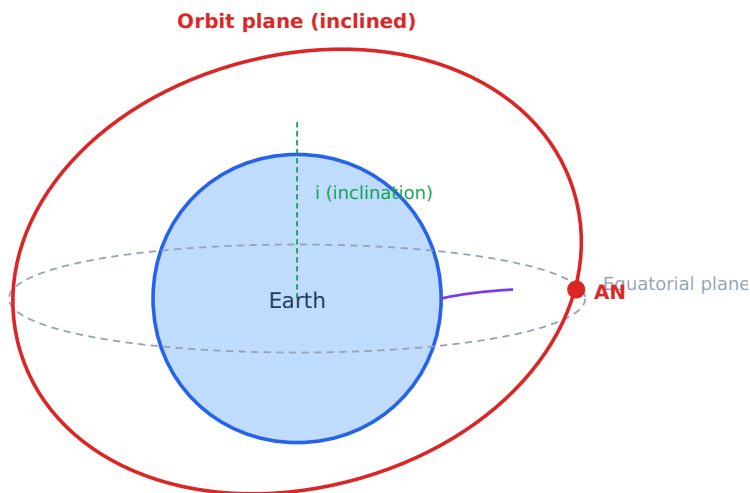
Six parameters (the *classical orbital elements*) fully describe the size, shape, and orientation of an orbit, plus the satellite's position on it.

[Source: Vallado, Ch. 2; Curtis, Ch. 2]

| Element             | Symbol   | Physical Meaning                                  | Typical Range               |
|---------------------|----------|---------------------------------------------------|-----------------------------|
| Semi-major axis     | $a$      | Size of the orbit                                 | 6571--42164 km (LEO to GEO) |
| Eccentricity        | $e$      | Shape (0 = circle, $0 < e < 1$ = ellipse)         | 0--0.001 for LEO circular   |
| Inclination         | $i$      | Tilt of orbit plane from equatorial plane         | 0--180 deg                  |
| RAAN                | $\Omega$ | Orientation of ascending node in equatorial plane | 0--360 deg                  |
| Argument of perigee | $\omega$ | Orientation of perigee within orbit plane         | 0--360 deg                  |
| True anomaly        | $\nu$    | Satellite position along the orbit                | 0--360 deg                  |

For **circular LEO** missions (the most common CubeSat orbit type), the key design parameters reduce to three: **altitude** ( $h$ ), **inclination** ( $i$ ), and **LTAN** (local time of ascending node, for Sun-synchronous orbits).




 $\Omega$  (RAAN) Satellite

## Key Equations

### Key Equations -- Orbital Mechanics Fundamentals

**Orbital period** (circular orbit):

$$T = 2\pi \sqrt{\frac{a^3}{\mu}}$$

where  $a = R_E + h$  is the semi-major axis (m),  $\mu = 3.986 \times 10^{14} \text{ m}^3/\text{s}^2$  is Earth's gravitational parameter, and  $R_E = 6371 \text{ km}$ .

**Orbital velocity** (circular):

$$v = \sqrt{\frac{\mu}{a}}$$

**Eclipse fraction** (maximum, circular orbit, cylindrical shadow):

$$f_{\text{eclipse}} = \frac{1}{\pi} \arccos \left( \frac{\sqrt{a^2 - R_E^2}}{a} \right)$$

**Sun-synchronous inclination** ( $J_2$ -driven RAAN precession = 360 deg/year):

$$\cos(i) = -\frac{2\dot{\Omega}_{\text{req}} a^{7/2}}{3R_E^2 J_2 \sqrt{\mu}}$$

where  $J_2 = 1.0826 \times 10^{-3}$ ,  $\dot{\Omega}_{\text{req}} = \frac{2\pi}{365.25 \times 86400} \text{ rad/s}$ .

## 2. Worked Examples: Orbital Computations

### Worked Example -- 500 km Circular LEO

**Given:**  $h = 500$  km, circular orbit.

**Step 1 -- Semi-major axis:**  $a = 6371 + 500 = 6871$  km  $= 6.871 \times 10^6$  m

**Step 2 -- Orbital period:**  $T = 2\pi\sqrt{\frac{(6.871 \times 10^6)^3}{3.986 \times 10^{14}}} = 2\pi\sqrt{8.137 \times 10^5} = 2\pi \times 5668 = 5669$  s  $\approx$  **94.5 min**

**Step 3 -- Orbital velocity:**  $v = \sqrt{\frac{3.986 \times 10^{14}}{6.871 \times 10^6}} = \sqrt{5.802 \times 10^7} =$  **7617 m/s**  $\approx$  **7.62 km/s**

**Step 4 -- Eclipse fraction (maximum):**

$$f = \frac{1}{\pi} \arccos\left(\frac{\sqrt{6871^2 - 6371^2}}{6871}\right) = \frac{1}{\pi} \arccos\left(\frac{2594}{6871}\right) = \frac{1}{\pi} \arccos(0.3774) = \frac{1}{\pi} \times 67.8^\circ = \mathbf{0.376} \text{ (37.6\%)}$$

**Step 5 -- Eclipse and sunlight duration:**  $t_{\text{eclipse}} = 94.5 \times 0.376 =$  **35.5 min**;  $t_{\text{sun}} = 94.5 - 35.5 =$  **59.0 min**

**Step 6 -- Sun-synchronous inclination:**  $\cos(i) = -\frac{2 \times 1.991 \times 10^{-7} \times (6.871 \times 10^6)^{3.5}}{3 \times (6.371 \times 10^6)^2 \times 1.0826 \times 10^{-3} \times \sqrt{3.986 \times 10^{14}}}$

Numerator:  $\approx -1.301 \times 10^{17}$ ; Denominator:  $\approx 1.006 \times 10^{18}$

$\cos(i) \approx -0.1293 \Rightarrow i \approx$  **97.4 deg**

## 3. Orbit Selection Trade-Offs

The orbit is the single most impactful early design decision. It cascades to every subsystem.

[Source: SMAD, Ch. 7; Wertz et al., "Reducing Space Mission Cost," Ch. 3]

| Parameter         | Lower Altitude (300--400 km)         | Higher Altitude (600--800 km)         |
|-------------------|--------------------------------------|---------------------------------------|
| GSD               | Better (shorter range to target)     | Worse (longer range)                  |
| Orbital lifetime  | Short (1--5 years, natural decay)    | Long (25--100+ years)                 |
| Debris compliance | Easy (FCC 5-year rule met naturally) | May require active deorbit propulsion |
| Radiation (TID)   | Lower (2--5 krad/yr)                 | Higher (10--20 krad/yr above 700 km)  |
| Launch cost       | Lower $\Delta V$ to orbit            | Slightly higher                       |
| Link budget       | Better (shorter slant range)         | Worse (longer path loss)              |
| Coverage/swath    | Narrower per pass                    | Wider per pass                        |
| Atmospheric drag  | Significant (limits lifetime)        | Negligible above ~700 km              |
| Eclipse fraction  | ~35--38%                             | ~33--35%                              |

Orbit Types and Applications

| Orbit         | Altitude       | Inclination | Best For                           | Real Example                   |
|---------------|----------------|-------------|------------------------------------|--------------------------------|
| LEO (non-SSO) | 300--600 km    | Any         | Technology demos, ISS deployment   | Many CubeSats                  |
| SSO           | 400--800 km    | 97--99 deg  | EO (consistent solar illumination) | Planet SuperDove (475 km)      |
| ISS orbit     | ~410 km        | 51.6 deg    | ISS-deployed CubeSats              | NanoRacks deployments          |
| MEO           | 2000--20200 km | Various     | Navigation                         | GPS (20200 km)                 |
| GEO           | 35786 km       | 0 deg       | Comms, weather                     | Anik F2 (Telesat)              |
| HEO (Molniya) | 500--40000 km  | 63.4 deg    | High-latitude coverage             | Meridian (Russia)              |
| Lunar NRHO    | 1500--70000 km | Lunar       | Cislunar operations                | CAPSTONE (NASA/Advanced Space) |

Debris Compliance Rules

| Rule                  | Requirement                                     | Applies To                             |
|-----------------------|-------------------------------------------------|----------------------------------------|
| IADC guideline (2021) | Post-mission disposal within 25 years           | International (voluntary but expected) |
| FCC rule (2024+)      | Post-mission disposal within <b>5 years</b>     | All FCC-licensed LEO satellites        |
| ECSS-U-AS-10C Rev.2   | Compliance with IADC + ESA Zero Debris Charter  | ESA missions                           |
| ISED (Canada)         | Currently 25-year rule; tightening under review | Canadian-licensed satellites           |

Critical altitude boundaries for FCC 5-year compliance:

| Altitude    | Natural Lifetime | FCC Compliant? | Action Needed                              |
|-------------|------------------|----------------|--------------------------------------------|
| < 450 km    | < 5 years        | Yes            | None (natural decay)                       |
| 450--550 km | 5--20 years      | Marginal       | Drag augmentation may suffice              |
| 550--650 km | 20--50 years     | <b>No</b>      | Active deorbit (propulsion or drag sail)   |
| > 700 km    | > 100 years      | <b>No</b>      | Propulsion mandatory; ESA Zero Debris zone |

[Source: IADC-02-01 Rev 3, Sec. 5.3.2; FCC 22-74; ECSS-U-AS-10C Rev.2]

Perturbation Effects

| Perturbation                      | Cause            | Effect on Orbit                            | Design Impact                                                 |
|-----------------------------------|------------------|--------------------------------------------|---------------------------------------------------------------|
| J <sub>2</sub> (Earth oblateness) | Equatorial bulge | RAAN precession, argument of perigee drift | Enables Sun-synchronous; frozen orbits at $\omega = 90^\circ$ |

| Perturbation             | Cause                        | Effect on Orbit                               | Design Impact                                          |
|--------------------------|------------------------------|-----------------------------------------------|--------------------------------------------------------|
| Atmospheric drag         | Residual atmosphere          | Semi-major axis decay, eventual re-entry      | Limits lifetime below ~600 km; drives propulsion need  |
| Solar radiation pressure | Photon momentum              | Eccentricity oscillations, orbit perturbation | Significant for large A/m ratio (drag sails, large SA) |
| Third-body (Moon/Sun)    | Gravitational pull           | Long-period oscillations in $e$ , $i$         | Significant for GEO and HEO; negligible for LEO        |
| Magnetic field           | Lorentz force on charged S/C | Very small drag-like effect                   | Negligible for most missions                           |

## 4. Ground Coverage and Revisit

### Key Equations -- Ground Coverage

**Swath width** (nadir-pointing sensor with half-cone angle  $\theta$ ):

$$W_{\text{swath}} = 2h \tan(\theta)$$

**Ground track spacing** (for a single satellite in LEO): The Earth rotates ~22.9 deg per orbit (for  $T \approx 95$  min). At the equator, this corresponds to:

$$\Delta_{\text{lon}} = \frac{360^\circ}{T_{\text{sidereal}}} \times T_{\text{orbit}} \approx 22.9^\circ \approx 2550 \text{ km (at equator)}$$

**Revisit time** (approximate, single satellite):

$$t_{\text{revisit}} \approx \frac{\Delta_{\text{lon}}}{W_{\text{swath}}} \times T_{\text{orbit}}$$

**Constellation revisit** (N identical satellites in same plane):

$$t_{\text{revisit,constellation}} \approx \frac{t_{\text{revisit,single}}}{N}$$

**Example -- SuperDove constellation:** Planet operates ~200 SuperDove satellites. With a ~24 km swath at 475 km and multiple orbital planes, the constellation achieves daily global revisit -- a dramatic improvement over a single satellite's ~7-day revisit.

### Ground Station Contact Geometry

**Contact time per pass** depends on the minimum elevation angle  $\epsilon_{\text{min}}$  (typically 5--10 deg):

$$t_{\text{contact}} \approx \frac{T}{\pi} \arccos\left(\frac{\cos(\rho)}{\cos(\epsilon_{\text{min}})}\right) - \text{geometric correction}$$

**Simplified rule of thumb** for LEO at 500 km,  $\epsilon_{\text{min}} = 10^\circ$ : - Maximum pass duration: ~10 min (overhead pass) - Average pass duration: ~6--7 min - Passes per day (mid-latitude station): ~4--6

## 5. Hohmann Transfer and Deorbit

### Key Equations -- Hohmann Transfer

The minimum-energy transfer between two circular orbits of radii  $r_1$  and  $r_2$ :

$$\Delta V_1 = \sqrt{\frac{\mu}{r_1}} \left( \sqrt{\frac{2r_2}{r_1 + r_2}} - 1 \right)$$

$$\Delta V_2 = \sqrt{\frac{\mu}{r_2}} \left( 1 - \sqrt{\frac{2r_1}{r_1 + r_2}} \right)$$

$$\Delta V_{\text{total}} = |\Delta V_1| + |\Delta V_2|$$

### Worked Example -- Deorbit from 600 km to 200 km (re-entry perigee)

$r_1 = 6971$  km,  $r_2 = 6571$  km

$$\begin{aligned} \Delta V_1 &= \sqrt{\frac{3.986 \times 10^5}{6971}} \left( \sqrt{\frac{2 \times 6571}{6971 + 6571}} - 1 \right) \\ &= 7.561 \times (\sqrt{0.9705} - 1) = 7.561 \times (-0.01498) = -0.1133 \text{ km/s} \end{aligned}$$

$\Delta V_{\text{total}} \approx \mathbf{113 \text{ m/s}}$  (only the first burn needed to lower perigee)

This deorbit  $\Delta V$  is a critical input to the propulsion sizing in Session 3.4.

## 6. Radiation Environment

[Source: ECSS-E-ST-10-04C Rev.1; SMAD, Ch. 8.1]

### Total Ionising Dose (TID) by Orbit

| Orbit                  | TID (krad/year behind 2mm Al) | Electronics Class         |
|------------------------|-------------------------------|---------------------------|
| ISS (410 km, 51.6 deg) | 2--5                          | Commercial COTS OK        |
| SSO 500 km             | 5--10                         | Commercial / rad-tolerant |
| SSO 800 km             | 10--20                        | Rad-tolerant recommended  |
| MEO 2000 km            | 50--100                       | Rad-hard required         |
| GEO 35786 km           | 10--20                        | Rad-hard required         |

**Rule of thumb:** Below 600 km in LEO, commercial COTS electronics can survive 3-year missions with modest shielding (2--3 mm Al equivalent). Above 600 km, radiation becomes a significant design driver and component cost escalates.

## South Atlantic Anomaly (SAA)

At 200--600 km altitude over South America, trapped protons from the inner Van Allen belt dip to lower altitudes. The SAA causes: - Single-event upsets (SEU) in memory - Single-event latch-ups (SEL) in CMOS - Increased background noise in optical detectors

Mitigation: error-correcting memory (EDAC), watchdog timers, latch-up protection circuits.

## 1U Worked Example: UniSat-1

### Orbit Selection: ISS Orbit (400 km, 51.6 deg) for Rideshare

UniSat-1 selects the ISS orbit not by optimisation but by **access**: deployment from the ISS via NanoRacks or a similar deployer is the cheapest and most accessible launch opportunity for a university 1U CubeSat.

#### Worked Example -- Orbital Parameters at 400 km, 51.6 deg

**Step 1 -- Semi-major axis:**  $a = 6371 + 400 = 6771 \text{ km} = 6.771 \times 10^6 \text{ m}$

**Step 2 -- Orbital period:**  $T = 2\pi\sqrt{\frac{(6.771 \times 10^6)^3}{3.986 \times 10^{14}}} = 2\pi \times 5564 = 5565 \text{ s} \approx \mathbf{92.4 \text{ min}}$

**Step 3 -- Orbital velocity:**  $v = \sqrt{\frac{3.986 \times 10^{14}}{6.771 \times 10^6}} = \mathbf{7672 \text{ m/s}} \approx 7.67 \text{ km/s}$

**Step 4 -- Eclipse fraction (maximum):**

$$f = \frac{1}{\pi} \arccos\left(\frac{\sqrt{6771^2 - 6371^2}}{6771}\right) = \frac{1}{\pi} \arccos(0.3423) = \frac{1}{\pi} \times 70.0^\circ \approx \mathbf{0.389 \text{ (38.9%)}}$$

**Step 5 -- Eclipse and sunlight duration:**  $t_{\text{eclipse}} = 92.4 \times 0.389 \approx \mathbf{36 \text{ min}}$ ;  $t_{\text{sun}} = 92.4 - 36 = \mathbf{56 \text{ min}}$

### Why this orbit works for UniSat-1:

| Factor           | 400 km / 51.6 deg                       | Impact on UniSat-1                                                         |
|------------------|-----------------------------------------|----------------------------------------------------------------------------|
| Launch cost      | Lowest (ISS resupply rideshare)         | Fits university budget                                                     |
| Orbital lifetime | ~1 year (natural decay)                 | Exceeds 6-month mission; compliant with FCC 5-year rule without propulsion |
| Radiation        | Low (~2--3 krad/yr behind 2 mm Al)      | COTS electronics safe for 6-month mission                                  |
| Inclination      | 51.6 deg                                | Adequate latitude coverage for space weather science                       |
| Eclipse          | ~36 min (~39% of orbit)                 | Manageable with 10 Wh battery                                              |
| Ground contacts  | Mid-latitude stations: ~4--6 passes/day | Sufficient for 9600 bps UHF downlink                                       |

**What this orbit does NOT provide:** - Sun-synchronous lighting (not needed -- magnetometer is not an optical instrument) - Polar coverage (acceptable -- 51.6 deg covers the majority of magnetic field variation) - Long lifetime (acceptable -- 6-month design life is well within ~1-year natural lifetime)

**No propulsion trade:** At 400 km, atmospheric drag causes natural re-entry within approximately 1 year (depending on solar activity and ballistic coefficient). UniSat-1 therefore needs no propulsion system for

either orbit maintenance or debris compliance. This eliminates an entire subsystem -- a major simplification for a 1U mission.

## 7. SpaceCDF Orbit Trade Exercise

### Instructions

1. Navigate to the **Mission Architecture** tab in SpaceCDF
2. Open the **Orbit Trade Advisor** panel
3. Enter your mission parameters: - GSD target (optical missions) or "N/A" for non-optical - Revisit target (days) - Mission lifetime (years) - Latitude band of interest - Cost ceiling (MEUR) - Aperture diameter (if known from optical sizing)
4. Click **"Compute Orbit Trade"**
5. Review the scored candidates: - Which orbit scores highest overall? - Is it debris-compliant (FCC 5-year rule)? - What is the natural orbital lifetime?
6. Click **"Use"** on your preferred orbit to populate the design parameters
7. Verify orbit fields updated in the dashboard (look for "Set from advisor" badge)

### Discussion Points

- Does the best-scoring orbit match your engineering intuition?
- What happens when you change scoring weights (prioritise GSD over cost, or vice versa)?
- For non-optical missions (comms, AIS), does the orbit scoring still make physical sense?
- If your selected orbit is above 600 km, what deorbit strategy will you use?
- Complete Worksheet 2.3

## Session Summary

| Topic              | Key Takeaway                                                                       |
|--------------------|------------------------------------------------------------------------------------|
| Keplerian elements | Six parameters define the orbit; for LEO: altitude + inclination + LTAN            |
| Key formulae       | $T = 2\pi\sqrt{a^3/\mu}$ ; $v = \sqrt{\mu/a}$ ; eclipse fraction from geometry     |
| Sun-synchronous    | Requires specific inclination (~97 deg at 500 km) using $J_2$ precession           |
| Perturbations      | $J_2$ enables SSO; drag limits lifetime; SRP affects high-A/m satellites           |
| Trade-offs         | Lower altitude = better GSD and link but shorter lifetime and more drag            |
| Debris rules       | FCC 5-year, IADC 25-year; above ~550 km requires active deorbit                    |
| Hohmann transfer   | $\Delta V_{\text{deorbit}} \approx 100$ m/s from 600 km; critical propulsion input |
| Radiation          | Below 600 km: COTS OK; above: rad-tolerant/hard needed; SAA causes SEU             |
| Cascade            | Orbit choice affects every subsystem -- highest-leverage design decision           |

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## Session 2.4: Mission Architecture -- Segments, Interfaces, and Budgets

**Expected reading before this session.** NASA SEH §4.4 (≈ 30 min).

**Prerequisites:** Sessions 2.1--2.3 (requirements, functions, orbit selected) **SpaceCDF Tabs:** Mission Architecture, System Architecture, Interfaces, Dashboard

---

### References

- NASA, *Systems Engineering Handbook*, 2016, Sec. 4.4 (Process 4) & Sec. 6.3 (Process 12: Interface Management)
- ECSS, *ECSS-E-ST-10-24C: Interface Management*, 2015
- ECSS, *ECSS-E-HB-10-02A: Verification Guidelines*, 2010, Sec. 5.2 (Mass Margins)
- Wertz, Everett & Puschell, *Space Mission Engineering: The New SMAD*, 2011, Ch. 10--11
- ECSS, *ECSS-E-ST-20C: Electrical and Electronic*, 2021 (Power Budgets)

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### Learning Objectives

By the end of this session, participants will be able to:

1. Decompose a space mission into its constituent segments (space, ground, launch, user)
2. Construct a system architecture block diagram with subsystem boundaries
3. Identify and classify all subsystem-to-subsystem interfaces using an N-squared matrix
4. Write formal interface requirements for critical subsystem pairs
5. Construct a mass budget with ECSS margin policy and a mode-based power budget
6. Interpret SpaceCDF's dashboard KPIs and budget displays

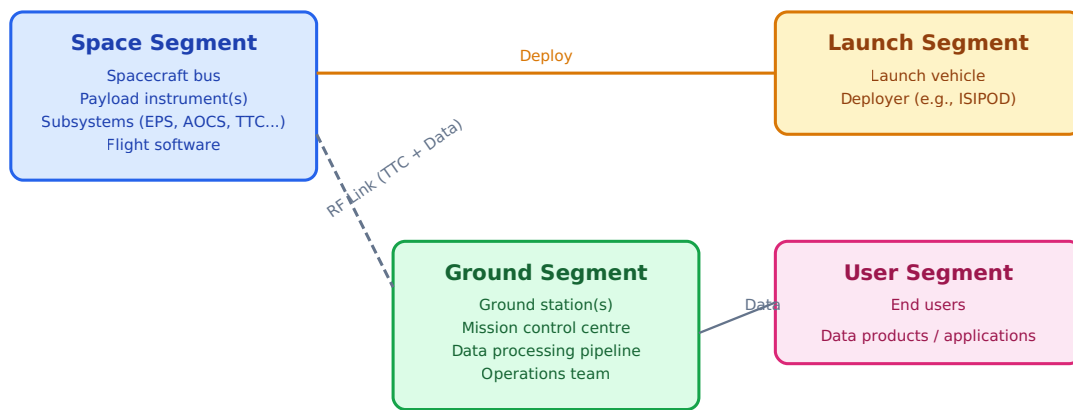
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### 1. Mission Segment Decomposition

[Source: NASA SEH Sec. 4.4; ECSS-E-ST-10C Sec. 5.4; SMAD, Ch. 1]

Every space mission decomposes into segments, each with distinct functions and interfaces:

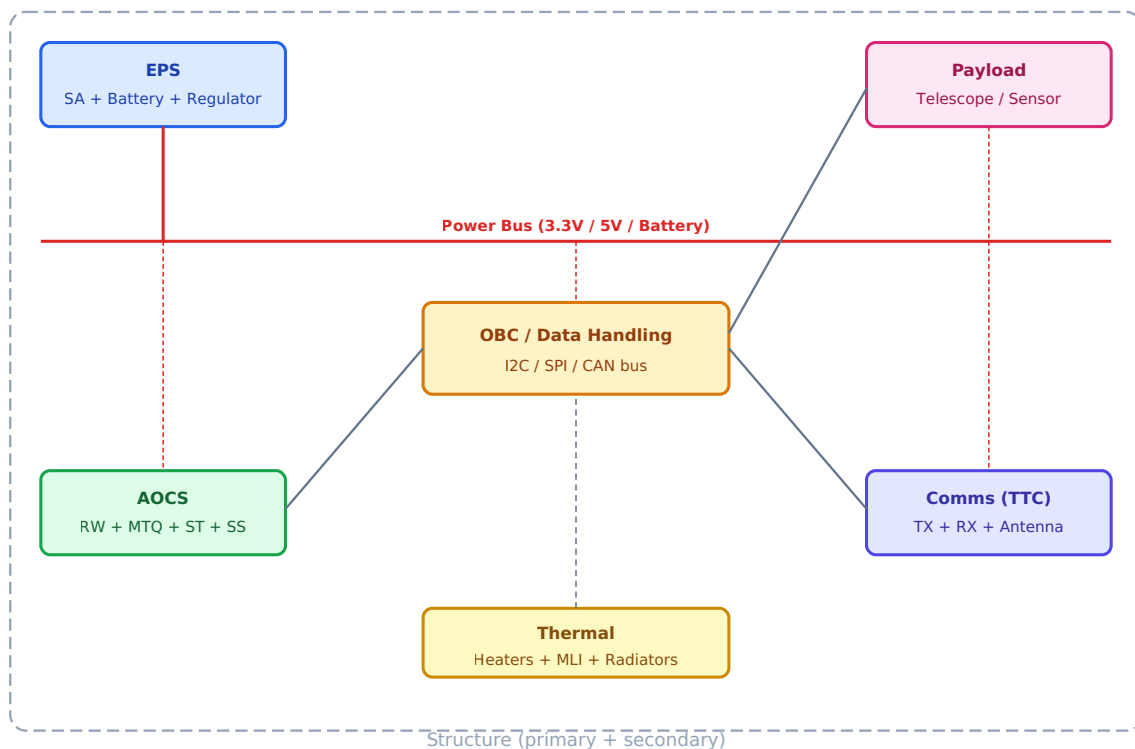




| Segment | Elements                                 | Key Interfaces                                    |
|---------|------------------------------------------|---------------------------------------------------|
| Space   | Spacecraft bus, payload, flight software | To ground (RF), to launch (mechanical/electrical) |
| Ground  | Ground station, MCC, data processing     | To space (RF), to user (network)                  |
| Launch  | Launch vehicle, deployer, adapter        | To space (mechanical, electrical inhibits)        |
| User    | End users, applications, data consumers  | To ground (data products)                         |

## 2. System Architecture Block Diagram

The system block diagram shows the internal architecture of the space segment -- all subsystems and their data/power/mechanical connections.



## Subsystem Roles (CubeSat Reference)

| Subsystem                | Abbreviation | Primary Function                                      | Typical Mass Fraction |
|--------------------------|--------------|-------------------------------------------------------|-----------------------|
| Payload                  | PL           | Mission-specific sensing or communication             | 25--40%               |
| Electrical Power System  | EPS          | Generate, store, distribute electrical power          | 15--25%               |
| Attitude & Orbit Control | AOCS         | Determine and control attitude; orbit knowledge       | 8--15%                |
| Communications (TTC)     | LINK / TTC   | Telemetry downlink, telecommand uplink, data downlink | 5--10%                |
| On-Board Computer        | OBC / C&DH   | Command execution, data handling, flight software     | 2--5%                 |
| Thermal Control          | TCS          | Maintain all components within temperature limits     | 1--5%                 |
| Structure                | STR          | Mechanical support, launch load path, CDS compliance  | 15--25%               |
| Propulsion               | PROP         | Orbit manoeuvres, deorbit (if required)               | 0--15%                |
| Harness                  | HAR          | Electrical interconnections                           | 3--7%                 |

### 3. The N-Squared Interface Matrix

[Source: NASA SEH Sec. 6.3 -- Process 12: Interface Management; ECSS-E-ST-10-24C]

Interface problems are the leading cause of integration failures. NASA SEH states: "Most system failures can be traced back to interface problems."

#### N-Squared Matrix Structure

The  $N^2$  matrix is a standard systems engineering tool for mapping all subsystem interactions:

- **Diagonal cells:** Subsystems (EPS, AOCS, Comms, Thermal, Structure, Propulsion, OBC, Payload)
- **Off-diagonal cells:** Interface between the row subsystem and the column subsystem
- **Upper triangle:** Outputs from row to column (data, power, commands)
- **Lower triangle:** Outputs from column to row

For 8 subsystems:  $8 \times 7/2 = 28$  potential interface pairs. A typical CubeSat has 18--22 active interfaces.

#### Interface Types

| Type       | Symbol | Description                                      | Example                            |
|------------|--------|--------------------------------------------------|------------------------------------|
| Mechanical | M      | Physical attachment, loads, alignment tolerances | Payload mounting to structure face |
| Electrical | E      | Power connections, bus voltage, switched lines   | EPS 5V bus to all subsystems       |

| Type    | Symbol | Description                                            | Example                                     |
|---------|--------|--------------------------------------------------------|---------------------------------------------|
| Thermal | T      | Heat transfer paths, thermal coupling, conduction      | Transponder waste heat to radiator panel    |
| Data    | D      | Digital bus (I <sup>2</sup> C, SPI, UART, CAN, RS-422) | OBC commands to AOCS controller             |
| RF      | R      | Electromagnetic coupling or intentional RF paths       | TX emissions coupling into payload receiver |
| Optical | O      | Light paths, field-of-view clearance, stray light      | Star tracker FOV clearance from solar array |

## Common CubeSat Interface Concerns

| Interface Pair        | Types | Key Concern                                                                 |
|-----------------------|-------|-----------------------------------------------------------------------------|
| EPS <-> AOCS          | E     | Bus voltage compatibility; reaction wheel peak power draw                   |
| EPS <-> Comms         | E     | TX peak power demand (~6--10 W); switched line allocation                   |
| EPS <-> Thermal       | E, T  | SA thermal coupling; radiator vs SA area competition on external faces      |
| EPS <-> Payload       | E     | Peak power switching; duty cycle coordination                               |
| Structure <-> AOCS    | M     | Reaction wheel and star tracker mounting alignment; vibration isolation     |
| Structure <-> Payload | M, O  | Payload alignment stability; optical FOV clearance                          |
| OBC <-> AOCS          | D     | Attitude data for payload pointing; mode transition commands                |
| OBC <-> Comms         | D     | Telemetry packet routing; telecommand distribution                          |
| OBC <-> Payload       | D     | Science data acquisition trigger; instrument commanding                     |
| Comms <-> Payload     | R     | <b>EMC:</b> TX conducted/radiated emissions vs payload receiver sensitivity |
| Comms <-> AOCS        | R, O  | Antenna pattern vs star tracker FOV; antenna pointing coordination          |
| Thermal <-> Payload   | T     | Detector cooling requirement; operating temperature limits                  |
| AOCS <-> Payload      | M     | Reaction wheel micro-vibration vs payload pointing stability (jitter)       |

## Conflict Detection and Resolution

Interface conflicts arise when two subsystems have incompatible requirements at their shared boundary.

### Severity Classification:

| Severity        | Description                            | Example                                            | Required Action                 |
|-----------------|----------------------------------------|----------------------------------------------------|---------------------------------|
| <b>Critical</b> | Design cannot close without resolution | EMC: TX radiation prevents payload operation       | Must resolve before PDR         |
| <b>Major</b>    | Significant impact on design margin    | Radiator area competes with SA area                | Mitigation plan required by PDR |
| <b>Minor</b>    | Manageable with minor adjustment       | Star tracker FOV partially blocked by antenna stow | Accommodation analysis          |

### Resolution Options:

1. **Relocate:** Move a component to avoid the conflict (e.g., star tracker to different face)
2. **Shield/Isolate:** Add EMC shielding, vibration isolators, thermal insulation
3. **Time-Division:** Schedule conflicting activities to avoid simultaneity (e.g., no TX during imaging)
4. **Accept Risk:** Document residual risk and margin impact in the risk register

### Writing Interface Requirements

For each significant interface, write formal requirements:

#### Example -- EPS <-> All Subsystems:

```
IR-PWR-001: "The EPS shall provide regulated bus voltages of
 3.3 V +/- 0.1 V and 5.0 V +/- 0.25 V to all subsystems."
IR-PWR-002: "Each subsystem shall not exceed its allocated power
 draw without EPS coordination."
```

#### Example -- Comms <-> Payload (EMC):

```
IR-EMC-001: "TX conducted emissions shall be below -60 dBm in the
 payload receiver band (1.5-1.6 GHz) during imaging mode."
IR-EMC-002: "TX and payload acquisition shall not operate simultaneously
 unless IR-EMC-001 is verified by test."
```

## 4. Engineering Budgets: Mass and Power

[Source: ECSS-E-HB-10-02A Sec. 5.2; SMAD, Ch. 10--11]

Engineering budgets are the quantitative backbone of the design. They answer: **"Will this design close?"**

### Mass Budget

#### Key Equations -- Mass Budget

Mass margin:

$$\text{Margin}_{\%} = \frac{M_{\text{allocation}} - M_{\text{MEV}}}{M_{\text{allocation}}} \times 100\%$$

where MEV = Maximum Expected Value = CBE + maturity margins.

**Status thresholds:** Green: > 20% | Amber: 10--20% | Red: < 10% | Exceeded: < 0%

| Term                                | Definition                                                 |
|-------------------------------------|------------------------------------------------------------|
| <b>CBE</b> (Current Best Estimate)  | Best estimate of actual mass based on current knowledge    |
| <b>MEV</b> (Maximum Expected Value) | CBE + equipment maturity margin = worst-case expected mass |
| <b>Equipment maturity margin</b>    | Applied per component based on design maturity (TRL)       |

| Term          | Definition                                    |
|---------------|-----------------------------------------------|
| System margin | Applied at system level as management reserve |

### ECSS Margin Policy by Phase

[Source: ECSS-E-HB-10-02A Sec. 5.2]

| Phase            | Equipment Margin | System Margin | Compound |
|------------------|------------------|---------------|----------|
| O/A (concept)    | 20%              | 20%           | ~44%     |
| B1 (preliminary) | 10%              | 20%           | ~32%     |
| B2 (detailed)    | 5%               | 15%           | ~21%     |
| C/D (build/test) | 3%               | 10%           | ~13%     |
| E (as-built)     | 0%               | 5%            | ~5%      |

#### Worked Example -- 3U CubeSat Mass Budget (Phase A)

| Subsystem                  | CBE (kg)    | Equip. Margin (20%) | MEV (kg)                        |
|----------------------------|-------------|---------------------|---------------------------------|
| Payload                    | 1.50        | 0.30                | 1.80                            |
| EPS                        | 0.75        | 0.15                | 0.90                            |
| AOCS                       | 0.55        | 0.11                | 0.66                            |
| Comms (TTC)                | 0.25        | 0.05                | 0.30                            |
| OBC                        | 0.08        | 0.02                | 0.10                            |
| Thermal                    | 0.05        | 0.01                | 0.06                            |
| Structure                  | 0.35        | 0.07                | 0.42                            |
| Harness                    | 0.15        | 0.03                | 0.18                            |
| <b>Dry Total</b>           | <b>3.68</b> |                     | <b>4.42</b>                     |
| System Margin (20%)        |             |                     | <b>0.88</b>                     |
| <b>Dry MEV</b>             |             |                     | <b>5.30</b>                     |
| Propellant                 |             |                     | 0.00                            |
| <b>Wet Mass</b>            |             |                     | <b>5.30</b>                     |
| <b>Launcher Allocation</b> |             |                     | <b>6.00</b> (3U limit)          |
| <b>Mass Margin</b>         |             |                     | <b>0.70 kg (11.7%)</b> -- Amber |

### Mode-Based Power Budget

The power budget is computed **per operational mode** because not all subsystems draw power simultaneously:

| Subsystem    | Safe (W)   | Idle (W)   | Imaging (W) | Downlink (W) | Eclipse (W) |
|--------------|------------|------------|-------------|--------------|-------------|
| OBC          | 1.0        | 1.0        | 1.0         | 1.0          | 1.0         |
| AOCS         | 0.5        | 1.0        | 3.0         | 2.0          | 0.5         |
| Payload      | 0          | 0          | 5.0         | 0            | 0           |
| Comms (TX)   | 0.5        | 0.5        | 0.5         | 6.0          | 0           |
| Thermal      | 0.5        | 0.5        | 0.5         | 0.5          | 2.0         |
| <b>Total</b> | <b>2.5</b> | <b>3.0</b> | <b>10.0</b> | <b>9.5</b>   | <b>3.5</b>  |

## Orbit-Average Power

### Key Equations -- Orbit-Average Power

$$P_{\text{avg}} = \sum_{\text{modes}} (P_{\text{mode}} \times \text{duty}_{\text{mode}})$$

Example for 95-min orbit (60 min sunlight, 35 min eclipse):

| Mode         | Power (W) | Duty (%)    | Contribution (W) |
|--------------|-----------|-------------|------------------|
| Idle         | 3.0       | 45%         | 1.35             |
| Imaging      | 10.0      | 10%         | 1.00             |
| Downlink     | 9.5       | 8%          | 0.76             |
| Eclipse      | 3.5       | 37%         | 1.30             |
| <b>Total</b> |           | <b>100%</b> | <b>4.41 W</b>    |

## Other Budget Types (Preview)

These budgets will be developed in detail during Week 2 Day 3--4 sessions:

| Budget          | Key Equation                                                                       | Session |
|-----------------|------------------------------------------------------------------------------------|---------|
| <b>Link</b>     | $\text{Margin} = \text{EIRP} - \text{FSPL} + G/T - k - 10\log_{10}(R_b) - E_b/N_0$ | 3.3     |
| <b>Pointing</b> | $\theta_{\text{total}} = \sqrt{\sum \theta_i^2} \text{ (RSS)}$                     | 3.2     |
| $\Delta V$      | $\Delta V = I_{\text{sp}} \cdot g_0 \cdot \ln(m_0/m_f) \text{ (Tsiolkovsky)}$      | 3.4     |
| <b>Data</b>     | Daily Downlink $\geq$ Daily Generation                                             | 3.3     |

## 1U Worked Example: UniSat-1

### Simplified Architecture: Only 5 Subsystems

UniSat-1 demonstrates that a 1U CubeSat can be built with a radically simplified architecture compared to a 3U EO mission. Several subsystems are eliminated entirely:

| Subsystem  | 3U EO CubeSat             | UniSat-1 (1U)                           | Rationale for Elimination                                                                         |
|------------|---------------------------|-----------------------------------------|---------------------------------------------------------------------------------------------------|
| EPS        | Required                  | Required                                | Always needed                                                                                     |
| OBC        | Required                  | Required                                | Always needed (minimal: MSP430 or ARM Cortex-M class)                                             |
| Comms      | S-band + X-band           | UHF only                                | 9600 bps is sufficient for < 1 kbps payload data                                                  |
| Structure  | Required                  | Required                                | Always needed (ISIS 1U or Pumpkin Rev C)                                                          |
| Payload    | Telescope (complex)       | MEMS magnetometer (simple: 50 g, 0.2 W) |                                                                                                   |
| AOCS       | Star tracker + RWs + MTQs | Eliminated                              | Passive magnetic only (permanent magnet + hysteresis rods, treated as structure, not a subsystem) |
| Thermal    | MLI + heaters             | Eliminated                              | Passive coatings only; 400 km LEO thermal environment is benign for 6-month mission               |
| Propulsion | May be needed             | Eliminated                              | 400 km orbit decays naturally in ~1 year; no orbit maintenance needed                             |
| Harness    | Significant               | Minimal                                 | Only 3--4 board-to-board connections via PC/104 stack                                             |

#### N-squared matrix -- drastically reduced:

For UniSat-1, the interface matrix shrinks from 28 potential pairs (8 subsystems) to just 10 pairs (5 subsystems). The active interfaces are:

| Interface Pair    | Types | Key Concern                                        |
|-------------------|-------|----------------------------------------------------|
| EPS <-> OBC       | E, D  | Power regulation, housekeeping telemetry           |
| EPS <-> Comms     | E     | TX peak power (~0.5 W -- small relative to budget) |
| EPS <-> Payload   | E     | Payload switching (0.2 W -- trivial)               |
| OBC <-> Comms     | D     | TM/TC packet routing                               |
| OBC <-> Payload   | D     | Magnetometer data acquisition (I2C or SPI)         |
| Structure <-> All | M     | Mounting, CDS rail compliance                      |

#### Mass budget (Phase A):

| Subsystem                                  | CBE (g) | Equip. Margin (20%) | MEV (g) |
|--------------------------------------------|---------|---------------------|---------|
| Payload (MEMS magnetometer)                | 50      | 10                  | 60      |
| EPS (board + battery + body-mounted cells) | 250     | 50                  | 300     |
| OBC (MSP430/Cortex-M board)                | 30      | 6                   | 36      |

| Subsystem                             | CBE (g)    | Equip. Margin (20%) | MEV (g)                       |
|---------------------------------------|------------|---------------------|-------------------------------|
| Comms (UHF transceiver + antenna)     | 80         | 16                  | 96                            |
| Structure (1U frame)                  | 200        | 40                  | 240                           |
| Passive magnetic AOCS (magnet + rods) | 30         | 6                   | 36                            |
| Harness                               | 50         | 10                  | 60                            |
| <b>Dry Total</b>                      | <b>690</b> |                     | <b>828</b>                    |
| System Margin (20%)                   |            |                     | <b>166</b>                    |
| <b>Dry MEV</b>                        |            |                     | <b>994</b>                    |
| <b>CDS Allocation</b>                 |            |                     | <b>1330</b>                   |
| <b>Mass Margin</b>                    |            |                     | <b>336 g (25.3%) -- Green</b> |

**Power budget (all modes use the same simple duty cycle):**

| Mode                     | Power (W) | Duty (%)    | Contribution (W) |
|--------------------------|-----------|-------------|------------------|
| Idle (OBC + beacon)      | 0.7       | 50%         | 0.35             |
| Science (+ magnetometer) | 0.9       | 10%         | 0.09             |
| Downlink (+ UHF TX)      | 1.2       | 5%          | 0.06             |
| Eclipse (OBC only)       | 0.5       | 35%         | 0.18             |
| <b>Orbit Average</b>     |           | <b>100%</b> | <b>0.68 W</b>    |

With ~2 W available from body-mounted cells, the power margin is substantial (~1.3 W, or 66%). This is one of the luxuries of a simple payload on a 1U bus.

**Key architectural insight:** The UniSat-1 architecture is so simple that a small team (3--5 people) can design, build, test, and operate it. This makes it ideal for a university or educational programme. The CDF process still applies -- but the sessions are shorter and the trade space is narrower.

## 5. SpaceCDF Exercise

### Instructions

1. **System Architecture tab:** Review or edit the subsystem block diagram for your mission
2. **Interfaces tab:** Review the  $N^2$  matrix - Click on 3 interface cells to examine types and concerns - For any red-bordered cell (conflict), use the "Resolve Conflict" workflow - Write interface requirements for your most critical pair
3. **Dashboard:** Examine all KPI cards: - Mass margin: green/amber/red? - Power margin per mode - Link margin (if computed) - Cost vs ceiling
4. **Budget Breakdown:** Open per-subsystem mass and power charts
5. Complete Worksheet 2.4



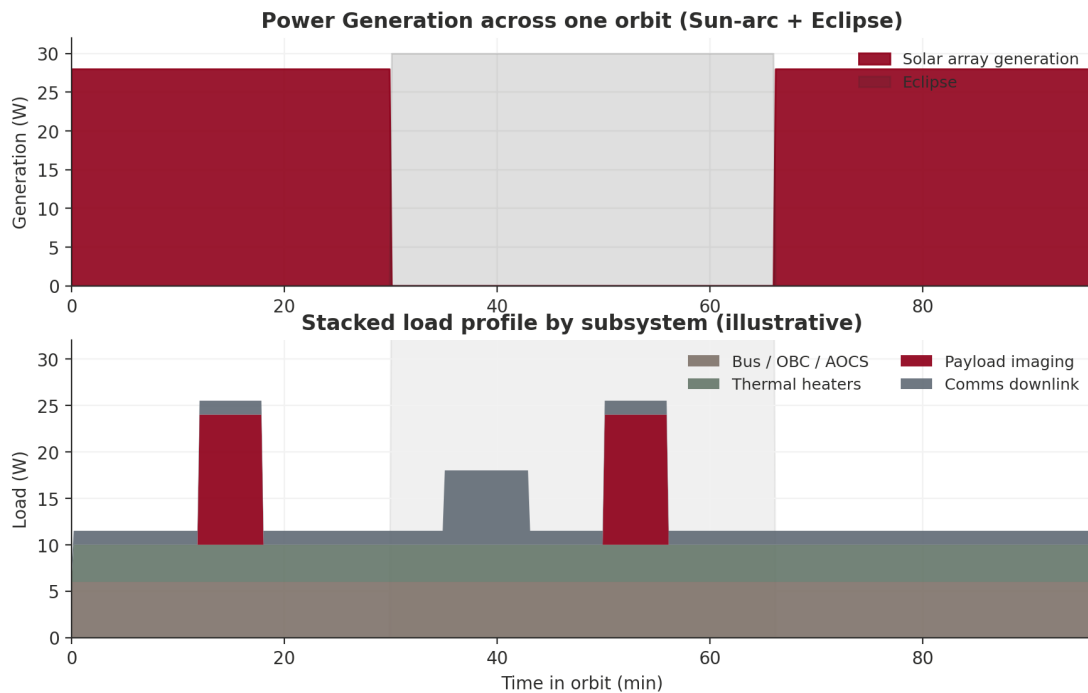
## Discussion Questions

- Which budget is tightest (closest to zero or negative margin)?
- What single design change would most improve the tightest budget?
- How does the ECSS margin policy affect your design freedom at Phase A vs Phase C?
- Which interface pair is most likely to cause integration problems?

## Session Summary

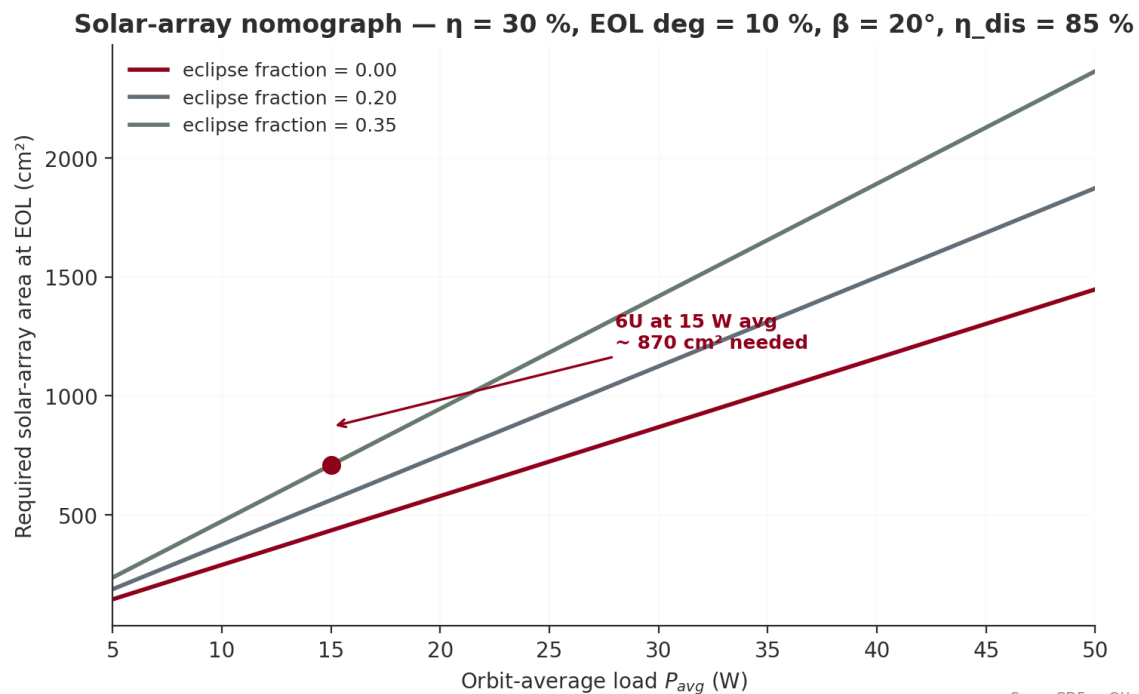
| Topic                  | Key Takeaway                                                                         |
|------------------------|--------------------------------------------------------------------------------------|
| Mission segments       | Space, Ground, Launch, User -- each with defined interfaces                          |
| System architecture    | Block diagram shows subsystems, data buses, power distribution                       |
| $N^2$ matrix           | Maps all 28 potential interface pairs; 6 types (M, E, T, D, R, O)                    |
| Interface conflicts    | Severity classification (critical/major/minor); 4 resolution options                 |
| Interface requirements | Formal boundary agreements; must be verifiable                                       |
| Mass budget            | $CBE + \text{equipment margin} + \text{system margin} = MEV$ ; compare to allocation |
| Power budget           | Mode-based; duty cycling gives orbit-average; SA must cover peak + recharge          |
| ECSS margins           | Decrease with maturity: ~44% at Phase A to ~13% at Phase C/D                         |
| Budget closure         | Negative margin = design does not close; reduce demand or increase allocation        |

## Session 3.1: Power System and Thermal Control Design



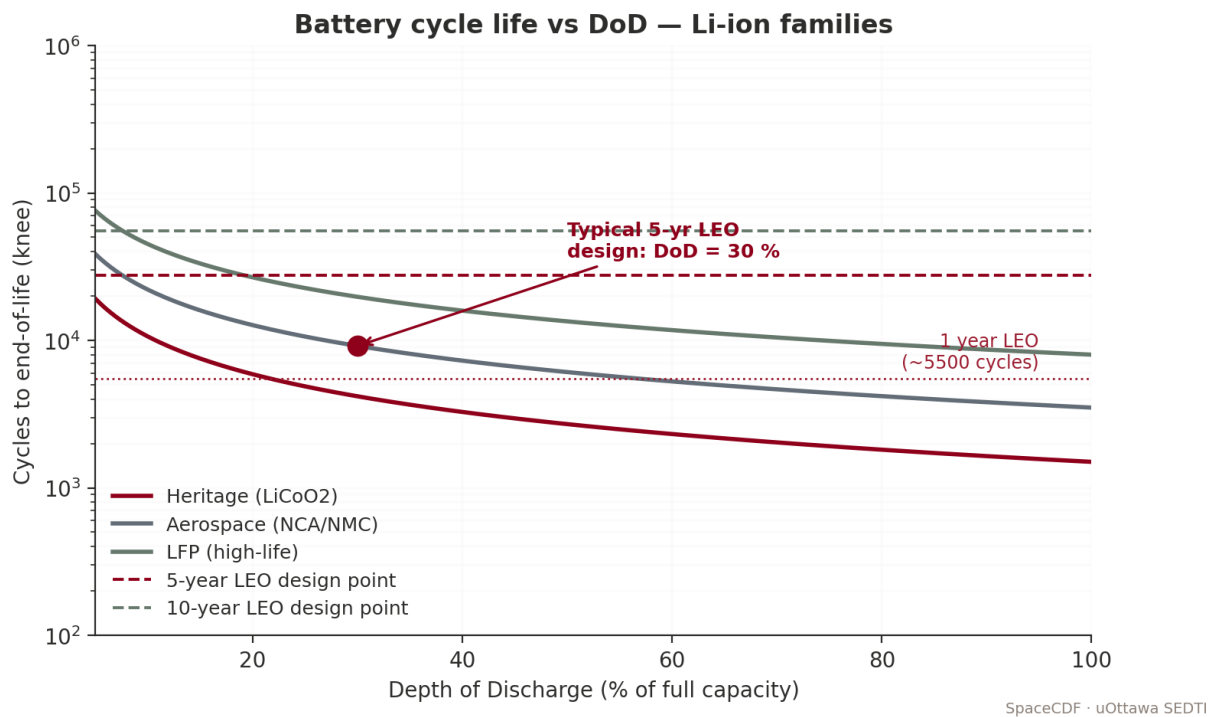
SpaceCDF · uOttawa SEDTI

Power generation and stacked load profile.



SpaceCDF · uOttawa SEDTI

Solar-array sizing nomograph.



Battery cycle life vs DoD.

**Expected reading before this session.** SMAD4 Ch. 11 (power); ECSS-E-ST-20C §5 ( $\approx 60$  min). Patel, *Spacecraft Power Systems*, Ch. 4.

**Prerequisites:** Sessions 2.1--2.4 (requirements, functions, orbit, architecture defined) **SpaceCDF Tabs:** Dashboard (Power KPI), Engineering Budgets, Timing Budget, Parametric

## References

- Wertz, Everett & Puschell, *Space Mission Engineering: The New SMAD*, 2011, Ch. 11.4 (EPS), Ch. 11.5 (Thermal)
- ECSS, *ECSS-E-ST-20C: Electrical and Electronic*, 2021
- ECSS, *ECSS-E-ST-31C: Thermal Control*, 2020
- Patel, *Spacecraft Power Systems*, 2005, Ch. 3--8
- Gilmore, *Spacecraft Thermal Control Handbook, Vol. 1*, 2002
- GomSpace, *P31u EPS Datasheet*, 2023
- MMA Design, *HaWK Solar Array Datasheet*, 2023
- Spectrolab, *30% Triple-Junction Solar Cell Datasheet*, 2020
- Ratnakumar et al., *Lithium-Ion Batteries for Space*, NASA JPL, 2003
- Gilmore, *Spacecraft Thermal Control Handbook, Vol. 2: Cryogenics*, 2003

## Learning Objectives

By the end of this session, participants will be able to:

1. Size a solar array from mission power demand, eclipse profile, and degradation

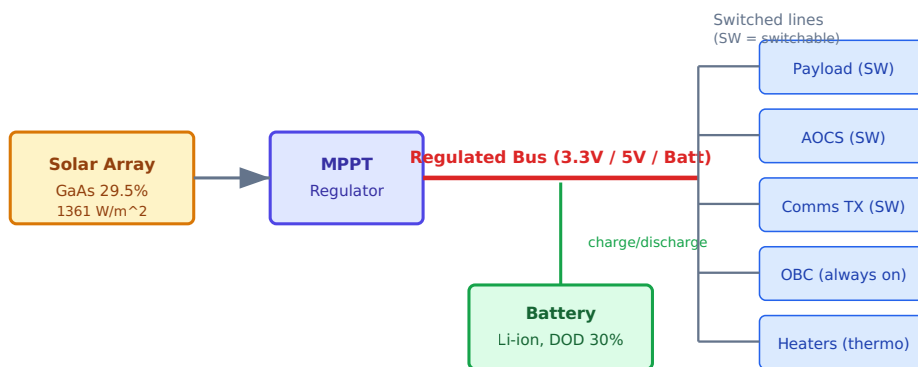
2. Size a battery from eclipse energy demand, depth-of-discharge, and cycle-life requirements
3. Compute orbit-average power using duty cycle analysis
4. Explain EPS architecture (DET vs PPT, MPPT, bus regulation) and articulate the physics of each
5. Perform first-order thermal balance analysis (hot case and cold case) with full radiative derivation
6. Select thermal control methods and apply ECSS thermal margins
7. Explain MLI construction, heat pipe operation, and heater sizing
8. Verify power and thermal budgets in SpaceCDF

## 1. Electrical Power System Architecture

[Source: SMAD, Ch. 11.4; ECSS-E-ST-20C; Patel, Ch. 3]

The EPS is the "utility company" of the spacecraft. It must continuously supply regulated power to all subsystems through every operational mode, including eclipse. Unlike terrestrial power systems, spacecraft EPS cannot draw from a grid -- the solar array, battery, and power conditioning electronics must form a fully self-contained, autonomous energy system with zero maintenance for the mission lifetime.

### EPS Block Diagram



### Solar Cell Physics

A photovoltaic cell converts photon energy into electrical energy via the photovoltaic effect. When a photon with energy  $E = h\nu \geq E_g$  (where  $E_g$  is the semiconductor bandgap) strikes the cell, it promotes an electron from the valence band to the conduction band, creating an electron-hole pair. The built-in electric field at the p-n junction sweeps carriers to opposite terminals, producing a voltage ( $V_{oc}$ ) and current ( $I_{sc}$ ).

#### Single-junction vs multi-junction cells:

A single-junction cell (e.g., silicon,  $E_g = 1.12$  eV) can only absorb photons with  $E \geq E_g$ . Photons with  $E < E_g$  pass through unabsorbed; photons with  $E \gg E_g$  lose excess energy as heat (thermalisation loss). The theoretical maximum efficiency for a single-junction cell under AM0 (space) illumination is approximately 31% (Shockley-Queisser limit).

Multi-junction (MJ) cells stack two or three p-n junctions of different semiconductor materials, each tuned to absorb a different portion of the solar spectrum:

| Cell Technology               | Structure                         | Bandgaps (eV)      | AM0 Efficiency | Temp Coefficient | Flight Heritage                             |
|-------------------------------|-----------------------------------|--------------------|----------------|------------------|---------------------------------------------|
| Monocrystalline Si            | Single junction                   | 1.12               | 16--18%        | −0.45 %/degC     | Extensive (ISS, many LEO)                   |
| GaAs single-junction          | Single junction                   | 1.42               | 22--24%        | −0.21 %/degC     | Moderate                                    |
| InGaP/GaAs dual-junction      | 2-junction                        | 1.86 / 1.42        | 26--28%        | −0.20 %/degC     | Moderate                                    |
| InGaP/GaAs/Ge triple-junction | 3-junction (lattice-matched)      | 1.86 / 1.42 / 0.67 | 28--30%        | −0.19 %/degC     | Extensive (>90% of modern S/C)              |
| InGaP/GaAs/InGaAs IMM         | 3-junction (inverted metamorphic) | 1.86 / 1.42 / 1.0  | 32--33%        | −0.18 %/degC     | Growing (SolAero ZTJ, Spectrolab XTJ Prime) |

[Source: Spectrolab XTJ Prime datasheet; SolAero ZTJ datasheet; Green et al., "Solar Cell Efficiency Tables," Progress in Photovoltaics, v.62, 2024]

**Why triple-junction GaAs dominates space:** The AM0 solar spectrum (above the atmosphere) is richer in UV and blue photons than the AM1.5 terrestrial spectrum. Triple-junction cells capture this energy across three bandgaps. The top cell (InGaP,  $E_g = 1.86$  eV) absorbs blue/UV, the middle cell (GaAs,  $E_g = 1.42$  eV) absorbs visible, and the bottom cell (Ge,  $E_g = 0.67$  eV) absorbs near-IR. Each junction contributes voltage in series, while current is limited by the lowest-current junction (current matching constraint).

**Temperature effects:** In orbit, solar cells operate at 40--80 degC depending on mounting and orbit. Cell efficiency decreases with temperature due to increased intrinsic carrier concentration (which reduces  $V_{oc}$ ). The temperature coefficient for triple-junction GaAs is approximately  $-0.19\%/degC$  relative -- meaning a cell rated at 29.5% at 28 degC drops to approximately 28.5% at 80 degC. This must be included in power budget calculations:

$$\eta_{cell}(T) = \eta_{ref} \times [1 + \beta(T - T_{ref})]$$

where  $\beta \approx -0.0019 / degC$  for triple-junction GaAs and  $T_{ref} = 28$  degC (standard test conditions).

**Degradation mechanisms:** Solar cells degrade in orbit due to: - **Radiation damage:** Energetic protons and electrons (trapped in the Van Allen belts and from solar particle events) displace atoms in the crystal lattice, creating recombination centres that reduce minority carrier lifetime and thus current. Degradation is characterised as equivalent 1 MeV electron fluence. Typical rates: 2--3%/year in LEO, 5--8%/year in MEO (through proton belt), 1--2%/year in GEO. - **UV darkening:** Ultraviolet radiation darkens cover glass adhesive over time, reducing transmission. - **Micrometeoroid erosion:** Gradual pitting of cover glass reduces optical transmission. - **Electrostatic discharge (ESD):** In GEO or polar orbits, differential charging can cause arcing between cells, permanently damaging interconnects.

Cover glass (typically ceria-doped borosilicate, 100--150 um thick) mitigates radiation and UV effects. The combined degradation factor is:

$$L_d = (1 - \delta)^n$$

where  $\delta = 0.025$  (2.5%/year for triple-junction GaAs in LEO with standard cover glass) and  $n$  = mission lifetime in years.

## Body-Mounted vs Deployable Solar Arrays

**Body-mounted cells** are bonded directly to the spacecraft's external panels. They are the simplest and most reliable option (no deployment mechanism, no hinges, no drive motors) but are severely area-limited.

| Mounting                   | Advantages                                                                            | Disadvantages                                                                                                   | Typical Power                 |
|----------------------------|---------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------|-------------------------------|
| <b>Body-mounted</b>        | No mechanism risk, no power for tracking, low mass, low cost                          | Limited area (satellite surface only), poor illumination geometry for nadir-pointing S/C, high cell temperature | 1U: ~2 W, 3U: ~7 W, 6U: ~12 W |
| <b>Fixed deployable</b>    | 2--4x more area, better sun angle, lower cell temp (radiative cooling from back side) | Deployment mechanism (single point of failure), aerodynamic drag increase, structural dynamics                  | 3U: ~15--25 W, 6U: ~30--48 W  |
| <b>Tracking deployable</b> | Optimal sun incidence ( $\cos \theta \approx 1$ ), maximum power                      | SADM (Solar Array Drive Mechanism) adds mass/cost/complexity, continuous power for motor                        | Large S/C: 100+ W             |

**Real mission examples:** - **Planet SuperDove (3U+):** Body-mounted + two fixed deployable wings. ~25 W BOL. The deployables are spring-hinged panels that fold against the 3U body during launch and deploy after ejection from the P-POD. - **Asteria (6U, JPL):** Dual deployable arrays, 48 W BOL. Used MMA Design HaWK panels with triple-junction GaAs cells. - **ISS CubeSats (various 1U):** Body-mounted only, 2--3 W. Adequate for simple sensor missions with low duty cycles.

## Architecture Types

| Architecture                            | How It Works (Physics)                                                                                                                                                                                                                                                                                                         | Efficiency                    | Complexity | Typical Use                            |
|-----------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------|------------|----------------------------------------|
| <b>DET</b> (Direct Energy Transfer)     | SA connects directly to bus; a shunt regulator diverts excess current to a resistor bank (dissipated as heat) when SA output exceeds load. No series regulator between SA and bus. Voltage varies with illumination.                                                                                                           | 80--85% (shunt losses)        | Low        | Heritage GEO spacecraft, some CubeSats |
| <b>PPT / MPPT</b> (Peak Power Tracking) | A DC-DC converter (typically boost or buck-boost) between SA and bus continuously adjusts its input impedance to operate the SA at its maximum power point (MPP). Uses perturb-and-observe or incremental conductance algorithm. Extracts 10--15% more power than DET, especially at off-nominal temperatures and end-of-life. | 90--95%                       | Medium     | Most modern CubeSats                   |
| <b>Unregulated bus</b>                  | Battery connects directly to the power bus through protection FETs only. Bus voltage equals battery voltage (varies from 3.0 V at empty to 4.2 V at full per cell, or 6.0--8.4 V for 2S configuration). Subsystems must tolerate voltage variation.                                                                            | Highest (no regulator losses) | Lowest     | Very simple CubeSats, 1U missions      |
| <b>Regulated bus</b>                    | DC-DC converters create fixed voltage rails (3.3 V, 5 V, 12 V) from battery/SA input. Subsystems see constant voltage regardless                                                                                                                                                                                               | Good (85--90% overall)        | Medium     | Most CubeSat COTS EPS (GomSpace)       |

| Architecture | How It Works (Physics)                                                                       | Efficiency | Complexity | Typical Use                 |
|--------------|----------------------------------------------------------------------------------------------|------------|------------|-----------------------------|
|              | of battery state. Adds ~5--10% losses in regulators but greatly simplifies subsystem design. |            |            | P31u, Endurosat, AAC Clyde) |

**MPPT physics:** A solar cell's I-V curve has a distinct "knee" where the maximum power ( $P = I \times V$ ) is extracted. At open-circuit ( $I = 0$ ), voltage is maximum but power is zero. At short-circuit ( $V = 0$ ), current is maximum but power is zero. The MPP sits at approximately 75--80% of  $V_{oc}$  and 90--95% of  $I_{sc}$ . Temperature shifts the I-V curve (higher T moves  $V_{oc}$  left), so the MPP moves. An MPPT controller dynamically tracks this point, typically updating every 0.1--1 s. Common CubeSat MPPT converters achieve 95--97% tracking efficiency.

**Battery charge regulation:** The EPS must prevent overcharging (which causes lithium plating, gas generation, and thermal runaway in Li-ion cells) and overdischarging (which causes copper dissolution from the negative current collector, permanently damaging the cell). Modern CubeSat EPS boards implement: - **CC-CV charging:** Constant-current charging until cell voltage reaches 4.2 V, then constant-voltage taper until current drops below C/20 - **Under-voltage lockout:** Bus disconnect when cell voltage falls below 3.0 V (or 2.8 V for emergency) - **Cell balancing:** For multi-cell series configurations (2S or higher), passive or active balancing circuits ensure cells remain within 50 mV of each other - **Temperature cutoffs:** Charging inhibited below 0 degC and above 45 degC (Li-ion charging below 0 degC causes lithium plating)

**CubeSat standard:** Most commercial EPS boards (GomSpace P31u, Endurosat, AAC Clyde) use MPPT + regulated bus with 3.3 V and 5 V rails, plus an unregulated battery rail (6.0--8.4 V for 2S Li-ion).

## 2. Solar Array Sizing

### Key Equations -- Solar Array Sizing (Full Derivation)

#### Step 1: Orbit-average power demand:

$$P_{avg} = \sum_{modes} P_{mode} \times f_{duty,mode}$$

This is computed from the ConOps mode table. For each mode (imaging, downlink, eclipse/safe, idle), multiply the mode power by the fraction of orbit spent in that mode.

#### Step 2: SA end-of-life power requirement:

During sunlight, the SA must simultaneously: (a) power all sunlit loads and (b) recharge the battery for the upcoming eclipse. The recharge power accounts for battery charge/discharge efficiency:

$$P_{SA,EOL} = P_{peak,sunlight} + \frac{P_{eclipse} \times t_{eclipse}}{t_{sunlight} \times \eta_{path,eclipse}}$$

where  $\eta_{path,eclipse} = \eta_{charge} \times \eta_{discharge} \times \eta_{regulator}$ .

For a typical CubeSat EPS:  $\eta_{charge} \approx 0.92$  (Li-ion coulombic efficiency  $\times$  charge regulator),  $\eta_{discharge} \approx 0.95$  (battery internal resistance losses),  $\eta_{regulator} \approx 0.90$  (DC-DC converter). Combined:  $\eta_{path,eclipse} \approx 0.79$ .

A simpler approximation uses  $\eta_{\text{charge}} \approx 0.90$  as a lumped path efficiency, which is common in textbooks but slightly optimistic.

**Step 3: Account for degradation and temperature:**

$$P_{\text{SA,BOL}} = \frac{P_{\text{SA,EOL}}}{L_d \times L_T}$$

where: -  $L_d = (1 - \delta)^n$  is the radiation degradation factor ( $\delta = 0.025/\text{yr}$  for TJ GaAs in LEO) -  $L_T = 1 + \beta(T_{\text{cell}} - T_{\text{ref}})$  is the temperature derating factor ( $\beta = -0.0019/\text{degC}$ ,  $T_{\text{ref}} = 28 \text{ degC}$ ,  $T_{\text{cell}}$  typically 60--80 degC in LEO)

For a cell operating at 65 degC:  $L_T = 1 + (-0.0019)(65 - 28) = 1 - 0.070 = 0.930$  (7% power loss from temperature).

**Step 4: Compute SA area:**

$$A_{\text{SA}} = \frac{P_{\text{SA,BOL}}}{\eta_{\text{cell}} \times S \times \cos(\theta) \times f_{\text{pack}} \times f_{\text{cover}}}$$

where: -  $\eta_{\text{cell}} = 0.295$  (triple-junction GaAs efficiency at AM0, 28 degC -- STC rating) -  $S = 1361 \text{ W/m}^2$  (solar constant at 1 AU, per [Kopp & Lean 2011](#)) -  $\theta$  = sun incidence angle (0 deg for ideal tracking; for body-mounted, use orbit-averaged  $\cos \theta$ ) -  $f_{\text{pack}} = 0.85$ -- $0.90$  (cell packing factor -- fraction of panel area covered by cells; gaps exist for cell interconnects, edge clearance, and harness routing) -  $f_{\text{cover}} = 0.97$  (cover glass transmission loss, typically 2--4%)

**Step 5: SA mass:**

$$m_{\text{SA}} = A_{\text{SA}} \times \sigma_{\text{SA}} + m_{\text{mechanism}}$$

where  $\sigma_{\text{SA}}$  = areal density of the panel: - Body-mounted (cells on Al substrate):  $\sigma \approx 2.0$ -- $2.5 \text{ kg/m}^2$  - Rigid deployable (cells on Al honeycomb/CFRP panel):  $\sigma \approx 1.5$ -- $2.0 \text{ kg/m}^2$  - Flexible deployable (roll-out or fold-out):  $\sigma \approx 0.8$ -- $1.2 \text{ kg/m}^2$  -  $m_{\text{mechanism}}$  = hinge, spring, hold-down mechanism: typically 0.1--0.3 kg per panel for CubeSats

**Worked Example -- 3U EO CubeSat (SuperDove-class) Solar Array**

**Given:**  $P_{\text{peak,sunlight}} = 10.0 \text{ W}$  (imaging mode),  $P_{\text{eclipse}} = 3.5 \text{ W}$ ,  $t_{\text{eclipse}} = 35 \text{ min}$ ,  $t_{\text{sunlight}} = 60 \text{ min}$ , mission lifetime = 3 years, cell temperature = 65 degC, single deployable panel with fixed sun angle  $\theta = 23 \text{ deg}$  (average over orbit for SSO with body-fixed panel).

**Step 2:** Recharge power (using detailed path efficiency):  $\eta_{\text{path}} = 0.92 \times 0.95 \times 0.90 = 0.786$

$$P_{\text{recharge}} = \frac{3.5 \times 35}{60 \times 0.786} = \frac{122.5}{47.2} = 2.60 \text{ W}$$

$$P_{\text{SA,EOL}} = 10.0 + 2.60 = 12.60 \text{ W}$$

**Step 3:** BOL accounting for 3-year degradation + temperature:  $L_d = (1 - 0.025)^3 = 0.9269$

$$L_T = 1 + (-0.0019)(65 - 28) = 0.930$$

$$P_{\text{SA,BOL}} = \frac{12.60}{0.9269 \times 0.930} = \frac{12.60}{0.862} = 14.62 \text{ W}$$

**Step 4:** SA area:  $A_{\text{SA}} = \frac{14.62}{0.295 \times 1361 \times \cos(23^\circ) \times 0.85 \times 0.97}$



$$= \frac{14.62}{0.295 \times 1361 \times 0.921 \times 0.85 \times 0.97}$$

$$= \frac{14.62}{305.0} = 0.0479 \text{ m}^2$$

This is approximately 22 cm x 22 cm -- achievable with a single deployable panel on a 3U CubeSat (the MMA HaWK panel provides up to 0.06 m<sup>2</sup> per wing).

**Step 5: SA mass (deployable, rigid):**  $m_{SA} = 0.0479 \times 1.8 + 0.15 = 0.086 + 0.15 = 0.24 \text{ kg}$  (panel + mechanism)

**Comparison to Planet SuperDove:** The actual SuperDove uses body-mounted cells plus two deployable wings, achieving ~25 W BOL. Our calculation (14.6 W BOL from one panel) is consistent -- SuperDove's higher power supports continuous imaging plus S-band downlink simultaneously.

### CubeSat SA Power Reference

| Configuration     | 1U   | 3U    | 6U    | 12U    |
|-------------------|------|-------|-------|--------|
| Body-mounted only | ~2 W | ~7 W  | ~12 W | ~20 W  |
| Single deployable | ~4 W | ~15 W | ~30 W | ~55 W  |
| Dual deployable   | --   | ~25 W | ~48 W | ~100 W |
| Quad deployable   | --   | --    | ~80 W | ~180 W |

[Source: GomSpace, ISIS, MMA Design vendor datasheets; ASTERIA 6U confirmed 48 W BOL; Dove/SuperDove confirmed ~25 W]

## 3. Battery Sizing

### Li-ion Cell Chemistry and Physics

All modern spacecraft batteries use lithium-ion (Li-ion) chemistry. During discharge, lithium ions migrate from the graphite anode (negative electrode) through an organic electrolyte and polymer separator to the lithium metal oxide cathode (positive electrode), while electrons flow through the external circuit doing work. During charging, the process reverses.

Common cathode chemistries used in space:

| Chemistry                            | Cathode                 | Nominal Voltage | Energy Density | Cycle Life (30% DOD) | Thermal Stability | Space Heritage                   |
|--------------------------------------|-------------------------|-----------------|----------------|----------------------|-------------------|----------------------------------|
| <b>LCO</b> (LiCoO <sub>2</sub> )     | Cobalt oxide            | 3.7 V           | 150--200 Wh/kg | ~10,000              | Moderate          | ISS, many CubeSats (18650 cells) |
| <b>NMC</b> (LiNiMnCoO <sub>2</sub> ) | Nickel-manganese-cobalt | 3.7 V           | 170--250 Wh/kg | ~5,000               | Moderate          | Growing heritage                 |
| <b>LFP</b> (LiFePO <sub>4</sub> )    | Iron phosphate          | 3.2 V           | 90--120 Wh/kg  | ~50,000              | Excellent         |                                  |

| Chemistry                               | Cathode                 | Nominal Voltage | Energy Density | Cycle Life (30% DOD) | Thermal Stability | Space Heritage                       |
|-----------------------------------------|-------------------------|-----------------|----------------|----------------------|-------------------|--------------------------------------|
|                                         |                         |                 |                |                      |                   | Niche (when cycle life is paramount) |
| <b>NCA</b><br>(LiNiCoAlO <sub>2</sub> ) | Nickel-cobalt-aluminium | 3.6 V           | 200--260 Wh/kg | ~3,000               | Lower             | Limited space heritage               |

[Source: Ratnakumar et al., NASA JPL; Saft VES-16 space cell datasheet; Samsung SDI 18650 specifications]

#### Cell form factors:

- **18650 cylindrical:** 18 mm diameter, 65 mm long. The workhorse of CubeSat missions. Common cells: Samsung 25R (2500 mAh, 20A continuous), Panasonic NCR18650B (3350 mAh, moderate rate), Sony VTC6 (3000 mAh, high rate). Energy: 9--12 Wh per cell.
- **Pouch cells:** Custom dimensions, higher energy density (~250 Wh/kg) but require external structural support. Used in some 6U+ missions and all large spacecraft. GomSpace NanoPower BPX uses pouch cells.
- **Prismatic cells:** Rigid case, intermediate between cylindrical and pouch. Used in some mission-specific designs.

#### Cell configuration:

Cells are arranged in series (S) to increase voltage and parallel (P) to increase capacity: - **1S (single cell):** 3.0--4.2 V bus. Used for very simple 1U CubeSats. - **2S (two in series):** 6.0--8.4 V bus. Standard for most CubeSats (matches GomSpace P31u default). - **2S2P (two series, two parallel):** 6.0--8.4 V bus, double capacity. Used for higher-energy missions (e.g., GomSpace BP4 pack: 4 cells, 2S2P, ~38 Wh).

**Protection circuits:** Every flight battery pack includes: - **Cell voltage monitoring:** Per-cell voltage measurement to detect over/under-voltage - **Over-current protection:** Current-sense resistors + MOSFET switches to disconnect at overcurrent (prevents short-circuit thermal runaway) - **Temperature monitoring:** Thermistors on each cell; inhibit charging below 0 degC and above 45 degC - **Heater circuit:** Kapton heater on battery pack, thermostatically controlled, to maintain cells above minimum charging temperature during eclipse

**Capacity fade model:** Li-ion cells lose capacity over time due to solid electrolyte interphase (SEI) growth on the anode. A simplified calendar + cycling fade model:

$$C(t, N) = C_0 \times (1 - \alpha\sqrt{t}) \times (1 - \beta \cdot N \cdot \text{DOD}^\gamma)$$

where  $C_0$  = initial capacity,  $t$  = time (years),  $N$  = number of cycles,  $\alpha \approx 0.02$  (calendar fade),  $\beta \approx 3 \times 10^{-6}$ ,  $\gamma \approx 2.1$  (cycling fade). For LEO at 30% DOD, this gives approximately 5--8% capacity loss per year -- consistent with flight data from ISS battery replacements and CubeSat fleet telemetry.

#### Key Equations -- Battery Sizing

##### Required battery energy:

$$E_{\text{bat}} = \frac{P_{\text{eclipse}} \times t_{\text{eclipse}}}{\text{DOD} \times \eta_{\text{discharge}}}$$

where: - DOD = maximum depth of discharge -  $\eta_{\text{discharge}} = 0.95$  (discharge efficiency, accounting for internal resistance  $I^2R$  losses) -  $t_{\text{eclipse}}$  in hours

**Battery mass:**

$$m_{\text{bat}} = \frac{E_{\text{bat}}}{e_{\text{specific}}}$$

where  $e_{\text{specific}} = 150\text{--}200\text{ Wh/kg}$  for packaged Li-ion 18650 cells (cell-level energy density is higher, but packaging adds ~30% mass).

**Cycle life vs DOD relationship** (Li-ion 18650, LCO chemistry):

| DOD | Typical Cycle Life | Suitable Mission Duration | Annual Eclipses (LEO, 15/day) |
|-----|--------------------|---------------------------|-------------------------------|
| 80% | ~500 cycles        | < 1 month                 | 450                           |
| 50% | ~2,000 cycles      | < 4 months                | 1,800                         |
| 30% | ~10,000 cycles     | 1--2 years                | 5,475                         |
| 20% | ~30,000 cycles     | 3--5 years                | 10,950                        |
| 10% | ~100,000 cycles    | > 7 years                 | 38,325                        |

**Design rule of thumb:** For a multi-year LEO mission, start with 20--30% DOD. For a short technology demonstration (< 6 months), 40--50% DOD is acceptable and significantly reduces battery size/mass/cost.

**Worked Example -- Battery for 3U EO CubeSat (SuperDove-class)**

**Given:**  $P_{\text{eclipse}} = 3.5\text{ W}$ ,  $t_{\text{eclipse}} = 35\text{ min} = 0.583\text{ h}$ ,  $\text{DOD} = 0.25$  (conservative for 3-year mission),  $\eta = 0.95$ .

**Step 1 -- Required battery energy:**  $E_{\text{bat}} = \frac{3.5 \times 0.583}{0.25 \times 0.95} = \frac{2.04}{0.2375} = 8.59\text{ Wh}$

**Step 2 -- Apply ECSS margin (20% at Phase A/B):**  $E_{\text{bat,spec}} \geq 8.59 \times 1.20 = 10.3\text{ Wh}$ . Specify **minimum 10 Wh**, ideally **20 Wh** for operational flexibility.

**Step 3 -- Verify cycle life:** 3-year mission at 15 orbits/day = 16,425 eclipses. At 25% DOD, Li-ion 18650 cells provide ~20,000 cycles. **Margin = 22%. Pass.**

Including capacity fade: after 3 years, ~15--20% capacity loss from calendar + cycling aging. Effective DOD increases to  $0.25/0.82 = 0.30$  -- still within the 10,000-cycle regime. **Acceptable with monitoring.**

**Step 4 -- Battery mass:**  $m_{\text{bat}} = \frac{20}{170} = 0.118\text{ kg}$  (using 170 Wh/kg for packaged 18650 cells)

**Step 5 -- Battery volume:** Two 18650 cells in 2S1P:  $2 \times 18\text{mm} \times 65\text{mm} =$  approximately 34 mL, fitting easily within a 3U stack.

**Comparison to Planet SuperDove:** SuperDove carries approximately 20 Wh in a 2S2P configuration (4 cells), consistent with our sizing. The actual operating DOD is estimated at 10--15% per eclipse, giving substantial cycle-life margin for the multi-year constellation replenishment cadence.

**Failure modes to watch for:**

- **Thermal runaway:** If a cell is overcharged, mechanically damaged, or experiences an internal short, the exothermic decomposition of the cathode material can lead to thermal runaway (self-heating > heat dissipation), potentially reaching 600+ degC. Mitigation: cell-level fuses, per-cell voltage monitoring, thermal cutoffs.
- **Lithium plating:** Charging below 0 degC causes metallic lithium to deposit on the anode surface rather than intercalating into graphite. This is irreversible, reduces capacity, and can cause internal shorts. Mitigation: battery heater + thermostat + software lockout.
- **Capacity imbalance:** In series configurations, the weakest cell limits the pack. If cells age at different rates, the weakest cell hits under-voltage lockout while others still have capacity. Mitigation: cell balancing circuits, matched cell lots.

## 4. Thermal Control System

[Source: ECSS-E-ST-31C; Gilmore, Ch. 1--4; SMAD, Ch. 11.5]

### The Physics of Spacecraft Thermal Control

Spacecraft thermal control is fundamentally different from terrestrial thermal engineering because **there is no convection in vacuum**. The only heat transfer mechanisms are:

1. **Conduction:** Heat flow through solid material, governed by Fourier's law:  $\dot{Q} = -kA \frac{dT}{dx}$ . Critical within the spacecraft structure and between components and mounting surfaces.
2. **Radiation:** Heat transfer via electromagnetic radiation, governed by the Stefan-Boltzmann law:  $\dot{Q} = \epsilon \sigma A T^4$ . This is the **only** mechanism for rejecting heat to the environment.

There is no convective cooling. A component that overheats cannot be cooled by a fan. All waste heat must be conducted to a radiating surface and then radiated to space. This is the central constraint of spacecraft thermal design.

### Thermal Environment in LEO

A spacecraft in LEO experiences four thermal inputs and one thermal sink:

| Source               | Flux                                                                   | Direction                       | Variability                                                                                                                                                              |
|----------------------|------------------------------------------------------------------------|---------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Direct solar         | $S = 1361 \pm 1 \text{ W/m}^2$ (at 1 AU)                               | Sun-facing surfaces only        | Seasonal ( $\pm 3.3\%$ due to Earth's orbital eccentricity: $S_{\text{perihelion}} = 1414 \text{ W/m}^2$ in January, $S_{\text{aphelion}} = 1322 \text{ W/m}^2$ in July) |
| Earth albedo         | $\alpha_E \times S \approx 0.30 \times 1361 \approx 408 \text{ W/m}^2$ | Earth-facing surfaces (nadir)   | Varies with cloud cover, surface type (0.06 for ocean to 0.80 for fresh snow); orbit-average range 0.25--0.35                                                            |
| Earth infrared       | $q_{\text{IR}} \approx 240 \text{ W/m}^2$ (orbit average)              | Earth-facing surfaces (nadir)   | Range 200--270 $\text{W/m}^2$ depending on latitude, season, cloud cover                                                                                                 |
| Internal dissipation | $Q_{\text{int}} = P_{\text{dissipated}}$                               | From electronics waste heat     | Varies with operational mode; nearly all electrical power eventually becomes heat                                                                                        |
| Deep space (sink)    | $T_{\text{space}} \approx 2.7 \text{ K}$ (CMB)                         | Zenith-facing radiator surfaces | Effectively 0 K for engineering purposes                                                                                                                                 |



## Hot Case and Cold Case

| Case             | Conditions                                                                                                                                                                                                                             | Design Concern                                      |
|------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------|
| <b>Hot case</b>  | Maximum solar exposure ( $S = 1414 \text{ W/m}^2$ , perihelion), all subsystems active (max $Q_{\text{int}}$ ), worst sun angle (max $A_{\text{sun}}$ ), BOL coatings ( $\alpha_s$ at minimum -- fresh white paint), max albedo (0.35) | Components exceed maximum operating temperature     |
| <b>Cold case</b> | Eclipse (no solar), minimum power dissipation (safe mode, min $Q_{\text{int}}$ ), EOL coatings ( $\alpha_s$ increased by UV darkening), minimum Earth IR ( $200 \text{ W/m}^2$ ), deep space view                                      | Components fall below minimum operating temperature |

**Design philosophy:** The thermal engineer designs to keep all components within their qualified temperature range under both worst-case hot and worst-case cold conditions, with ECSS-mandated margins applied.

## Surface Coatings -- The Passive Thermal Toolbox

| Coating                             | $\alpha_s$ (BOL) | $\alpha_s$ (EOL, 5 yr LEO) | $\varepsilon$ | $\alpha_s/\varepsilon$ (BOL) | Use Case                                                              |
|-------------------------------------|------------------|----------------------------|---------------|------------------------------|-----------------------------------------------------------------------|
| White paint (AZ-93, S13G-LO)        | 0.14--0.20       | 0.25--0.35                 | 0.89--0.92    | 0.16--0.22                   | Radiator surfaces (stay cool; low solar absorption, high IR emission) |
| Black paint (Aeroglaze Z306)        | 0.95             | 0.95                       | 0.89          | 1.07                         | Internal surfaces (maximise radiative exchange between components)    |
| Gold tape (2 mil Kapton + VDA)      | 0.22--0.25       | 0.25--0.30                 | 0.03--0.05    | 5.0--7.5                     | MLI outer layer, thermal isolation                                    |
| Bare aluminium (polished)           | 0.10--0.15       | 0.15--0.20                 | 0.03--0.05    | 2.5--4.0                     | Reflective surfaces, low emissivity                                   |
| Alodine (chromate conversion on Al) | 0.35--0.40       | 0.40--0.50                 | 0.12--0.16    | 2.5--3.0                     | Moderate thermal control, structural surfaces                         |
| Anodised aluminium (clear or black) | 0.30--0.50       | 0.35--0.55                 | 0.75--0.86    | 0.4--0.65                    | CubeSat external structure (standard finish per CDS)                  |
| MLI blanket (effective)             | 0.05--0.15       | 0.10--0.20                 | 0.02--0.05    | ~3                           | Thermal isolation of sensitive components                             |
| Solar cells (with cover glass)      | 0.75--0.92       | 0.78--0.92                 | 0.80--0.85    | ~1.0                         | SA surfaces (high absorption is unavoidable; cells get hot)           |
| OSR (Optical Solar Reflector)       | 0.05--0.08       | 0.08--0.12                 | 0.78--0.80    | 0.06--0.10                   | High-performance radiators (large S/C, GEO)                           |

**Key design insight:** A surface with low  $\alpha_s/\varepsilon$  (e.g., white paint: 0.2) stays cool because it reflects most solar energy but efficiently radiates thermal IR. A surface with high  $\alpha_s/\varepsilon$  (e.g., gold tape: 6.0) stays warm because it absorbs solar energy but barely radiates. This is why white paint is used on radiators and gold/MLI is used for insulation.

**Coating degradation:** UV radiation darkens most white paints over time, increasing  $\alpha_s$  while leaving  $\varepsilon$  nearly unchanged. This means  $\alpha_s/\varepsilon$  increases, and the surface gets hotter at EOL. The thermal engineer must design the hot case with BOL coatings (which give the highest  $\alpha_s$  ... wait -- actually BOL white paint has the *lowest*  $\alpha_s$ , so the **cold case** is more conservative at BOL, and the **hot case** is more conservative at EOL when  $\alpha_s$  has increased. This is a common source of confusion: BOL coatings give a colder cold case; EOL coatings give a hotter hot case.

## MLI (Multi-Layer Insulation) -- Construction and Physics

MLI blankets are the most common thermal insulation on spacecraft. They work by minimising both radiation and conduction heat transfer through multiple reflective layers separated by low-conductance spacers.

**Construction (typical MLI blanket):** 1. **Outer cover:** 1 mil (25  $\mu\text{m}$ ) aluminised Kapton (VDA -- Vapour Deposited Aluminium on one side). Provides mechanical protection and low solar absorptance. 2. **Inner reflective layers:** 10--20 layers of 0.25 mil (6  $\mu\text{m}$ ) double-aluminised Mylar (DAM). Each layer reflects IR radiation, and the vacuum gaps between layers have zero convection. 3. **Spacer material:** Dacron or Nomex netting between each DAM layer, preventing conductive contact between adjacent reflective sheets. 4. **Inner cover:** 1 mil aluminised Kapton, protecting the inner layers.

**Effective emissivity:** An ideal MLI blanket with  $N$  reflective layers has an effective emissivity of:

$$\varepsilon_{\text{eff}} = \frac{1}{2/\varepsilon_{\text{inner}} + (N-1)(2/\varepsilon_{\text{layer}} - 1)}$$

For 20 layers of DAM ( $\varepsilon_{\text{layer}} = 0.03$ ):  $\varepsilon_{\text{eff}} \approx 0.002$ . In practice, real MLI achieves  $\varepsilon_{\text{eff}} = 0.01$  --  $0.03$  due to seams, penetrations (harness, mounting), and edge effects. The ratio of actual to theoretical performance is typically 2--5x worse.

**CubeSat MLI challenges:** CubeSats have limited surface area and many penetrations (connectors, antennas, sensors, solar cells), making it difficult to achieve good MLI performance. Most CubeSats in LEO do not use MLI -- they rely on the moderate thermal environment (Earth IR provides a "warm floor") and surface coatings. MLI becomes essential for deep-space CubeSats (e.g., MarCO, CAPSTONE) or missions with sensitive payloads (IR detectors, laser systems).

## Heater Sizing

When passive thermal control cannot prevent a component from falling below its minimum temperature (typically during eclipse or safe mode), electrical heaters are required.

### Key Equations -- Heater Sizing

**Required heater power** (to maintain minimum temperature during worst cold case):

$$P_{\text{heater}} = \varepsilon \sigma A_{\text{rad}} T_{\text{min}}^4 - Q_{\text{environment,cold}} - Q_{\text{internal,cold}}$$

where  $T_{\text{min}}$  is the minimum allowable temperature of the component (converted to Kelvin).

**Heater types for CubeSats:** - **Kapton foil heaters:** Etched-foil resistance elements laminated between Kapton sheets. Flexible, thin (0.2--0.5 mm), lightweight (2--10 g each). Typical power: 0.5--5 W per heater. Bond directly to component surface with pressure-sensitive adhesive. - **Cartridge heaters:** Cylindrical, inserted into drilled holes. Higher power density but heavier and less common on CubeSats.

**Thermostat control:** Simple bimetallic thermostats (e.g., Honeywell Klaxon) switch heaters on/off at set temperatures (e.g., on at -5 degC, off at +5 degC). Mass: ~2 g each. For higher reliability, software-controlled heaters using temperature sensor feedback and EPS switches are preferred on modern CubeSats -- but this requires OBC to be running, which may not be the case in safe mode.

## Heat Pipes and Thermal Straps

**Heat pipes** are passive two-phase heat transfer devices that transport large amounts of thermal energy with very small temperature differences. They are widely used on larger spacecraft and are beginning to appear on 6U+ CubeSats.

**How a heat pipe works:** 1. Working fluid (ammonia, methanol, or water) evaporates at the hot end (evaporator), absorbing latent heat 2. Vapour travels through the hollow pipe core to the cold end (condenser) 3. At the condenser, vapour releases latent heat and condenses back to liquid 4. Liquid returns to the evaporator via capillary action in a wick structure (sintered metal, axial grooves, or screen mesh) 5. The process is continuous, passive (no moving parts, no power), and can transport 10--100 W across 20--50 cm with < 5 degC temperature difference

**Thermal conductance of a heat pipe:** Effective thermal conductivity is 10,000--100,000 W/m/K (compared to copper at 400 W/m/K and aluminium at 237 W/m/K). A 6 mm diameter ammonia heat pipe can transport ~30 W over 30 cm with < 3 degC gradient.

**Thermal straps** are flexible conductive links (braided copper, graphite fibre, or pyrolytic graphite sheet) used to conduct heat between components that cannot be rigidly connected (e.g., across a hinge or between a vibration-isolated payload and the spacecraft bus). Typical conductance: 0.5--5 W/K.

| Heat Transport Method | Conductance          | Mass              | Power   | Orientation Sensitivity                               | CubeSat Use         |
|-----------------------|----------------------|-------------------|---------|-------------------------------------------------------|---------------------|
| Aluminium conduction  | ~0.5--2 W/K per path | Part of structure | 0 W     | None                                                  | Always (inherent)   |
| Copper thermal strap  | 1--5 W/K             | 10--50 g          | 0 W     | None                                                  | Occasional (6U+)    |
| Heat pipe (ammonia)   | 5--50 W/K            | 20--100 g         | 0 W     | Gravity-dependent (must test in relevant orientation) | Rare (6U+, some 3U) |
| Pumped fluid loop     | 50--500 W/K          | 500+ g            | 5--20 W | None                                                  | Large S/C only      |

## Thermal Control Methods Summary

| Method           | Type    | Mass Impact                | Typical Use                                            | Key Design Parameter                         |
|------------------|---------|----------------------------|--------------------------------------------------------|----------------------------------------------|
| Surface coatings | Passive | Negligible                 | Always -- select $\alpha_s/\varepsilon$ ratio per face | $\alpha_s/\varepsilon$ ratio                 |
| MLI blankets     | Passive | 0.5--2.0 kg/m <sup>2</sup> | Insulate sensitive components from environment         | Number of layers, $\varepsilon_{\text{eff}}$ |
| Radiators        | Passive | Part of structure          | Reject internal waste heat to deep space               | Radiator area, $\varepsilon$ , view to space |



| Method         | Type    | Mass Impact         | Typical Use                                       | Key Design Parameter                    |
|----------------|---------|---------------------|---------------------------------------------------|-----------------------------------------|
| Heaters        | Active  | 0.005--0.02 kg each | Maintain minimum temp during eclipse/safe mode    | Power, thermostat set point             |
| Heat pipes     | Passive | 0.02--0.10 kg each  | Transport heat from source to radiator            | Working fluid, $Q_{\max}$ , orientation |
| Thermal straps | Passive | 0.01--0.05 kg each  | Flexible conductive link across joints/hinges     | Conductance (W/K)                       |
| Louvers        | Active  | 0.1--0.5 kg         | Variable-conductance radiators (rare on CubeSats) | Open/close temperature range            |

## ECSS Thermal Margins

[Source: ECSS-E-ST-31C, Table 5-1]

| Phase         | Hot Margin (above predicted max) | Cold Margin (below predicted min) |
|---------------|----------------------------------|-----------------------------------|
| Qualification | Predicted + 15 degC              | Predicted - 15 degC               |
| Acceptance    | Predicted + 10 degC              | Predicted - 10 degC               |
| Operating     | Predicted + 5 degC               | Predicted - 5 degC                |

These margins ensure that thermal model uncertainties (typically  $\pm 5$  -- 10 degC for simplified models,  $\pm 2$  -- 5 degC for detailed correlated models) do not cause in-orbit temperature exceedances.

### Worked Example -- 3U CubeSat Full Thermal Analysis

**Hot case (sunlit, all systems active, perihelion, EOL coatings):**

Simplified single-node model. 3U CubeSat, nadir-pointing.

- Sun-facing area ( +Z, zenith-facing 3U panel):  $A_{\text{sun}} = 0.034 \text{ m}^2$
- SA surfaces (sun-facing):  $\alpha_s = 0.85$ ,  $\varepsilon = 0.82$  (solar cell properties)
- Nadir face ( -Z):  $A_{\text{nadir}} = 0.034 \text{ m}^2$ , anodised Al ( $\alpha_s = 0.45$  EOL,  $\varepsilon = 0.82$ )
- Side faces (4x):  $A_{\text{side}} = 4 \times 0.01 \text{ m}^2 = 0.04 \text{ m}^2$ , anodised Al
- Internal dissipation (imaging mode):  $Q_{\text{int}} = 10.0 \text{ W}$

$Q_{\text{solar}} = 0.85 \times 0.034 \times 1414 = 40.9 \text{ W}$  (ouch -- but much of this is captured by the SA and converted to electricity, so the net thermal input from solar cells is

$$Q_{\text{solar,thermal}} = \alpha_s \times A \times S \times (1 - \eta_{\text{cell}}) = 0.85 \times 0.034 \times 1414 \times 0.705 = 28.8 \text{ W}$$

$$Q_{\text{albedo}} = 0.45 \times 0.034 \times 0.35 \times 1414 = 7.6 \text{ W}$$

$$Q_{\text{Earth IR}} = 0.82 \times 0.034 \times 270 = 7.5 \text{ W (nadir face, hot case Earth IR)}$$

$Q_{\text{int}} = 10.0 \text{ W}$  (but ~12.6 W of the 10 W load power ultimately becomes heat after doing useful work)

$$\text{Total } Q_{\text{in}} \approx 28.8 + 7.6 + 7.5 + 10.0 = 53.9 \text{ W}$$

Total radiating area (all 6 faces, minus SA area which is a net absorber):  $A_{\text{rad}} \approx 0.066 \text{ m}^2$   
(accounting for partial Earth blockage on nadir face)

Average emissivity:  $\varepsilon_{\text{avg}} \approx 0.82$

$$T_{\text{hot}} = \left( \frac{53.9}{0.82 \times 5.67 \times 10^{-8} \times 0.066} \right)^{0.25} = \left( \frac{53.9}{3.07 \times 10^{-9}} \right)^{0.25} \\ = (1.756 \times 10^{10})^{0.25} = 364 \text{ K} = +91 \text{ degC}$$

**This exceeds most component limits!** However, this simplified calculation overestimates temperature because it treats the satellite as a single isothermal node and includes solar cell thermal absorption on the zenith face. In practice: - The zenith face (solar cells) runs hotter than the bus - Internal components are conductively coupled to all faces, including the cold nadir face - A multi-node model typically predicts peak internal temperatures of +40 to +55 degC for this scenario

**Thermal engineer's response:** If the single-node calculation exceeds 60 degC, a detailed multi-node thermal model is required. The simplified calculation is a screening tool, not a design tool.

**Cold case (eclipse, safe mode, aphelion, BOL coatings):**

- No solar flux, no albedo
- Earth IR only:  $Q_{\text{Earth IR}} = 0.82 \times 0.034 \times 200 = 5.58 \text{ W}$  (cold case:  $200 \text{ W/m}^2$ )
- Internal dissipation (safe mode):  $Q_{\text{int}} = 1.5 \text{ W}$  (OBC + heater)
- Total  $Q_{\text{in}} = 5.58 + 1.5 = 7.08 \text{ W}$

$$T_{\text{cold}} = \left( \frac{7.08}{0.82 \times 5.67 \times 10^{-8} \times 0.070} \right)^{0.25} \\ = \left( \frac{7.08}{3.26 \times 10^{-9}} \right)^{0.25} = (2.172 \times 10^9)^{0.25} = 216 \text{ K} = -57 \text{ degC}$$

**This is too cold** for Li-ion batteries (min -20 degC operating, min 0 degC charging) and most COTS electronics (min -40 degC).

**Action:** Add battery heater. To maintain battery at  $T_{\text{min}} = -10 \text{ degC} = 263 \text{ K}$ :

Need additional heat input:  $Q_{\text{heater}} = \varepsilon \sigma A_{\text{rad}} T_{\text{min}}^4 - Q_{\text{other}}$

$$= 0.82 \times 5.67 \times 10^{-8} \times 0.070 \times 263^4 - 7.08 = 3.26 \times 10^{-9} \times 4.78 \times 10^9 - 7.08 = 15.6 - 7.08 = 8.5 \text{ W}$$

**Problem:** 8.5 W heater in eclipse exceeds the battery capacity. **Resolution:** This is an isothermal whole-spacecraft calculation. In reality, the battery is inside the bus, partially insulated by the structure and surrounding boards. A targeted heater of 0.5--1.0 W directly on the battery pack, with some MLI wrapping, is typically sufficient to keep the battery above -10 degC in a 35-minute eclipse. The structure's thermal mass (aluminium at  $c_p = 900 \text{ J/kg/K}$ ) provides significant thermal inertia -- a 1 kg 1U CubeSat cooling from +20 degC at 7 W net loss drops only about 16 degC in 35 minutes.

**Transient check:**  $\Delta T = \frac{Q_{\text{net}} \times t}{m \times c_p} = \frac{(7.08 - 0) \times 35 \times 60}{5.0 \times 900} = \frac{14,868}{4500} = 3.3 \text{ degC per 35-min eclipse}$

Wait -- in eclipse  $Q_{\text{out}} > Q_{\text{in}}$ : net cooling rate =  $\varepsilon \sigma A_{\text{rad}} T^4 - Q_{\text{in,eclipse}}$ . At  $T = 293 \text{ K}$  (20 degC):

$$Q_{\text{out}} = 0.82 \times 5.67 \times 10^{-8} \times 0.070 \times 293^4 = 3.26 \times 10^{-9} \times 7.37 \times 10^9 = 24.0 \text{ W}$$

$$Q_{\text{net cooling}} = 24.0 - 7.08 = 16.9 \text{ W}$$

$$\Delta T = \frac{16.9 \times 2100}{5.0 \times 900} = \frac{35,490}{4500} = 7.9 \text{ degC drop in 35 minutes}$$

So from +20 degC, the satellite cools to about +12 degC after one eclipse -- well within limits. **No heater needed for the bus; battery heater only if battery is thermally isolated from bus.**

**ECSS margin check -- Payload CCD:** Predicted maximum temperature of payload CCD = 42 degC. - Operating limit = 50 degC. Margin = 50 - 42 = 8 degC > 5 degC. **Pass.** - Qualification test: must test at 42 + 15 = 57 degC. If qualification limit is 60 degC: **Pass.**

Predicted minimum temperature of battery = -8 degC during worst eclipse. - Operating limit = -10 degC. Margin = -8 - (-10) = 2 degC < 5 degC. **Fail -- inadequate margin.** - **Action:** Add heater (0.5 W survival heater with thermostat set to -5 degC), or add MLI around battery pack.

## 1U Worked Example: UniSat-1

### Power Sizing: Body-Mounted Only

UniSat-1 uses body-mounted solar cells on all five sun-exposed faces (the sixth face mounts the deployment switch interface). With no deployable panels, the power system is simpler, lighter, and cheaper -- but severely power-limited.

#### Worked Example -- UniSat-1 Solar Array Sizing

**Given:** Body-mounted cells on 5 faces of a 1U (100 x 100 mm each). ISS orbit: 400 km, 51.6 deg inclination, 92.4 min period, 56 min sunlight, 36 min eclipse.  $P_{\text{eclipse}} = 0.5 \text{ W}$  (OBC only),  $P_{\text{peak,sunlight}} = 1.2 \text{ W}$  (science + downlink overlap avoided by scheduling). Mission lifetime = 6 months. Cell temperature = 55 degC (body-mounted cells run cooler on 1U due to better thermal coupling to bus mass).

**Step 1 -- Effective illuminated area:** At any given time in LEO with passive magnetic attitude (slow tumble ~1 deg/s), on average only ~1.5 faces are well-illuminated. Effective average area:  
 $A_{\text{eff}} \approx 1.5 \times (0.10 \times 0.10) = 0.015 \text{ m}^2$

**Step 2 -- SA BOL power (with temperature derating):**  $L_T = 1 + (-0.0019)(55 - 28) = 0.949$   
 $P_{\text{SA,BOL}} = \eta_{\text{cell}} \times L_T \times S \times A_{\text{eff}} \times f_{\text{pack}} = 0.295 \times 0.949 \times 1361 \times 0.015 \times 0.80 = 4.57 \text{ W}$   
 (illuminated peak)

**Step 3 -- Orbit-average power available:**

$$P_{\text{avg,avail}} = P_{\text{SA,BOL}} \times \frac{t_{\text{sun}}}{T} \times \eta_{\text{EPS}} = 4.57 \times \frac{56}{92.4} \times 0.85 = 2.35 \text{ W}$$

After 6-month degradation ( $(1 - 0.025)^{0.5} = 0.987$ ):  $P_{\text{avg,EOL}} = 2.35 \times 0.987 = 2.32 \text{ W}$

**Step 4 -- Power demand (orbit-average):** From Session 2.4:  $P_{\text{avg,demand}} = 0.68 \text{ W}$ .

**Power margin:**  $2.32 - 0.68 = 1.64 \text{ W}$  (**71% margin**). Even with conservative geometry assumptions, the link closes comfortably.

**Note on body-mounted vs tumbling:** The key uncertainty in 1U body-mounted power is the attitude. With passive magnetic stabilisation, the satellite aligns roughly with Earth's magnetic field, providing more predictable illumination than a random tumble. However, the effective area varies significantly around the orbit as the B-field direction changes with latitude. The 1.5-face average is conservative. A Monte Carlo simulation of illumination geometry over many orbits typically yields a more optimistic 1.7--1.9 effective face average.

#### Worked Example -- UniSat-1 Battery Sizing

**Given:**  $P_{\text{eclipse}} = 0.5 \text{ W}$ ,  $t_{\text{eclipse}} = 36 \text{ min} = 0.60 \text{ h}$ ,  $\text{DOD} = 0.50$  (acceptable for 6-month mission: ~2,740 cycles),  $\eta = 0.95$ .

$$E_{\text{bat}} = \frac{0.5 \times 0.60}{0.50 \times 0.95} = \frac{0.30}{0.475} = 0.63 \text{ Wh}$$

With margin: specify minimum **10 Wh** (standard GomSpace NanoPower P31u battery pack -- this is the smallest available COTS battery with flight heritage).

**Cycle count check:** 6 months at 15 orbits/day = 2,740 eclipses. At 50% DOD, Li-ion cells comfortably survive > 2,000 cycles. **Pass.**

**Actual operating DOD:** With 10 Wh battery and 0.63 Wh per eclipse demand, actual DOD =  $0.63 / 10 = 6.3\%$  per eclipse. At this DOD, cycle life exceeds 100,000 cycles. **Battery degradation is negligible** over the 6-month mission.

**Key insight:** The 10 Wh battery is massively oversized for the actual eclipse demand. This is common in 1U missions -- the minimum COTS battery available provides far more capacity than needed. The excess capacity provides excellent margin and enables recovery from anomalies (multiple missed sunlit periods).

### Thermal: Passive Only

UniSat-1 uses no heaters, no MLI, and no active thermal control. This is justified by three factors:

1. **Low altitude (400 km):** Strong Earth IR flux ( $\sim 240 \text{ W/m}^2$ ) provides a warm floor, preventing extreme cold cases
2. **Short mission (6 months):** No long-term coating degradation to worry about
3. **Tolerant components:** COTS electronics typically operate from -20 degC to +60 degC; the 400 km LEO thermal environment stays within -10 degC to +45 degC for a 1U with standard aluminium/anodised surfaces

### Quick Thermal Check -- UniSat-1 Cold Case (Transient)

Worst eclipse, all subsystems off except OBC (0.5 W internal dissipation): - Earth IR absorbed:

$$\epsilon \times A_{\text{nadir}} \times q_{\text{IR}} = 0.85 \times 0.01 \times 240 = 2.04 \text{ W} - \text{Internal dissipation: } 0.5 \text{ W} - \text{Total heat in: } 2.54 \text{ W}$$

Steady-state temperature (if eclipse were infinite):

$$T_{\text{steady}} = (2.54 / (0.85 \times 5.67 \times 10^{-8} \times 0.045))^{0.25} = 195 \text{ K} = -78 \text{ degC}$$

**This looks alarming** -- but the eclipse is only 36 minutes, and the thermal mass prevents the satellite from reaching steady state.

**Transient analysis:** Starting at  $T_0 = +15 \text{ degC}$  (293 K) at eclipse entry:

Net cooling rate at 293 K:

$$Q_{\text{rad,out}} = 0.85 \times 5.67 \times 10^{-8} \times 0.045 \times 293^4 = 2.17 \times 10^{-9} \times 7.37 \times 10^9 = 16.0 \text{ W}$$

$$\text{Net cooling: } 16.0 - 2.54 = 13.5 \text{ W}$$

$$\text{Temperature drop: } \Delta T = \frac{Q_{\text{net}} \times t}{m \times c_p} = \frac{13.5 \times 36 \times 60}{1.0 \times 900} = \frac{29,160}{900} = 32 \text{ degC}$$

But this is a linear approximation -- as the satellite cools, the radiation rate drops as  $T^4$ , so the actual cooling slows. A more accurate estimate gives  $\Delta T \approx 20$  -- 25 degC, resulting in a minimum temperature of about -5 to -10 degC.

**Conclusion:** The 1U at 400 km reaches approximately -5 to -10 degC during worst-case eclipse, starting from a warm sunlit entry. This is within COTS operating limits (-20 degC to +60 degC for most components) and within battery operating range (-20 degC to +60 degC for discharge). **No heaters**

**needed.** The ISS orbit's relatively short eclipse (36 min vs 35 min for SSO) and the strong Earth IR flux at 400 km make passive thermal control viable for a 1U mission.

## 5. Real Mission Examples

### Planet SuperDove EPS

| Parameter        | Value                                              | Design Rationale                                                    |
|------------------|----------------------------------------------------|---------------------------------------------------------------------|
| Form factor      | 3U+, ~5 kg                                         | Flock constellation; P-POD compatible                               |
| SA configuration | Body-mounted + two deployable wings                | ~25 W BOL needed for continuous imaging + S-band downlink           |
| SA cells         | Triple-junction GaAs (Spectrolab or SolAero)       | Standard space-grade, 29.5% AM0                                     |
| Battery          | Li-ion, ~20 Wh (2S2P 18650)                        | Supports ~3.5 W eclipse load for 35 min at < 20% DOD                |
| Bus voltage      | Unregulated 7.2--8.4 V (2S) + regulated 3.3 V, 5 V | Standard CubeSat EPS architecture                                   |
| Peak demand      | ~18 W (imaging mode)                               | Multi-spectral imager + star tracker + reaction wheels + OBC        |
| Orbit            | 475 km SSO, ~94 min period                         | Optimal for EO: sun-synchronous for consistent lighting             |
| Eclipse          | ~35 min max, ~3 W demand                           | OBC + AOCS only during eclipse; no imaging or downlink              |
| Thermal          | Body-mounted radiator panels + battery heater      | Passive thermal control; battery heater for eclipse charging margin |

[Source: Planet Labs conference presentations; Salas et al., "SuperDove Constellation," SSC 2021]

### CAPSTONE Thermal Design

NASA's CAPSTONE (12U, 25 kg) operates in a near-rectilinear halo orbit (NRHO) around the Moon with extreme thermal cycling:

- Perilune: strong Earth/Moon IR + solar
- Apolune: deep space cold, long shadow periods (up to 12+ hours)
- Thermal control: MLI wrapping (10-layer DAM blankets), heaters on propulsion lines (to prevent propellant freezing), passive radiator panels with white paint (AZ-93)
- Battery heaters: 5 W total, thermostatically controlled, critical for survival during long eclipses
- Operating temperature range: -20 degC to +50 degC for electronics; propulsion lines maintained above +5 degC

[Source: Advanced Space, "CAPSTONE Design Overview," SmallSat Conference 2022]

## 6. SpaceCDF Exercise

### Instructions

1. **Run the design** in SpaceCDF if not already converged
2. **Dashboard** -- review power KPIs: - Is power margin positive in **all** modes (sunlight, eclipse, safe)? - What is the orbit-average power demand? - What SA configuration was selected (body/deployable)?
3. **Timing Budget** card -- review mode durations: - Does the duty cycle match your ConOps modes? - Is eclipse time consistent with your orbit calculation from Session 2.3?
4. **Engineering Budgets** -- review the power waterfall: - Which subsystem is the largest power consumer in each mode? - Where could power be reduced if the budget is tight?
5. **Parametric** tab -- review thermal predictions: - Hot case temperature prediction - Cold case temperature prediction - Any exceedances?

### Worksheet 3.1 Tasks

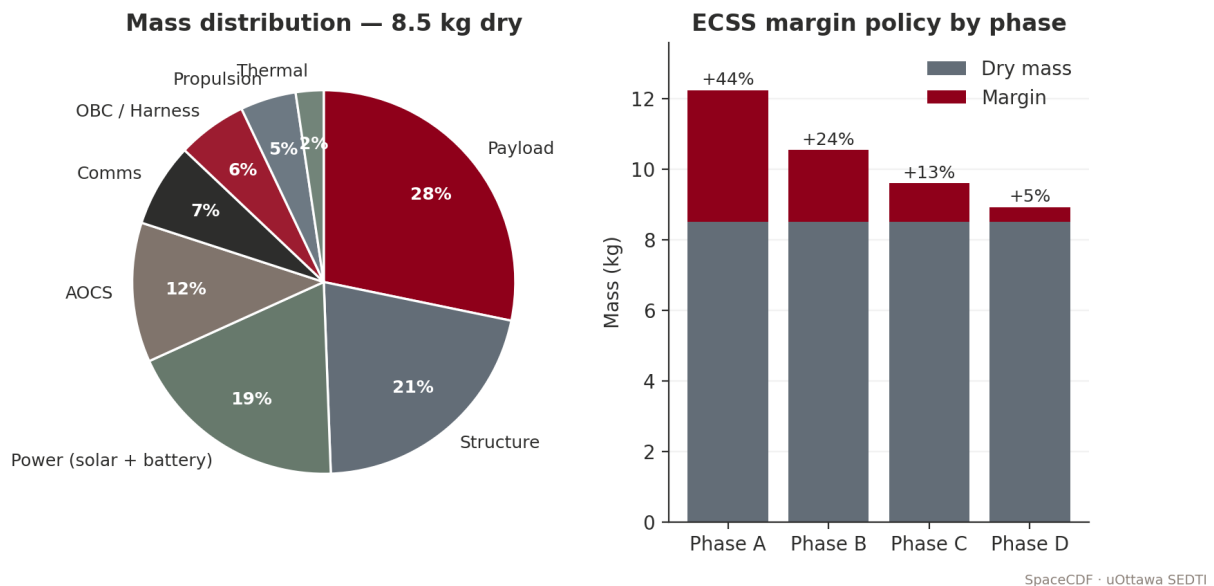
1. Size the solar array for your mission (show all 5 calculation steps, including temperature derating)
2. Size the battery (show calculation with DOD justification and cycle-life verification)
3. Compute orbit-average power using duty cycle table
4. Identify hot case and cold case conditions for your orbit
5. Check thermal margins against ECSS requirements
6. Identify the dominant thermal concern for your mission (hot case or cold case) and propose a mitigation

## Session Summary

| Topic                 | Key Takeaway                                                                                                       |
|-----------------------|--------------------------------------------------------------------------------------------------------------------|
| Solar cell physics    | Triple-junction GaAs (InGaP/GaAs/Ge) achieves 28--30% AM0; multi-junction stacking captures broader solar spectrum |
| Temperature effects   | Cells lose ~0.19%/degC (relative); operating at 65 degC costs ~7% power vs STC                                     |
| Degradation           | Radiation damage: $(1 - \delta)^n$ , $\delta \approx 2.5\%/yr$ LEO; cover glass mitigates proton/electron damage   |
| EPS architecture      | MPPT + regulated bus is CubeSat standard; MPPT extracts 10--15% more power than DET                                |
| SA sizing             | $P_{SA} = P_{peak} + P_{recharge}$ ; derate for degradation $(1 - \delta)^n$ and temperature $L_T$                 |
| SA area               | $A = P_{BOL} / (\eta \cdot S \cdot \cos \theta \cdot f_{pack} \cdot f_{cover})$                                    |
| Battery chemistry     | Li-ion (LCO/NMC): 150--200 Wh/kg packaged; 3.0--4.2 V per cell; CC-CV charging                                     |
| Battery sizing        | $E = P_{ecl} \cdot t_{ecl} / (DOD \cdot \eta)$ ; DOD 20--30% for multi-year LEO                                    |
| Battery failure modes | Thermal runaway (overcharge), lithium plating (cold charge), capacity imbalance (series cells)                     |
| SA power reference    | Body-mounted: 2--12 W; single deploy: 4--30 W; dual deploy: 25--48 W                                               |

| Topic                        | Key Takeaway                                                                                                                          |
|------------------------------|---------------------------------------------------------------------------------------------------------------------------------------|
| Thermal physics              | No convection in vacuum; radiation ( $\varepsilon\sigma T^4$ ) is only mechanism for heat rejection                                   |
| Thermal balance              | $Q_{\text{in}} = Q_{\text{out}}$ ; solve for $T = (Q/\varepsilon\sigma A)^{1/4}$ (single-node); multi-node for real design            |
| $\alpha_s/\varepsilon$ ratio | Low ratio = cold surface (radiator); high ratio = warm surface (insulation)                                                           |
| MLI construction             | VDA Kapton outer + 10--20 DAM layers + Dacron spacers; $\varepsilon_{\text{eff}} = 0.01 \text{ -- } 0.03$                             |
| Heater sizing                | $P_{\text{heater}} = \varepsilon\sigma AT_{\text{min}}^4 - Q_{\text{env}} - Q_{\text{int}}$ ; Kapton foil heaters, thermostat control |
| Thermal margins              | ECSS: +/-5 degC operating, +/-10 degC acceptance, +/-15 degC qualification                                                            |
| Transient effects            | Thermal mass ( $mc_p$ ) prevents reaching steady state during short eclipses; 1U at 400 km cools ~20--25 degC in 36 min eclipse       |

## Session 3.2: Attitude and Orbit Control System (AOCS)



Mass distribution and ECSS margin policy.

**Expected reading before this session.** Cal Poly CDS Rev 14 §3 (≈ 30 min); ECSS-E-ST-32C §4. Sarafin Ch. 3 – 5.

**Prerequisites:** Sessions 2.1--2.4 and 3.1 (requirements, orbit, power defined) **SpaceCDF Tabs:** Dashboard (AOCS KPI), Pointing Budget, Architecture (AOCS)

### References

- Wertz, Everett & Puschell, *Space Mission Engineering: The New SMAD*, 2011, Ch. 11.1 (ADCS)
- Sidi, *Spacecraft Dynamics and Control*, 1997, Ch. 4--9
- Markley & Crassidis, *Fundamentals of Spacecraft Attitude Determination and Control*, 2014
- ECSS, *ECSS-E-ST-60-10C: Control Performance*, 2008
- ECSS, *ECSS-E-ST-60-20C: Star Tracker Performance Testing*, 2019
- Wertz, *Space Mission Analysis and Design*, 3rd ed., 1999, Ch. 11 (ADCS)
- Hughes, *Spacecraft Attitude Dynamics*, 1986
- Blue Canyon Technologies, *XACT ADCS Datasheet*, 2023
- CubeSpace, *ADCS Product Catalogue*, 2023

### Learning Objectives

By the end of this session, participants will be able to:

1. Explain the distinction between attitude determination and attitude control and the physics of each sensor/actuator type
2. Select AOCS hardware architecture based on pointing requirements



3. Compute a pointing error budget using root-sum-square (RSS) combination
4. Calculate disturbance torques from gravity gradient, aerodynamic drag, solar radiation pressure, and residual magnetic dipole
5. Size actuators (reaction wheels, magnetorquers) for the computed disturbance environment with worked equations
6. Explain the physics of momentum storage, saturation, and desaturation
7. Compare CMGs vs reaction wheels and articulate when each is appropriate
8. Verify AOCS design against requirements using SpaceCDF's pointing budget tool

## 1. AOCS Fundamentals

[Source: SMAD, Ch. 11.1; Markley & Crassidis, Ch. 1--3]

The AOCS performs two distinct functions:

| Function                           | Purpose                                                         | Hardware                                                                                |
|------------------------------------|-----------------------------------------------------------------|-----------------------------------------------------------------------------------------|
| <b>Attitude Determination (AD)</b> | Know the spacecraft's orientation relative to a reference frame | Sensors: star tracker, sun sensors, magnetometer, gyroscope, Earth sensor, GPS receiver |
| <b>Attitude Control (AC)</b>       | Change or maintain the spacecraft's orientation                 | Actuators: reaction wheels, magnetorquers, thrusters, control moment gyros              |

**The attitude state:** A spacecraft's attitude is its orientation in 3D space relative to a reference frame (typically J2000 Earth-centred inertial, or the local-vertical/local-horizontal frame for nadir-pointing missions). The attitude is described by a rotation (e.g., quaternion, direction cosine matrix, or Euler angles) plus angular velocity  $\vec{\omega}$ . Euler's equation of rotational motion governs the dynamics:

$$\mathbf{I}\dot{\vec{\omega}} + \vec{\omega} \times (\mathbf{I}\vec{\omega}) = \vec{T}_{\text{external}} + \vec{T}_{\text{control}}$$

where  $\mathbf{I}$  is the spacecraft inertia tensor,  $\vec{T}_{\text{external}}$  is the sum of disturbance torques, and  $\vec{T}_{\text{control}}$  is the actuator torque. The  $\vec{\omega} \times (\mathbf{I}\vec{\omega})$  term is the gyroscopic coupling -- it means that rotating about one axis can induce motion about other axes if the inertia tensor is not spherically symmetric.

### Attitude Sensors -- How They Work

#### Star Trackers

A star tracker is the most accurate attitude sensor available, providing absolute attitude knowledge to arcsecond-level accuracy.

**How it works:** 1. A CMOS or CCD detector (typically 1024x1024 to 2048x2048 pixels) images a patch of sky through a wide-angle lens (typically 15--25 deg FOV) 2. The onboard processor detects bright point sources (stars) in the image, computing their centroid positions to sub-pixel accuracy using Gaussian fitting 3. The processor matches the observed pattern of star positions against an onboard star catalogue (typically Hipparcos or Tycho-2, containing 3,000--10,000 stars, stored in a k-d tree or hash table for fast lookup) 4. The "lost in space" algorithm identifies the star pattern without any prior attitude knowledge (first acquisition), typically taking 1--5 seconds 5. Once identified, the processor computes a quaternion rotation from the catalogue (inertial) frame to the camera (body) frame 6. In tracking mode, the processor tracks known stars frame-to-frame, providing updates at 1--10 Hz with 3--10 arcsec accuracy (1-sigma, per axis)

**Key specifications:**

| Parameter                     | Typical CubeSat Star Tracker                                            | Large S/C Star Tracker        |
|-------------------------------|-------------------------------------------------------------------------|-------------------------------|
| Accuracy (1-sigma, boresight) | 5--15 arcsec                                                            | 0.5--3 arcsec                 |
| Accuracy (1-sigma, roll)      | 30--100 arcsec                                                          | 5--20 arcsec                  |
| FOV                           | 10--15 deg circular                                                     | 15--25 deg circular           |
| Update rate                   | 2--5 Hz                                                                 | 5--20 Hz                      |
| Sensitivity                   | Stars to magnitude 6--7                                                 | Stars to magnitude 8--10      |
| Mass                          | 50--350 g                                                               | 1--5 kg                       |
| Power                         | 0.5--2 W                                                                | 5--15 W                       |
| Products                      | Blue Canyon NST (350g), Sinclair SS-411 (90g), CubeSpace CubeStar (50g) | Sodern Hydra, Leonardo AA-STR |

[Source: Blue Canyon NST datasheet; CubeSpace CubeStar datasheet; ECSS-E-ST-60-20C]

**Exclusion zones:** Star trackers cannot operate when bright objects are in or near the FOV: - **Sun:** Exclusion angle typically 25--45 deg (direct sunlight saturates the detector and can cause permanent damage to some sensor types) - **Earth (illuminated limb):** Exclusion angle typically 25--35 deg (Earth's brightness overwhelms star signals) - **Moon:** Exclusion angle typically 10--15 deg - **Stray light:** Internal reflections from nearby spacecraft structure can create false stars

**Implication for mission design:** A nadir-pointing spacecraft in LEO always has the Earth within ~65 deg of one hemisphere. The star tracker must be mounted on a face that never points toward Earth (typically the zenith or anti-velocity face). If the Sun is near the orbital plane (beta angle near 0), the star tracker may be periodically blinded. Two star trackers mounted on different faces provide redundancy and eliminate single-axis exclusion zone gaps.

**Why star trackers are the most accurate:** Other sensors measure vectors to specific objects (Sun, Earth, magnetic field) -- each provides only 2 of the 3 attitude degrees of freedom (direction but not roll around that vector). A star tracker measures multiple star directions simultaneously, providing a full 3-axis attitude fix from a single measurement. The accuracy is limited by optical diffraction, centroiding noise, and catalogue accuracy -- all of which are extremely well characterised.

### Sun Sensors

Sun sensors determine the direction to the Sun, providing 2-axis attitude information (the Sun vector in the body frame). They are simple, reliable, radiation-tolerant, and low-power.

**Coarse sun sensors (photodiode-based):** A coarse sun sensor consists of one or more photodiodes behind a mask or window. The photocurrent is proportional to the cosine of the incidence angle:

$$I = I_0 \cos(\theta)$$

A set of 6 coarse sun sensors (one per face of the spacecraft) determines the Sun direction to 2--5 deg accuracy by comparing the photocurrents from each face. The face with the highest current is sun-facing; the ratio between adjacent faces gives the angle.

**Fine sun sensors (linear array or quadrant detector):** A fine sun sensor uses a slit mask above a linear photodiode array (similar to a miniature sundial). Sunlight passes through the slit and illuminates a specific

position on the array, which is proportional to the incidence angle. Fine sun sensors achieve 0.1–1.0 deg accuracy.

| Type                          | Accuracy      | FOV         | Mass    | Power     | Products                                    |
|-------------------------------|---------------|-------------|---------|-----------|---------------------------------------------|
| Coarse (photodiode)           | 2–5 deg       | ~hemisphere | 1–5 g   | < 1 mW    | NewSpace NCSS-SA05, Solar MEMS nanoSSOC-A60 |
| Fine (analog, slit+array)     | 0.1–0.5 deg   | 60–120 deg  | 5–30 g  | 10–50 mW  | NewSpace NFSS-411, Solar MEMS nanoSSOC-D60  |
| Digital (APS detector + slit) | 0.01–0.05 deg | 60–120 deg  | 30–50 g | 50–200 mW | TNO micro digital sun sensor                |

[Source: NewSpace Systems NFSS-411 datasheet; Solar MEMS nanoSSOC datasheets]

**Why every spacecraft needs sun sensors:** Sun sensors are the only sensor guaranteed to work in any attitude, at any rate, in any environment (LEO, GEO, deep space). They are the primary sensor for safe mode: when the spacecraft tumbles, the OBC reboots, or the star tracker is blinded, the sun sensors can determine the Sun direction and enable the spacecraft to orient its solar arrays for power generation. Without sun sensors, a spacecraft that enters safe mode may not recover power.

### Magnetometers

A magnetometer measures the local geomagnetic field vector  $\vec{B}$  in the body frame. By comparing the measured  $\vec{B}$  to a model of Earth's magnetic field (IGRF -- International Geomagnetic Reference Field) at the known orbit position, the spacecraft attitude can be determined.

**How it works:** - **Fluxgate magnetometer:** A ferromagnetic core is periodically driven into saturation by an excitation coil. The presence of an external magnetic field creates an asymmetry in the saturation waveform, which is detected by a sense coil. Three orthogonal fluxgate elements provide the 3-axis field vector. Resolution: 1–10 nT. Accuracy:  $\pm 200$ –500 nT (limited by spacecraft residual magnetic dipole contamination). - **Magnetoresistive (AMR/GMR):** Thin-film magnetic sensors. Smaller and cheaper than fluxgates but noisier and less stable. Common in COTS CubeSat magnetometers.

**Attitude determination from magnetometers:** - A single magnetometer measurement provides the magnetic field direction in the body frame (2 DOF, analogous to a sun sensor providing the Sun direction) - Comparing with the IGRF model gives attitude, but accuracy is limited by: - IGRF model uncertainty (~100 nT in LEO) - Spacecraft residual magnetic dipole contamination (can be 1000+ nT close to electronics) - Only 2 DOF per measurement (no roll determination around the field vector) - Magnetometer-only attitude determination: ~5–10 deg accuracy - Magnetometer + sun sensor (combined): ~1–3 deg accuracy (the two independent vectors provide a full 3-axis solution via TRIAD or q-method)

**Residual dipole contamination:** Every electronic circuit creates a magnetic field. Current loops, permanent magnets in motors/speakers, magnetised structural components, and battery cells all contribute to the spacecraft's residual magnetic dipole. This contaminates the magnetometer reading. Mitigation: mount the magnetometer on a deployable boom (10–30 cm from the bus), use magnetically clean design practices (twisted pairs, balanced current loops), and perform a residual dipole calibration in orbit by comparing magnetometer readings during eclipse (no solar array current) vs sunlit.

| Parameter | Typical CubeSat Magnetometer                              |
|-----------|-----------------------------------------------------------|
| Range     | $\pm 60,000$ nT (sufficient for LEO: B ~20,000–50,000 nT) |

| Parameter                    | Typical CubeSat Magnetometer                  |
|------------------------------|-----------------------------------------------|
| Resolution                   | 10--100 nT                                    |
| Accuracy (after calibration) | 200--1000 nT                                  |
| Mass                         | 5--30 g                                       |
| Power                        | 10--50 mW                                     |
| Products                     | NewSpace NMAG, PNI RM3100, Honeywell HMC5883L |

### Gyroscopes (Rate Sensors)

Gyroscopes measure angular velocity  $\vec{\omega}$  rather than absolute attitude. They provide high-bandwidth rate information for control loops and bridge the gap between star tracker updates.

**Types used on CubeSats:** - **MEMS gyroscope:** Vibrating structure (tuning fork or ring) whose Coriolis force is proportional to rotation rate. Small, cheap (< 50 EUR per axis), low power. Bias drift: 1--10 deg/hr. Products: InvenSense MPU-6050, Analog Devices ADIS16265. - **MEMS IMU (6-axis):** Combined 3-axis accelerometer + 3-axis gyroscope. Mass: 5--15 g. Very common in CubeSats. Products: Sensoror STIM300 (high-end), InvenSense ICM-20948 (COTS). - **Fibre optic gyroscope (FOG):** Much better stability (bias drift 0.01--1 deg/hr) but larger, heavier (200+ g), and more expensive. Used on high-end CubeSats and small satellites.

**Why gyroscopes drift:** MEMS gyroscopes have a non-zero bias (a constant offset in the measured rate even when stationary) that changes with temperature and time. Integrating angular rate to get attitude

( $\theta = \int \omega dt$ ) accumulates this bias error linearly. A 1 deg/hr bias drift means the attitude estimate drifts 1

deg per hour. Star trackers provide absolute attitude corrections that reset this drift -- the combination of star tracker (low rate, absolute) + gyroscope (high rate, relative) via a Kalman filter is the standard approach for high-performance attitude determination.

### GPS Receivers for Orbit Determination

GPS receivers determine the spacecraft's **position and velocity** (orbit determination), not attitude (though multi-antenna GPS can provide coarse attitude).

**How it works in LEO:** - GPS satellites orbit at ~20,200 km altitude; LEO spacecraft orbit below them at 300--800 km - GPS signals travel downward through the ionosphere to the LEO receiver - The receiver must use a specialised correlator that handles Doppler shifts up to ~40 kHz (LEO relative velocity ~7.5 km/s vs GPS satellite velocity ~3.9 km/s) - Typical accuracy: 5--20 m position, 0.1--0.5 m/s velocity (single frequency, C/A code) - Dual-frequency GPS with carrier phase: sub-meter position accuracy

**Why GPS matters for AOCS:** Accurate orbit knowledge is needed for: - Nadir pointing (must know where "down" is, which requires knowing position) - Ground target tracking (must know position to compute pointing angles) - IGRF evaluation (magnetometer attitude determination needs position input) - Orbit manoeuvre planning

| Parameter         | Typical CubeSat GPS Receiver                             |
|-------------------|----------------------------------------------------------|
| Position accuracy | 5--20 m (C/A code)                                       |
| Velocity accuracy | 0.1--0.5 m/s                                             |
| Time accuracy     | 100 ns                                                   |
| Altitude limit    | Typically 600 km (must verify ITAR/COCOM limits removed) |

| Parameter | Typical CubeSat GPS Receiver                          |
|-----------|-------------------------------------------------------|
| Mass      | 15--30 g                                              |
| Power     | 0.5--1.0 W                                            |
| Products  | SkyFox Labs piNAV-NG, NovAtel OEM719, Hemisphere V200 |

[Source: SkyFox Labs piNAV datasheet; NovAtel OEM7 specifications]

### AOCS Architecture Selection by Pointing Requirement

| Pointing Requirement | Architecture                       | Sensors                                                          | Actuators                          | Mass     | Power  | Cost     |
|----------------------|------------------------------------|------------------------------------------------------------------|------------------------------------|----------|--------|----------|
| > 5 deg              | Passive magnetic                   | None (or 1 magnetometer)                                         | Permanent magnet + hysteresis rods | ~0.05 kg | 0 W    | ~2 kEUR  |
| 2--5 deg             | B-dot detumble + magnetic pointing | 3-axis magnetometer + coarse sun sensors                         | 3-axis magnetorquers               | ~0.10 kg | 0.2 W  | ~8 kEUR  |
| 0.1--2 deg           | Active 3-axis (RW + sensors)       | Fine sun sensors + magnetometer + (optional gyro)                | 3--4 RW + 3 MTQ for desaturation   | ~0.50 kg | 2--4 W | ~35 kEUR |
| < 0.1 deg            | Fine pointing (RW + star tracker)  | Star tracker + fine sun sensors + magnetometer + gyro            | 4 RW + 3 MTQ                       | ~0.80 kg | 3--5 W | ~55 kEUR |
| < 0.01 deg           | Very fine pointing                 | Dual star trackers + MEMS gyro + fine sun sensors + magnetometer | 4 RW + 3 MTQ                       | ~1.20 kg | 4--6 W | ~80 kEUR |

#### Real mission examples:

| Mission                             | Form Factor | Pointing Req           | Architecture                      | AOCS Mass | Key Sensor                   |
|-------------------------------------|-------------|------------------------|-----------------------------------|-----------|------------------------------|
| <b>Astrocast</b> (3U, IoT)          | 3U          | ~5 deg                 | MTQ + sun sensors + magnetometer  | 0.1 kg    | Sun sensors                  |
| <b>Planet SuperDove</b> (3U+, EO)   | 3U+         | ~0.1 deg               | 4 RW + ST + 3 MTQ + 6 sun sensors | 0.8 kg    | Blue Canyon NST star tracker |
| <b>CAPSTONE</b> (12U, cislunar nav) | 12U         | ~0.05 deg              | 4 RW + ST + sun sensors + IMU     | ~1.0 kg   | Star tracker + IMU           |
| <b>ASTERIA</b> (6U, exoplanet)      | 6U          | ~0.003 deg (10 arcsec) | 4 RW + ST + fine guidance sensor  | ~1.2 kg   | Custom fine guidance camera  |

[Source: Pong et al., "ASTERIA: Achieving 10-arcsecond Pointing on a 6U CubeSat," SSC 2018]

## 2. Disturbance Torques

In LEO, four external torques disturb the spacecraft attitude. The AOCS must counteract them continuously. Understanding the source and magnitude of each disturbance is essential for sizing actuators.

[Source: SMAD, Ch. 11.1; Sidi, Ch. 5; Wertz 1999, Ch. 11]

### Gravity Gradient Torque

**Physics:** A spacecraft in orbit experiences a non-uniform gravitational field -- the side closer to Earth is pulled more strongly than the far side. For an elongated body, this differential pull creates a torque that tends to align the long axis with the local vertical (nadir direction). This is the principle behind gravity gradient stabilisation.

#### Key Equations -- Gravity Gradient Torque

$$T_{gg} = \frac{3\mu}{2a^3} |I_z - I_x| \sin(2\theta)$$

where: -  $\mu = 3.986 \times 10^{14} \text{ m}^3/\text{s}^2$  (Earth's gravitational parameter) -  $a = R_E + h$  (semi-major axis in metres) -  $I_z, I_x$  are principal moments of inertia about the maximum and minimum axes ( $\text{kg m}^2$ ) -  $\theta$  = angle between the long axis and the local vertical - Worst case occurs at  $\theta = 45^\circ$ , where  $\sin(2\theta) = 1$

For a body with  $I_z \approx I_x$  (a cube),  $T_{gg} \approx 0$  -- this is why 1U CubeSats experience minimal gravity gradient torque.

### Aerodynamic Torque

**Physics:** At LEO altitudes (200--600 km), residual atmospheric molecules collide with the spacecraft surface. The force acts through the centre of pressure (cp), which generally does not coincide with the centre of mass (cm). The offset creates a torque.

#### Key Equations -- Aerodynamic Torque

$$T_{aero} = \frac{1}{2} \rho v^2 C_D A_{ref} d_{cp - cm}$$

where: -  $\rho$  = atmospheric density ( $\text{kg/m}^3$ ) -- varies by orders of magnitude with altitude, solar activity (F10.7), and local time: - 300 km:  $\rho \approx 2 \times 10^{-11}$  (solar min) to  $3 \times 10^{-10}$  (solar max) - 400 km:  $\rho \approx 4 \times 10^{-12}$  to  $1 \times 10^{-11}$  - 500 km:  $\rho \approx 6 \times 10^{-13}$  to  $5 \times 10^{-12}$  - 600 km:  $\rho \approx 1 \times 10^{-13}$  to  $8 \times 10^{-13}$  -  $v$  = orbital velocity ( $\sim 7.6 \text{ km/s}$  at 500 km) -  $C_D \approx 2.0$ -- $2.3$  (molecular flow drag coefficient; 2.2 is standard for flat plates in free molecular flow) -  $A_{ref}$  = cross-sectional area perpendicular to velocity ( $\text{m}^2$ ) -  $d_{cp - cm}$  = offset between centre of pressure and centre of mass (m); typically 0.5--5 cm for CubeSats depending on deployable configuration

### Solar Radiation Pressure Torque

**Physics:** Sunlight carries momentum. When photons strike a surface, they transfer momentum ( $p = E/c$  for absorption,  $p = 2E/c$  for specular reflection). The resulting force acts through the centre of solar pressure, which may not coincide with the cm.

### Key Equations -- SRP Torque

$$T_{\text{SRP}} = \frac{S}{c} A_s (1 + q) d_{\text{sp} - \text{cm}}$$

where: -  $S = 1361 \text{ W/m}^2$  (solar constant at 1 AU) -  $c = 3 \times 10^8 \text{ m/s}$  (speed of light) -  $S/c = 4.54 \times 10^{-6} \text{ N/m}^2$  (solar radiation pressure at 1 AU) -  $A_s$  = illuminated area ( $\text{m}^2$ ) -  $q$  = surface reflectance (0 for perfect absorber, 1 for perfect specular reflector) -  $d_{\text{sp} - \text{cm}}$  = offset between solar pressure centre and centre of mass (m)

**Note:** SRP torque is tiny in LEO compared to aero and magnetic torques. It becomes the dominant disturbance in GEO and deep space (where there is no atmosphere and the magnetic field is weak).

### Residual Magnetic Dipole Torque

**Physics:** A spacecraft with a net magnetic dipole moment  $\vec{M}$  (from current loops in wiring, permanent magnets in reaction wheel motors, magnetised ferromagnetic components) interacts with Earth's magnetic field  $\vec{B}$  to produce a torque:

### Key Equations -- Magnetic Dipole Torque

$$\vec{T}_{\text{mag}} = \vec{M} \times \vec{B}$$

Magnitude:  $T_{\text{mag}} = M \cdot B \cdot \sin(\alpha)$

where: -  $M$  = spacecraft residual magnetic dipole moment ( $\text{A m}^2$ ) -  $B$  = local geomagnetic field strength (T): - LEO (400--600 km):  $B \approx 2\text{--}5 \times 10^{-5} \text{ T}$  (varies with latitude; strongest near poles, weakest near equator) - GEO (35,786 km):  $B \approx 1 \times 10^{-7} \text{ T}$  -  $\alpha$  = angle between  $\vec{M}$  and  $\vec{B}$

**Typical CubeSat residual dipole moments:**

| Source                    | Dipole Moment ( $\text{A m}^2$ ) | Notes                                |
|---------------------------|----------------------------------|--------------------------------------|
| Reaction wheel motor      | 0.005--0.02 per wheel            | Permanent magnets in brushless motor |
| Solar array wiring        | 0.001--0.01                      | Current loops from SA to EPS         |
| Battery cells             | 0.001--0.005                     | Nickel in cell casing                |
| Unshielded cables         | 0.005--0.05                      | Depends on routing and length        |
| <b>Total (typical 3U)</b> | <b>0.01--0.10</b>                | Varies significantly with design     |

**Why magnetic torque dominates for CubeSats:** COTS electronics are not designed for magnetic cleanliness. Short wiring runs create small but unbalanced current loops. Reaction wheel motors contain permanent magnets. The result is a residual dipole of  $0.01\text{--}0.1 \text{ A m}^2$ , which in a  $30 \text{ uT}$  field produces  $3 \times 10^{-7}$  to  $3 \times 10^{-6} \text{ N m}$  -- often larger than gravity gradient or SRP torques.

**Worked Example -- Disturbance Torques for 3U CubeSat at 500 km (SuperDove-class)**

**Spacecraft properties:** 3U (100 x 100 x 340 mm), mass = 5 kg,  $I_z = 0.035 \text{ kg m}^2$  (long axis),  $I_x = 0.007 \text{ kg m}^2$  (short axis),  $A_{\text{ref}} = 0.034 \text{ m}^2$  (3U face),  $d_{\text{cp} - \text{cm}} = 0.02 \text{ m}$  (deployable panels offset cm from geometric centre).



**Gravity gradient** (worst case,  $\theta = 45^\circ$ ):  $T_{\text{gg}} = \frac{3 \times 3.986 \times 10^{14}}{2 \times (6871 \times 10^3)^3} \times |0.035 - 0.007| \times 1$   
 $= \frac{1.196 \times 10^{15}}{6.494 \times 10^{20}} \times 0.028 = 1.84 \times 10^{-6} \times 0.028 = 5.2 \times 10^{-8} \text{ N m}$

**Aerodynamic** (at 500 km, solar minimum,  $\rho \approx 6 \times 10^{-13} \text{ kg/m}^3$ ):

$$F_{\text{aero}} = 0.5 \times 6 \times 10^{-13} \times 7617^2 \times 2.2 \times 0.034 = 1.30 \times 10^{-6} \text{ N}$$

$$T_{\text{aero}} = F_{\text{aero}} \times d_{\text{cp-cm}} = 1.30 \times 10^{-6} \times 0.02 = 2.6 \times 10^{-8} \text{ N m}$$

Note: at solar maximum ( $\rho \approx 5 \times 10^{-12}$ ), this increases by ~8x to  $2.1 \times 10^{-7} \text{ N m}$ .

**Solar radiation pressure:**  $F_{\text{SRP}} = \frac{1361}{3 \times 10^8} \times 0.034 \times 1.5 = 2.31 \times 10^{-7} \text{ N}$

$$T_{\text{SRP}} = F_{\text{SRP}} \times d_{\text{sp-cm}} = 2.31 \times 10^{-7} \times 0.02 = 4.6 \times 10^{-9} \text{ N m}$$

**Residual magnetic dipole** ( $M = 0.05 \text{ A m}^2$ ,  $B = 3 \times 10^{-5} \text{ T}$ ):

$$T_{\text{mag}} = 0.05 \times 3 \times 10^{-5} = 1.5 \times 10^{-6} \text{ N m}$$

**Summary:**

| Source                        | Torque (N m)                 | Rank       | Notes                                  |
|-------------------------------|------------------------------|------------|----------------------------------------|
| Gravity gradient              | $5.2 \times 10^{-8}$         | 3rd        | Small because 3U is not very elongated |
| Aerodynamic (solar min)       | $2.6 \times 10^{-8}$         | 4th        | Increases 8x at solar max              |
| Solar radiation pressure      | $4.6 \times 10^{-9}$         | 5th        | Negligible at LEO distances            |
| Residual magnetic dipole      | $1.5 \times 10^{-6}$         | <b>1st</b> | <b>Dominates by &gt;10x</b>            |
| <b>Total (worst-case sum)</b> | $\approx 1.6 \times 10^{-6}$ |            | Conservative estimate                  |
| <b>Total (RSS)</b>            | $\approx 1.5 \times 10^{-6}$ |            | More realistic (uncorrelated sources)  |

**Key finding:** The residual magnetic dipole dominates for CubeSats due to COTS electronics and short wiring runs. **Magnetic cleanliness matters.** Reducing the residual dipole from  $0.05$  to  $0.01 \text{ A m}^2$  (achievable with careful wire routing, twisted pairs, and degaussing) would reduce the total disturbance by 5x.

### 3. Attitude Actuators -- Physics and Sizing

#### Reaction Wheels -- Physics of Angular Momentum Storage

**How reaction wheels work:**

A reaction wheel is a flywheel (typically a brass or steel ring, 20--200 g for CubeSats) spun by a brushless DC motor. By Newton's third law, changing the wheel's angular momentum produces an equal and opposite torque on the spacecraft:

$$\vec{H}_{\text{total}} = \vec{H}_{\text{spacecraft}} + \vec{H}_{\text{wheels}} = \text{constant}$$

If the wheel speeds up ( $\Delta H_{\text{wheel}} > 0$ ), the spacecraft receives an equal and opposite angular momentum change ( $\Delta H_{\text{SC}} = -\Delta H_{\text{wheel}}$ ), causing it to rotate. The control torque is:

$$T_{\text{control}} = \frac{dH_{\text{wheel}}}{dt} = I_{\text{wheel}} \cdot \dot{\omega}_{\text{wheel}}$$



where  $I_{\text{wheel}}$  is the wheel's moment of inertia and  $\dot{\omega}_{\text{wheel}}$  is the wheel's angular acceleration.

**Momentum storage:** The maximum angular momentum a wheel can store is:

$$H_{\text{max}} = I_{\text{wheel}} \times \omega_{\text{max}}$$

For a Blue Canyon RW210:  $I_{\text{wheel}} \approx 1.5 \times 10^{-5} \text{ kg m}^2$ ,  $\omega_{\text{max}} \approx 6000 \text{ RPM} = 628 \text{ rad/s}$ , giving  $H_{\text{max}} = 0.0094 \text{ N m s} \approx 10 \text{ mN m s}$ .

**Saturation:** As disturbance torques act on the spacecraft, the reaction wheels absorb angular momentum. Over time, the wheel speed increases until it reaches  $\omega_{\text{max}}$  (saturation). At saturation, the wheel can no longer absorb momentum in that direction, and control authority is lost. The time to saturation from zero speed is:

$$t_{\text{sat}} = \frac{H_{\text{max}}}{T_{\text{disturbance}}} = \frac{10 \times 10^{-3}}{1.5 \times 10^{-6}} = 6667 \text{ s} \approx 111 \text{ minutes}$$

This is approximately 1.2 orbits -- so the wheels would saturate after about one orbit without desaturation. This is why magnetorquers are essential companions to reaction wheels.

**The zero-crossing problem:** When a reaction wheel passes through zero speed (reversing direction), the static friction in the bearings creates a "dead zone" where the wheel cannot produce smooth, continuous torque. This causes a brief loss of control authority and increased jitter. Mitigations: - **Bias momentum:** Operate all wheels with a positive bias speed (e.g., 500 RPM), so they never cross zero during normal operations - **4-wheel pyramid configuration:** The skewed geometry means individual wheels reverse less frequently - **Lubrication:** Space-rated bearings use solid or vapour-deposited lubricants ( $\text{MoS}_2$ , Braycote) that minimise static friction

**Jitter:** Reaction wheel imbalance (mass asymmetry in the flywheel) creates vibrations at the spin frequency and its harmonics. For imaging missions, this jitter degrades image quality. Jitter amplitude depends on wheel speed, imbalance mass, and the spacecraft's structural transfer function. Typical CubeSat reaction wheel jitter: 5--20 arcsec at the payload, depending on isolation.

## Reaction Wheel Configurations

| Configuration          | Description                                                        | Pros                                                                                | Cons                                                         | Use                                   |
|------------------------|--------------------------------------------------------------------|-------------------------------------------------------------------------------------|--------------------------------------------------------------|---------------------------------------|
| <b>3 orthogonal</b>    | One wheel per body axis (X, Y, Z)                                  | Minimum mass, simple control                                                        | No redundancy; single wheel failure = loss of 1-axis control | Low-cost missions with short lifetime |
| <b>3 + 1 skew</b>      | 3 orthogonal + 1 on a skew axis (e.g., [1,1,1] direction)          | Single-fault tolerant; the skew wheel + remaining 2 provide 3-axis control          | Slightly more complex control law distribution               | <b>Standard for CubeSats</b>          |
| <b>4-wheel pyramid</b> | 4 wheels tilted ~20--30 deg from body axes, symmetrically arranged | Optimal torque/momentum distribution; single-fault tolerant; reduced zero-crossings | More complex mounting, heavier                               | High-performance missions, agile S/C  |

The **distribution matrix** maps wheel torques to body-frame torques:

$$\vec{T}_{\text{body}} = \mathbf{D} \cdot \vec{T}_{\text{wheels}}$$

For a 4-wheel pyramid with cant angle  $\beta$ :

$$\mathbf{D} = \begin{bmatrix} \cos\beta & 0 & -\cos\beta & 0 \\ 0 & \cos\beta & 0 & -\cos\beta \\ \sin\beta & \sin\beta & \sin\beta & \sin\beta \end{bmatrix}$$

### Key Equations -- Reaction Wheel Sizing

**Torque requirement** (to counteract disturbances + provide slewing capability):

$$T_{\text{RW,min}} \geq k \times T_{\text{disturbance,total}}$$

where  $k \geq 2$  is the control margin factor (typically 2--5 to provide adequate control bandwidth and slewing performance).

**Momentum storage requirement** (accumulation between desaturation cycles):

$$H_{\text{required}} = T_{\text{disturbance}} \times \frac{t_{\text{desat}}}{2}$$

where  $t_{\text{desat}}$  is the time between magnetorquer desaturation events (typically one half-orbit to one orbit). The factor of 1/2 accounts for the average (sinusoidal disturbance torques average to half their peak over a quarter orbit).

**Slew rate** (for agile/imaging missions):

$$\dot{\theta}_{\text{max}} = \frac{H_{\text{RW,max}}}{I_{\text{axis}}}$$

For a 3U CubeSat with RW210 ( $H = 10 \text{ mN m s}$ ) and  $I_{\text{axis}} = 0.035 \text{ kg m}^2$ :

$\dot{\theta}_{\text{max}} = 0.010/0.035 = 0.286 \text{ rad/s} = 16.4 \text{ deg/s}$  -- more than adequate for target-to-target slewing.

**Slew time for a given angle** (acceleration-limited, trapezoidal profile):

$$t_{\text{slew}} = 2\sqrt{\frac{\theta_{\text{slew}} \cdot I_{\text{axis}}}{T_{\text{RW}}}}$$

For a 90 deg (1.57 rad) slew with RW210 ( $T = 1.0 \text{ mN m}$ ) and  $I = 0.035$ :

$$t_{\text{slew}} = 2\sqrt{\frac{1.57 \times 0.035}{0.001}} = 2\sqrt{54.95} = 14.8 \text{ s}$$

### Worked Example -- Reaction Wheel Sizing for 3U CubeSat (SuperDove-class)

**Given:**  $T_{\text{disturbance}} = 1.5 \times 10^{-6} \text{ N m}$  (from Section 2), desaturation interval = 1 orbit (5670 s), pointing requirement = 0.1 deg, slew requirement = 90 deg in < 60 s.

**Torque requirement:**  $T_{\text{RW,min}} = 3 \times 1.5 \times 10^{-6} = 4.5 \times 10^{-6} \text{ N m} = 0.0045 \text{ mN m}$

This is a very low torque requirement. The minimum available CubeSat wheel (RW-0.01 at 0.23 mN m) exceeds this by 50x. The sizing driver is actually the slew rate and momentum storage, not the disturbance rejection torque.

**Momentum storage:**  $H_{\text{required}} = 1.5 \times 10^{-6} \times \frac{5670}{2} = 4.25 \times 10^{-3} \text{ N m s} = 4.25 \text{ mN m s}$

**Slew time check with candidate wheels:**

| Product | Torque (mN m) | Momentum (mN m s) | Mass (g) | 90 deg Slew (s) | Momentum Margin      | Manufacturer |
|---------|---------------|-------------------|----------|-----------------|----------------------|--------------|
| RW-0.01 | 0.23          | 1.0               | 30       | 69 s            | -3.25 mN m s (FAIL)  | Hyperion     |
| RW210   | 1.0           | 10                | 55       | 14.8 s          | +5.75 mN m s (135%)  | Blue Canyon  |
| RW3-1.0 | 1.0           | 15                | 50       | 14.8 s          | +10.75 mN m s (253%) | CubeSpace    |
| RW400   | 4.0           | 40                | 120      | 7.4 s           | +35.75 mN m s (841%) | Blue Canyon  |

The RW-0.01 fails the momentum storage requirement (would saturate in < 1 orbit). The RW210 (10 mN m s) provides 135% margin and 14.8 s slew time. **Selected: RW210 (or CubeSpace RW3-1.0).**

**Configuration:** 4 wheels (3+1 skew) for single-fault tolerance. Total AOCS actuator mass: 4 x 55 g = 220 g.

**Magnetorquers -- Physics of Magnetic Torque Generation****How magnetorquers work:**

A magnetorquer (MTQ) is simply a coil of wire (or a ferromagnetic rod wrapped with wire). When current flows through the coil, it creates a magnetic dipole moment:

$$\vec{m} = N \cdot I \cdot A \cdot \hat{n}$$

where  $N$  = number of turns,  $I$  = current (A),  $A$  = coil cross-sectional area ( $\text{m}^2$ ), and  $\hat{n}$  is the coil normal direction.

This magnetic dipole interacts with Earth's geomagnetic field  $\vec{B}$  to produce a torque:

$$\vec{T}_{\text{MTQ}} = \vec{m} \times \vec{B}$$

The torque is **perpendicular** to both the dipole moment and the magnetic field. This has a critical implication: **a magnetorquer cannot produce torque parallel to the local magnetic field vector.** At any instant, only 2 of 3 axes can be torqued. Over a full orbit, as the field direction rotates, all 3 axes become accessible -- but not simultaneously.

**Magnetorquer types for CubeSats:**

| Type                             | Dipole Moment             | Mass     | Power      | Form Factor                                     | Products                               |
|----------------------------------|---------------------------|----------|------------|-------------------------------------------------|----------------------------------------|
| <b>Air-core coil</b> (PCB trace) | 0.01--0.05 A $\text{m}^2$ | 1--5 g   | 0.1--0.3 W | Flat PCB, integrates into solar panel substrate | ZARM Technik MTC-1, custom             |
| <b>Air-core rod</b> (wound wire) | 0.05--0.50 A $\text{m}^2$ | 10--30 g | 0.2--0.5 W | Cylindrical, 60--100 mm long                    | CubeSpace<br>CubeMAG,<br>NewSpace NTQS |

| Type                   | Dipole Moment             | Mass      | Power      | Form Factor                                     | Products                      |
|------------------------|---------------------------|-----------|------------|-------------------------------------------------|-------------------------------|
| Ferromagnetic core rod | 0.2--5.0 A m <sup>2</sup> | 20--100 g | 0.3--1.0 W | Cylindrical with mu-metal core, 60--100 mm long | ZARM Technik MTQ-1, ISIS iMTQ |

The ferromagnetic core concentrates the magnetic flux, providing 5--20x more dipole moment per unit current than an air-core coil of the same size. However, ferromagnetic cores can retain residual magnetism after power-off, contributing to the spacecraft's residual magnetic dipole.

#### Why magnetorquers cannot point (only detumble and desaturate):

The torque  $\vec{T} = \vec{m} \times \vec{B}$  is always perpendicular to  $\vec{B}$ . This means: 1. You cannot generate torque about the  $\vec{B}$  direction at any given instant 2. The achievable torque direction changes continuously as the spacecraft orbits (because  $\vec{B}$  rotates) 3. Pointing control requires torque in any direction at any time -- magnetorquers cannot provide this

Magnetorquers are excellent for: - **Detumbling:** The B-dot controller ( $\vec{m} = -k\dot{\vec{B}}$ ) brakes spacecraft rotation by opposing the change in the measured B-field. Works regardless of attitude. - **Desaturation:**

Systematically dumping momentum from reaction wheels by applying the correct dipole moment:

$\vec{m} = -k(\vec{H}_{\text{wheel}} \times \hat{B})$  - **Coarse pointing** (2--10 deg): Possible over time using model-predictive control that plans the dipole commands over a full orbit, exploiting the field rotation. But accuracy is limited.

#### Key Equations -- Magnetorquer Sizing

Desaturation torque:

$$T_{\text{MTQ}} = m_{\text{dipole}} \times B \times \sin(\alpha)$$

Average torque over an orbit (accounting for varying  $\alpha$ ):  $T_{\text{MTQ,avg}} \approx 0.7 \times m_{\text{dipole}} \times B_{\text{avg}}$

Desaturation time (to dump one wheel from full momentum):

$$t_{\text{dump}} = \frac{H_{\text{wheel}}}{T_{\text{MTQ,avg}}}$$

**Design requirement:**  $t_{\text{dump}} < t_{\text{shadow}}$  (must complete desaturation during the portion of the orbit where the field geometry is favourable).

#### Worked Example -- Magnetorquer Sizing for 3U CubeSat

**Given:** RW210 momentum = 10 mN m s,  $B_{\text{avg}} = 3 \times 10^{-5}$  T.

**Option 1: CubeMAG rod** ( $m = 0.2$  A m<sup>2</sup>):  $T_{\text{MTQ,avg}} = 0.7 \times 0.2 \times 3 \times 10^{-5} = 4.2 \times 10^{-6}$  N m

$$t_{\text{dump}} = \frac{10 \times 10^{-3}}{4.2 \times 10^{-6}} = 2381 \text{ s} \approx 40 \text{ min}$$

This is approximately 42% of one orbit period. **Acceptable** -- desaturation can be scheduled once per orbit during the non-imaging portion.

**Disturbance rejection check:** The MTQ average torque ( $4.2 \times 10^{-6}$  N m) is 2.8x the total disturbance torque ( $1.5 \times 10^{-6}$  N m). The MTQ can dump momentum faster than it accumulates. **Pass.**

**Configuration:** 3 MTQ rods, one per body axis (X, Y, Z). This ensures torque can be generated about any two axes at any given time. Total MTQ mass:  $3 \times 30 \text{ g} = 90 \text{ g}$ .

## Control Moment Gyroscopes (CMGs) vs Reaction Wheels

**CMGs** are an alternative momentum exchange device used on large, agile spacecraft. A CMG consists of a spinning flywheel mounted on a gimbal. Instead of changing the wheel speed (as in a reaction wheel), the CMG changes the direction of the angular momentum vector by rotating the gimbal. This produces a gyroscopic output torque:

$$\mathbf{T}_{\text{CMG}} = \mathbf{H}_{\text{wheel}} \times \dot{\boldsymbol{\delta}}$$

where  $\mathbf{H}_{\text{wheel}}$  is the constant wheel momentum and  $\dot{\boldsymbol{\delta}}$  is the gimbal rate.

**The torque amplification effect:** For a CMG wheel spinning at high speed ( $H = 1 \text{ -- } 100 \text{ N m s}$ ), even a slow gimbal rate ( $\dot{\delta} = 1 \text{ rad/s}$ ) produces a large output torque ( $T = 1 \text{ -- } 100 \text{ N m}$ ). This is orders of magnitude more than a reaction wheel of similar mass. CMGs are "torque machines"; reaction wheels are "momentum machines."

| Parameter          | Reaction Wheel                                | CMG (Single Gimbal)                                                   |
|--------------------|-----------------------------------------------|-----------------------------------------------------------------------|
| Output torque      | $T = I_w \dot{\omega}_w$ (low, 0.001--10 N m) | $T = H_w \dot{\delta}$ (high, 0.1--1000+ N m)                         |
| Control complexity | Simple (speed command)                        | Complex (gimbal singularity avoidance)                                |
| Mass efficiency    | Lower torque/kg                               | Higher torque/kg (10--100x)                                           |
| Failure mode       | Bearing wear, motor failure                   | Gimbal lock (singularity), bearing wear                               |
| Typical use        | CubeSats, small satellites, non-agile         | Large agile satellites, ISS, Earth observation with rapid retargeting |
| CubeSat status     | Standard (many COTS products)                 | Emerging (Honeybee Robotics microCMG, some research prototypes)       |

**When to use CMGs:** - Spacecraft requiring rapid slewing ( $> 3 \text{ deg/s}$  for large spacecraft) - Large moments of inertia where reaction wheel torque is insufficient - Missions requiring frequent retargeting (e.g., video from orbit, rapid revisit EO)

**When to use reaction wheels:** - CubeSats and small satellites (adequate torque, simpler control, more COTS options) - Missions with modest agility requirements ( $< 10 \text{ deg/s}$  slew for CubeSats) - Cost-constrained missions (CMGs are significantly more expensive)

The ISS uses four 4600 kg CMGs, each storing 3500 N m s of momentum, to maintain attitude without propellant. The Pleiades Neo Earth observation satellite uses CMGs for rapid retargeting between imaging strips.

## 4. Pointing Error Budget

[Source: ECSS-E-ST-60-10C; SMAD, Ch. 11.1]

The pointing error budget combines all independent error sources using root-sum-square (RSS) to determine the total pointing uncertainty. This is a statistical combination assuming errors are uncorrelated and normally distributed.

#### ECSS pointing performance taxonomy:

| Term                                    | Definition                                                 | Measured Over                               |
|-----------------------------------------|------------------------------------------------------------|---------------------------------------------|
| <b>APE</b> (Absolute Performance Error) | Total error between actual pointing and commanded pointing | Single measurement                          |
| <b>RPE</b> (Relative Performance Error) | Variation in pointing over a short time (jitter/stability) | Measurement window (e.g., integration time) |
| <b>MPE</b> (Mean Performance Error)     | Systematic bias in pointing                                | Long-term average                           |

For most CubeSat missions, APE is the primary requirement. RPE matters for long-exposure imaging (e.g., ASTERIA's 10-arcsec stability over 20-minute exposures).

#### Key Equations -- Pointing Error Budget (RSS)

$$\theta_{\text{APE}} = \sqrt{\theta_{\text{sensor}}^2 + \theta_{\text{actuator}}^2 + \theta_{\text{alignment}}^2 + \theta_{\text{thermal}}^2 + \theta_{\text{jitter}}^2 + \theta_{\text{orbit}}^2 + \theta_{\text{timing}}^2}$$

The result must satisfy:

$$\theta_{\text{APE}} \leq \theta_{\text{requirement}}$$

#### Error Source Definitions and Physics

| Source                     | Description                                                                      | Physics                                                                    | Typical Values (Star Tracker)   | Typical Values (Sun Sensor) |
|----------------------------|----------------------------------------------------------------------------------|----------------------------------------------------------------------------|---------------------------------|-----------------------------|
| <b>Sensor accuracy</b>     | Intrinsic measurement noise of attitude sensor                                   | Photon noise, centroiding error, optical distortion                        | 3--15 arcsec (0.001--0.004 deg) | 0.5--2 deg                  |
| <b>Actuator resolution</b> | Minimum controllable torque step; control loop dead band                         | Motor cogging torque, driver quantisation, control bandwidth               | 2--5 arcsec (0.001 deg)         | N/A (MTQ: 1--5 deg)         |
| <b>Alignment knowledge</b> | Misalignment between sensor boresight and payload boresight; measured during I&T | Mechanical tolerances, shimming, bonding accuracy, measurement uncertainty | 30--60 arcsec (0.01--0.02 deg)  | 0.5 deg                     |
| <b>Thermal distortion</b>  | Structural deformation with temperature changes; orbital thermal cycling         | CTE mismatch between materials, temperature gradients across structure     | 10--30 arcsec (0.003--0.01 deg) | 0.1 deg                     |
| <b>Jitter</b>              |                                                                                  |                                                                            |                                 | N/A                         |

| Source                 | Description                                                                   | Physics                                                                       | Typical Values (Star Tracker)   | Typical Values (Sun Sensor) |
|------------------------|-------------------------------------------------------------------------------|-------------------------------------------------------------------------------|---------------------------------|-----------------------------|
|                        | High-frequency vibration from reaction wheels, mechanisms                     | Wheel imbalance forces at spin frequency and harmonics, structural resonances | 5--20 arcsec (0.001--0.006 deg) |                             |
| <b>Orbit knowledge</b> | Uncertainty in satellite position (affects nadir pointing vector computation) | GPS accuracy, propagation error between GPS fixes                             | 1--5 arcsec (< 0.001 deg)       | 0.05 deg                    |
| <b>Timing</b>          | Time-stamping error between sensor read and actuator command                  | Clock synchronisation, interrupt latency, bus communication delay             | 1--3 arcsec (< 0.001 deg)       | 0.01 deg                    |

#### Worked Example -- Pointing Budget for 3U EO CubeSat (Star Tracker + RW)

| Error Source              | Value (deg)                                      | Value (arcsec)   | Value (deg <sup>2</sup> ) | Notes                                  |
|---------------------------|--------------------------------------------------|------------------|---------------------------|----------------------------------------|
| Star tracker accuracy     | 0.003                                            | 10.8             | $9.0 \times 10^{-6}$      | Blue Canyon NST, 1-sigma boresight     |
| Reaction wheel resolution | 0.001                                            | 3.6              | $1.0 \times 10^{-6}$      | Motor cogging + control dead band      |
| Alignment knowledge       | 0.020                                            | 72               | $4.0 \times 10^{-4}$      | <b>Dominant</b> -- shimmed to 1 arcmin |
| Thermal distortion        | 0.010                                            | 36               | $1.0 \times 10^{-4}$      | Al structure, 40 degC orbital range    |
| RW jitter                 | 0.005                                            | 18               | $2.5 \times 10^{-5}$      | At 3000 RPM, no isolation mount        |
| Orbit knowledge (GPS)     | 0.001                                            | 3.6              | $1.0 \times 10^{-6}$      | GPS fix every 10 s                     |
| Timing error              | 0.001                                            | 3.6              | $1.0 \times 10^{-6}$      | < 1 ms timestamp sync                  |
| <b>RSS Total</b>          | $\sqrt{5.37 \times 10^{-4}} = 0.023 \text{ deg}$ | <b>83 arcsec</b> |                           |                                        |

**Requirement:** 0.1 deg (3-sigma) -- this is typical for a 5 m GSD imager at 500 km (where 0.1 deg corresponds to ~870 m pointing error on ground, or ~175 pixels for a 5 m GSD sensor).

**Margin:**  $0.1 - 0.023 = 0.077 \text{ deg}$  (77% margin) -- **comfortable**.

**Key insight:** Alignment knowledge (0.020 deg = 72 arcsec) dominates the budget at 74% of the RSS. Improving the star tracker accuracy from 10 to 3 arcsec would change the RSS total from 83 to 82.3 arcsec -- negligible improvement. **Budget-driven design** means investing effort in the dominant term:

better alignment calibration (e.g., on-orbit calibration using ground targets) would have far more impact than upgrading any sensor.

To achieve < 0.01 deg (36 arcsec) pointing, the alignment must be improved to < 0.005 deg (18 arcsec), which requires precision optical alignment during I&T and/or on-orbit alignment calibration.

## 5. Momentum Management and Desaturation

Disturbance torques cause angular momentum to accumulate in reaction wheels over time. Without management, wheels saturate and lose control authority. Understanding this cycle is essential for AOCS design.

**The momentum lifecycle:**

1. **Accumulation:** External disturbance torques (gravity gradient, aero, SRP, magnetic) act on the spacecraft body. The control loop commands the reaction wheels to counteract these torques, absorbing the angular momentum. Wheel speed increases (or decreases) at a rate of  $\dot{H} = T_{\text{disturbance}}$ .
2. **Monitoring:** The OBC monitors wheel speeds. When any wheel exceeds a threshold (typically 80% of maximum), a desaturation manoeuvre is triggered.
3. **Desaturation:** The OBC activates the magnetorquers to generate a torque that opposes the stored wheel momentum. The algorithm computes the optimal dipole command:

$$\vec{m}_{\text{cmd}} = -k_d (\vec{H}_{\text{wheel}} \times \hat{B})$$

where  $k_d$  is the desaturation gain and  $\hat{B}$  is the unit magnetic field vector. This produces a torque  $\vec{T} = \vec{m} \times \vec{B}$  that is in the direction to reduce  $\vec{H}_{\text{wheel}}$ .

1. **Completion:** Wheel speeds return to near-zero (or bias speed). The cycle repeats.

**Desaturation constraints:** - MTQs can only generate torque **perpendicular** to the local magnetic field vector. They cannot dump momentum parallel to  $\vec{B}$ . - Near the magnetic equator,  $\vec{B}$  is nearly horizontal (north-pointing). MTQs can effectively dump momentum about the pitch and roll axes but not yaw. - Near the magnetic poles,  $\vec{B}$  is nearly vertical. MTQs can dump pitch and yaw but not roll. - Over a full orbit, the field direction rotates sufficiently to dump all three axes. But at any instant, one axis is poorly controllable. - A multi-pass desaturation strategy (spreading the dump over a full orbit) is more efficient than a single-point dump.

**Typical desaturation frequency for CubeSats:** - At 500 km with 1.5 uN m total disturbance and 10 mN m s wheels: one desaturation per orbit (every ~95 minutes) - Duration: 5--15 minutes per cycle - Power: 0.5--1.5 W during desaturation (3 MTQ rods active) - During desaturation, pointing accuracy degrades slightly (the MTQ torques perturb the attitude). Imaging should be inhibited during desaturation.

### Wheel Configurations -- 3+1 Redundancy

The standard 4-wheel configuration provides full 3-axis control with one spare: - **3 wheels** in the body X, Y, Z axes provide minimum control - **4th wheel** on a skew axis (typically [1,1,1] normalised, or cant angle 20--30 deg from each axis) provides redundancy and enhanced torque distribution - If one wheel fails, the remaining three (including the skew wheel) maintain 3-axis control with reduced but adequate authority



**4-wheel torque envelope:** With 4 wheels in a pyramid configuration, the maximum torque in any body direction is:

$$T_{\max, \text{body}} = \sqrt{2} \cdot T_{\text{wheel}} \approx 1.41 \times T_{\text{wheel}}$$

for the optimal distribution. This is better than the 3-orthogonal configuration where  $T_{\max, \text{body}} = T_{\text{wheel}}$  along any axis.

## 1U Worked Example: UniSat-1

### Passive Magnetic Attitude Stabilisation

UniSat-1 does not have an active AOCS. Instead, it uses **passive magnetic stabilisation** -- the simplest and cheapest attitude control method, requiring zero power and minimal mass.

**How it works -- physics:**

1. **Permanent magnet:** A small bar magnet (typically AlNiCo or NdFeB, ~10--20 g, dipole moment  $M_p = 0.1 \text{ -- } 1.0 \text{ A m}^2$ ) is embedded along one body axis (say, the Z-axis). In Earth's magnetic field  $\vec{B}$ , the magnet experiences a restoring torque:

$$\vec{T}_{\text{restoring}} = \vec{M}_p \times \vec{B}$$

This torque acts to align the magnet axis with the local field direction, analogous to a compass needle aligning with magnetic north. The restoring torque is maximum when the magnet is perpendicular to the field ( $\alpha = 90^\circ$ ) and zero when aligned ( $\alpha = 0^\circ$ ).

For  $M_p = 0.5 \text{ A m}^2$  and  $B = 3 \times 10^{-5} \text{ T}$ :  $T_{\max} = 0.5 \times 3 \times 10^{-5} = 1.5 \times 10^{-5} \text{ N m}$ . This is ~10x larger than any environmental disturbance torque, ensuring stable alignment.

1. **Hysteresis rods:** Two or more strips of magnetically soft material (e.g., Permalloy, HyMu-80, ~5--10 g each, dimensions ~60 x 1 x 1 mm) are mounted perpendicular to the permanent magnet. As the satellite oscillates around the field-aligned equilibrium, the external field component along the hysteresis rod alternates, driving the rod material around its B-H hysteresis loop. The area enclosed by the hysteresis loop represents energy dissipated per cycle as heat in the rod material. This extracts kinetic energy from the satellite's oscillation, damping it over time.

The energy dissipated per oscillation cycle is:

$$E_{\text{dissipated}} = V_{\text{rod}} \times \oint H dB$$

where  $V_{\text{rod}}$  is the rod volume and  $\oint H dB$  is the area of the hysteresis loop. For HyMu-80 material, typical energy density is ~100 J/m<sup>3</sup> per cycle.

**Damping time constant:** From initial tumble (~10 deg/s after deployment) to settled oscillation (~1--5 deg amplitude), the damping process typically takes hours to days, depending on the hysteresis rod material, volume, and the initial tumble rate.

**Performance:**

| Parameter            | Passive Magnetic                            | Active (RW + ST)                           |
|----------------------|---------------------------------------------|--------------------------------------------|
| Pointing accuracy    | ~10--15 deg (to local B-field)              | < 0.1 deg (to inertial frame)              |
| Settling time        | Hours to days after deployment              | Minutes after mode transition              |
| Residual tumble rate | ~1--5 deg/s (damped from initial ~10 deg/s) | < 0.01 deg/s                               |
| Power                | 0 W                                         | 3--5 W                                     |
| Mass                 | ~30--50 g                                   | 500--800 g                                 |
| Cost                 | ~2 kEUR                                     | ~55 kEUR                                   |
| Failure modes        | Demagnetisation (radiation, temperature)    | Motor failure, bearing wear, software bugs |

### Why this works for UniSat-1:

The MEMS magnetometer payload does not require accurate pointing. In fact, it benefits from being in a slowly rotating/tumbling state because this provides magnetic field measurements across multiple directions, improving the scientific data quality. The magnetometer can measure the field vector regardless of spacecraft orientation.

**No pointing budget needed:** Since there is no payload pointing requirement, there is no need for a pointing error budget. This eliminates the star tracker, reaction wheels, magnetorquers (as actuators), sun sensors, and gyroscopes -- removing the most expensive and power-hungry subsystem from the design.

### Disturbance environment for 1U at 400 km:

| Source                       | Torque (N m)            | Calculation                                                                     | Notes                                                  |
|------------------------------|-------------------------|---------------------------------------------------------------------------------|--------------------------------------------------------|
| Gravity gradient             | $\sim 1 \times 10^{-8}$ | $\frac{3\mu}{2a^3} \Delta I$ ; $\Delta I \approx 0.001 \text{ kg m}^2$ for 1U   | Small because nearly cubic shape ( $I_z \approx I_x$ ) |
| Aerodynamic                  | $\sim 3 \times 10^{-8}$ | $\frac{1}{2} \rho v^2 C_D A d$ ; $\rho(400\text{km}) \approx 5 \times 10^{-12}$ | Higher $\rho$ at 400 km than 500 km                    |
| Solar radiation pressure     | $\sim 1 \times 10^{-9}$ | $\frac{S}{c} A (1 + q) d$ ; $A = 0.01 \text{ m}$                                | Small area                                             |
| Permanent magnet (restoring) | $\sim 1 \times 10^{-5}$ | $M_p \times B \times \sin(\alpha)$                                              | <b>Dominant</b> -- this IS the control torque          |

The permanent magnet restoring torque ( $\sim 10^{-5} \text{ N m}$ ) is three orders of magnitude larger than all disturbances combined. This ensures the satellite remains approximately field-aligned.

**Limitation:** Passive magnetic stabilisation provides alignment to the *local* magnetic field, which rotates as the satellite orbits. The satellite does not point at nadir, the Sun, or any fixed direction. For missions requiring Earth-pointing or Sun-tracking, active AOCS is mandatory. For UniSat-1's magnetometer mission, this is not a limitation -- it is a feature.

## 6. SpaceCDF Exercise

### Instructions

1. **Architecture tab (AOCS):** Select the AOCS architecture appropriate for your pointing requirement - Review the derived requirements that appear - Check that the selected hardware fits within your power and mass budgets
2. **Pointing Budget card on the Dashboard:** - Review the RSS error tree - Identify the largest contributor - Verify the total is within your pointing requirement
3. **Dashboard AOCS KPI:** - Pointing accuracy achieved - Margin to requirement - AOCS power demand by mode
4. **Equipment Browser (if time permits):** - Browse reaction wheels: compare mass, torque, momentum, cost - Browse star trackers: compare accuracy, mass, FOV, exclusion zones - Note that star tracker exclusion zones constrain mounting face options

### Worksheet 3.2 Tasks

1. Select AOCS architecture and justify based on pointing requirement
2. Calculate all 4 disturbance torques for your orbit and spacecraft configuration
3. Size the reaction wheel (torque, momentum storage, and slew time)
4. Size the magnetorquer for desaturation (verify dump time < 1 orbit)
5. Complete the pointing error budget table (RSS of all 7 sources)
6. Verify margin to pointing requirement and identify the dominant error source

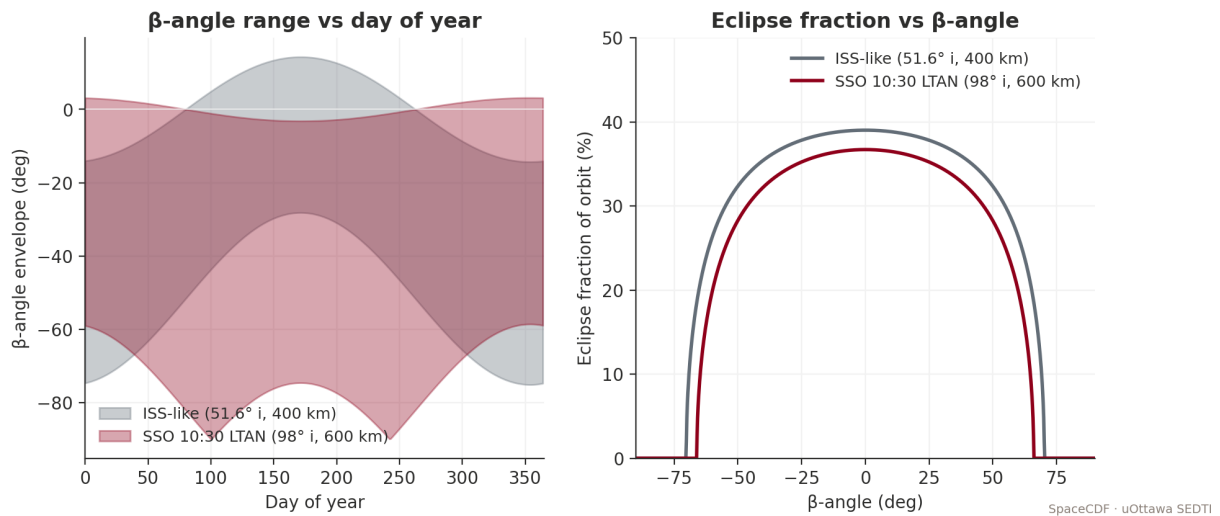
### Session Summary

| Topic                  | Key Takeaway                                                                                                                              |
|------------------------|-------------------------------------------------------------------------------------------------------------------------------------------|
| AD vs AC               | Determination = know orientation (sensors); Control = change orientation (actuators)                                                      |
| Star trackers          | Pattern-match stars against catalogue; 3--15 arcsec accuracy; exclusion zones (Sun 25--45 deg, Earth 25--35 deg); most accurate sensor    |
| Sun sensors            | Coarse (photodiode, 2--5 deg) or fine (slit+array, 0.1 deg); essential for safe mode; every S/C needs them                                |
| Magnetometers          | Measure $\vec{B}$ field; compare with IGRF model; 5--10 deg accuracy alone; residual dipole contamination is key error source             |
| GPS receivers          | Orbit determination (position/velocity), not attitude; 5--20 m accuracy in LEO; needed for nadir pointing computation                     |
| Gyroscopes             | Measure angular rate $\vec{\omega}$ ; MEMS drift 1--10 deg/hr; combined with star tracker via Kalman filter for high-bandwidth estimation |
| Architecture selection | Driven by pointing requirement: passive magnetic for > 5 deg; MTQ for 2--5 deg; RW+ST for < 0.1 deg                                       |
| Disturbance torques    | Gravity gradient, aero, SRP, magnetic dipole; <b>magnetic dipole dominates for CubeSats</b>                                               |
| Reaction wheel physics | $H = I_w \omega_w$ ; $T = dH/dt$ ; saturation occurs when $\omega \rightarrow \omega_{\max}$ ; zero-crossing jitter                       |

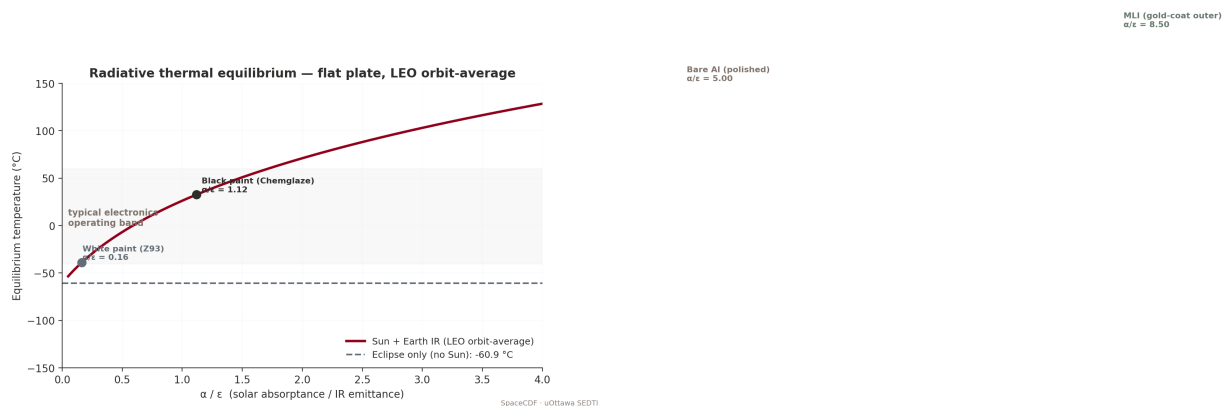
| Topic                | Key Takeaway                                                                                                                       |
|----------------------|------------------------------------------------------------------------------------------------------------------------------------|
| RW sizing            | Torque $\geq 2 \times$ disturbance; momentum $\geq$ half-orbit accumulation; check slew time                                       |
| Magnetorquer physics | $\vec{T} = \vec{m} \times \vec{B}$ ; torque always perpendicular to $\vec{B}$ ; cannot point, only detumble/desaturate             |
| MTQ sizing           | Desaturation torque must exceed disturbance accumulation rate; dump time $< 1$ orbit                                               |
| CMGs vs RWs          | CMGs: torque amplification ( $\vec{T} = \vec{H} \dot{\delta}$ ), for large/agile S/C; RWs: simpler, cheaper, standard for CubeSats |
| Pointing budget      | RSS of 7 independent sources; alignment knowledge typically dominates; improve the dominant term                                   |
| Budget-driven design | Investing in the smallest error source has negligible impact on total; focus on the dominant term                                  |
| Redundancy           | 4-wheel (3+1 skew) provides single-fault tolerance; distribution matrix maps wheels to body torques                                |
| Momentum management  | MTQs dump momentum against Earth's $\vec{B}$ -field; ~once per orbit; 5--15 min; imaging inhibited during dump                     |

## Session 3.3: Communications and Link Budget Design

$\beta$ -angle behaviour over a year — and eclipse fraction analytics



$\beta$ -angle envelope and eclipse fraction analytics.



Radiative thermal equilibrium vs  $\alpha/\epsilon$ .

**Expected reading before this session.** SMAD4 Ch. 12 (thermal); ECSS-E-ST-31C §5 ( $\approx 60$  min). Gilmore Ch. 1.

**Prerequisites:** Sessions 2.1--3.2 (requirements, orbit, power, AOCS defined) **SpaceCDF Tabs:** Link Budget, Spectrum Selector, Equipment Browser (Comms)

## References

- Wertz, Everett & Puschell, *Space Mission Engineering: The New SMAD*, 2011, Ch. 13 (Communications)
- ECSS, *ECSS-E-ST-50-05C: Radio Frequency and Modulation*, 2011
- CCSDS, *131.0-B-4: TM Synchronization and Channel Coding*, 2023
- ITU, *Radio Regulations*, 2020 (Articles 5 and 22)

- Maral & Bousquet, *Satellite Communications Systems*, 6th ed., 2020, Ch. 5
- Roddy, *Satellite Communications*, 4th ed., 2006, Ch. 4--6
- Haykin, *Communication Systems*, 5th ed., 2009
- Sklar, *Digital Communications: Fundamentals and Applications*, 2nd ed., 2001
- IARU, *Amateur Satellite Frequency Coordination*, 2023

## Learning Objectives

By the end of this session, participants will be able to:

1. Construct a complete link budget from first principles (in decibels)
2. Compute free space path loss for any frequency and slant range
3. Select an appropriate frequency band based on data rate, licensing, and equipment availability
4. Explain the physics of each antenna type and compute antenna gain and beamwidth
5. Choose modulation and coding scheme based on required  $E_b/N_0$  and spectral efficiency
6. Explain the physical basis for coding gain and why FEC is essential for space links
7. Size an antenna (gain, beamwidth, mass) for the selected frequency
8. Determine data throughput and verify the data budget closes
9. Identify ground station options and compute contact geometry
10. Use SpaceCDF's link budget tool and spectrum selector

## 1. The Link Budget Concept

[Source: SMAD, Ch. 13; ECSS-E-ST-50-05C; Roddy, Ch. 4]

The link budget is the accounting statement for the communication link. Every gain and every loss from transmitter to receiver is tallied in **decibels (dB)** to determine whether the link "closes" -- meaning the received signal is strong enough to decode with acceptable error rate.

The fundamental question: **does the received signal have enough energy per bit, relative to the noise, to achieve the required bit error rate?** This is quantified by  $E_b/N_0$  -- the ratio of energy per information bit to noise spectral density.

### Link Budget Flow



### Decibel Refresher

All link budget terms are in decibels to convert multiplication/division into addition/subtraction:

| Conversion        | Formula                                       | Example             |
|-------------------|-----------------------------------------------|---------------------|
| Watts to dBW      | $P_{\text{dBW}} = 10\log_{10}(P_{\text{W}})$  | 2 W = +3.0 dBW      |
| Milliwatts to dBm | $P_{\text{dBm}} = 10\log_{10}(P_{\text{mW}})$ | 1 W = +30 dBm       |
| dBW to Watts      | $P_{\text{W}} = 10^{P_{\text{dBW}}/10}$       | -3 dBW = 0.5 W      |
| Ratio to dB       | $G_{\text{dB}} = 10\log_{10}(G)$              | Gain of 100 = 20 dB |
| dBW vs dBm        | $\text{dBm} = \text{dBW} + 30$                | 0 dBW = 30 dBm      |

**Common power values:**

| Power | dBW   | dBm   |
|-------|-------|-------|
| 0.1 W | -10.0 | +20.0 |
| 0.5 W | -3.0  | +27.0 |
| 1 W   | 0.0   | +30.0 |
| 2 W   | +3.0  | +33.0 |
| 5 W   | +7.0  | +37.0 |
| 10 W  | +10.0 | +40.0 |

## 2. Complete Link Budget Equation

### Key Equations -- Link Budget (dB form)

**EIRP** (Effective Isotropic Radiated Power):

$$\text{EIRP} = P_{\text{TX}} + G_{\text{TX}} - L_{\text{TX}} \quad (\text{dBW})$$

EIRP represents the power that an isotropic antenna would need to radiate to produce the same signal strength in the direction of maximum antenna gain. It combines the transmitter power, antenna gain (directivity), and cable/filter losses.

**Free Space Path Loss:**

$$\text{FSPL} = 20\log_{10}\left(\frac{4\pi d}{\lambda}\right) = 20\log_{10}\left(\frac{4\pi df}{c}\right) \quad (\text{dB})$$

where  $d$  = slant range (m),  $f$  = frequency (Hz),  $c = 3 \times 10^8$  m/s.

**Physical interpretation:** FSPL is not "energy absorption" -- no energy is lost. It is the geometric spreading of radiated power over a sphere of radius  $d$ . An isotropic antenna radiates equally in all directions, so the power per unit area at distance  $d$  is  $P/(4\pi d^2)$ . The  $\lambda^2$  dependence arises because a receive antenna's effective aperture scales as  $A_{\text{eff}} = G\lambda^2/(4\pi)$  -- at higher frequencies the receive antenna "captures" less of the available flux (unless the antenna is physically larger).

**Receiver figure of merit:**

$$G/T = G_{RX} - 10\log_{10}(T_{sys}) \quad (\text{dB/K})$$

$G/T$  is the single most important parameter of any receive station. It combines the antenna gain (signal capture) with the system noise temperature (noise floor). A high  $G/T$  means good sensitivity.

**System noise temperature**  $T_{sys}$  includes: - Antenna noise temperature  $T_A$  (depends on what the antenna "sees": sky ~10--50 K at zenith, ~150--300 K looking at Earth/ground) - Feed/cable losses:  $T_{feed} = T_{physical}(L - 1)$  where  $L$  = loss factor - LNA noise temperature:  $T_{LNA} = T_0(F - 1)$  where  $F$  = noise figure,  $T_0 = 290$  K - Subsequent stages (reduced by LNA gain)

Rule of thumb:  $T_{sys} \approx 100$  -- 200 K for a professional ground station with cryogenic or low-noise LNA;  $T_{sys} \approx 400$  -- 800 K for an amateur station with COTS LNA.

**Carrier-to-noise density ratio:**

$$C/N_0 = \text{EIRP} - \text{FSPL} - L_{\text{atm}} - L_{\text{point}} - L_{\text{pol}} + G/T - k \quad (\text{dBHz})$$

where  $k = -228.6$  dBW/K/Hz (Boltzmann constant:  $k = 1.381 \times 10^{-23}$  J/K).

**Energy per bit to noise density:**

$$E_b/N_0 = C/N_0 - 10\log_{10}(R_b) \quad (\text{dB})$$

where  $R_b$  = data rate (bps). This is the fundamental quality metric: each bit needs a certain amount of energy ( $E_b$ ) relative to the noise floor ( $N_0$ ) to be correctly demodulated.

**Link margin:**

$$\text{Margin} = E_b/N_{0, \text{available}} - E_b/N_{0, \text{required}} - L_{\text{implementation}} \quad (\text{dB})$$

**Requirement:** Margin  $\geq 3$  dB for Phase B+ (per ECSS-E-ST-50-05C). This 3 dB margin covers: - Transmitter power variation (aging, temperature) - Antenna gain uncertainties - Atmospheric scintillation - Pointing error variations - Ground station performance variation

**Complete Link Budget Table**

| Line | Parameter            | Formula / Typical Value                | Unit | Physical Meaning                       |
|------|----------------------|----------------------------------------|------|----------------------------------------|
| 1    | TX Power             | $P_{TX}$ (e.g., 2 W = +3.0)            | dBW  | RF power from amplifier output         |
| 2    | TX Antenna Gain      | $G_{TX}$ (e.g., +6.0 for patch)        | dBi  | Directivity relative to isotropic      |
| 3    | TX Line Losses       | $L_{TX}$ (cables, filters: -1.5)       | dB   | Ohmic loss in RF cables and connectors |
| 4    | <b>EIRP</b>          | $= P_{TX} + G_{TX} - L_{TX}$           | dBW  | Effective radiated power               |
| 5    | Free Space Path Loss | $\text{FSPL} = 20\log_{10}(4\pi df/c)$ | dB   | Geometric spreading + aperture effect  |



| Line | Parameter                                    | Formula / Typical Value                                             | Unit     | Physical Meaning                                             |
|------|----------------------------------------------|---------------------------------------------------------------------|----------|--------------------------------------------------------------|
| 6    | Atmospheric Loss                             | $L_{\text{atm}}$ (-0.3 to -3.0)                                     | dB       | Molecular absorption ( $\text{O}_2$ , $\text{H}_2\text{O}$ ) |
| 7    | Pointing Loss                                | $L_{\text{point}}$ (-0.5 to -3.0)                                   | dB       | Signal reduction from antenna mispointing                    |
| 8    | Polarisation Loss                            | $L_{\text{pol}}$ (-0.1 to -3.0)                                     | dB       | Mismatch between TX and RX polarisation                      |
| 9    | RX Antenna Gain                              | $G_{\text{RX}}$ (e.g., +35 for 3 m dish)                            | dBi      | Ground antenna directivity                                   |
| 10   | System Noise Temp                            | $T_{\text{sys}}$ (e.g., 150 K = 21.8 dBK)                           | dBK      | Total noise temperature of receive chain                     |
| 11   | <b>G/T</b>                                   | $= G_{\text{RX}} - 10\log_{10}(T_{\text{sys}})$                     | dB/K     | Receiver figure of merit                                     |
| 12   | Boltzmann Constant                           | $k = -228.6$                                                        | dBW/K/Hz | Thermal noise power spectral density                         |
| 13   | <b>C/N<sub>0</sub></b>                       | $= \text{EIRP} - \text{FSPL} + \text{G/T} - k - L_{\text{losses}}$  | dBHz     | Signal-to-noise density ratio                                |
| 14   | Data Rate                                    | $10\log_{10}(R_b)$                                                  | dBbps    | Information throughput                                       |
| 15   | <b>E<sub>b</sub>/N<sub>0</sub> available</b> | $= C/N_0 - 10\log_{10}(R_b)$                                        | dB       | Available energy per bit                                     |
| 16   | <b>E<sub>b</sub>/N<sub>0</sub> required</b>  | From modulation/coding selection                                    | dB       | Minimum needed for target BER                                |
| 17   | Implementation Loss                          | Typically 1.5--2.5 dB                                               | dB       | Real vs ideal demodulator performance                        |
| 18   | <b>LINK MARGIN</b>                           | $= E_b/N_{0,\text{avail}} - E_b/N_{0,\text{req}} - L_{\text{impl}}$ | dB       | Must be $\geq 3$ dB                                          |

### 3. Antenna Types -- Physics and Selection

[Source: SMAD, Ch. 13; Balanis, Antenna Theory: Analysis and Design, 4th ed., 2016]

The antenna converts guided RF energy (in cables/waveguides) into radiated electromagnetic waves (and vice versa for receive). The key performance parameters are:

- **Gain (G)**: How much more signal the antenna concentrates in its beam direction compared to an isotropic radiator. Higher gain = narrower beam = more signal but requires more precise pointing.
- **Beamwidth ( $\theta_{3\text{dB}}$ )**: The angular width of the main beam between the -3 dB (half-power) points.
- **Polarisation**: Linear (vertical/horizontal), circular (RHCP/LHCP), or elliptical.

**Fundamental trade-off:**  $G \propto 1/\theta_{3\text{dB}}^2$ . A higher-gain antenna has a narrower beam and requires more precise pointing. For a spacecraft without accurate pointing, a low-gain omnidirectional antenna is mandatory. For a spacecraft with 0.1 deg pointing, a high-gain directional antenna is feasible.

## Antenna Types for Spacecraft

| Antenna Type              | Gain (dBi) | Beamwidth                              | Pointing Needed?           | Mass (CubeSat)               | Physics                                                                                                      |
|---------------------------|------------|----------------------------------------|----------------------------|------------------------------|--------------------------------------------------------------------------------------------------------------|
| Monopole/<br>Dipole       | 0--2       | ~omnidirectional<br>(toroidal pattern) | No                         | 5--20 g<br>(deployable wire) | Current distribution on a wire radiates; $\lambda/4$ monopole on ground plane or $\lambda/2$ dipole          |
| Turnstile                 | 0--3       | Hemispherical                          | No                         | 10--30 g                     | Two crossed dipoles fed 90 deg apart; produces circular polarisation                                         |
| Patch<br>(microstrip)     | 5--8       | 60--90 deg                             | Loose (~10 deg)            | 10--50 g                     | Resonant conducting patch on dielectric substrate over ground plane; $\lambda/2$ patch at resonance          |
| Patch array<br>(2x2, 4x4) | 10--18     | 15--40 deg                             | Moderate (~5 deg)          | 50--200 g                    | Multiple patch elements with corporate feed network; gain = $G_{\text{element}} + 10\log_{10}(N)$            |
| Horn                      | 15--25     | 10--25 deg                             | Yes (~2 deg)               | 100--500 g                   | Flared waveguide; smooth aperture illumination gives low side lobes                                          |
| Parabolic reflector       | 25--45     | 1--5 deg                               | Yes (< 1 deg)              | 500 g -- 5 kg                | Feed at focal point illuminates parabolic dish; $G = \eta_a(\pi D/\lambda)^2$                                |
| Phased array              | 15--35     | Electronically steered, 1--30 deg      | Electronic (no mechanical) | 200 g -- 2 kg                | Array of elements with individual phase shifters; beam steered by adjusting relative phases; no moving parts |

**Monopole/dipole physics:** A quarter-wave monopole ( $L = \lambda/4$ ) is the simplest antenna. At 437 MHz (UHF),  $\lambda = 0.686$  m, so a monopole is 17.2 cm long -- easily deployable from a CubeSat using a spring-loaded tape measure or nitinol wire. The radiation pattern is omnidirectional in the azimuthal plane (perpendicular to the wire) and null along the wire axis. Gain is approximately 0--2 dBi depending on the ground plane size.

**Patch antenna physics:** A microstrip patch antenna consists of a conducting patch (typically rectangular,  $L \approx \lambda/2$  at the desired frequency) printed on a dielectric substrate (e.g., Rogers RT/duroid,  $\epsilon_r = 2.2$  -- 10.2, thickness 1--3 mm) above a ground plane. The patch resonates at the frequency where its length equals  $\lambda_{\text{eff}}/2$  (where  $\lambda_{\text{eff}} = \lambda_0/\sqrt{\epsilon_r}$ ). At S-band (2.25 GHz), a patch is approximately 40 x 40 mm -- small enough to mount on a CubeSat face. Gain is typically 5--8 dBi with a hemispherical pattern. Circular polarisation is achieved by feeding two orthogonal modes 90 deg apart (dual-feed or corner-truncated patch).

**Parabolic reflector physics:** A parabolic dish focuses incoming plane waves to its focal point (for receive) or collimates spherical waves from a feed at the focus into a parallel beam (for transmit). The gain depends on the dish diameter  $D$  and the wavelength  $\lambda$ :

#### Key Equations -- Antenna

**Gain of a parabolic antenna:**

$$G = \eta_a \left( \frac{\pi D}{\lambda} \right)^2$$

In dBi:  $G_{\text{dBi}} = 10 \log_{10} \left[ \eta_a \left( \frac{\pi D}{\lambda} \right)^2 \right]$

where  $\eta_a \approx 0.55$ -- $0.65$  (aperture efficiency, accounting for illumination taper, spillover, blockage, and surface errors),  $D$  = antenna diameter (m),  $\lambda = c/f$ .

**Half-power beamwidth (HPBW):**

$$\theta_{3\text{dB}} \approx \frac{70\lambda}{D} \quad (\text{degrees})$$

**Patch antenna gain** (single element): typically 5--8 dBi, beamwidth ~60--90 deg

**Patch array gain** ( $N$  elements):  $G_{\text{array}} = G_{\text{element}} + 10 \log_{10} (N)$

**Pointing loss** (when antenna is mispointed by angle  $\Delta\theta$ ):

$$L_{\text{point}} \approx -12 \left( \frac{\Delta\theta}{\theta_{3\text{dB}}} \right)^2 \quad (\text{dB})$$

This shows that a narrower beam (smaller  $\theta_{3\text{dB}}$ ) is more sensitive to pointing errors.

**Worked example -- antenna gain at different bands for a 30 cm dish:**

| Band    | Frequency | $\lambda$ (mm) | $D/\lambda$ | Gain (dBi) | HPBW (deg) | Pointing req |
|---------|-----------|----------------|-------------|------------|------------|--------------|
| S-band  | 2.25 GHz  | 133            | 2.3         | 14.3       | 30         | ~5 deg       |
| X-band  | 8.4 GHz   | 35.7           | 8.4         | 25.4       | 8.3        | ~1 deg       |
| Ka-band | 26 GHz    | 11.5           | 26.1        | 35.2       | 2.7        | ~0.3 deg     |

This table illustrates the fundamental link between frequency, antenna size, gain, and pointing requirement. At Ka-band, even a small dish provides high gain -- but the beamwidth is so narrow that sub-degree pointing accuracy is essential. This is why Ka-band CubeSat links require star tracker + reaction wheel AOCs.

**Phased array physics:** A phased array consists of multiple antenna elements (patch, dipole, or slot) arranged in a grid. Each element has an individual phase shifter (and sometimes amplitude control). By adjusting the relative phase between elements, the beam can be electronically steered without moving the antenna. The beam direction  $\theta_s$  for element spacing  $d$  and phase increment  $\Delta\phi$ :

$$\sin(\theta_s) = \frac{\Delta\phi \cdot \lambda}{2\pi d}$$

Advantages: no moving parts, fast beam steering (microseconds), multiple simultaneous beams possible.  
 Disadvantages: complex, expensive, high power consumption for active arrays, scan loss at wide angles ( $\cos \theta_s$  factor). Phased arrays are used on Starlink satellites (Ka-band), Iridium NEXT (L-band), and are emerging for CubeSats in Ka-band.

## 4. Free Space Path Loss by Band

FSPL increases with both frequency and distance. At a fixed slant range, higher-frequency bands lose more signal -- but this is offset by the ability to use smaller, higher-gain antennas at higher frequencies.

### Key Equations -- FSPL (expanded form)

$$\text{FSPL (dB)} = 20\log_{10}(4\pi) + 20\log_{10}(d) + 20\log_{10}(f) - 20\log_{10}(c)$$

$$= 21.98 + 20\log_{10}(d_m) + 20\log_{10}(f_{\text{Hz}}) - 169.54$$

### Practical form (with km and GHz):

$$\text{FSPL (dB)} = 92.45 + 20\log_{10}(d_{\text{km}}) + 20\log_{10}(f_{\text{GHz}})$$

### Slant Range Geometry

The slant range  $d$  from a LEO spacecraft to a ground station depends on the orbit altitude  $h$  and the elevation angle  $\varepsilon$  above the horizon:

$$d = R_E \left[ \sqrt{\left(\frac{R_E + h}{R_E}\right)^2 - \cos^2(\varepsilon)} - \sin(\varepsilon) \right]$$

For typical LEO orbits:

| Altitude | Elevation 90 deg (nadir) | Elevation 30 deg | Elevation 10 deg | Elevation 5 deg |
|----------|--------------------------|------------------|------------------|-----------------|
| 400 km   | 400 km                   | 723 km           | 1150 km          | 1500 km         |
| 500 km   | 500 km                   | 875 km           | 1300 km          | 1650 km         |
| 600 km   | 600 km                   | 1020 km          | 1460 km          | 1820 km         |

**Design rule:** Always compute the link budget at the **minimum elevation angle** (worst case), typically 5--10 deg. Below 5 deg, atmospheric losses increase sharply, ground clutter enters the antenna sidelobes, and the link is generally unusable.

### FSPL by Band and Geometry

| Band   | Centre Frequency | FSPL at 500 km (nadir) | FSPL at 1300 km (10 deg elev) | Difference |
|--------|------------------|------------------------|-------------------------------|------------|
| VHF    | 146 MHz          | 139.0 dB               | 147.3 dB                      | 8.3 dB     |
| UHF    | 437 MHz          | 148.3 dB               | 157.6 dB                      | 9.3 dB     |
| S-band | 2250 MHz         | 162.5 dB               | 170.8 dB                      | 8.3 dB     |

| Band    | Centre Frequency | FSPL at 500 km (nadir) | FSPL at 1300 km (10 deg elev) | Difference |
|---------|------------------|------------------------|-------------------------------|------------|
| X-band  | 8200 MHz         | 173.8 dB               | 182.1 dB                      | 8.3 dB     |
| Ka-band | 26 GHz           | 183.8 dB               | 192.1 dB                      | 8.3 dB     |

**The elevation angle penalty:** Going from nadir to 10 deg elevation increases FSPL by ~8 dB (distance increases by ~2.6x; FSPL scales as  $20\log_{10}(2.6) = 8.3$  dB). This is a significant loss and is why contact time at high elevations is much more valuable than contact time at low elevations.

## 5. Frequency Band Selection and Licensing

[Source: ITU Radio Regulations, Articles 5 and 22; ISSED RSS-SAT; FCC Part 25; IARU Satellite Frequency Coordination]

Band selection is a **design constraint** that affects data rate, antenna size, atmospheric losses, equipment availability, licensing cost, and data policy. The choice of band is one of the earliest and most consequential decisions in mission design.

### Band Comparison Table

| Band    | Frequency Range | Allocation         | Max BW  | Practical Data Rate | Atmospheric Loss (10 deg) | Rain Fade                     | Licensing                          | Equipment Availability |
|---------|-----------------|--------------------|---------|---------------------|---------------------------|-------------------------------|------------------------------------|------------------------|
| VHF     | 144--146 MHz    | Amateur            | 15 kHz  | < 9.6 kbps          | 0.1 dB                    | None                          | IARU coord (free, 3--6 mo)         | Many COTS, low cost    |
| UHF     | 435--438 MHz    | Amateur            | 20 kHz  | < 19.2 kbps         | 0.2 dB                    | Negligible                    | IARU coord (free, 3--6 mo)         | Many COTS, low cost    |
| S-band  | 2200--2290 MHz  | Space research/EES | 5 MHz   | 0.1--10 Mbps        | 0.5 dB                    | < 0.5 dB                      | ISED/FCC (\$30--45K, 6--12 mo)     | Many COTS              |
| X-band  | 8025--8400 MHz  | EES                | 375 MHz | 10--400 Mbps        | 1.0 dB                    | 1--3 dB                       | ISED/FCC + ITU (\$50--80K, 12+ mo) | Growing COTS           |
| Ka-band | 25.5--27.0 GHz  | EES/FSS            | 1.5 GHz | 100--2000+ Mbps     | 2--5 dB                   | 3--15 dB (location-dependent) | Complex ITU (\$100K+, 18+ mo)      | Emerging COTS          |

EES = Earth Exploration Satellite; FSS = Fixed Satellite Service

**Atmospheric attenuation physics:** The atmosphere absorbs and scatters RF energy. Molecular oxygen has a strong absorption line at 60 GHz (used for inter-satellite links where atmospheric penetration is not needed). Water vapour absorbs at 22.2 GHz (near Ka-band). At S-band and below, atmospheric losses are

minimal ( $< 1$  dB). At Ka-band, losses of 2--5 dB are typical at 10 deg elevation, and rain fade can add 3--15 dB in tropical regions.

**Rain fade:** Raindrops scatter and absorb RF energy. The effect scales approximately as  $f^2$  -- negligible below 4 GHz, significant above 10 GHz, and severe above 20 GHz. Rain fade is characterised by the rain rate (mm/hr) and the path length through rain. In tropical regions, rain rates of 50+ mm/hr can cause 10+ dB additional loss at Ka-band. Mitigation: adaptive data rate (reduce rate during rain), site diversity (multiple ground stations), power control (increase TX power during fade). For Ka-band links, a rain fade margin of 5--10 dB is typically included.

**Amateur band regulations (IARU):** - Non-commercial, educational, and experimental use only - Open data policy: all transmissions must be unencrypted and the protocol must be published - No commercial data or imagery downlink - Coordination through IARU (International Amateur Radio Union): free but takes 3--6 months - Very popular for university CubeSats (low cost, no license fees, large amateur community provides free ground station support via SatNOGS network)

**Commercial band licensing:** - Requires national filing (FCC in US,ISED in Canada, Ofcom in UK, CNES/ANFR in France) + ITU coordination for frequencies above 1 GHz - Costs: \$30--45K for S-band (typical), \$50--80K for X-band, \$100K+ for Ka-band - Timeline: 6--12 months for S-band, 12--18 months for X-band, 18+ months for Ka-band - Commercial data can be encrypted and proprietary - Bandwidth allocation may be limited by the national authority

## Band Selection Decision Tree

Required data rate?

- |                                                |                                                      |
|------------------------------------------------|------------------------------------------------------|
| $\leq 9.6$ kbps AND non-commercial/educational | -> VHF/UHF amateur (IARU, free, 3-6 mo)              |
| $\leq 10$ Mbps                                 | -> S-band commercial (ISED/FCC, \$30-45K, 6-12 mo)   |
| $\leq 400$ Mbps                                | -> X-band (ISED/FCC + ITU, \$50-80K, 12+ mo)         |
| $> 400$ Mbps                                   | -> Ka-band (complex ITU, \$100K+, 18+ mo, rain fade) |

## 6. Modulation and Coding -- Physics and Selection

[Source: CCSDS 131.0-B-4; Sklar, Ch. 7--8; Haykin, Ch. 10; DVB-S2 standard]

### Modulation -- How Information Becomes RF

Modulation encodes digital data onto an RF carrier by varying the carrier's amplitude, frequency, or phase.

**Phase Shift Keying (PSK):** The most common modulation family for space communications. The carrier phase is shifted by discrete amounts to represent different bit patterns:

- **BPSK (Binary PSK):** 2 phase states (0 deg and 180 deg). Each symbol carries 1 bit. Spectral efficiency: 1 bps/Hz. Most robust to noise.
- **QPSK (Quadrature PSK):** 4 phase states (0, 90, 180, 270 deg). Each symbol carries 2 bits. Spectral efficiency: 2 bps/Hz. Requires the same  $E_b/N_0$  as BPSK but doubles the data rate for the same bandwidth. **The standard choice for most space links.**
- **8PSK:** 8 phase states, 3 bits/symbol, spectral efficiency 3 bps/Hz. Requires ~3.5 dB more  $E_b/N_0$  than QPSK for the same BER.
- **16APSK:** 16 states (amplitude + phase), 4 bits/symbol, spectral efficiency 4 bps/Hz. Requires ~7 dB more  $E_b/N_0$  than QPSK.

**Frequency Shift Keying (FSK):** The carrier frequency is shifted between discrete values. Less spectrally efficient than PSK but more tolerant of amplifier nonlinearity. Variants: - **FSK:** Simple frequency switching. - **GFSK (Gaussian FSK):** Gaussian filter smooths frequency transitions, reducing spectral spreading. Used by many CubeSat UHF radios (AX.25 protocol at 9600 bps). - **MSK (Minimum Shift Keying):** A special case of FSK with minimum frequency deviation that still allows coherent detection. Constant envelope (important for nonlinear amplifiers). Spectral efficiency  $\sim 1$  bps/Hz.

**Why QPSK is preferred over BPSK for space links:** QPSK transmits 2 bits per symbol while requiring the same energy per bit as BPSK. This means QPSK achieves twice the data rate for the same bandwidth and the same  $E_b/N_0$  performance. The only additional complexity is that the receiver must resolve 4 phase states instead of 2, which is trivial for modern digital receivers.

### Forward Error Correction (FEC) -- Coding Gain

FEC adds redundant bits to the data stream before transmission. The receiver uses these redundant bits to detect and correct bit errors without retransmission. The improvement in  $E_b/N_0$  requirement (compared to uncoded) is called the **coding gain**.

**Why coding is essential for space links:** Space links operate at very low received signal power (femtowatts to picowatts). Without coding, the required  $E_b/N_0$  for BER  $10^{-6}$  is 10.5 dB. With LDPC coding, this drops to 2.0 dB -- a coding gain of 8.5 dB. This is equivalent to either: increasing the TX power by 7x, or increasing the antenna diameter by 2.7x, or reducing the data rate by 7x. Coding achieves the same benefit at the cost of only a few watts of digital processing power.

**Code types used in space communications:**

| Code Type                                | Code Rate   | Coding Gain at BER $10^{-6}$ | Decoding Complexity                | Standard      | Use                                                     |
|------------------------------------------|-------------|------------------------------|------------------------------------|---------------|---------------------------------------------------------|
| <b>Convolutional (K=7)</b>               | 1/2         | $\sim 5.5$ dB                | Low (Viterbi decoder, hardware)    | CCSDS         | Legacy telecommand, AX.25                               |
| <b>Convolutional + RS (concatenated)</b> | $\sim 0.44$ | $\sim 7.5$ dB                | Medium                             | CCSDS         | Standard CCSDS telemetry (many heritage missions)       |
| <b>Turbo code</b>                        | 1/2 to 1/6  | $\sim 8$ – $10$ dB           | High (iterative decoder)           | CCSDS         | Deep space links (Mars, Jupiter)                        |
| <b>LDPC (Low-Density Parity Check)</b>   | 1/2         | $\sim 8.5$ dB                | Medium-High (iterative BP decoder) | CCSDS, DVB-S2 | Modern space downlinks, <b>recommended for CubeSats</b> |
| <b>LDPC</b>                              | 3/4         | $\sim 6.5$ dB                | Medium-High                        | CCSDS, DVB-S2 | Balanced performance and throughput                     |
| <b>LDPC</b>                              | 7/8         | $\sim 5.0$ dB                | Medium-High                        | DVB-S2        | Maximum throughput, strong signal                       |

[Source: CCSDS 131.0-B-4; DVB-S2 ETSI EN 302 307]

**Code rate  $r$ :** The ratio of information bits to total transmitted bits. A rate-1/2 code transmits 1 information bit for every 2 transmitted bits (50% overhead). This means the channel data rate must be

$R_{\text{channel}} = R_{\text{info}}/r$  -- a rate-1/2 code requires twice the channel bandwidth for a given information rate.

### Complete Modulation + Coding Table

| Modulation + Coding       | $E_b/N_0$ Required (BER $10^{-6}$ ) | Spectral Efficiency (bps/Hz) | Typical Use                    | Implementation   |
|---------------------------|-------------------------------------|------------------------------|--------------------------------|------------------|
| GMSK uncoded              | 10.5 dB                             | ~1.0                         | AX.25 amateur, legacy          | Simple radio IC  |
| BPSK uncoded              | 10.5 dB                             | 1.0                          | Legacy telecommand             | Simple           |
| QPSK uncoded              | 10.5 dB                             | 2.0                          | Simple telemetry               | Moderate         |
| QPSK + conv (r=1/2, K=7)  | 5.0 dB                              | 1.0                          | Standard CCSDS TM              | Hardware Viterbi |
| QPSK + conv + RS (concat) | 3.0 dB                              | 0.88                         | Heritage CCSDS                 | Hardware         |
| QPSK + LDPC (r=1/2)       | 2.0 dB                              | 1.0                          | High-efficiency downlink       | FPGA-based       |
| QPSK + LDPC (r=3/4)       | 4.0 dB                              | 1.5                          | <b>Balanced -- recommended</b> | FPGA-based       |
| QPSK + LDPC (r=7/8)       | 5.5 dB                              | 1.75                         | Bandwidth-limited              | FPGA-based       |
| 8PSK + LDPC (r=3/4)       | 6.5 dB                              | 2.25                         | High-rate downlink             | FPGA-based       |
| 16APSK + LDPC (r=3/4)     | 8.5 dB                              | 3.0                          | Maximum throughput             | FPGA-based       |

**Design guidance for CubeSats:** - **UHF amateur:** GMSK or AFSK (AX.25 protocol) at 1.2--9.6 kbps channel rate. Add convolutional coding if link margin is tight. - **S-band:** QPSK + LDPC (r=1/2 or r=3/4). This is the sweet spot: 5.5--8.5 dB coding gain with manageable complexity. Most COTS S-band CubeSat transmitters (Endurosat, NanoAvionics, AAC Clyde) support DVB-S2 or CCSDS LDPC natively. - **X-band / Ka-band:** 8PSK or 16APSK + LDPC for maximum spectral efficiency when bandwidth is limited.

## 7. Ground Stations

The ground station is half of the communication link. Its performance ( $G/T$ ) directly determines the achievable data rate. Upgrading the ground station is often the cheapest way to improve link performance (compared to upgrading the spacecraft transmitter or antenna).



## Ground Station Types

| Type                      | Antenna         | G/T (S-band)    | Cost                           | Availability                  | Examples                                          |
|---------------------------|-----------------|-----------------|--------------------------------|-------------------------------|---------------------------------------------------|
| <b>Amateur (SatNOGS)</b>  | 10--15 dBi Yagi | -15 to -10 dB/K | Free (volunteer-operated)      | Global network, 200+ stations | SatNOGS network                                   |
| <b>University</b>         | 2--3 m dish     | +10 to +15 dB/K | \$50--200K (build)             | Limited availability          | Many universities have S-band stations            |
| <b>Commercial (small)</b> | 3--5 m dish     | +15 to +25 dB/K | \$500/pass or \$5--20K/month   | KSAT Lite, AWS Ground Station | KSAT, Amazon, Leaf Space                          |
| <b>Commercial (large)</b> | 5--13 m dish    | +25 to +35 dB/K | \$1000/pass or \$20--50K/month | SSC, KSAT, ATLAS              | SSC (Esrange), KSAT (Svalbard), ATLAS (Fairbanks) |
| <b>Deep Space Network</b> | 34--70 m dish   | +45 to +60 dB/K | NASA-funded only               | DSN (3 sites globally)        | Goldstone, Canberra, Madrid                       |

[Source: KSAT Lite pricing 2024; AWS Ground Station pricing; SSC SmallSat ground segment]

### Contact geometry and pass duration:

A LEO satellite is in view of a ground station for a limited time per orbit. The pass duration depends on the orbit altitude and the maximum elevation angle:

$$t_{\text{pass}} \approx \frac{2}{\pi} \arccos \left( \frac{\cos(\epsilon_{\text{max}})}{\cos(\epsilon_{\text{min}})} \right)$$

For a simplified estimate:

| Altitude | Min Elevation | Max Pass Duration | Typical Usable Duration | Passes/Day (mid-latitude) |
|----------|---------------|-------------------|-------------------------|---------------------------|
| 400 km   | 10 deg        | ~7 min            | ~5 min                  | 3--4                      |
| 500 km   | 10 deg        | ~8 min            | ~6 min                  | 4--5                      |
| 600 km   | 10 deg        | ~9 min            | ~7 min                  | 4--5                      |
| 800 km   | 10 deg        | ~11 min           | ~9 min                  | 5--6                      |

The "usable duration" is shorter than the total pass because the link only closes above the minimum elevation angle, and the first/last 30--60 seconds are used for signal acquisition and link setup.

**Ground station selection criteria:**

- **Location:** Polar stations (Svalbard at 78 degN, McMurdo at 78 degS) see every orbit of a polar/SSO satellite, providing 12+ contacts per day. Mid-latitude stations see only 3--5 passes. Equatorial stations see even fewer passes of polar satellites.
- **G/T:** Determines the achievable data rate. A 3 dB improvement in G/T allows doubling the data rate.
- **Licensing:** Must be compatible with the spacecraft frequency allocation
- **Cost:** Ranges from free (SatNOGS, university) to \$1000+/pass (commercial)
- **Reliability:** SLA (service-level agreement) for commercial stations; university stations may have limited operator availability

**EIRP (ground station transmit, for uplink):** For telecommand uplink, the ground station transmits to the spacecraft. Ground station EIRP is typically 40--60 dBW for commercial stations, sufficient for robust command links even with low-gain spacecraft receive antennas.

## 8. Worked Examples: Complete Link Budgets

### 3U EO CubeSat -- S-band Downlink

#### Worked Example -- S-band Downlink for 3U EO CubeSat (SuperDove-class)

**Scenario:** 500 km SSO, S-band (2250 MHz), 1 Mbps downlink, 10 deg minimum elevation (slant range 1300 km), 3 m ground station dish ( $T_{\text{sys}} = 150$  K,  $G_{\text{RX}} = 35$  dBi at S-band).

| Line | Parameter                                                   | Value        | Unit     | Calculation/Source                                 |
|------|-------------------------------------------------------------|--------------|----------|----------------------------------------------------|
| 1    | TX Power (2 W)                                              | +3.0         | dBW      | COTS S-band TX (Endurosat)                         |
| 2    | TX Antenna Gain (patch)                                     | +6.0         | dBi      | Single-element S-band patch                        |
| 3    | TX Line Losses                                              | -1.5         | dB       | 15 cm cable + connector                            |
| 4    | <b>EIRP</b>                                                 | <b>+7.5</b>  | dBW      | $3.0 + 6.0 - 1.5$                                  |
| 5    | FSPL (2250 MHz, 1300 km)                                    | -170.8       | dB       | $92.45 + 20\log_{10}(1300) + 20\log_{10}(2.25)$    |
| 6    | Atmospheric Loss                                            | -0.5         | dB       | S-band, 10 deg elevation                           |
| 7    | Pointing Loss                                               | -1.0         | dB       | 5 deg mispoint, 80 deg beamwidth patch             |
| 8    | Polarisation Loss (RHCP-RHCP)                               | -0.3         | dB       | Minor axial ratio mismatch                         |
| 9    | RX Antenna Gain (3 m dish)                                  | +35.0        | dBi      | $10\log_{10}(0.6 \times (\pi \times 3.0/0.133)^2)$ |
| 10   | System Noise Temp (150 K)                                   | 21.8         | dBK      | Professional LNA + sky temp                        |
| 11   | <b>G/T</b>                                                  | <b>+13.2</b> | dB/K     | $35.0 - 21.8$                                      |
| 12   | Boltzmann Constant                                          | +228.6       | dBW/K/Hz | — $k$ in link equation                             |
| 13   | <b>C/N<sub>0</sub></b>                                      | <b>+76.7</b> | dBHz     | $7.5 - 170.8 - 0.5 - 1.0 - 0.3 + 13.2 + 228.6$     |
| 14   | Data Rate (1 Mbps)                                          | 60.0         | dBbps    | $10\log_{10}(10^6)$                                |
| 15   | <b>E<sub>b</sub>/N<sub>0</sub> available</b>                | <b>+16.7</b> | dB       | $76.7 - 60.0$                                      |
| 16   | E <sub>b</sub> /N <sub>0</sub> required (QPSK + LDPC r=3/4) | 4.0          | dB       | From modulation/coding table                       |
| 17   | Implementation Loss                                         | 2.0          | dB       | Real demodulator vs ideal                          |
| 18   | <b>LINK MARGIN</b>                                          | <b>+10.7</b> | dB       | $16.7 - 4.0 - 2.0$                                 |

**Result:** Link closes with 10.7 dB margin (requirement:  $\geq 3$  dB). **Pass.**

**Design insight:** The generous 10.7 dB margin suggests the link is over-designed for 1 Mbps. The team could increase the data rate:

Maximum data rate at 3 dB margin:  $E_b/N_0$  available at max rate =  $4.0 + 2.0 + 3.0 = 9.0$  dB

$C/N_0 = 76.7$  dBHz, so  $R_{b, \max} = 10^{(76.7 - 9.0)/10} = 10^{6.77} \approx \mathbf{5.9 \text{ Mbps}}$  at 3 dB margin.

Alternatively, at 5 Mbps ( $10 \log_{10}(5 \times 10^6) = 67.0$  dBbps):  $E_b/N_0 = 76.7 - 67.0 = 9.7$  dB. Margin =  $9.7 - 4.0 - 2.0 = 3.7$  dB. **Pass** (barely).

## 1U Worked Example: UniSat-1

### UHF Link Budget: 437 MHz at 9600 bps

UniSat-1 uses the UHF amateur band at 437 MHz with a ground station equipped with a 10 dBi Yagi antenna. This is the lowest-cost and simplest communication architecture available to CubeSat missions.

#### Worked Example -- UHF Downlink Link Budget for UniSat-1

**Scenario:** 400 km orbit, UHF (437 MHz), 9600 bps downlink (GMSK), 10 deg minimum elevation angle, amateur ground station with 10 dBi Yagi antenna,  $T_{\text{sys}} = 600$  K (COTS LNA, coax cable losses, sky noise near horizon).

**Slant range at 10 deg elevation:** From 400 km altitude, the worst-case slant range at 10 deg elevation is approximately 1150 km.

| Line | Parameter                         | Value        | Unit     | Notes                                                                             |
|------|-----------------------------------|--------------|----------|-----------------------------------------------------------------------------------|
| 1    | TX Power (0.5 W)                  | -3.0         | dBW      | Standard CubeSat UHF radio                                                        |
| 2    | TX Antenna Gain (monopole)        | 0.0          | dBi      | Quarter-wave monopole, ~omnidirectional                                           |
| 3    | TX Line Losses                    | -0.5         | dB       | Short cable to antenna                                                            |
| 4    | <b>EIRP</b>                       | <b>-3.5</b>  | dBW      | Low EIRP is the fundamental UHF challenge                                         |
| 5    | FSPL (437 MHz, 1150 km)           | -155.5       | dB       | Lower than S-band (good)                                                          |
| 6    | Atmospheric Loss                  | -0.3         | dB       | UHF atmospheric loss is minimal                                                   |
| 7    | Pointing Loss (omni antenna)      | -0.5         | dB       | Omni pattern -- negligible                                                        |
| 8    | Polarisation Loss (linear-linear) | -3.0         | dB       | <b>Major loss</b> -- Faraday rotation in ionosphere rotates polarisation randomly |
| 9    | RX Antenna Gain (10 dBi Yagi)     | +10.0        | dBi      | 5-element Yagi, manually tracked                                                  |
| 10   | System Noise Temp (600 K)         | 27.8         | dBK      | Amateur station: warm LNA + cable loss                                            |
| 11   | <b>G/T</b>                        | <b>-17.8</b> | dB/K     | Low G/T is the ground station limitation                                          |
| 12   | Boltzmann Constant                | +228.6       | dBW/K/Hz |                                                                                   |
| 13   | <b>C/N<sub>0</sub></b>            | <b>+48.0</b> | dBHz     |                                                                                   |

| Line | Parameter                         | Value | Unit  | Notes |
|------|-----------------------------------|-------|-------|-------|
| 14   | Data Rate (9600 bps)              | 39.8  | dBbps |       |
| 15   | $E_b/N_0$ available               | +8.2  | dB    |       |
| 16   | $E_b/N_0$ required (GMSK uncoded) | 10.5  | dB    |       |
| 17   | Implementation Loss               | -2.0  | dB    |       |

**Margin = 8.2 - 10.5 - 2.0 = -4.3 dB. The link does NOT close!**

**Fix 1: Add FEC.** Convolutional coding ( $r=1/2$ ,  $K=7$ ): -  $E_b/N_0$  required drops to **5.0 dB** (5.5 dB coding gain) - Channel rate remains 9600 bps, but useful throughput is 4800 bps (half is redundancy) - **Margin = 8.2 - 5.0 - 2.0 = +1.2 dB** -- still below 3 dB requirement.

**Fix 2: Upgrade ground antenna to cross-Yagi (13 dBi) with circular polarisation:** - RX gain: +13.0 dBi (was +10.0) -> +3.0 dB improvement - Polarisation loss: -0.5 dB (was -3.0 dB, now RHCP-to-RHCP) -> +2.5 dB improvement - Net improvement: +5.5 dB - New  $C/N_0$ : 53.5 dBHz - New  $E_b/N_0$  available: 53.5 - 39.8 = 13.7 dB - **Margin = 13.7 - 5.0 - 2.0 = +6.7 dB** -- **Pass** (> 3 dB).

**Polarisation loss physics:** At UHF (437 MHz), the ionosphere causes Faraday rotation of linearly polarised signals. The rotation angle depends on the total electron content (TEC) along the path and varies with time of day, solar activity, and signal path. If the satellite transmits linear polarisation and the ground station receives with linear polarisation, the polarisation planes may be orthogonal at times, causing up to complete signal loss (theoretically infinite loss, practically 20+ dB fades). Using circular polarisation on at least one end (preferably both) eliminates Faraday rotation loss, reducing it to ~0.3--0.5 dB axial ratio mismatch. **This is why circular polarisation is mandatory for reliable UHF satellite links.**

**Final link budget summary (with FEC + cross-Yagi):**

| Parameter         | Value                       |
|-------------------|-----------------------------|
| TX power          | 0.5 W                       |
| TX antenna        | Monopole (0 dBi)            |
| Frequency         | 437 MHz                     |
| Channel rate      | 9600 bps                    |
| Useful throughput | 4800 bps (with $r=1/2$ FEC) |
| Ground antenna    | 13 dBi cross-Yagi, RHCP     |
| Link margin       | <b>+6.7 dB</b>              |

**Data throughput for UniSat-1:** At 4800 bps useful throughput, a 7-minute pass delivers:

$$V_{\text{pass}} = 4800 \times 420 \times 0.85 = 1.71 \text{ Mbit} = 214 \text{ kB per pass.}$$

The 0.85 factor accounts for protocol overhead (packet headers, acknowledgements, retransmissions, link setup time).

With 4 passes/day:  $V_{\text{daily}} = 856 \text{ kB/day} \approx \mathbf{0.84 \text{ MB/day}}$ . The magnetometer generates  $< 1 \text{ kbps} \times 600 \text{ s/orbit} \times 15 \text{ orbits} = 9 \text{ Mbit/day} = 1.13 \text{ MB/day}$ . This is **marginal** -- the team may need to prioritise data or add a second ground station. Using the SatNOGS network (200+ volunteer stations worldwide) could provide 10+ additional contacts per day at no cost.

**Key lesson from UniSat-1 link budget:** UHF links are power-starved compared to S-band or X-band. The lower FSPL at 437 MHz (~15 dB less than S-band) does not compensate for the low TX power (0.5 W vs 2 W = 6 dB less), low antenna gain (0 dBi vs 6 dBi = 6 dB less), and higher system noise temperature of amateur stations (600 K vs 150 K = 6 dB worse). FEC coding and circular polarisation are essential for closing a UHF CubeSat link.

## 9. Data Budget

The data budget determines whether the communication system can deliver all mission data to the ground. Even if the link budget closes, the mission fails if the total data volume exceeds the downlink capacity.

### Key Equations -- Data Budget

#### Daily data generation:

$$V_{\text{gen}} = R_{\text{payload}} \times t_{\text{imaging}} \times N_{\text{orbits}} \times f_{\text{compression}}$$

#### Daily downlink capacity:

$$V_{\text{DL}} = R_{\text{downlink}} \times t_{\text{contact}} \times N_{\text{passes}} \times \eta_{\text{protocol}}$$

where  $\eta_{\text{protocol}} = 0.80\text{--}0.90$  accounts for packet overhead, retransmissions, link setup time, and handshaking.

#### Data budget closure:

$$V_{\text{DL}} \geq V_{\text{gen}} \quad (\text{data budget closes})$$

**Backlog clearance time:** If  $V_{\text{DL}} < V_{\text{gen}}$  per day, data accumulates in onboard storage. The backlog clearance time is:

$$t_{\text{clear}} = \frac{V_{\text{stored}}}{V_{\text{DL}} - V_{\text{gen}}}$$

If  $V_{\text{DL}} < V_{\text{gen}}$ , the backlog grows forever -- the mission cannot sustain its data generation rate.

### Worked Example -- Data Budget for 3U EO CubeSat

**Generation:** 240 Mbps raw imaging data x 5 min/orbit x 15 orbits/day x 0.25 (4:1 JPEG2000 compression) =  $240 \times 300 \times 15 \times 0.25 = 270,000 \text{ Mbit/day} = \mathbf{33.75 \text{ GB/day}}$

Wait -- that's extremely high. Let's be more realistic about imaging time. Not every orbit has a target. Assume 4 imaging passes per day, 5 minutes each:

$$V_{\text{gen}} = 240 \times 10^6 \times 300 \times 4 \times 0.25 = 72,000 \text{ Mbit} = \mathbf{9.0 \text{ GB/day}}$$

Still high. Planet SuperDove images approximately 1--2 minutes per orbit over priority targets, compresses heavily (10:1+), and downlinks selectively.

**Revised:**  $240 \text{ Mbps} \times 1 \text{ min/pass} \times 4 \text{ passes/day} \times 0.10 \text{ (10:1 compression)} = 240 \times 60 \times 4 \times 0.10 = 5,760 \text{ Mbit} = \mathbf{720 \text{ MB/day}}$

**Downlink (S-band at 5 Mbps):**  $5 \text{ Mbps} \times 6 \text{ min/pass} \times 5 \text{ passes/day} \times 0.85 = 153 \text{ Mbit/pass} \times 5 = 765 \text{ Mbit} = \mathbf{95.6 \text{ MB/day} \times 5 = 478 \text{ MB/day}}$

Hmm, let's recompute carefully:  $V_{DL} = 5 \times 10^6 \times 360 \times 5 \times 0.85 = 7,650 \text{ Mbit} = 956 \text{ MB/day}$

**Result:**  $956 \text{ MB/day} > 720 \text{ MB/day}$ . **Data budget closes** with 33% margin.

**Sensitivity:** If imaging duty cycle doubles (2 min/pass), generation rises to  $1440 \text{ MB/day} > 956 \text{ MB/day}$ . Options: (a) X-band for higher data rate, (b) additional ground stations, (c) more aggressive compression, (d) onboard data prioritisation/selection.

## 10. SpaceCDF Exercise

### Instructions

1. **Spectrum Selector** (Dashboard): Select your license type and frequency band
2. **Link Budget** tab: - Enter TX power, antenna type, frequency, ground station parameters - Review the computed link margin - Verify it meets  $\geq 3 \text{ dB}$  requirement
3. **Compare** to your hand calculation from the worked example
4. **Equipment Browser:** Select a transponder and antenna that match your band choice - Note RF compatibility warnings (band mismatch between transponder and antenna)
5. Complete Worksheet 3.3

### Discussion Questions

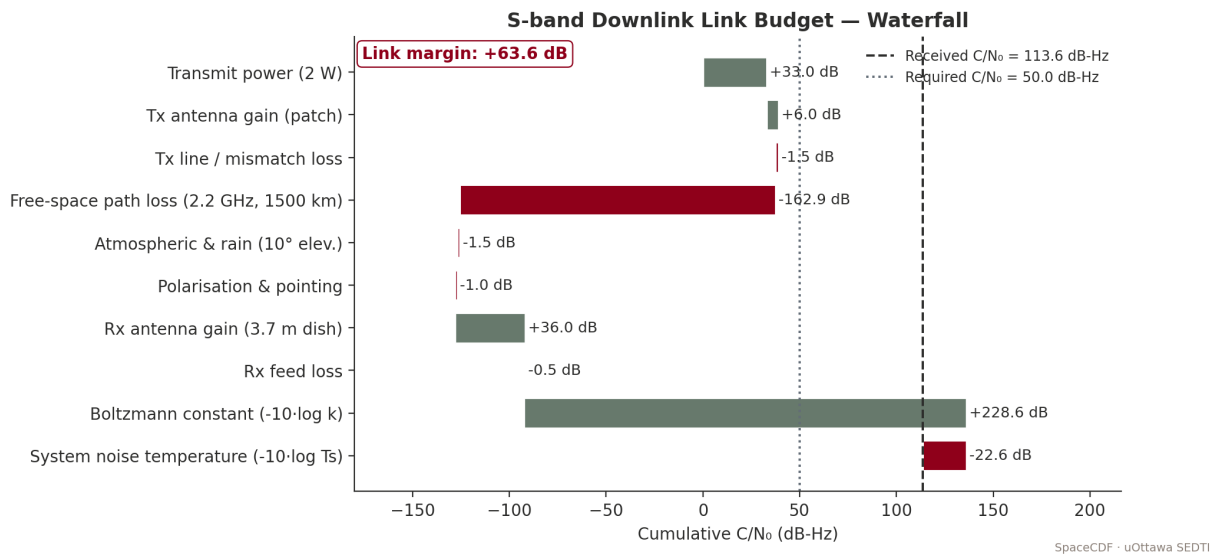
- What is the most impactful parameter in your link budget? (Usually FSPL or G/T)
- How does doubling the data rate affect link margin? (Reduces by 3 dB -- because  $10\log_{10}(2) = 3 \text{ dB}$ )
- Could you use a lower-power transmitter and still close the link? What is the minimum TX power?
- Does your data budget close? If not, what is the cheapest fix? (Usually: more ground station passes, or lower imaging duty cycle)
- What is the effect of Faraday rotation on your UHF link? (If applicable)

## Session Summary

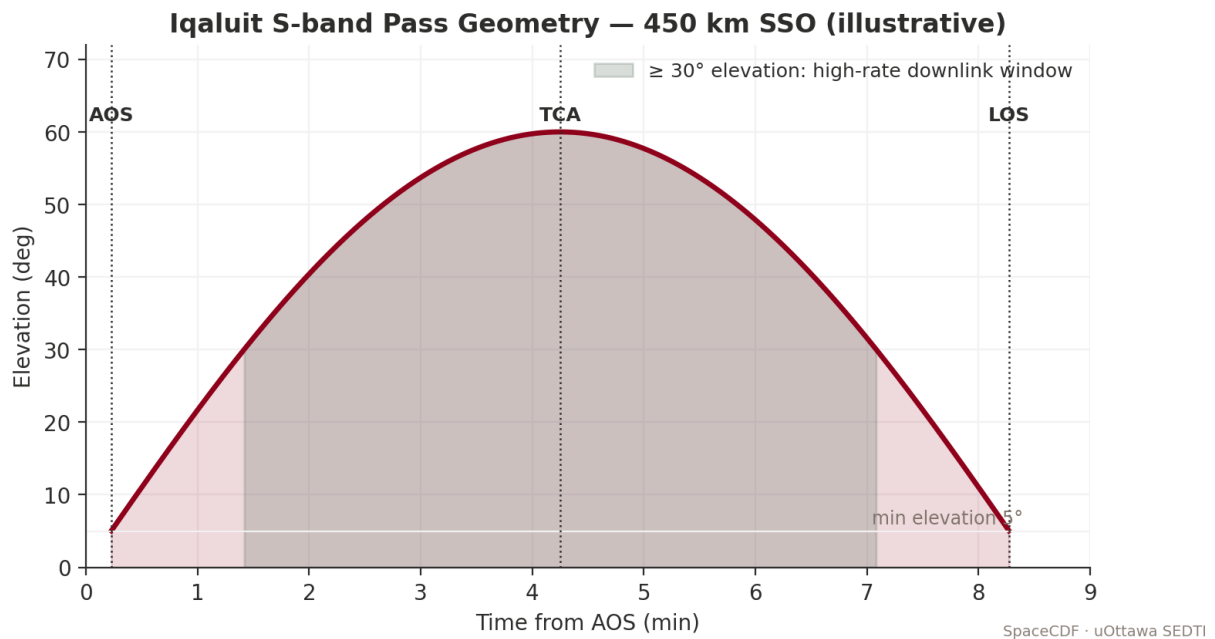
| Topic        | Key Takeaway                                                                                          |
|--------------|-------------------------------------------------------------------------------------------------------|
| Link budget  | $\text{Margin} = E_b/N_{0, \text{avail}} - E_b/N_{0, \text{req}} - L_{\text{impl}} \geq 3 \text{ dB}$ |
| EIRP         | $EIRP = P_{TX} + G_{TX} - L_{TX}$ -- transmitter's effective power in beam direction                  |
| FSPL physics | Geometric spreading ( $1/d^2$ ) + aperture scaling ( $\lambda^2$ ); not energy absorption             |
| FSPL formula |                                                                                                       |

| Topic           | Key Takeaway                                                                                                              |
|-----------------|---------------------------------------------------------------------------------------------------------------------------|
|                 | $92.45 + 20\log_{10}(d_{\text{km}}) + 20\log_{10}(f_{\text{GHz}})$ ; increases 6 dB per doubling of frequency or distance |
| G/T             | Receiver figure of merit: antenna gain minus noise temperature; most important ground station parameter                   |
| Antenna types   | Monopole (0 dBi, omni) to parabolic (25--45 dBi, narrow beam); gain vs pointing trade-off                                 |
| Antenna gain    | $G = \eta_a (\pi D / \lambda)^2$ ; patch ~6 dBi; 3 m dish: 35 dBi at S-band, 25 dBi at X-band                             |
| Pointing loss   | $-12(\Delta\theta/\theta_{3\text{dB}})^2$ dB; narrower beam = more sensitive to pointing errors                           |
| Modulation      | QPSK: 2 bits/symbol, same $E_b/N_0$ as BPSK but 2x spectral efficiency; <b>standard choice</b>                            |
| FEC coding      | LDPC (r=3/4): $E_b/N_0 = 4.0$ dB, coding gain ~6.5 dB; <b>essential for space links</b>                                   |
| Coding gain     | 5--10 dB improvement for free (just digital processing); equivalent to 3--10x power increase                              |
| UHF challenges  | Low EIRP, Faraday rotation (use circular polarisation), high ground station noise; FEC mandatory                          |
| Band selection  | UHF for < 9.6 kbps (free, amateur); S-band for < 10 Mbps; X-band for < 400 Mbps; Ka-band for > 400 Mbps                   |
| Rain fade       | Negligible below 4 GHz; 1--3 dB at X-band; 3--15 dB at Ka-band; add margin or adaptive rate                               |
| Ground stations | G/T: amateur -15 dB/K, university +12 dB/K, commercial +25 dB/K; polar stations see every orbit                           |
| Data budget     | Daily downlink capacity must exceed daily data generation; compression ratio is a key lever                               |
| Licensing       | Amateur (free, IARU, 3--6 mo); S-band (\$30--45K, 6--12 mo); X-band (\$50--80K, 12+ mo)                                   |

## Session 3.4: Structure, Propulsion, and Equipment Selection

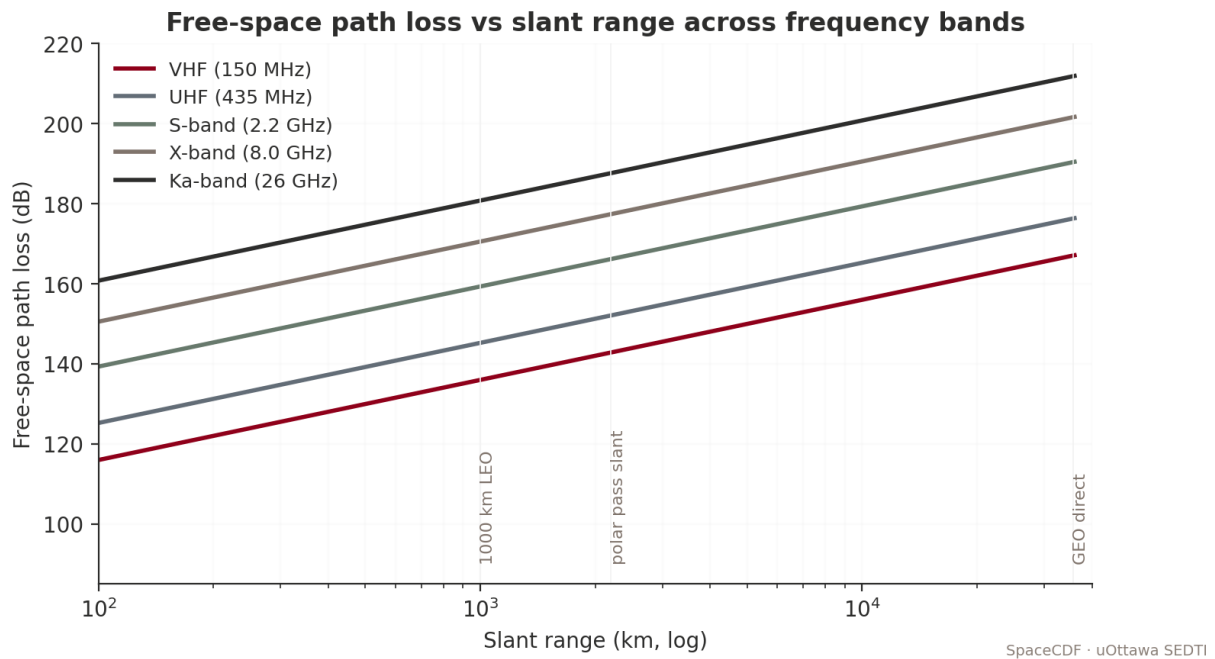


*S-band downlink link-budget waterfall.*

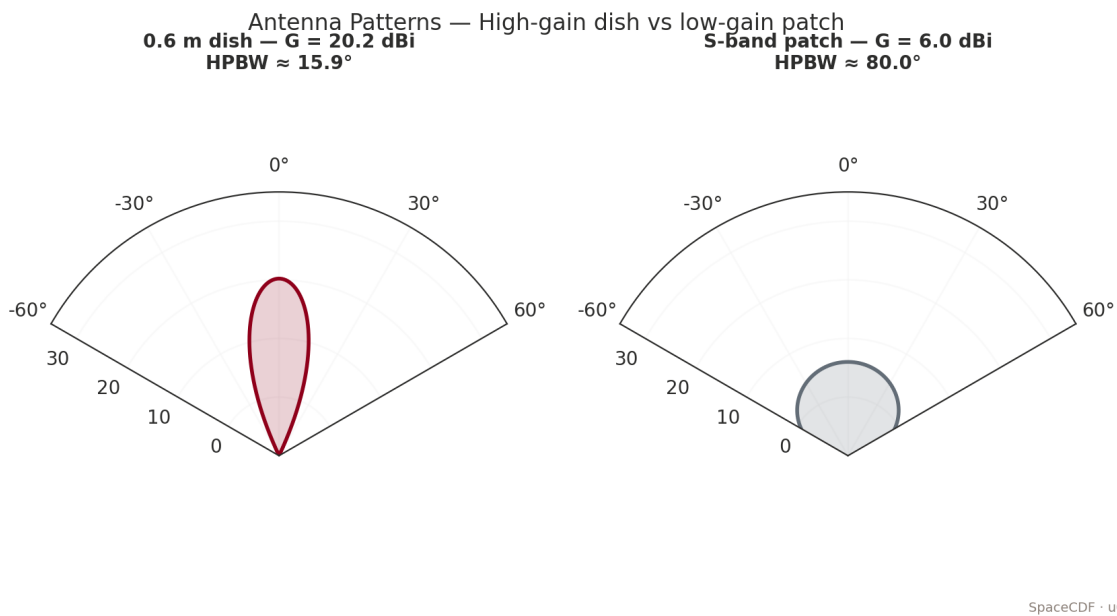


*Iqaluit S-band pass geometry.*

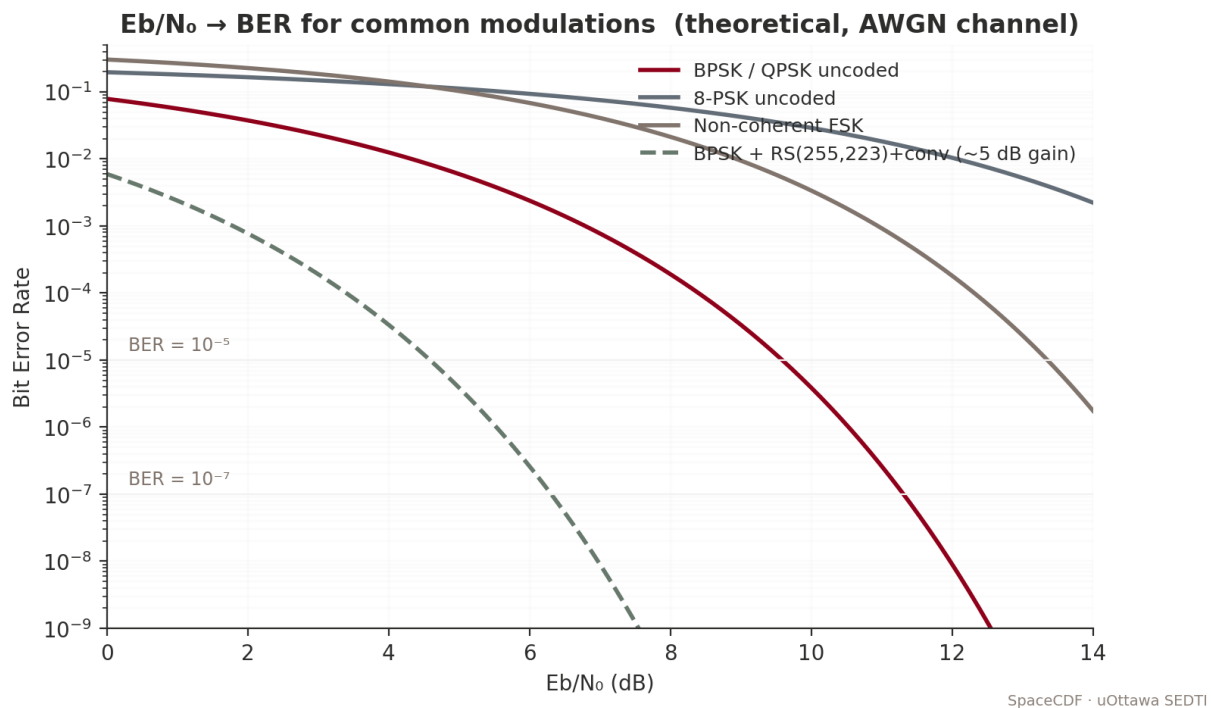




Free-space path loss vs slant range across bands.



Antenna patterns: high-gain dish vs patch.



$E_b/N_0$  vs BER for common modulations.

**Expected reading before this session.** SMAD4 Ch. 13 (comms); CCSDS PUS §1; Pratt et al. Ch. 4.

**Prerequisites:** Sessions 2.1--3.3 (all subsystem sizing complete) **SpaceCDF Tabs:** Equipment Browser, Dashboard, Trade Studies, Budget Breakdown

## References

- Cal Poly, *CubeSat Design Specification (CDS) Rev 14.1*, February 2022
- Wertz, Everett & Puschell, *Space Mission Engineering: The New SMAD*, 2011, Ch. 11.3 (Structure), Ch. 17 (Propulsion)
- ECSS, *ECSS-E-ST-32C Rev.1: Structural General Requirements*, 2008
- ECSS, *ECSS-E-ST-35C: Propulsion General Requirements*, 2008
- Sarafin, *Spacecraft Structures and Mechanisms*, 1995
- Sutton & Biblarz, *Rocket Propulsion Elements*, 9th ed., 2017, Ch. 2--4
- Enpulsion, *NANO R3 Thruster Datasheet*, 2023
- VACCO, *MiPS Propulsion System Datasheet*, 2023
- Tyvak, *Structure Specifications*, 2023
- ECSS, *ECSS-E-ST-10-03C: Testing*, 2012

## Learning Objectives

By the end of this session, participants will be able to:

1. Verify CubeSat Design Specification (CDS) compliance for 1U--12U form factors
2. Explain the structural load environment (quasi-static, vibration, shock) and its physical origins

3. Compute structural margin of safety for quasi-static launch loads
4. Estimate the fundamental frequency requirement and verify it against deployer specifications
5. Apply the Tsiolkovsky rocket equation to compute propellant mass for a given  $\Delta V$
6. Explain the physics of each propulsion technology and select based on mission requirements
7. Select OBC architecture based on processing, radiation, and interface requirements
8. Size onboard data storage from the data budget
9. Select flight hardware using SpaceCDF's equipment browser with budget tracking

## 1. CubeSat Structure and CDS Compliance

[Source: Cal Poly CDS Rev 14.1, February 2022; ECSS-E-ST-32C]

### CDS Dimensional Specifications

| Form Factor | Dimensions (mm)       | Max Mass (kg) | Internal Volume (cm <sup>3</sup> ) | Typical Deployer                 |
|-------------|-----------------------|---------------|------------------------------------|----------------------------------|
| 1U          | 100 x 100 x 113.5     | 2.0           | ~1000                              | ISIPOD, P-POD                    |
| 1.5U        | 100 x 100 x 170.2     | 3.0           | ~1500                              | ISIPOD                           |
| 2U          | 100 x 100 x 227.0     | 4.0           | ~2000                              | ISIPOD, P-POD                    |
| 3U          | 100 x 100 x 340.5     | 6.0           | ~3000                              | P-POD, ISIPOD, NanoRacks NRCSD   |
| 6U          | 100 x 226.3 x 340.5   | 12.0          | ~6000                              | 6U deployer (Exolaunch, D-Orbit) |
| 12U         | 226.3 x 226.3 x 340.5 | 24.0          | ~12000                             | 12U deployer (Exolaunch)         |

### Structural Materials

#### CubeSat rail material: Aluminium 7075-T6

This is the most commonly specified structural aluminium alloy for CubeSat rails. The CDS mandates hard-anodised aluminium for the rails (the four load-bearing edges that slide along the deployer guide channels).

| Property                           | Al 7075-T6 | Al 6061-T6 | Ti-6Al-4V | CFRP (quasi-isotropic)       |
|------------------------------------|------------|------------|-----------|------------------------------|
| Density (kg/m <sup>3</sup> )       | 2810       | 2700       | 4430      | 1600                         |
| Yield strength $\sigma_y$ (MPa)    | 503        | 276        | 880       | N/A (use ultimate)           |
| Ultimate strength $\sigma_u$ (MPa) | 572        | 310        | 950       | 500–800                      |
| Young's modulus $E$ (GPa)          | 71.7       | 68.9       | 114       | 70–150 (direction-dependent) |
| CTE (ppm/degC)                     | 23.6       | 23.1       | 8.6       | 0–2 (tuneable)               |
| Thermal conductivity (W/m/K)       | 130        | 167        | 6.7       | 3–10                         |

[Source: MMPDS / ASM Handbook; Hexcel HexPly datasheets]

**Why Al 7075-T6 for rails:** - High strength-to-weight ratio (superior to 6061-T6) - Hard anodisation provides a durable, low-friction surface finish for deployer guide channel contact (reduces galling, provides electrical insulation) - Good machinability - Extensive flight heritage (virtually every CubeSat ever launched) - CTE-matched to Al deployer structure (prevents differential thermal expansion binding)

**Why NOT other materials for rails:** - **Titanium:** Excellent strength but poor thermal conductivity (thermal hot spots), difficult to machine, risk of galling against aluminium deployer - **CFRP:** Cannot be anodised; CTE mismatch with Al deployer causes binding at temperature extremes; poor electrical conductivity (grounding/bonding issues) - **Stainless steel:** Too heavy; poor CTE match

**Anodisation physics:** Anodisation is an electrochemical process that grows a hard aluminium oxide ( $\text{Al}_2\text{O}_3$ ) layer on the surface. Hard anodisation (Type III) produces a 25--75  $\mu\text{m}$  thick oxide layer with hardness of 60--70 HRC (harder than most steel). This layer provides: wear resistance against deployer contact, electrical insulation (prevents arcing between satellite and deployer), corrosion resistance, and controlled surface optical properties ( $\alpha_s \approx 0.3\text{--}0.5$ ,  $\varepsilon \approx 0.8\text{--}0.85$  for clear anodise;  $\alpha_s \approx 0.9$ ,  $\varepsilon \approx 0.85$  for black anodise).

### Key CDS Requirements

| Requirement                    | Specification                                            | Physical Rationale                                                                                                                                      |
|--------------------------------|----------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------|
| Rail material                  | Hard anodised aluminium (7075-T6 or 6061-T6)             | Wear resistance, CTE match to deployer, electrical isolation                                                                                            |
| Rail cross-section             | 8.5 x 8.5 mm minimum contact area                        | Adequate bearing area for launch loads; prevents rail yielding under quasi-static acceleration                                                          |
| Surface finish                 | All external surfaces anodised or non-outgassing coating | Prevent contamination of other payloads on launch vehicle (molecular outgassing deposits on optics)                                                     |
| Deployment switches            | Minimum 1 on each accessible rail face (+X, -X)          | Inhibit all spacecraft activity until fully deployed from deployer (prevents inadvertent deployment, RF emissions in fairing)                           |
| Remove Before Flight (RBF) pin | Required; physically disables all power systems          | Final safety inhibit; removed at launch pad after integration; ensures zero RF emissions and zero deployment actuator current until intentional removal |
| Protrusions                    | None beyond rail envelope in stowed configuration        | Ensures clean ejection from deployer; prevents snagging on guide rails or adjacent CubeSat                                                              |
| Centre of gravity              | Within 2 cm of geometric centre (per deployer ICD)       | Prevents wobble during deployment ejection; ensures all CubeSats eject with similar tip-off rates                                                       |
| Fundamental frequency          | > 40 Hz first mode (typical deployer requirement)        | Prevents dynamic coupling between satellite and launch vehicle structural modes (which cluster at 10--30 Hz)                                            |

### PC/104 Stack Architecture

Most CubeSat avionics use the PC/104-compatible stack architecture, a heritage from the industrial embedded computing standard adapted for space:

**Physical specifications:** - **Board size:** 96 x 90 mm (standard) or 90 x 96 mm - **Connector:** 104-pin stack-through header (2 x 52 pins, 2.54 mm pitch) -- original PC/104 pinout carries power + I2C + SPI + UART + GPIO - **Stack spacing:** Typically 10--15 mm between boards (constrained by component height and

connector mating height) - **Stack capacity:** 1U accommodates ~4 boards; 3U accommodates ~12 boards (340 mm / ~28 mm per board slot)

**What rides on the stack:** - EPS board (battery management, MPPT, power distribution) - OBC board (processor, memory, interfaces) - Communications board (UHF radio, or S-band transponder) - AOCS board (if integrated -- some vendors combine IMU + magnetorquer driver + RW interface on one PCB) - Payload interface board (ADC, sensor interfaces)

**Mechanical concerns:** - Solder joints are the weakest point; random vibration causes fatigue cracking at heavy component leads (especially tall electrolytic capacitors, large connectors, and crystal oscillators) - Board-to-board connectors must be properly preloaded (too loose = intermittent contact; too tight = difficult assembly/disassembly during I&T) - Standoffs and spacers must be correctly torqued; Loctite 222 (low-strength threadlocker) is standard

## Launch Load Environment

The launch environment subjects the satellite to loads from engine thrust, aerodynamic buffeting, stage separation, and pyrotechnic events. The satellite must survive all of these without structural failure or functional degradation.

| Load Type                        | Physical Source                                                                 | Typical Level                              | Duration           | Frequency Range       | Verification Method                        |
|----------------------------------|---------------------------------------------------------------------------------|--------------------------------------------|--------------------|-----------------------|--------------------------------------------|
| <b>Quasi-static acceleration</b> | Engine thrust + aeroloading                                                     | 6--12 g axial, 2--4 g lateral              | Seconds to minutes | 0 (static equivalent) | Analysis + sine vibration test             |
| <b>Sine vibration</b>            | Low-frequency vehicle dynamics                                                  | 0.5--3 g (5--100 Hz)                       | Minutes            | 5--100 Hz             | Sine sweep test (3 axes)                   |
| <b>Random vibration</b>          | Acoustic noise + turbulent boundary layer                                       | 5--15 grms (20--2000 Hz)                   | 60--120 s per axis | 20--2000 Hz           | Random vibration test (3 axes)             |
| <b>Shock</b>                     | Pyrotechnic separation events (stage sep, fairing sep, deployer spring release) | 500--2000 g at separation (high frequency) | < 10 ms            | 100--10,000 Hz        | Shock response spectrum (SRS) test         |
| <b>Acoustic</b>                  | Sound pressure from engine exhaust, aerodynamic noise                           | 120--140 dB (20--10,000 Hz)                | Minutes            | 20--10,000 Hz         | Usually covered by random vib for CubeSats |

[Source: ECSS-E-ST-10-03C; NASA GEVS (GSFC-STD-7000B); Falcon 9 Payload User's Guide; PSLV User's Guide]

**Random vibration PSD profile (typical CubeSat deployer level):**

| Frequency (Hz) | ASD Level ( $g^2/Hz$ ) | Notes                         |
|----------------|------------------------|-------------------------------|
| 20             | 0.01                   | Low-frequency start (ramp up) |
| 50             | 0.04                   | Ramp up at +6 dB/oct          |
| 100            | 0.04                   | Flat region start             |

| Frequency (Hz) | ASD Level ( $g^2/Hz$ ) | Notes                                            |
|----------------|------------------------|--------------------------------------------------|
| 800            | 0.04                   | Flat region end                                  |
| 2000           | 0.01                   | Roll off at -6 dB/oct                            |
| <b>Overall</b> | <b>~7 grms</b>         | Typical for CubeSat deployer qualification level |

The flat region at  $0.04 g^2/Hz$  from 100--800 Hz is where most structural damage occurs, because this is where PCB resonances and solder joint fatigue are excited.

**What fails during vibration testing:**

- Solder joints:** Heavy components (connectors, tall capacitors, transformers) with long lever arms crack at their solder joints. Mitigation: use surface-mount components, stake tall components with adhesive (Loctite 4860 or similar), use conformal coating.
- Deployable mechanisms:** Antenna hinges, solar panel hold-down mechanisms, and deployment springs can fail if not properly constrained. Mitigation: adequate preload on hold-down mechanisms, shock testing of pyrotechnic release devices.
- Optical components:** Lenses and mirrors can shift or crack if not properly mounted with strain-relief. Mitigation: RTV potting, flexure mounts.
- Wire harness:** Chafing against structure edges. Mitigation: edge radii  $> 1$  mm, harness tie-downs every 50 mm, protective sleeving.

## Structural Margin of Safety

### Key Equations -- Structural Margin of Safety

$$MoS = \frac{\sigma_{allowable}}{\sigma_{design} \times FoS} - 1$$

where: -  $\sigma_{allowable}$  = material yield or ultimate strength (MPa) -  $\sigma_{design}$  = computed stress under design loads (MPa) - FoS = factor of safety

**Requirement:**  $MoS \geq 0$  for all load cases.

**Factors of safety (ECSS-E-ST-32C):**

| Material / Joint        | Yield FoS | Ultimate FoS | Rationale                                 |
|-------------------------|-----------|--------------|-------------------------------------------|
| Metallic (Al 7075-T6)   | 1.25      | 1.5          | Standard structural metals                |
| Composite (CFRP)        | 1.5       | 2.0          | Higher variability in laminate properties |
| Bonded joints           | 1.5       | 2.0          | Bond strength is highly process-dependent |
| Pressurised systems     | 1.5       | 2.0          | Burst hazard to other payloads            |
| Mechanisms (single-use) | --        | 2.0          | Must work first time; no test opportunity |

**Design loads:** The design load includes the quasi-static acceleration (from the launch vehicle user guide), multiplied by a dynamic amplification factor ( $DAF \approx 1.25 - 1.5$ ) if the satellite's natural frequency is near any launch vehicle forcing frequency.

### Worked Example -- Axial Load on 3U CubeSat Rail (SuperDove-class)

**Given:** 3U CubeSat, mass = 5 kg, axial launch load = 9 g (Falcon 9 typical), 4 rails (load shared equally), rail cross-section =  $8.5 \times 8.5$  mm, material = Al 7075-T6 ( $\sigma_y = 503$  MPa,  $\sigma_u = 572$  MPa).

**Step 1 -- Design load per rail:**  $F = \frac{m \times n \times g_0}{4} = \frac{5 \times 9 \times 9.81}{4} = \frac{441.5}{4} = 110.4$  N

**Step 2 -- Compressive stress:**  $\sigma = \frac{F}{A_{\text{rail}}} = \frac{110.4}{8.5 \times 10^{-3} \times 8.5 \times 10^{-3}} = \frac{110.4}{7.225 \times 10^{-5}} = 1.53 \text{ MPa}$

**Step 3 -- Margin of safety (yield):**  $\text{MoS}_y = \frac{503}{1.53 \times 1.25} - 1 = \frac{503}{1.91} - 1 = 262 \gg 0$  **Pass** (by a very large margin)

**Step 4 -- Margin of safety (ultimate):**  $\text{MoS}_u = \frac{572}{1.53 \times 1.5} - 1 = \frac{572}{2.30} - 1 = 248 \gg 0$  **Pass**

**Key insight:** For CubeSats, quasi-static axial stress on the rails is never the critical load case. The rails are massively over-designed for direct compression. The critical structural design drivers are usually:

1. **Stiffness** (fundamental frequency > 40 Hz) -- driven by internal board/component mounting, not rail strength
2. **Random vibration fatigue** on PCB solder joints -- the real failure mode
3. **Deployment mechanism reliability** -- spring force, latch engagement, alignment tolerances
4. **CG location** -- difficult to achieve with asymmetric payloads or propulsion tanks

## Fundamental Frequency

### Key Equations -- Fundamental Frequency (simplified beam model)

For a cantilevered beam (simplified CubeSat model, clamped at deployer interface):

$$f_1 = \frac{1.875^2}{2\pi L^2} \sqrt{\frac{EI}{\rho A_{\text{cross}}}}$$

where  $E$  = Young's modulus (Pa),  $I$  = second moment of area ( $\text{m}^4$ ),  $\rho$  = linear density (kg/m),  $A_{\text{cross}}$  = cross-section area ( $\text{m}^2$ ),  $L$  = length (m).

**Requirement:**  $f_1 > 40 \text{ Hz}$  (from deployer ICD). Some deployers require  $> 90 \text{ Hz}$  (e.g., NanoRacks NRCSD).

**For a 3U CubeSat modelled as a cantilevered Al box beam:** -  $L = 0.34 \text{ m}$ ,  $E = 72 \text{ GPa}$ , box wall thickness  $t = 1.5 \text{ mm}$  -  $I \approx \frac{b^4 - (b - 2t)^4}{12} = \frac{0.10^4 - 0.097^4}{12} \approx 1.36 \times 10^{-6} \text{ m}^4$  - Linear mass:  $\rho_L = m/L = 5/0.34 = 14.7 \text{ kg/m}$

$$f_1 = \frac{3.516}{2\pi \times 0.34^2} \sqrt{\frac{72 \times 10^9 \times 1.36 \times 10^{-6}}{14.7}} = \frac{3.516}{0.726} \sqrt{6666} = 4.84 \times 81.6 = 395 \text{ Hz}$$

**This easily exceeds 40 Hz.** The structure itself is very stiff. However, the actual first mode is usually determined by: (a) the heaviest internal component on its mounting bracket (e.g., a 350 g star tracker cantilevered on a bracket), or (b) a deployable mechanism in its stowed configuration (e.g., a folded solar panel constrained only by a hold-down pin). **These local modes, not the overall structural mode, are typically the design concern.**

## 2. Propulsion System Design

[Source: SMAD, Ch. 17; Sutton & Biblarz, Ch. 2--4; Goebel & Katz, Fundamentals of Electric Propulsion, 2008]

## When Propulsion is Required

| Need                                         | Typical $\Delta V$ | Example Scenario                                    | Timeline                |
|----------------------------------------------|--------------------|-----------------------------------------------------|-------------------------|
| <b>Orbit maintenance</b> (drag compensation) | 5--15 m/s per year | LEO below 400 km in solar maximum                   | Continuous low-thrust   |
| <b>Deorbit</b> (active disposal)             | 50--150 m/s        | Active disposal from > 600 km (FCC 5-year rule)     | End of mission          |
| <b>Collision avoidance</b>                   | 1--5 m/s per event | Conjunction avoidance, 2--5 events per year for LEO | On-demand, within hours |
| <b>Constellation phasing</b>                 | 10--50 m/s         | Spreading satellites into operational orbit slots   | Weeks to months         |
| <b>Orbit raising</b>                         | 50--200 m/s        | Transfer from deployment orbit to operational orbit | Weeks to months         |
| <b>Station-keeping</b>                       | 1--10 m/s per year | Maintain orbit altitude and phase                   | Periodic                |

## When NO Propulsion is Needed

- **Orbit < 500 km:** Natural atmospheric decay provides FCC 5-year deorbit compliance (depends on ballistic coefficient and solar activity)
- **Low-cost technology demonstration:** Limited lifetime acceptable, no orbit maintenance needed
- **Constellation using differential drag for phasing** (e.g., Planet SuperDove adjusts its cross-section area to create differential drag, enabling free phasing manoeuvres)
- **Budget-constrained missions** where propulsion cost/risk/mass exceeds benefit

## The Tsiolkovsky Rocket Equation -- Physics

The rocket equation is the fundamental relationship governing all propulsive manoeuvres. It derives from conservation of momentum: the momentum of the exhaust equals the momentum change of the spacecraft.

### Key Equations -- Tsiolkovsky Rocket Equation

Starting from  $F = \dot{m} v_e$  (thrust = mass flow rate x exhaust velocity) and integrating:

$$\Delta V = v_e \ln \left( \frac{m_0}{m_f} \right) = I_{sp} \cdot g_0 \cdot \ln \left( \frac{m_0}{m_f} \right)$$

Rearranged for propellant mass:

$$m_{\text{propellant}} = m_{\text{dry}} \times (e^{\Delta V / (I_{sp} \cdot g_0)} - 1)$$

where: -  $v_e = I_{sp} \times g_0$  = effective exhaust velocity (m/s) -  $I_{sp}$  = specific impulse (s) -- the "fuel efficiency" of the thruster. Physically: how many seconds a thruster can produce 1 N of thrust from 1 kg of propellant under standard gravity. Higher  $I_{sp}$  = less propellant needed. -  $g_0 = 9.80665 \text{ m/s}^2$  -- standard gravitational acceleration (conversion factor) -  $m_0$  = initial (wet) mass (kg) =  $m_f + m_{\text{propellant}}$  -  $m_f$  = final (dry) mass (kg) = spacecraft mass after all propellant is consumed



**The tyranny of the rocket equation:** The propellant mass grows exponentially with  $\Delta V/v_e$ . For  $\Delta V = v_e$  (one exhaust velocity worth of  $\Delta V$ ), 63% of the initial mass must be propellant. For  $\Delta V = 2v_e$ , 86% must be propellant. This is why high- $I_{sp}$  systems are so valuable for large  $\Delta V$  missions -- they move the  $v_e$  in the denominator, dramatically reducing the mass ratio.

## CubeSat Propulsion Technologies -- Physics and Comparison

### Cold Gas Propulsion

**Physics:** Compressed gas ( $N_2$ , xenon, R-236fa refrigerant, or other) is stored in a tank at 1--30 MPa. When a valve opens, gas expands through a nozzle, converting thermal/pressure energy to kinetic energy. No combustion, no heating, no chemical reaction.

**Thrust:**  $F = \dot{m}v_e + (p_e - p_a)A_e$ . For a small converging nozzle,  $v_e \approx \sqrt{2c_p T_0}$  where  $T_0$  is the tank temperature. For  $N_2$  at 300 K:  $v_e \approx 500$ --750 m/s, giving  $I_{sp} \approx 50$ --75 s.

**Advantages:** Simplest system (no ignition, no power except valve solenoid), fast response (ms-level valve opening), high reliability, no plume contamination concerns.

**Disadvantages:** Low  $I_{sp}$  (large propellant mass for given  $\Delta V$ ), bulky high-pressure tank, limited total impulse.

**Products:** VACCO MiPS (R-236fa,  $I_{sp} = 40$  s, 4 thrusters, 0.3 kg dry), VACCO ArgoMoon (Xe cold gas), Bradford ECAPS cold gas.

**Missions using cold gas:** MarCO (6U, JPL -- used R-236fa cold gas for trajectory correction and attitude control en route to Mars), many ISS-deployed CubeSats for collision avoidance.

### Resistojet / Warm Gas

**Physics:** Similar to cold gas, but the propellant is electrically heated before expansion through the nozzle. Heating increases the gas temperature  $T_0$ , which increases the exhaust velocity ( $v_e \propto \sqrt{T_0}$ ). Common propellants: butane ( $C_4H_{10}$ ), water ( $H_2O$ ), ammonia ( $NH_3$ ).

**Butane propulsion:** Butane is stored as a liquid at its saturation pressure (~2 atm at 20 degC). When heated to 200--400 degC and expanded through a nozzle, it achieves  $I_{sp} \approx 80$ --100 s. The liquid storage is much denser than compressed gas, allowing more propellant in a smaller tank.

**Advantages:** Higher  $I_{sp}$  than cold gas (~2x), dense liquid storage, moderate complexity.

**Disadvantages:** Requires electrical power for heating (5--15 W during firing), lower thrust than cold gas, potential for nozzle clogging if propellant decomposes.

**Products:** Busek BGT-X5 (butane,  $I_{sp} = 80$  s), NanoAvionics EPSS (butane,  $I_{sp} = 85$  s), Pale Blue water resistojet ( $I_{sp} = 70$ --80 s).

### Green Monopropellant

**Physics:** A liquid propellant is injected into a catalyst bed where it decomposes exothermically, producing hot gases that expand through a nozzle. "Green" propellants are alternatives to hydrazine ( $N_2H_4$ ) that are less toxic and easier to handle.

| Propellant | Chemical | $I_{sp}$ (s) | Density (kg/m <sup>3</sup> ) | Toxicity | TRL | Heritage |
|------------|----------|--------------|------------------------------|----------|-----|----------|
|            |          | 220--230     | 1010                         |          | 9   |          |

| Propellant                                             | Chemical                               | $I_{sp}$ (s) | Density ( $\text{kg/m}^3$ ) | Toxicity                               | TRL | Heritage                               |
|--------------------------------------------------------|----------------------------------------|--------------|-----------------------------|----------------------------------------|-----|----------------------------------------|
| <b>Hydrazine</b><br>( $\text{N}_2\text{H}_4$ )         | Monopropellant,<br>Shell 405 catalyst  |              |                             | <b>Extremely toxic</b><br>(carcinogen) |     | 50+ years,<br>thousands of<br>missions |
| <b>AF-M315E</b><br>(ASCENT)                            | HAN-based ionic<br>liquid              | 235--250     | 1460                        | Low toxicity                           | 8   | GPIM demo<br>(2019, NASA)              |
| <b>LMP-103S</b>                                        | ADN-based ionic<br>liquid              | 225--235     | 1240                        | Low toxicity                           | 8   | PRISMA demo<br>(2010, SSC),<br>SkySat  |
| <b>HTP (<math>\text{H}_2\text{O}_2</math><br/>90%)</b> | Hydrogen peroxide +<br>silver catalyst | 150--165     | 1400                        | Moderate<br>(oxidiser)                 | 7   | Various small<br>satellites            |

[Source: Masse et al., "GPIM AF-M315E Propulsion System," AIAA 2019; Anflo et al., "Flight Demonstration of LMP-103S," AIAA 2011]

**Advantages:** High thrust (0.1--1 N for CubeSat systems), good  $I_{sp}$  (220+ s), proven technology (heritage from hydrazine systems).

**Disadvantages:** Requires catalyst preheating (2--10 W, 10--30 min warmup), higher system mass (tank + catalyst bed + valves + feed system), propellant handling safety requirements (even "green" propellants require PPE), higher cost (\$100K+ for flight units).

**Products:** Aerojet MPS-130 (AF-M315E, 1 N thrust,  $I_{sp} = 235$  s, 3 kg system mass), Bradford HPGP (LMP-103S, 1 N,  $I_{sp} = 230$  s).

### Electric Propulsion -- Electrospray (FEEP)

**Physics:** Field Emission Electric Propulsion (FEEP) uses a strong electric field ( $\sim 10^9$  V/m) at the tip of a needle or along the edge of a slit to ionise and extract metal atoms (typically indium or gallium) or ionic liquid droplets. The ions are accelerated by an electric field to high velocities (10--50 km/s), producing very high  $I_{sp}$ .

**How it works (indium FEEP):** 1. Solid indium is heated to just above its melting point (157 degC) to form a liquid reservoir 2. Capillary action draws liquid indium to an array of sharp emitter tips 3. A high voltage (1--10 kV) between the emitter tips and an extractor grid creates an intense electric field at the tip apex 4. The field ionises individual indium atoms via field evaporation 5. The ions are accelerated through the extractor grid, creating a beam of  $\text{In}^+$  ions at 20--40 km/s 6. A neutraliser (typically a carbon nanotube or thermionic emitter) emits electrons to neutralise the beam and prevent spacecraft charging

**Performance:**  $I_{sp} = 500$  -- 5000 s (adjustable by varying acceleration voltage), thrust = 0.01--1 mN, power = 20--60 W.

**Advantages:** Extremely high  $I_{sp}$  (minimal propellant consumption), no pressurised tanks, compact solid propellant storage, precise thrust control (useful for formation flying).

**Disadvantages:** Very low thrust (months-long burn times for significant  $\Delta V$ ), requires significant power (20--60 W for ~0.5 mN thrust), plume contamination from metal ions (indium deposition on surfaces), limited heritage.

**Products:** Enpulsion NANO R3 (indium FEEP, 0.35 mN,  $I_{sp} = 1000$  -- 5000 s, 0.9 kg dry mass, < 40 W), Accion TILE (ionic liquid electrospray, 0.1 mN,  $I_{sp} = 1500$  s).

**Missions using FEEP/electrospray:** LISA Pathfinder (ESA, precision formation flying, used colloid thrusters), SSTL NovaSAR-1 (Enpulsion NANO for orbit maintenance).

### Electric Propulsion -- Hall Effect Thruster

**Physics:** A Hall effect thruster uses crossed electric and magnetic fields to ionise a neutral propellant gas (xenon, krypton, or iodine) and accelerate the resulting ions to high velocity.

**How it works:** 1. Neutral propellant gas ( $\text{Xe}$  or  $\text{I}_2$ ) is injected into an annular discharge channel 2. A radial magnetic field (from permanent magnets or electromagnets) traps electrons in a Hall current loop, preventing them from reaching the anode 3. The trapped electrons collide with neutral gas atoms, ionising them 4. The ions, being much heavier, are not significantly deflected by the magnetic field and are accelerated axially by the electric field (100--500 V) between anode and cathode 5. An external cathode (hollow cathode or RF cathode) provides electrons for beam neutralisation and to sustain the discharge

**Performance:**  $I_{sp} = 800\text{--}3000\text{ s}$ , thrust = 1--50 mN for CubeSat-scale systems, power = 50--300 W.

**Advantages:** Higher thrust than FEEP (N range for larger systems), excellent  $I_{sp}$ , well-proven technology (GEO station-keeping heritage: Aerojet PPS-1350, Busek BHT-200).

**Disadvantages:** Requires significant power (> 50 W for CubeSat-scale), heavy cathode assembly, xenon storage requires high-pressure tanks (100--300 bar), channel erosion limits lifetime.

**Products:** Exotrail ExoMG-nano (40 mN,  $I_{sp} = 800\text{ s}$ , 1.5 kg, 60 W), Busek BHT-200 (13 mN,  $I_{sp} = 1370\text{ s}$ , 1.0 kg, 200 W), Enpulsion MICRO (Hall thruster, iodine propellant, 1 mN,  $I_{sp} = 1000\text{ s}$ ).

**Iodine propulsion:** Iodine ( $\text{I}_2$ ) is emerging as an alternative to xenon for Hall thrusters and gridded ion engines. Iodine is solid at room temperature (stored without a pressure vessel), has a density of  $4940\text{ kg/m}^3$  (vs xenon at  $\sim 1600\text{ kg/m}^3$  at 100 bar), and has similar atomic mass (127 vs 131). The Busek BIT-3 (iodine Hall thruster) and ThrustMe NPT30-I2 have demonstrated iodine propulsion in orbit.

### No Propulsion -- Passive Deorbit Strategies

For missions below  $\sim 500\text{ km}$  where propulsion is not needed for operations, passive deorbit can satisfy the FCC 5-year or IADC 25-year guidelines:

| Method                            | Mechanism                                                            | Mass        | Volume   | Effectiveness                                          | TRL  |
|-----------------------------------|----------------------------------------------------------------------|-------------|----------|--------------------------------------------------------|------|
| <b>Atmospheric drag (natural)</b> | Below 500 km, atmospheric drag naturally decays the orbit            | 0           | 0        | Depends on ballistic coefficient and solar cycle       | 9    |
| <b>Drag sail</b>                  | Deployable membrane increases cross-section area by 10--100x         | 0.1--0.5 kg | 0.25--1U | Very effective above 600 km                            | 7--8 |
| <b>Drag chute (tether)</b>        | Electrodynamic tether interacts with geomagnetic field to decelerate | 0.2--0.5 kg | 0.5U     | Moderate effectiveness; depends on orbital inclination | 6--7 |

[Source: Cranfield Icarus drag sail, 0.1 kg; NanoSail-D2, NASA; InflateSail, SSC]

## Propulsion System Comparison Table

| Parameter                           | Cold Gas<br>(N <sub>2</sub> ) | Warm Gas<br>(Butane) | Green Monoprop<br>(AF-M315E) | Electrospray<br>(FEFP) | Hall Effect<br>(Xe) |
|-------------------------------------|-------------------------------|----------------------|------------------------------|------------------------|---------------------|
| $I_{sp}$ (s)                        | 40--75                        | 80--100              | 230--250                     | 500--5000              | 800--3000           |
| Thrust                              | 10--100 mN                    | 5--50 mN             | 0.1--1 N                     | 0.01--1 mN             | 1--50 mN            |
| Propellant mass (100 m/s, 5 kg S/C) | 0.87 kg                       | 0.44 kg              | 0.18 kg                      | 0.042 kg               | 0.055 kg            |
| System dry mass                     | 0.3 kg                        | 0.5 kg               | 3.0 kg                       | 0.9 kg                 | 1.5 kg              |
| <b>Total system mass</b>            | <b>1.17 kg</b>                | <b>0.94 kg</b>       | <b>3.18 kg</b>               | <b>0.94 kg</b>         | <b>1.55 kg</b>      |
| Burn time (100 m/s)                 | Minutes                       | Minutes              | Seconds                      | <b>Months</b>          | Days--weeks         |
| Power during firing                 | < 1 W (valve)                 | 5--15 W              | 2--10 W (preheat)            | 20--60 W               | 50--300 W           |
| Complexity                          | Low                           | Low-medium           | High                         | Medium                 | High                |
| Cost                                | ~15 kEUR                      | ~30 kEUR             | ~120 kEUR                    | ~50 kEUR               | ~80 kEUR            |
| TRL                                 | 9                             | 7--8                 | 7--8                         | 7--8                   | 6--8 (CubeSat)      |

### Worked Example -- Propellant Mass Comparison for 100 m/s Deorbit

**Scenario:** 3U CubeSat,  $m_{dry} = 5.0$  kg, deorbit from 600 km ( $\Delta V = 113$  m/s).

**Cold gas** ( $I_{sp} = 60$  s,  $v_e = 589$  m/s):

$$m_{prop} = 5.0 \times (e^{113/589} - 1) = 5.0 \times (e^{0.192} - 1) = 5.0 \times 0.212 = \mathbf{1.06 \text{ kg}}$$

Total system:  $1.06 + 0.3 = 1.36$  kg = **23% of 6 kg CubeSat mass limit**

**Green monopropellant** ( $I_{sp} = 235$  s,  $v_e = 2305$  m/s):

$$m_{prop} = 5.0 \times (e^{113/2305} - 1) = 5.0 \times (e^{0.0490} - 1) = 5.0 \times 0.0502 = \mathbf{0.251 \text{ kg}}$$

Total system:  $0.251 + 3.0 = 3.25$  kg = **54% of mass limit** (system is heavy even though propellant is light)

**Electrospray (FEFP)** ( $I_{sp} = 1200$  s,  $v_e = 11,772$  m/s):

$$m_{prop} = 5.0 \times (e^{113/11772} - 1) = 5.0 \times (e^{0.00960} - 1) = 5.0 \times 0.00965 = \mathbf{0.048 \text{ kg}}$$

Total system:  $0.048 + 0.9 = 0.95$  kg = **16% of mass limit** (lightest total, but takes months to execute)

**Trade-off summary:** The electrospray system is the lightest overall because the high  $I_{sp}$  minimises propellant mass, and the dry mass is moderate. Cold gas uses the most propellant but has the lowest dry mass. Green monopropellant is dominated by its heavy feed system. **The optimal choice depends on the mission timeline:** if deorbit must happen quickly (days), cold gas or monoprop; if months are acceptable, electric propulsion wins on mass.

### 3. On-Board Data Handling

#### OBC Architecture -- Processor Selection

The OBC is the spacecraft's brain, managing all data handling, commanding, telemetry generation, and FDIR (Fault Detection, Isolation, and Recovery). Processor selection involves a fundamental trade between radiation tolerance, processing power, power consumption, and cost.

#### Flight-Heritage Processors

| Processor                                            | Architecture                 | Clock (MHz)        | RAM                  | Rad Tolerance                                 | Power (W)   | TRL (Space) | Cost         | Typical Use                                                                      |
|------------------------------------------------------|------------------------------|--------------------|----------------------|-----------------------------------------------|-------------|-------------|--------------|----------------------------------------------------------------------------------|
| <b>TI MSP430</b>                                     | 16-bit RISC                  | 16--25             | 2--10 kB             | Moderate (COTS, tested to 30 krad)            | 0.005--0.01 | 9           | < 10 EUR     | Ultra-low-power housekeeping safe mode OBC                                       |
| <b>ARM Cortex-M4</b> (STM32F4)                       | 32-bit ARM                   | 168                | 192 kB + ext         | Low-moderate (COTS, 10--30 krad)              | 0.05--0.20  | 8--9        | 10--20 EUR   | <b>Standard CubeSat OBC</b> , TM/TC handling, ADCS control loop                  |
| <b>ARM Cortex-M7</b> (STM32H7)                       | 32-bit ARM                   | 400--480           | 1 MB + ext           | Low-moderate (COTS, 10--20 krad)              | 0.1--0.5    | 7--8        | 15--30 EUR   | Higher-performance CubeSat OBC, onboard image processing                         |
| <b>ARM Cortex-A</b> (Linux-capable, e.g., NXP i.MX6) | 32/64-bit ARM                | 500--1200          | 256 MB--1 GB DDR     | Low (COTS, < 10 krad)                         | 1--3        | 6--7        | 20--50 EUR   | Payload processing, AI/ML inference, Linux OS                                    |
| <b>Xilinx Zynq</b> (SoC: ARM + FPGA)                 | ARM Cortex-A9 + Artix-7 FPGA | 667 + programmable | 512 MB + FPGA fabric | Moderate (Zynq-7000) to High (Kintex radhard) | 2--5        | 7--8        | 100--500 EUR | High-throughput data processing, SDR (software-defined radio), image compression |
| <b>LEON3/4</b> (rad-hard SPARC)                      | 32-bit SPARC                 | 50--250            | External             | High (100--300 krad, SEL immune)              | 1--3        | 9           | 10K--50K EUR | ESA heritage missions, GEO, deep space                                           |

| Processor                      | Architecture   | Clock (MHz) | RAM    | Rad Tolerance                  | Power (W) | TRL (Space) | Cost      | Typical Use                                   |
|--------------------------------|----------------|-------------|--------|--------------------------------|-----------|-------------|-----------|-----------------------------------------------|
| <b>RAD750</b><br>(BAE Systems) | 32-bit PowerPC | 200         | 128 MB | Very high (1 Mrad, SEL immune) | 5--10     | 9           | 200K+ EUR | NASA flagship missions (MRO, Curiosity, JWST) |

[Source: ST Microelectronics STM32 datasheets; Xilinx Zynq-7000 datasheet; Cobham Gaisler LEON3 datasheet]

### RTOS vs Bare-Metal vs Linux

| Approach                                | OS                       | Pros                                                                                    | Cons                                                                                                  | When to Use                                                             |
|-----------------------------------------|--------------------------|-----------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------|
| <b>Bare-metal</b>                       | None (custom event loop) | Minimum overhead, deterministic timing, smallest code size                              | Hard to maintain, no task isolation, no file system                                                   | Ultra-simple 1U missions (MSP430)                                       |
| <b>RTOS</b> (FreeRTOS, ChibiOS, Zephyr) | Real-time OS             | Deterministic scheduling, task isolation, mature ecosystem, small footprint (10--50 kB) | More complex than bare-metal; requires task priority design                                           | <b>Standard for CubeSat C&amp;DH:</b> ADCS loop, TM/TC, mode management |
| <b>Linux</b> (Yocto, Buildroot)         | Full OS                  | Rich ecosystem (Python, networking, file system, device drivers), easy development      | Non-deterministic (not suitable for hard real-time), large footprint (50+ MB), power-hungry processor | Payload data processing, AI/ML, onboard image analysis                  |

### Radiation effects on processors:

The space radiation environment causes two categories of effects:

1. **Total Ionising Dose (TID):** Accumulated radiation damage from trapped protons/electrons and solar particles. Measured in rad(Si) or gray. Causes threshold voltage shifts in CMOS transistors, increasing leakage current and eventually causing functional failure. - LEO (500 km, 51.6 deg): ~1--5 krad/year behind 2 mm Al shielding - LEO polar/SSO (800 km): ~5--10 krad/year - MEO (through proton belt): ~50--100 krad/year - GEO: ~10--30 krad/year
2. **Single Event Effects (SEE):** A single energetic particle (proton or heavy ion) deposits enough charge in a transistor to flip a bit (SEU -- Single Event Upset), latch a transistor (SEL -- Single Event Latchup, potentially destructive), or burn out a power device (SEB -- Single Event Burnout). - **SEU rate in LEO:** ~1--10 bit flips per day per GB of SRAM (highly variable with orbit and shielding) - **SEL mitigation:** Current-limiting resistors on power lines, latchup detection circuits, watchdog resets - **SEU mitigation:** Error Detection and Correction (EDAC) on memory (Hamming codes, TMR -- Triple Modular Redundancy)

**CubeSat approach to radiation:** Most CubeSats in LEO (< 600 km, < 3-year mission) use COTS processors with EDAC on memory and a watchdog timer. Total dose over a 3-year LEO mission is typically 3--15 krad,

which most COTS ARM Cortex-M processors survive (tested and characterised, even if not guaranteed). For longer missions, higher orbits, or critical applications, rad-tolerant or rad-hard processors are needed.

## Data Storage Sizing

### Key Equations -- Data Storage

$$S_{\text{required}} = V_{\text{daily}} \times N_{\text{days}} \times f_{\text{safety}}$$

where  $V_{\text{daily}}$  = daily data generation,  $N_{\text{days}}$  = days between full downlinks (typically 1--3 for LEO),  $f_{\text{safety}} = 2$  (to handle missed passes, ground station outages, and safe mode periods).

**Storage technologies:**

| Technology           | Capacity   | Write Speed  | Radiation Tolerance       | Power    | CubeSat Use                                                    |
|----------------------|------------|--------------|---------------------------|----------|----------------------------------------------------------------|
| NOR flash            | 4--256 MB  | 1--5 MB/s    | Moderate (10--50 krad)    | Low      | Code storage, boot ROM, critical parameters                    |
| NAND flash (SLC)     | 1--128 GB  | 10--50 MB/s  | Low-moderate (5--20 krad) | Low      | <b>Primary data storage</b>                                    |
| NAND flash (MLC/TLC) | 32--512 GB | 50--200 MB/s | Low (< 10 krad)           | Low      | Maximum capacity (use EDAC and scrubbing)                      |
| SD card (industrial) | 4--128 GB  | 10--50 MB/s  | Low (< 5 krad)            | Very low | Budget missions (risk: wear levelling + radiation = data loss) |
| MRAM                 | 1--64 MB   | 10--50 MB/s  | High (> 100 krad)         | Very low | Critical parameters, non-volatile log                          |

### Worked Example -- Storage for 3U EO CubeSat

**Given:** Daily generation = 720 MB (from Session 3.3 data budget), daily downlink = 480 MB, days to clear backlog =  $720/480 = 1.5$  days.

$$S_{\text{required}} = 720 \times 3 \times 2 = 4320 \text{ MB} \approx \mathbf{4.3 \text{ GB}}$$

**Specify:**  $\geq 8$  GB NAND flash storage (next standard size). The GomSpace A3200 OBC includes 4 GB NAND flash; adding an 8 GB SD card or additional NAND chip provides adequate capacity.

For a high-resolution imager generating 4.5 GB/day, storage requirement is:  $4500 \times 3 \times 2 = 27$  GB. Specify  $\geq 32$  GB flash storage.

## Flight Software Functions

| Function        | Description | Typical Execution Rate | Criticality                                           |
|-----------------|-------------|------------------------|-------------------------------------------------------|
| Mode management |             | Event-driven           | <b>Critical</b> (incorrect transition = mission loss) |

| Function                 | Description                                                                                                                                | Typical Execution Rate       | Criticality                                                        |
|--------------------------|--------------------------------------------------------------------------------------------------------------------------------------------|------------------------------|--------------------------------------------------------------------|
|                          | Transition between Safe, Idle, Imaging, Downlink, Eclipse modes based on state machine                                                     |                              |                                                                    |
| <b>ADCS control loop</b> | Read sensors (star tracker, gyro, magnetometer), compute attitude estimate (Kalman filter), command actuators (PID controller)             | 1--10 Hz                     | <b>Critical</b> (loss of pointing = loss of mission)               |
| <b>TM/TC handling</b>    | Generate CCSDS telemetry packets, parse and execute telecommands                                                                           | 1 Hz (TM), event-driven (TC) | <b>Critical</b> (loss of communication = loss of mission)          |
| <b>Data handling</b>     | Payload data acquisition, compression (JPEG2000, CCSDS 122.0), buffering, downlink queue management                                        | On-demand                    | Important                                                          |
| <b>FDIR</b>              | Fault Detection, Isolation, and Recovery: watchdog timer, over-current protection, sensor consistency checks, autonomous safe mode trigger | 1--10 Hz                     | <b>Critical</b> (must detect and recover from faults autonomously) |
| <b>Housekeeping</b>      | Monitor temperatures, voltages, currents, wheel speeds; log to non-volatile memory                                                         | 0.1--1 Hz                    | Important                                                          |
| <b>Scheduling</b>        | Time-tagged command execution: autonomous imaging over target, downlink preparation before ground pass, desat scheduling                   | 1 Hz                         | Important                                                          |
| <b>Thermal control</b>   | Read temperature sensors, control heaters (on/off thermostat or PID)                                                                       | 0.1 Hz                       | Moderate (for heater-equipped missions)                            |

## 4. Equipment Selection Exercise

### Instructions

This is the primary hands-on session for Day 4 of the design week. Teams select actual hardware components.

1. **Open the Equipment Browser** (button in header bar)
2. The sidebar shows categories **annotated by need**: - Blue dot = Required for your mission - Circle = Optional - Dimmed = Not applicable
3. **For each required category, select a component**: - Check the quantity needed (e.g., 4 reaction wheels, 3 magnetorquers) - Note any RF compatibility warnings (transponder band must match antenna band) - Watch the **live budget bar** showing running mass / power / cost totals
4. **For each selection, verify**: - Does it fit within the subsystem mass allocation? - Is power draw within the power budget for its operational mode? - Is the interface compatible (PC/104? I<sup>2</sup>C? SPI? CAN?) - Is the component qualified for the launch vibration environment?



- 
5. **Review the Budget Breakdown** on the Dashboard: - Has per-subsystem mass changed from the parametric estimate? - Is the overall mass margin still positive? - Is the power budget still positive in all modes?

### Component Trade Study

For at least one subsystem, select 2--3 candidate components and run a formal tabular trade:

1. Navigate to the **Trade Studies** tab
2. Load or create a "Component Selection Trade" study
3. Define criteria: mass, power, cost, TRL, heritage, performance
4. Score each candidate (1--5 scale)
5. Apply weights and compute weighted scores
6. Document the winner and rationale

### Real Mission Example: Iridium NEXT Equipment Selection

Iridium NEXT (Thales Alenia Space, 2017--2019) serves as a large-scale example of rigorous equipment selection. For the phased-array antenna:

| Criterion       | Weight | Candidate A (Thales) | Candidate B (Raytheon) |
|-----------------|--------|----------------------|------------------------|
| Performance     | 0.30   | 4.5                  | 4.0                    |
| Mass            | 0.20   | 3.5                  | 4.0                    |
| Cost            | 0.20   | 3.0                  | 4.5                    |
| TRL             | 0.15   | 5.0                  | 4.0                    |
| Schedule        | 0.15   | 4.0                  | 3.5                    |
| <b>Weighted</b> |        | <b>3.95</b>          | <b>4.00</b>            |

The selection was ultimately Candidate A (Thales) due to contractual considerations beyond the numerical trade -- illustrating that trade studies inform but do not dictate decisions.

---

## 5. SpaceCDF Budget Closure Check

### Instructions

After equipment selection, perform a final budget health check:

1. **Dashboard KPIs:** Record all margins - Mass margin (%) -- green/amber/red? - Power margin per mode (W) - Link margin (dB) - Cost vs ceiling (MEUR) - Pointing accuracy vs requirement (deg)
2. **Budget Comparison:** Compare parametric estimates to equipment-based totals
3. **Identify any negative margins** -- these must be resolved before proceeding to integration (Week 3)

## If a Budget Does Not Close

| Budget                            | Common Fix                                                                                                             | Impact                   | Typical Mass/Power/Cost Trade                                        |
|-----------------------------------|------------------------------------------------------------------------------------------------------------------------|--------------------------|----------------------------------------------------------------------|
| <b>Mass</b><br>(negative)         | Remove propulsion; select lighter components; reduce redundancy; move to larger form factor                            | Risk / performance trade | Removing propulsion saves 0.5--1.5 kg                                |
| <b>Power</b><br>(negative)        | Add deployable SA; reduce payload duty cycle; select lower-power AOCS; schedule operations to avoid simultaneous loads | Cost / schedule trade    | Deployable panel adds ~15 W but costs 0.3 kg + 25 kEUR               |
| <b>Link</b> (negative)            | Increase TX power; use higher-gain antenna; reduce data rate; upgrade coding; upgrade ground station                   | Mass / power trade       | 3 dB gain from coding is "free"; 3 dB from bigger antenna costs mass |
| <b>Cost</b> (over ceiling)        | Use COTS instead of rad-hard; remove propulsion; reduce ground segment; use SatNOGS instead of commercial ground       | Risk / capability trade  | COTS vs rad-hard saves 10--100x on processor cost                    |
| <b>Pointing</b><br>(insufficient) | Upgrade star tracker; improve alignment calibration; add vibration isolation for RW; reduce thermal gradients          | Cost / complexity trade  | Alignment improvement is usually cheapest                            |

## 1U Worked Example: UniSat-1

### CDS Compliance for 1U Form Factor

The CubeSat Design Specification (CDS Rev 14) defines the 1U envelope:

| Parameter             | 1U Specification                   | UniSat-1 Design                                                         | Compliance |
|-----------------------|------------------------------------|-------------------------------------------------------------------------|------------|
| Dimensions            | 100.0 x 100.0 x 113.5 mm           | 100.0 x 100.0 x 113.5 mm (ISIS 1U frame)                                | Pass       |
| Maximum mass          | 2.0 kg (CDS Rev 14, ISIPOD)        | 1.0 kg target (50% margin to 2 kg limit)                                | Pass       |
| Rail material         | Hard anodised Al 7075-T6           | Standard (part of ISIS frame, 7075-T6, Type III anodise)                | Pass       |
| Rail cross-section    | 8.5 x 8.5 mm minimum               | Standard (ISIS: 8.5 x 8.5 mm)                                           | Pass       |
| Deployment switches   | Min 1 per accessible face          | 2 switches (ISIS standard, on +X/-X rail faces)                         | Pass       |
| RBF pin               | Required                           | Included (ISIS standard, on -Z face)                                    | Pass       |
| CG offset             | <= 2 cm from geometric centre      | < 1 cm (symmetric PCB stack layout, battery centred)                    | Pass       |
| Protrusions           | None beyond rail envelope (stowed) | UHF monopole antenna stowed along rail (spring-loaded, within envelope) | Pass       |
| Fundamental frequency | > 40 Hz first mode                 | ~600 Hz (1U Al structure is extremely stiff)                            | Pass       |

**Note on CDS mass limit:** The CDS Rev 14 specifies 2.0 kg as the 1U deployer limit for the ISIPOD. However, many deployer providers (e.g., NanoRacks, Exolaunch) specify 1.33 kg for 1U. Always check the specific deployer ICD. UniSat-1 targets 1.0 kg, well within either limit.

**No propulsion:** At 400 km altitude, atmospheric drag provides natural deorbit. The ballistic coefficient for a 1U is:

$$BC = \frac{m}{C_D \times A} = \frac{1.0}{2.2 \times 0.01} = 45.5 \text{ kg/m}^2$$

This gives an orbital lifetime of approximately 8--14 months depending on solar activity (F10.7 index). At solar maximum (F10.7 > 200), the denser atmosphere deorbits the satellite in ~6 months. At solar minimum (F10.7 ~ 70), lifetime extends to ~18 months. Both are within the FCC 5-year rule and IADC 25-year guideline without any propulsion system.

**OBC selection rationale:** UniSat-1 uses a custom board based on the TI MSP430 (safe mode / housekeeping) + STM32F4 Cortex-M4 (main OBC). The MSP430 runs bare-metal firmware handling the watchdog, power monitoring, and safe-mode recovery. The STM32F4 runs FreeRTOS handling TM/TC, magnetometer data acquisition, and scheduling. Dual-processor architecture provides redundancy: if the STM32F4 fails (SEU, latchup), the MSP430 can maintain safe-mode operations and respond to ground commands.

**Data storage:** With 0.84 MB/day of magnetometer data and 4800 bps downlink, onboard storage is not a bottleneck. A 4 MB NOR flash is more than adequate (stores ~4 days of data as buffer). No NAND flash, no SD card needed.

#### Complete 1U Equipment List:

| # | Category    | Component                                                       | Mass (g) | Power (W)            | Cost (kEUR) | Qty | Interface                   |
|---|-------------|-----------------------------------------------------------------|----------|----------------------|-------------|-----|-----------------------------|
| 1 | Structure   | ISIS 1U CubeSat structure (Al 7075-T6)                          | 200      | --                   | 4.0         | 1   | Mechanical                  |
| 2 | EPS         | GomSpace NanoPower P31us (EPS board + 2S Li-ion battery, 10 Wh) | 200      | 0.3 (quiescent)      | 12.0        | 1   | I <sup>2</sup> C            |
| 3 | Solar cells | Body-mounted triple-junction GaAs cells (5 faces)               | 50       | -- (generates power) | 8.0         | 5   | Direct to EPS               |
| 4 | OBC         | Custom MSP430 + STM32F4 board (with 4 MB NOR flash)             | 30       | 0.3                  | 3.0         | 1   | I <sup>2</sup> C, SPI, UART |
| 5 | Comms       | UHF transceiver (NanoCom AX100, 0.5 W TX, 9600 bps)             | 60       | 0.5 (TX) / 0.1 (RX)  | 8.0         | 1   | SPI                         |
| 6 | Antenna     | UHF monopole (deployable, $\lambda/4 = 17$ cm nitinol)          | 20       | --                   | 2.0         | 1   | RF coax                     |
| 7 | Payload     |                                                                 | 50       | 0.2                  | 5.0         | 1   | SPI                         |

| # | Category       | Component                                                                     | Mass (g)   | Power (W)             | Cost (kEUR) | Qty | Interface      |
|---|----------------|-------------------------------------------------------------------------------|------------|-----------------------|-------------|-----|----------------|
|   |                | MEMS magnetometer (custom PCB, PNI RM3100 sensor)                             |            |                       |             |     |                |
| 8 | AOCS (passive) | Permanent magnet (AlNiCo, $0.5 \text{ A m}^2$ ) + 2 hysteresis rods (HyMu-80) | 30         | 0                     | 1.0         | 1   | None (passive) |
| 9 | Harness        | Internal cables, connectors, PC/104 stack header                              | 50         | --                    | 1.0         | 1   | Various        |
|   | <b>TOTAL</b>   |                                                                               | <b>690</b> | <b>~1.3 (peak TX)</b> | <b>~44</b>  |     |                |

**Mass budget:**

| Level                               | Mass (g)            |
|-------------------------------------|---------------------|
| CBE (Current Best Estimate)         | 690                 |
| + 20% equipment margin              | 828                 |
| + 20% system margin                 | 994                 |
| <b>MEV (Maximum Expected Value)</b> | <b>994</b>          |
| Deployer limit (ISIPOD)             | 2000                |
| <b>Margin to limit</b>              | <b>1006 g (50%)</b> |

**Power budget (worst case: TX mode in sunlight):**

| Load                          | Power (W)          |
|-------------------------------|--------------------|
| OBC (STM32F4 + MSP430)        | 0.3                |
| EPS quiescent                 | 0.3                |
| UHF TX                        | 0.5                |
| Magnetometer                  | 0.2                |
| <b>Total peak</b>             | <b>1.3</b>         |
| SA available (orbit avg, EOL) | 2.3                |
| <b>Margin</b>                 | <b>1.0 W (43%)</b> |

**Key insight:** The entire UniSat-1 BOM is 5 COTS components plus 2 custom boards (OBC and magnetometer PCB). Total hardware cost is ~44 kEUR -- an order of magnitude less than a typical 3U mission. With labour, I&T, launch, and operations, the total mission cost is 80--150 kEUR. This demonstrates that a useful space mission can be built for less than the cost of a mid-range car.

## Worked Example: Complete 3U EO CubeSat Equipment List

| Category       | Component                                             | Mass (kg)   | Power (W)                 | Cost (kEUR) | Qty | Interface                    |
|----------------|-------------------------------------------------------|-------------|---------------------------|-------------|-----|------------------------------|
| EPS Board      | GomSpace P31u (MPPT, 3.3V/5V/batt rails)              | 0.10        | 0.5                       | 8           | 1   | I <sup>2</sup> C             |
| Battery        | GomSpace BP4 (2S2P, 38 Wh, Li-ion 18650)              | 0.20        | --                        | 5           | 1   | I <sup>2</sup> C (telemetry) |
| Solar Panels   | MMA HaWK deployable (TJ GaAs, ~12 W/panel BOL)        | 0.45        | --                        | 25          | 2   | Direct to EPS                |
| OBC            | GomSpace A3200 (ARM Cortex-A, Linux, 4 GB NAND)       | 0.08        | 1.0                       | 12          | 1   | I <sup>2</sup> C, SPI, UART  |
| Reaction Wheel | Blue Canyon RW210 (1 mN m torque, 10 mN m s momentum) | 0.055       | 0.6                       | 8           | 4   | SPI                          |
| Magnetorquer   | CubeSpace CubeMAG (0.2 A m <sup>2</sup> dipole)       | 0.03        | 0.1                       | 3           | 3   | I <sup>2</sup> C             |
| Star Tracker   | Blue Canyon NST (10 arcsec accuracy, 2 Hz update)     | 0.35        | 1.5                       | 35          | 1   | SPI/UART                     |
| Sun Sensor     | NewSpace NFSS-411 (0.5 deg accuracy, fine analog)     | 0.005       | 0.01                      | 1           | 6   | Analog/I <sup>2</sup> C      |
| Transponder    | Endurosat S-band TX/RX (2 W, QPSK+LDPC, 1--5 Mbps)    | 0.10        | 6.0 (TX)                  | 15          | 1   | SPI                          |
| Antenna        | Endurosat S-band patch (6 dBi, RHCP)                  | 0.02        | --                        | 3           | 1   | RF coax                      |
| Payload        | Custom telescope (multispectral, 5 m GSD)             | 1.50        | 5.0                       | 150         | 1   | LVDS/SPI                     |
| Structure      | ISIS 3U frame (Al 7075-T6, hard anodised)             | 0.30        | --                        | 8           | 1   | Mechanical                   |
| Harness        | Custom cables, connectors, PC/104 stack               | 0.15        | --                        | 5           | 1   | Various                      |
| <b>TOTAL</b>   |                                                       | <b>3.57</b> | <b>~10 (imaging mode)</b> | <b>~290</b> |     |                              |

**Parametric estimate from Session 2.4:** 3.68 kg CBE. **Equipment total:** 3.57 kg. Difference: -3% (within expected accuracy of parametric models).

**Mass budget:** - CBE: 3.57 kg - + 20% equipment margin: 4.28 kg - + 20% system margin: 5.14 kg (MEV)  
 - CDS limit: 6.0 kg - **Margin to limit: 0.86 kg (14%)** -- amber, acceptable for Phase A but tight for Phase B+.

## Session Summary

| Topic                   | Key Takeaway                                                                                                             |
|-------------------------|--------------------------------------------------------------------------------------------------------------------------|
| CDS compliance          | Standard dimensions, Al 7075-T6 anodised rails (8.5 mm), deployment switches, RBF pin, CG limits                         |
| Structural materials    | Al 7075-T6: $\sigma_y = 503$ MPa, $E = 72$ GPa; anodisation for wear/insulation; CFRP not suitable for rails             |
| Launch loads            | 6--12 g axial QS, 7 grms random vib (20--2000 Hz), 500--2000 g shock; PCB solder joints are the weak point               |
| Structural MoS          | $MoS = \sigma_{allow} / (\sigma_{design} \times FoS) - 1 \geq 0$ ; FoS 1.25 yield / 1.5 ultimate (metallic)              |
| Frequency req           | First mode > 40 Hz; CubeSat Al structures easily exceed this; local modes (PCBs, deployables) are the risk               |
| Tsiolkovsky equation    | $m_{prop} = m_{dry} \times (e^{\Delta V / (I_{sp} g_0)} - 1)$ ; exponential growth with $\Delta V / v_e$                 |
| Cold gas                | $I_{sp}$ 40--75 s; simple, fast, reliable; heavy propellant penalty for $\Delta V > 30$ m/s                              |
| Green monoprop          | $I_{sp}$ 225--250 s; high thrust (0.1--1 N); heavy feed system; AF-M315E and LMP-103S flight-proven                      |
| Electrospray (FEED)     | $I_{sp}$ 500--5000 s; minimal propellant; months-long burns; 20--60 W power; indium or ionic liquid                      |
| Hall thruster           | $I_{sp}$ 800--3000 s; moderate thrust (1--50 mN); 50--300 W; iodine emerging as Xe alternative                           |
| Propulsion trades       | High- $I_{sp}$ : less propellant, more dry mass, long burns; Low- $I_{sp}$ : more propellant, lighter system, fast burns |
| When to skip propulsion | Below 500 km (natural deorbit); tech demo; differential drag constellation                                               |
| OBC processors          | MSP430 (0.01 W, safe mode), Cortex-M4 (0.2 W, standard CubeSat), Zynq (3 W, high-throughput), LEON3 (rad-hard, ESA)      |
| RTOS vs Linux           | FreeRTOS for real-time C&DH (standard); Linux for payload processing (data-intensive)                                    |
| Radiation effects       | TID: 1--10 krad/yr LEO; SEU: 1--10 bit flips/day/GB; mitigate with EDAC, watchdog, redundancy                            |
| Data storage            | $S \geq 2 \times$ daily generation; SLC NAND flash for primary storage; NOR flash for code/critical params               |
| Equipment selection     | Live budget tracking; RF compatibility check; interface compatibility; trade study for contested selections              |

| Topic          | Key Takeaway                                                                                     |
|----------------|--------------------------------------------------------------------------------------------------|
| Budget closure | All margins must be positive before proceeding to integration week; mass margin > 10% at Phase A |

## Session 4.1: Equipment Selection & Bill of Materials

### Technology Readiness Level (TRL) Ladder

|              |                                 |                                                                                        |
|--------------|---------------------------------|----------------------------------------------------------------------------------------|
| <b>TRL 9</b> | Flight-proven                   | <b>TRL 1-3</b><br>Discussed in Course Plan §2 — research / IRAD context                |
| <b>TRL 8</b> | Qualified system                |                                                                                        |
| <b>TRL 7</b> | Operational demo                |                                                                                        |
| <b>TRL 6</b> | Relevant-environment demo       | <b>TRL 4-6</b><br>CubeSat platforms typically buy at TRL $\geq 6$ (commercial COTS)    |
| <b>TRL 5</b> | Relevant-environment validation |                                                                                        |
| <b>TRL 4</b> | Lab validation                  | <b>TRL 7</b><br>First in-orbit demonstration — relevant for new instruments            |
| <b>TRL 3</b> | Proof of concept                |                                                                                        |
| <b>TRL 2</b> | Concept formulated              | <b>TRL 8-9</b><br>Operational missions — flight heritage drives selection in CDF Day 3 |
| <b>TRL 1</b> | Basic principles                |                                                                                        |

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Technology Readiness Level ladder.

**Expected reading before this session.** SMAD4 Ch. 14 (parts); NASA TRL definitions (NPR 7123.1D Appendix E).

**Prerequisites:** Day 3 complete (subsystems sized, components identified via parametric agents) **References:** ECSS-Q-ST-20C (Quality Assurance), ECSS-E-ST-10-24C (Interfaces), ITAR/EAR (22 CFR 120-130 / 15 CFR 730-774), CDS Rev 14.1, NASA SEH Rev 2 section 6.8

### Learning Objectives

By the end of this session, participants will be able to: 1. Apply a structured make/buy/reuse decision framework to component selection 2. Evaluate COTS components against requirements using TRL and heritage criteria 3. Construct a Bill of Materials (BOM) with full traceability to requirements 4. Identify export control constraints (ITAR, EAR, Canadian Controlled Goods) for selected equipment 5. Verify interface compatibility (RF, electrical, mechanical) during component selection 6. Track cumulative mass, power, and cost budgets as selections are made

### 1. The Make/Buy/Reuse Decision

Before selecting any hardware, the team must decide the procurement strategy for each subsystem. This is a fundamental systems engineering decision that affects cost, schedule, risk, and performance.



[Source: NASA SEH Rev 2 section 6.8 "Decision Analysis"; ECSS-M-ST-10C Rev.1 section 5.4]

## Decision Framework

For each subsystem or component:

1. Does a COTS product exist that meets requirements?
  - > Yes: BUY (lowest risk, fastest schedule)
  - > No: Continue
2. Does flight-proven hardware from a previous mission exist?
  - > Yes: REUSE (low risk, may need delta-qualification)
  - > No: Continue
3. Can the requirement be met by modifying existing hardware?
  - > Yes: MODIFY (moderate risk, moderate schedule)
  - > No: MAKE (highest risk, longest schedule, most expensive)

## Make/Buy/Reuse Trade Matrix

| Factor           | Buy (COTS)           | Reuse (Heritage)    | Modify          | Make (Custom)      |
|------------------|----------------------|---------------------|-----------------|--------------------|
| Cost (NRE)       | None                 | None-Low            | Moderate        | High               |
| Cost (Recurring) | Vendor price         | Reproduction cost   | Vendor + delta  | Full development   |
| Schedule         | 4-16 weeks lead      | 8-24 weeks          | 12-36 weeks     | 12-48 months       |
| Risk             | Low (if TRL >= 7)    | Low (flight-proven) | Medium          | High               |
| Performance      | Fixed by vendor      | Fixed by heritage   | Tuneable        | Fully customisable |
| IP ownership     | Vendor retains       | May be shared       | Negotiated      | Full ownership     |
| Qualification    | Vendor-provided data | Delta-qual only     | Partial re-qual | Full qualification |

## Key Equations

### Non-Recurring Engineering (NRE) Cost Estimate:

$$\text{NRE} = \text{Labour\_hours} \times \text{Hourly\_rate} + \text{Material\_cost} + \text{Facility\_cost} + \text{Testing\_cost}$$

For COTS: NRE is near zero (vendor absorbs development cost). For custom: NRE can be 3-10x the unit recurring cost.

## Worked Example

Problem: A 6U CubeSat mission needs a fine sun sensor with 0.1 degree accuracy. Three options:

| Option                    | Type   | Accuracy | Mass | Cost                      | TRL | Lead Time |
|---------------------------|--------|----------|------|---------------------------|-----|-----------|
| Bradford SSOC-D60         | COTS   | 0.1 deg  | 35 g | EUR 15K                   | 9   | 8 weeks   |
| In-house photodiode array | Custom | 0.05 deg | 20 g | EUR 8K + 400 hr NRE       | 4   | 12 months |
| Modified heritage sensor  | Reuse  | 0.08 deg | 40 g | EUR 6K + 80 hr delta-qual | 7   | 16 weeks  |

**Decision:** The COTS option (Bradford SSOC-D60) meets the requirement, has TRL 9, lowest schedule risk, and total cost of EUR 15K vs EUR 8K + ~EUR 40K NRE for custom. **Buy.**

## 2. Technology Readiness Level (TRL) Assessment

TRL is the standard metric for technology maturity. It was developed by NASA in the 1970s and is now used universally in space programmes.

[Source: NASA NPR 7123.1D Appendix E; ECSS-E-HB-11A "Technology Readiness Level (TRL) Guidelines"]

[URL: <https://www.nasa.gov/directorates/somd/space-communications-navigation-program/technology-readiness-levels/>]

### TRL Scale with Decision Criteria

| TRL | Definition                                                  | Evidence Required                 | CubeSat Decision                                |
|-----|-------------------------------------------------------------|-----------------------------------|-------------------------------------------------|
| 1   | Basic principles observed                                   | Published research                | Do not select                                   |
| 2   | Technology concept formulated                               | Analytical studies                | Do not select                                   |
| 3   | Experimental proof of concept                               | Lab measurements                  | Do not select                                   |
| 4   | Component validated in lab                                  | Breadboard tested                 | High risk -- avoid unless no alternative        |
| 5   | Component validated in relevant environment                 | Engineering model tested          | Acceptable for technology demonstrator missions |
| 6   | System/subsystem model demonstrated in relevant environment | Prototype in relevant environment | Acceptable with risk mitigation                 |
| 7   | System prototype demonstrated in operational environment    | Prototype tested in space         | Preferred minimum for CubeSats                  |
| 8   | Actual system completed and qualified                       | Flight-qualified hardware         | Strong preference                               |
| 9   | Actual system flight-proven                                 | Successful on-orbit operation     | Lowest risk                                     |

### TRL and Risk Relationship

#### Risk Reduction Factor (empirical):

$$\text{Risk\_factor} = 10^{-(\text{TRL} - 1) / 3}$$

| TRL | Risk Factor | Interpretation                       |
|-----|-------------|--------------------------------------|
| 3   | 0.22        | High probability of failure/redesign |
| 5   | 0.046       | Moderate risk                        |
| 7   | 0.010       | Low risk                             |
| 9   | 0.002       | Very low risk                        |

*This empirical relationship (from SMAD4 Table 20-12) illustrates why TRL  $\geq 6$  is the typical threshold for mission-critical components.*

### Real Mission Example: MarCO (Mars Cube One)

NASA's MarCO A and B (launched May 2018) were the first interplanetary CubeSats. They used: - Iris transponder (JPL, TRL 6 at selection -- first deep-space CubeSat radio) - COTS reaction wheels (Blue Canyon Technologies, TRL 9) - Custom deployable reflectarray antenna (JPL, TRL 5 at selection)

The custom antenna was the highest-risk item. It required extensive development testing and was the last component to reach TRL 6 for CDR. This delayed the schedule by 4 months but ultimately succeeded.

*Lesson: If you must fly a low-TRL component, it will dominate your schedule and risk register.*

*[Source: "Mars Cube One (MarCO) -- Lessons Learned", A. Klesh et al., 33rd Annual Small Satellite Conference, 2019]*

## 3. Bill of Materials (BOM) Construction

The BOM is the definitive list of all hardware, software, and consumables required to build the satellite. It is the bridge between design and procurement.

### BOM Structure

A properly structured BOM follows the WBS hierarchy:

```
BOM Level 0: Spacecraft (S/C-001)
 BOM Level 1: Subsystem (e.g., EPS-000)
 BOM Level 2: Assembly (e.g., EPS-SA-000 Solar Array Assembly)
 BOM Level 3: Component (e.g., EPS-SA-001 Solar Cell String)
 BOM Level 4: Part (e.g., EPS-SA-001-01 Azur 3G30C cell)
```

### BOM Fields

| Field           | Description                    | Example                 |
|-----------------|--------------------------------|-------------------------|
| Item ID         | Unique hierarchical identifier | EPS-BAT-001             |
| Description     | Component name and model       | GomSpace NanoPower P31u |
| Manufacturer    | Vendor name                    | GomSpace A/S            |
| Part Number     | Manufacturer part number       | P31U-9-30               |
| Quantity        | Number required                | 1                       |
| Unit Mass (g)   | Per-item mass                  | 94                      |
| Unit Power (W)  | Peak / average power draw      | 0.5 / 0.2               |
| Unit Cost (EUR) | Procurement cost               | 8,500                   |
| TRL             | Technology readiness level     | 9                       |
| Heritage        | Previous mission(s) flown      | GOMX-3, GOMX-4          |

| Field     | Description                            | Example  |
|-----------|----------------------------------------|----------|
| Lead Time | Procurement lead time                  | 12 weeks |
| ECCN/USML | Export classification                  | EAR99    |
| Status    | Selected / Ordered / Received / Tested | Selected |

## SpaceCDF BOM Generation

In SpaceCDF, the BOM is built automatically from the Equipment Browser selections: 1. Each component selected in the Equipment Browser creates a BOM entry 2. Quantities are set using the quantity selector (e.g., x4 for reaction wheels) 3. The **Exports** tab generates a formatted BOM spreadsheet 4. The BOM includes both parametric estimates and actual COTS data for comparison

## Key Equations

### BOM Mass Total:

$$M_{\text{BOM}} = \sum_i (m_i \times q_i) + M_{\text{harness}} + M_{\text{fasteners}} + M_{\text{margin}}$$

Where: -  $m_i$  = unit mass of component  $i$  -  $q_i$  = quantity of component  $i$  -  $M_{\text{harness}}$  = harness mass (typically 5-8% of dry mass for CubeSats) -  $M_{\text{fasteners}}$  = mechanical fasteners, standoffs, thermal hardware (~3-5%) -  $M_{\text{margin}}$  = system margin (typically 20% at Phase A)

## Worked Example

3U CubeSat BOM Summary:

| Subsystem       | Components                         | Total Mass (g) | Total Power (W)    | Total Cost (kEUR) |
|-----------------|------------------------------------|----------------|--------------------|-------------------|
| Structure       | Rails, panels, fasteners           | 350            | 0                  | 8.0               |
| EPS             | SA + battery + board               | 420            | 0.5                | 18.0              |
| OBC             | Flight computer + storage          | 80             | 1.2                | 10.0              |
| AOCS            | Star tracker + magnetorquers + RWs | 550            | 4.5                | 45.0              |
| TTC             | S-band transceiver + patch antenna | 180            | 8.0 (TX)           | 22.0              |
| Thermal         | MLI + heaters                      | 60             | 2.0 (peak)         | 3.0               |
| Payload         | Multispectral imager               | 800            | 12.0               | 65.0              |
| Harness         | Cables, connectors                 | 180            | 0                  | 2.5               |
| <b>Total</b>    |                                    | <b>2620</b>    | <b>28.2 (peak)</b> | <b>173.5</b>      |
| Allocation (6U) |                                    | 12000          | 40.0 (SA EOL)      | 250.0             |
| Margin          |                                    | 9380 (78%)     | 11.8 (30%)         | 76.5 (31%)        |

1U Worked Example: UniSat-1

Complete Bill of Materials

UniSat-1's BOM is remarkably short -- only 5--7 line items plus harness. This simplicity is a major advantage for university teams with limited procurement experience.

UniSat-1 BOM (Phase B -- vendor quotes obtained):

| Item ID     | Component                            | Manufacturer | Part Number   | Qty | Unit Mass (g) | Unit Cost (kEUR) | TRL | ECCN  | Lead (wks) |
|-------------|--------------------------------------|--------------|---------------|-----|---------------|------------------|-----|-------|------------|
| STR-001     | 1U CubeSat Structure                 | ISIS         | ISIS-1U-STR   | 1   | 200           | 4.0              | 9   | EAR99 | 8          |
| EPS-001     | NanoPower P31us (EPS + 10Wh battery) | GomSpace     | P31US-10      | 1   | 200           | 12.0             | 9   | EAR99 | 12         |
| EPS-SA-001  | Body-mounted GaAs solar cells        | AzurSpace    | 3G30C         | 5   | 10            | 1.5              | 9   | EAR99 | 10         |
| OBC-001     | Custom flight computer (Cortex-M)    | In-house     | UNISAT-OBC-01 | 1   | 30            | 3.0              | 5   | EAR99 | --         |
| COM-001     | UHF Transceiver                      | GomSpace     | NanoCom AX100 | 1   | 55            | 8.0              | 8   | EAR99 | 12         |
| COM-ANT-001 | UHF Deployable Antenna               | Endurosat    | UHF-ANT-S     | 1   | 25            | 2.5              | 8   | EAR99 | 8          |
| PL-001      | MEMS Magnetometer Board              | In-house     | UNISAT-MAG-01 | 1   | 50            | 5.0              | 4   | EAR99 | --         |
| AOCS-001    | Passive magnetic kit (magnet + rods) | NewSpace     | PMAG-1U       | 1   | 30            | 1.0              | 9   | EAR99 | 6          |
| HAR-001     | Internal harness                     | Custom       | --            | 1   | 50            | 1.0              | N/A | EAR99 | --         |
|             | TOTALS                               |              |               |     | 690 g         | ~44 kEUR         |     |       |            |

Cost summary (total mission, hardware + services):

| WBS Element    | Cost (kEUR) | Notes                                   |
|----------------|-------------|-----------------------------------------|
| Hardware (BOM) | 44          | All COTS except OBC and payload         |
| OBC software   | 5           | Student labour (costed at stipend rate) |

| WBS Element         | Cost (kEUR)     | Notes                                            |
|---------------------|-----------------|--------------------------------------------------|
| Payload calibration | 3               | University magnetometer lab                      |
| I&T                 | 8               | Assembly + vibration test (university facility)  |
| Ground station      | 5               | SatNOGS (free) + dedicated Yagi antenna purchase |
| Launch (ISS deploy) | 15              | NanoRacks 1U deployment fee                      |
| PM/SE/QA            | 5               | Faculty supervision                              |
| <b>TOTAL</b>        | <b>~85 kEUR</b> |                                                  |

#### Comparison to 3U EO mission:

| Metric             | UniSat-1 (1U) | 3U EO CubeSat |
|--------------------|---------------|---------------|
| BOM line items     | 9             | ~15--20       |
| Hardware cost      | ~44 kEUR      | ~290 kEUR     |
| Total mission cost | ~85 kEUR      | ~490 kEUR     |
| Development time   | 6--12 months  | 18--24 months |
| Team size          | 3--5 people   | 8--15 people  |

**Export control:** All UniSat-1 components are classified EAR99 (no licence required). There are no ITAR-controlled items because the mission uses no star trackers, no radiation-hardened processors, and no propulsion with ITAR-restricted technology. This is a significant advantage for international university collaborations.

#### Make/Buy/Reuse decisions:

| Component    | Decision      | Rationale                                                                                |
|--------------|---------------|------------------------------------------------------------------------------------------|
| Structure    | Buy (COTS)    | ISIS 1U frame is flight-proven, TRL 9, low cost                                          |
| EPS          | Buy (COTS)    | GomSpace P31us is the de facto standard, TRL 9                                           |
| OBC          | Make (custom) | Minimal board using university lab; lower cost than COTS OBC for this simple application |
| Comms        | Buy (COTS)    | GomSpace AX100, TRL 8, well-documented                                                   |
| Payload      | Make (custom) | Novel MEMS sensor -- this IS the technology demonstration                                |
| Passive AOCS | Buy (COTS)    | Standard magnetic stabilisation kit                                                      |

## 4. Export Control and Procurement

Export control is one of the most commonly overlooked aspects of CubeSat missions, especially for international teams. Violations carry severe penalties (criminal prosecution, programme cancellation).

[Source: ITAR -- 22 CFR Parts 120-130; EAR -- 15 CFR Parts 730-774; Canadian Controlled Goods Program -- Defence Production Act] [URL: [https://www.pmddtc.state.gov/ddtc\\_public](https://www.pmddtc.state.gov/ddtc_public) (US ITAR); <https://www.bis.doc.gov> (US EAR)]

## Export Control Regimes

| Regime       | Governing Law              | Applies To                                    | Key Concern for CubeSats                               |
|--------------|----------------------------|-----------------------------------------------|--------------------------------------------------------|
| ITAR         | US Arms Export Control Act | Defence articles (USML Categories IV, XI, XV) | Star trackers, rad-hard processors, some GPS receivers |
| EAR          | Export Administration Act  | Dual-use items (CCL)                          | Most COTS space components (ECCN 9A515)                |
| Canadian CGP | Defence Production Act     | Controlled goods in Canada                    | Handling US-origin ITAR items in Canadian facilities   |
| Wassenaar    | Wassenaar Arrangement      | Multilateral export controls                  | Encryption, high-accuracy GNSS, imaging sensors        |

## Component Classification Decision

```

Is the component on the US Munitions List (USML)?
-> Yes: ITAR-controlled. Need DSP-5 license for export.
 Categories: IV (launch vehicles), XI (military electronics),
 XV (spacecraft systems)
-> No: Check Commerce Control List (CCL)
 Is it classified under ECCN 9A515 (spacecraft)?
 -> Yes: EAR-controlled. May need BIS license.
 -> No: Likely EAR99 (no license required for most destinations)

```

## Common ITAR-Controlled CubeSat Components

| Component Type                           | Why Controlled                            | Alternative                                        |
|------------------------------------------|-------------------------------------------|----------------------------------------------------|
| Radiation-hardened processors            | Military-grade rad tolerance              | COTS processors with software mitigation           |
| High-accuracy star trackers (< 1 arcsec) | Missile guidance applicability            | Lower-accuracy models (> 5 arcsec)                 |
| Certain GPS/GNSS receivers               | Above COCOM limits (>60,000 ft, >1000 kt) | Space-rated receivers with COCOM compliance        |
| Propulsion systems (some)                | Missile technology                        | Cold-gas or water-based systems                    |
| Encryption modules                       | Signals intelligence                      | Open-source encryption (may still need EAR review) |

## Procurement Workflow

```

1. Requirements -> Derive component specification
2. Market survey -> Identify candidate COTS products
3. Export classification -> Request ECCN from vendor

```

4. Trade study -> Score and rank candidates
5. Request for Quote (RFQ) -> Obtain pricing and lead times
6. Purchase Order (PO) -> Commit to procurement
7. Incoming inspection -> Verify against PO and datasheet
8. Integration -> Install and functionally test

### Real Mission Example: Export Control Impact

The BRITE-Constellation (Austria/Canada/Poland) was a series of nanosatellites for stellar photometry. The Canadian BRITE satellites (UniBRITE and BRITE-Austria, built by UTIAS/SFL) required careful export control management: - US-origin ITAR components required Technical Assistance Agreements (TAAs) - Controlled Goods registration required for the Canadian team - Export permits needed for shipping between Canada, Austria, and the US launch site - Total regulatory compliance effort: ~6 person-months and 12+ months lead time

*Lesson: Start export classification immediately when components are identified. A single ITAR component can add 6-12 months to the schedule.*

[Source: Sarda, K. et al., "BRITE-Constellation Mission and Spacecraft", AIAA/USU Small Satellite Conference, 2014]

## 5. Interface Compatibility Verification

As components are selected, every interface must be verified for compatibility. The three interface categories are RF, electrical, and mechanical.

### RF Chain Compatibility

The RF chain (transponder, cable, antenna) must be frequency-matched. This is the most common equipment incompatibility for CubeSat newcomers.

| Rule                   | Correct Example                                     | Incorrect Example                          |
|------------------------|-----------------------------------------------------|--------------------------------------------|
| <b>Band match</b>      | S-band transponder + S-band patch antenna           | S-band transponder + X-band horn           |
| <b>Impedance match</b> | 50 ohm transponder + 50 ohm cable + 50 ohm antenna  | Mixed impedances cause reflections         |
| <b>Connector match</b> | SMA on transponder + SMA-SMA cable + SMA on antenna | SMA to N-type needs adapter (loss)         |
| <b>Polarisation</b>    | RHCP antenna + RHCP ground station                  | RHCP to LHCP causes > 20 dB cross-pol loss |

#### Impedance Mismatch Loss:

$$\text{Return\_Loss (dB)} = -20 * \log_{10}(|\Gamma|)$$

$$\text{Where } \Gamma = (Z_{\text{load}} - Z_{\text{source}}) / (Z_{\text{load}} + Z_{\text{source}})$$

$$\text{For a 50 ohm source into a 75 ohm load: } \Gamma = (75 - 50) / (75 + 50) = 0.2 \text{ Return\_Loss} = -20 * \log_{10}(0.2) = 14 \text{ dB}$$

$$\text{Mismatch\_Loss} = -10 * \log_{10}(1 - |\Gamma|^2) = -10 * \log_{10}(1 - 0.04) = 0.18 \text{ dB}$$



SpaceCDF checks RF compatibility automatically: if you select a transponder in one band and an antenna in another, a warning dialog appears.

## Electrical Interface Verification

| Parameter     | Typical CubeSat                          | What to Verify                                     |
|---------------|------------------------------------------|----------------------------------------------------|
| Bus voltage   | 3.3V, 5V, or unregulated (6-8.4V Li-ion) | Component input voltage range covers bus voltage   |
| Peak current  | Per switched line limit (typically 1-3A) | Component inrush current does not exceed limit     |
| Data protocol | I2C, SPI, UART, CAN, RS-422              | All devices on same bus use compatible protocol    |
| Connector     | PC/104, Hirose, Harwin                   | Physical connector type matches or adapter planned |

## Mechanical Interface Verification

| Parameter              | CDS 3U Requirement                                     | Verification                |
|------------------------|--------------------------------------------------------|-----------------------------|
| Dimensions             | 100.0 +/- 0.1 mm x 100.0 +/- 0.1 mm x 340.5 +/- 0.5 mm | Caliper measurement         |
| Rail profile           | 8.5 x 8.5 mm +/- 0.1 mm                                | Profile gauge               |
| Component stack height | <= 83 mm internal width per U                          | 3D model check              |
| CG location            | Within 2 cm of geometric centre                        | Mass properties measurement |

## 6. Equipment Selection Exercise

### Instructions

#### Part A: Equipment Selection

1. Open the **Equipment Browser** in SpaceCDF
2. For each **required** category (blue dot), select at least one component: - Start with EPS (batteries + solar panels + EPS board) - Then OBC, AOCS sensors/actuators, TTC, structure - Use quantity selectors (e.g., x4 for reaction wheels, x3 for magnetorquers)
3. Watch the **live budget bar** as you select -- keep mass under allocation
4. When selecting TTC: verify the spectrum band matches your earlier selection
5. After all selections, review the **Budget Breakdown** on the Dashboard

#### Part B: BOM Review and Export Check

1. Navigate to the **Exports** tab
2. Generate the **BOM** -- review all entries
3. For each component, check: Is the ECCN listed? Is it EAR99, ECCN 9A515, or ITAR?
4. Flag any components that may require export licences
5. Compute: BOM total mass vs parametric estimate vs launcher allocation

## Worksheet 4.1 Tasks

1. Complete the full BOM table (component, manufacturer, part number, mass, power, cost, TRL, ECCN)
2. Document the make/buy/reuse decision for each subsystem with rationale
3. Compute total BOM mass vs parametric estimate vs allocation -- state margin
4. Identify any export-controlled components and note the required licensing action
5. Describe one interface incompatibility found during selection and how it was resolved

## Cited References

| # | Reference                             | URL                                                                                                                                                                                                                                   |
|---|---------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1 | NASA Technology Readiness Levels      | <a href="https://www.nasa.gov/directorates/somd/space-communications-navigation-program/technology-readiness-levels/">https://www.nasa.gov/directorates/somd/space-communications-navigation-program/technology-readiness-levels/</a> |
| 2 | ECSS-Q-ST-20C Quality Assurance       | <a href="https://ecss.nl/standard/ecss-q-st-20c-rev-2-quality-assurance-1-march-2023/">https://ecss.nl/standard/ecss-q-st-20c-rev-2-quality-assurance-1-march-2023/</a>                                                               |
| 3 | US DDTC (ITAR)                        | <a href="https://www.pmddtc.state.gov/">https://www.pmddtc.state.gov/</a>                                                                                                                                                             |
| 4 | US BIS (EAR)                          | <a href="https://www.bis.doc.gov/">https://www.bis.doc.gov/</a>                                                                                                                                                                       |
| 5 | CubeSat Design Specification Rev 14.1 | <a href="https://www.cubesat.org/s/CDS-REV14_1-2022-02-09.pdf">https://www.cubesat.org/s/CDS-REV14_1-2022-02-09.pdf</a>                                                                                                               |
| 6 | SMAD4 Chapter 20 (Cost Modelling)     | Wertz, Everett, Puschell (eds.), Space Mission Engineering, Microcosm 2011                                                                                                                                                            |
| 7 | MarCO Lessons Learned (Klesh et al.)  | SSC19-WKII-07, 33rd Small Sat Conference, 2019                                                                                                                                                                                        |

## Session Summary

| Topic            | Key Takeaway                                                                          |
|------------------|---------------------------------------------------------------------------------------|
| Make/Buy/Reuse   | Buy COTS first; Reuse heritage second; Make custom only when necessary                |
| TRL              | Minimum TRL 6 for mission-critical components; TRL 7+ preferred for CubeSats          |
| BOM              | Hierarchical list: Subsystem -> Assembly -> Component -> Part, with full traceability |
| Export control   | Classify every component (EAR99 / ECCN / ITAR); start early -- delays programme       |
| RF compatibility | Band, impedance, connector, polarisation must all match across RF chain               |
| Electrical       | Bus voltage, data protocol, connector type verified per component                     |
| Budget tracking  | Live totals during selection; stop if allocation exceeded                             |

## Session 4.2: Verification & Validation Matrix and Compliance

**Expected reading before this session.** ECSS-E-ST-10-02C — Verification ( $\approx 60$  min); NASA SEH §6.4.

**Prerequisites:** Session 4.1 (equipment selected, BOM constructed) **References:** ECSS-E-ST-10-02C Rev.1 (Verification), ECSS-E-ST-10-03C (Testing), NASA SEH Rev 2 sections 5.3-5.4, MIL-STD-1540E, MIL-STD-461G

### Learning Objectives

By the end of this session, participants will be able to: 1. Distinguish between verification and validation using the ECSS/NASA definitions 2. Assign appropriate verification methods (IADT) to each requirement using a decision logic 3. Define environmental test levels derived from launch vehicle ICD specifications 4. Construct a compliance verification matrix mapping every requirement to its method, level, and phase 5. Identify when waivers are appropriate and document the waiver rationale

### 1. Verification vs Validation: Precise Definitions

[Source: ECSS-E-ST-10-02C Rev.1 section 4; NASA SEH Rev 2 section 5.3 (Process 7: Product Verification) and section 5.4 (Process 8: Product Validation)]

|                       | Verification                                     | Validation                                         |
|-----------------------|--------------------------------------------------|----------------------------------------------------|
| <b>Question</b>       | "Did we build the system right?"                 | "Did we build the right system?"                   |
| <b>Checks against</b> | Requirements (shall statements)                  | Stakeholder needs and operational expectations     |
| <b>Standard</b>       | ECSS-E-ST-10-02C                                 | ECSS-E-ST-10-06C                                   |
| <b>Performed by</b>   | Engineering team                                 | Users, operators, stakeholders                     |
| <b>When</b>           | Throughout development (Phases B-D)              | End of development + early operations (Phase D-E)  |
| <b>Methods</b>        | IADT (Inspection, Analysis, Demonstration, Test) | Operational demonstration, user acceptance testing |
| <b>Output</b>         | Verification Control Document (VCD)              | Validation report                                  |

### The Critical Distinction

A system can be **verified** (every requirement marked "pass") but still **fail validation** if the requirements were written incorrectly. This happens when: - Requirements were derived from misunderstood stakeholder needs - The operational environment differs from assumptions - Edge cases were not captured in requirements

## Example

- **Requirement:** "The system shall achieve GSD  $\leq 10$  m from 500 km altitude"
- **Verification (Test):** Calibration target imagery at 500 km -> measured GSD = 9.2 m -> **verified**
- **Validation:** Users attempt crop assessment with 9.2 m imagery -> "We can see field boundaries but cannot distinguish crop types; we needed GSD  $\leq 5$  m" -> **not validated**

*The requirement was met, but the stakeholder need was not satisfied because the requirement was insufficiently stringent.*

## 2. IADT Verification Methods in Detail

ECSS-E-ST-10-02C Rev.1 section 5.3 defines four verification methods. Note: ECSS uses "IADT" ordering; some NASA documents use "ATRI" -- the methods are identical.

[Source: ECSS-E-ST-10-02C Rev.1 section 5.3; freely available from <https://ecss.nl>]

### Inspection (I)

**Definition:** Visual or physical examination of the product, without operation or stimulation, to verify conformance to requirements.

**When to use:** - Physical characteristics (dimensions, mass, surface finish, labelling) - Manufacturing workmanship (solder joints, cable routing, MLI installation) - Markings, identification, and safety labels - Mechanical fit (deployer rail compliance)

**Evidence produced:** Inspection report with photographs, measurements, pass/fail checklist.

**Examples for CubeSats:** - Mass measurement on calibrated scale -> verifies "M<sub>dry</sub>  $\leq 4.0$  kg" - Caliper measurement of rail profile -> verifies CDS dimensional compliance - Visual inspection of antenna stowage -> verifies no protrusions beyond deployer envelope - Workmanship inspection of PCB solder joints -> verifies IPC-A-610 Class 3

### Analysis (A)

**Definition:** Mathematical models, simulations, statistical analysis, or heritage comparison demonstrate compliance with acceptable confidence.

**When to use:** - Physical testing is impossible or impractical (orbital lifetime, radiation total ionising dose) - Early in development (Phase B) before hardware exists - To demonstrate margin (link budget, thermal model, structural FEA) - Statistical parameters (reliability prediction, debris casualty risk) - Requirements involving orbital mechanics or mission-level performance

**Evidence produced:** Analysis report with model description, assumptions, inputs, results, uncertainty assessment, and sensitivity analysis.

**Examples for CubeSats:** - Orbital mechanics analysis -> verifies revisit time requirement - Thermal model (finite difference/finite element) -> verifies temperature within limits - Link budget calculation -> verifies margin  $\geq 3$  dB at worst case - Structural FEA -> verifies positive Margin of Safety under launch loads

#### Margin of Safety (structural):

$$\text{MoS} = (\text{Allowable\_stress} / (\text{Factor\_of\_Safety} \times \text{Applied\_stress})) - 1$$

MoS must be  $\geq 0$  for all load cases. Factor\_of\_Safety: 1.25 (yield), 1.5 (ultimate) per ECSS-E-ST-32-10C.

## Demonstration (D)

**Definition:** The system is operated under controlled conditions to show that it performs as required, without requiring precise measurement of performance parameters.

**When to use:** - Functional requirements ("the system shall deploy the antenna within 30 minutes of separation") - Operational procedures ("the system shall respond to a mode change command within 5 seconds") - Software functional requirements ("FDIR shall autonomously switch to safe mode on loss of attitude")

**Evidence produced:** Demonstration report with procedure, observations, photographs/video, pass/fail.

**Examples for CubeSats:** - Antenna deployment test -> demonstrates deployment within time limit - Safe mode entry test -> demonstrates autonomous FDIR response - Ground station commanding test -> demonstrates full command chain

## Test (T)

**Definition:** Physical hardware subjected to controlled conditions; measured performance compared quantitatively to requirement thresholds.

**When to use:** - Physical properties must be proven with quantitative measurement - Environmental survivability must be demonstrated (vibration, thermal-vacuum, EMC) - End-to-end functional chains must be proven under representative conditions - Performance must be measured, not just observed

**Test types by purpose:**

| Test Type            | Purpose                                     | Level       | Phase | Loads/Conditions                 |
|----------------------|---------------------------------------------|-------------|-------|----------------------------------|
| <b>Development</b>   | Explore design space, find problems early   | Unit/BB     | B     | As needed                        |
| <b>Qualification</b> | Prove design withstands worst-case + margin | Unit/System | C     | Qual levels (MPE + margin)       |
| <b>Acceptance</b>    | Prove flight hardware is defect-free        | System      | D     | Acceptance levels (MPE)          |
| <b>Proto-flight</b>  | Combined qual + acceptance (single unit)    | System      | C/D   | Qual levels, acceptance duration |

## Proto-Flight Approach

For CubeSats, the **proto-flight** approach is standard because building a separate qualification unit is prohibitively expensive. The flight unit is tested to **qualification levels** but for **acceptance duration**.

[Source: ECSS-E-ST-10-03C section 5.4.2.3; MIL-STD-1540E section 6.2.4]

**Rationale:** Qualification duration is longer and more stressful -- this is acceptable for a dedicated qualification model that will not fly. For proto-flight, the shorter duration reduces fatigue life consumption while still screening for workmanship defects.

| Parameter       | Qualification | Acceptance | Proto-flight |
|-----------------|---------------|------------|--------------|
| Vibration level | MPE + 3 dB    | MPE        | MPE + 3 dB   |

| Parameter              | Qualification      | Acceptance         | Proto-flight       |
|------------------------|--------------------|--------------------|--------------------|
| Vibration duration     | 2 min/axis         | 1 min/axis         | 1 min/axis         |
| TVAC temperature range | Predicted +/- 15 C | Predicted +/- 10 C | Predicted +/- 10 C |
| TVAC cycles            | 8                  | 4                  | 4-8                |

### 3. Environmental Test Programme

[Source: ECSS-E-ST-10-03C (Testing); Launch vehicle Payload User's Guides; MIL-STD-1540E]

#### Standard Environmental Test Sequence

The test sequence is designed so that each test stresses the hardware in a specific way, and functional tests between environmental tests detect any damage.

```

+-----+
| Receive Flight Hardware |
+-----+
 |
+-----v-----+
| Initial Functional Test | <- Reference baseline
+-----+
 |
+-----v-----+
| Sine Vibration (3 axes) | <- Low-frequency loads
+-----+
 |
+-----v-----+
| Random Vibration (3 axes)| <- Broadband acoustic loads
+-----+
 |
+-----v-----+ +-----+
| Post-Vibe Functional Test| ----->| Anomaly? |
+-----+ Fail | Investigate & |
 | Pass | resolve |
+-----v-----+ +-----+
| Thermal Vacuum Testing | <- 4-8 cycles, func at extremes
+-----+
 |
+-----v-----+
| Post-TVAC Functional Test|
+-----+
 |
+-----v-----+
| EMC Test (if required) |
+-----+
 |
+-----v-----+
| Mass + CG Measurement |
+-----+

```

```

 |
+-----v-----+
| Deployer Fit Check |
+-----+-----+
 |
+-----v-----+
| Pack & Ship to Launch |
+-----+-----+

```

## Sine Vibration Test Levels

[Source: Typical Falcon 9 Transporter Rideshare PUG]

Sine vibration simulates the low-frequency launch vehicle structural modes (typically 5-100 Hz). These are quasi-static loads.

| Frequency Range (Hz) | Level                         | Notes                          |
|----------------------|-------------------------------|--------------------------------|
| 5 - 8                | 12.5 mm displacement (0-peak) | Displacement-controlled        |
| 8 - 100              | 1.25 g (0-peak)               | Acceleration-controlled        |
| Sweep rate           | 2 octaves/minute              | Standard sweep rate            |
| Axes                 | 3 (X, Y, Z)                   | Sequential, one axis at a time |

### Sine Vibration Acceleration at Crossover Frequency:

$$a = (2 * \pi * f)^2 * d$$

$$\text{At } f = 8 \text{ Hz, } d = 12.5 \text{ mm} = 0.0125 \text{ m: } a = (2 * \pi * 8)^2 * 0.0125 = (50.27)^2 * 0.0125 = 2526.6 * 0.0125 = 31.6 \text{ m/s}^2 = 3.22 \text{ g}$$

*Note: The transition from displacement to acceleration control occurs where the displacement limit would exceed the acceleration limit. The actual crossover depends on the specific profile.*

## Random Vibration Test Levels

Random vibration simulates the broadband acoustic and mechanical environment during launch. Levels are specified as Power Spectral Density (PSD) in  $\text{g}^2/\text{Hz}$ .

| Frequency (Hz) | PSD Level ( $\text{g}^2/\text{Hz}$ ) | Notes                 |
|----------------|--------------------------------------|-----------------------|
| 20             | 0.01                                 | Start of profile      |
| 20 - 50        | +3 dB/octave ramp                    | Rising                |
| 50 - 800       | 0.04 (flat)                          | Maximum level plateau |
| 800 - 2000     | -6 dB/octave ramp                    | Falling               |
| 2000           | 0.004                                | End of profile        |
| Overall gRMS   | ~7.0 gRMS                            | Qualification level   |

### Computing Overall gRMS from PSD Profile:

$g_{RMS} = \sqrt{\int_{f1}^{f2} PSD(f) df}$

For a flat PSD region:  $g_{RMS\_flat} = \sqrt{PSD\_flat \times (f2 - f1)}$

Example:  $PSD = 0.04 \text{ g}^2/\text{Hz}$  from 50-800 Hz:  $g_{RMS\_flat} = \sqrt{0.04 \times 750} = \sqrt{30} = 5.48 \text{ gRMS}$

The ramp sections and flat section are summed in quadrature:  $g_{RMS\_total} = \sqrt{g_{RMS\_ramp1}^2 + g_{RMS\_flat}^2 + g_{RMS\_ramp2}^2}$

## Thermal Vacuum Test Profile

| Parameter                   | Proto-flight Level                        | Notes                                                  |
|-----------------------------|-------------------------------------------|--------------------------------------------------------|
| Hot case temperature        | Predicted max + 10 C                      | e.g., if thermal model predicts +55 C -> test to +65 C |
| Cold case temperature       | Predicted min - 10 C                      | e.g., if thermal model predicts -20 C -> test to -30 C |
| Number of cycles            | 4 minimum (proto-flight)                  | Start and end at hot; functional test at each extreme  |
| Vacuum level                | $< 10^{-5}$ mbar                          | Space-representative vacuum                            |
| Dwell time at extreme       | $\geq 1$ hour                             | Allow thermal stabilisation ( $< 1$ C/hr gradient)     |
| Temperature transition rate | 1-3 C/min                                 | Limited by chamber capability                          |
| Functional test             | Full performance at hot and cold extremes | Subset functional at intermediate transitions          |

### TVAC Cycle Duration Estimate:

$t_{\text{cycle}} = 2 \times (\Delta T / \text{ramp\_rate}) + 2 \times t_{\text{dwell}} + 2 \times t_{\text{functional}}$

Example:  $\Delta T = 95$  C (from -30 to +65 C), ramp = 2 C/min, dwell = 1 hr, functional = 2 hr:  $t_{\text{cycle}} = 2 \times (95/2 \text{ min}) + 2 \times 60 \text{ min} + 2 \times 120 \text{ min} = 95 + 120 + 240 = 455 \text{ min} = 7.6 \text{ hr}$

Total TVAC campaign (4 cycles + setup):  $\sim 4 \times 7.6 + 8 = 38.4 \text{ hr} \sim 5 \text{ working days}$

## EMC Testing

[Source: MIL-STD-461G; ECSS-E-ST-20-07C]

| Test                           | Standard     | Purpose                                             |
|--------------------------------|--------------|-----------------------------------------------------|
| CE102 Conducted Emissions      | MIL-STD-461G | Measure spurious conducted emissions on power lines |
| RE102 Radiated Emissions       | MIL-STD-461G | Measure unintentional RF radiation from satellite   |
| CS101 Conducted Susceptibility | MIL-STD-461G | Verify immunity to power line transients            |
| RS103 Radiated Susceptibility  | MIL-STD-461G | Verify immunity to external RF fields               |



*Note: EMC testing is not always required for CubeSats. It depends on the launch provider and co-manifested payloads. Check the rideshare ICD.*

## 4. Compliance Verification Matrix Construction

The Verification Matrix (also called Compliance Matrix or Verification Cross-Reference Matrix, VCRM) is the central document linking every requirement to its verification evidence.

[Source: ECSS-E-ST-10-02C Rev.1 section 6; DRD in Annex A]

### Matrix Structure

| Req ID  | Requirement Text         | Method | Phase | Level  | Responsible | Facility   | Status  | Evidence                 |
|---------|--------------------------|--------|-------|--------|-------------|------------|---------|--------------------------|
| SYS-001 | M_dry <= 4.0 kg          | I      | D     | System | Structures  | Clean room | Planned | Mass measurement report  |
| SYS-002 | Survive 7 gRMS random    | T      | C/D   | System | Structures  | Vibe lab   | Planned | Vibration test report    |
| SYS-003 | GSD <= 10 m at 500 km    | A      | B     | System | Payload     | N/A        | Planned | Optical analysis report  |
| SYS-004 | Link margin >= 3 dB      | A      | B     | System | Comms       | N/A        | Planned | Link budget analysis     |
| SYS-005 | Antenna deploys < 30 min | D      | D     | System | Mechanisms  | Clean room | Planned | Deployment demonstration |

### IADT Assignment Decision Logic

Is it a physical property (mass, dimensions, surface finish)?

-> Inspection (I), Phase D, System level

Can it only be proven by quantitative measurement under controlled conditions?

-> Test (T), Phase C/D, Unit or System level

Can it be demonstrated by operating the system (functional, not precision)?

-> Demonstration (D), Phase D, System level

Can it be shown with acceptable confidence through modelling?

-> Analysis (A), Phase B/C, System level

Is it a document, process, or plan?

-> Review of Design (R\*), Phase B

(\* Note: ECSS uses IADT; "Review" is sometimes used informally but is not a formal ECSS method. Reviews are part of the verification process but

requirements verified by examining design documentation fall under "Analysis" in strict ECSS usage.)

## Multiple Methods

Some requirements need **both** analysis (early confidence) and test (final proof):

| Requirement                            | Analysis (Phase B)       | Test (Phase C/D)                    |
|----------------------------------------|--------------------------|-------------------------------------|
| Structural survival under launch loads | FEA with positive MoS    | Vibration test -- no damage         |
| Pointing accuracy $\leq 0.1$ deg       | Error budget analysis    | On-orbit calibration measurement    |
| Thermal within limits (-20 to +55 C)   | Thermal model simulation | TVAC functional test at extremes    |
| RF link margin $\geq 3$ dB             | Link budget calculation  | RF compatibility test (if feasible) |

In the matrix, list the **primary** method and note "confirmed by [secondary method] in Phase [X]".

## Waiver Process

When a requirement cannot be verified by the prescribed method, a **waiver** is needed:

| Waiver Type             | When Used                                                           | Approval Level             |
|-------------------------|---------------------------------------------------------------------|----------------------------|
| <b>Non-compliance</b>   | Requirement cannot be met; accept deviation                         | Project manager + customer |
| <b>Method deviation</b> | Different verification method used (e.g., analysis instead of test) | Systems engineer + quality |
| <b>Tailoring</b>        | Standard requirement not applicable to this mission                 | Systems engineer           |

**Waiver documentation must include:** Requirement ID, deviation description, technical justification, risk assessment, and compensating measures.

## 5. SpaceCDF V&V Matrix Exercise

### Instructions

1. Navigate to the **V&V Matrix** tab in SpaceCDF
2. Requirements are auto-populated from the design
3. For each requirement, assign: - **Method:** I, A, D, or T (use the decision logic above) - **Phase:** When will verification occur (B, C, D)? - **Level:** Unit, subsystem, or system level? - **Status:** All should be "Planned" at this stage
4. Use the **filter buttons** to view by method type: - Count: How many are Analysis? Test? Demonstration? Inspection? - Are there any requirements with no method assigned?
5. Identify requirements that need **two methods** (analysis in Phase B + test in Phase D)
6. Discuss with your team: which tests are most critical? Which drive the test campaign schedule?

### Worksheet 4.2 Tasks

1. Assign IADT method to 10 key requirements (justify each choice)

2. Identify 3 requirements that need both analysis AND test
3. Specify the complete environmental test sequence for your mission
4. Determine which test levels apply from your selected launch vehicle PUG
5. Identify any requirements that may need waivers -- document the rationale

## Cited References

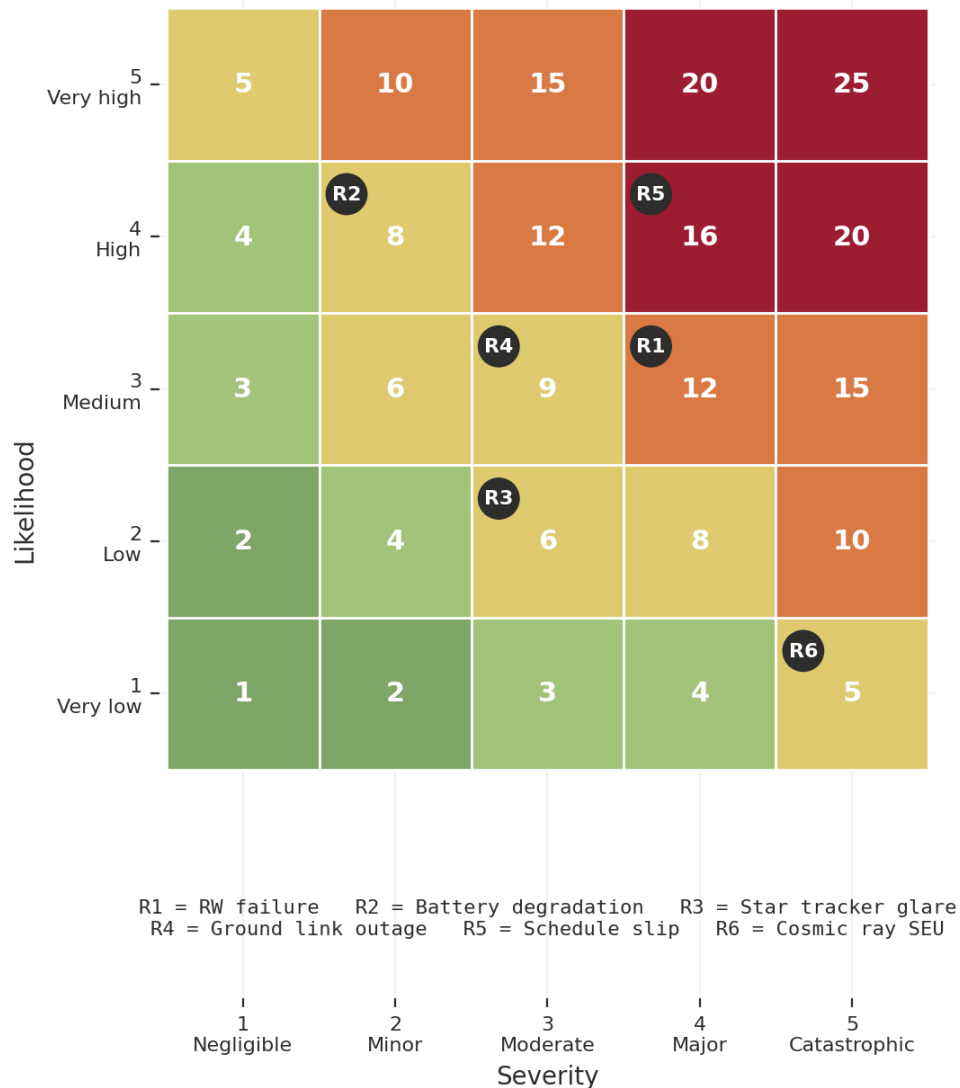
| # | Reference                                                               | URL                                                                                                                                             |
|---|-------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------|
| 1 | ECSS-E-ST-10-02C Rev.1 (Verification)                                   | <a href="https://ecss.nl/standard/ecss-e-st-10-02c-rev-1-verification/">https://ecss.nl/standard/ecss-e-st-10-02c-rev-1-verification/</a>       |
| 2 | ECSS-E-ST-10-03C (Testing)                                              | <a href="https://ecss.nl/standard/ecss-e-st-10-03c-testing/">https://ecss.nl/standard/ecss-e-st-10-03c-testing/</a>                             |
| 3 | NASA SEH Rev 2 sections 5.3-5.4                                         | <a href="https://www.nasa.gov/reference/systems-engineering-handbook/">https://www.nasa.gov/reference/systems-engineering-handbook/</a>         |
| 4 | MIL-STD-1540E (Test Requirements for Launch/Upper Stage/Space Vehicles) | <a href="https://quicksearch.dla.mil/qsDocDetails.aspx?ident_number=36197">https://quicksearch.dla.mil/qsDocDetails.aspx?ident_number=36197</a> |
| 5 | MIL-STD-461G (EMI/EMC)                                                  | <a href="https://quicksearch.dla.mil/qsDocDetails.aspx?ident_number=35789">https://quicksearch.dla.mil/qsDocDetails.aspx?ident_number=35789</a> |
| 6 | CubeSat Design Specification Rev 14.1                                   | <a href="https://www.cubesat.org/s/CDS-REV14_1-2022-02-09.pdf">https://www.cubesat.org/s/CDS-REV14_1-2022-02-09.pdf</a>                         |

## Session Summary

| Topic             | Key Takeaway                                                                              |
|-------------------|-------------------------------------------------------------------------------------------|
| V vs V            | Verification = meets requirements (IADT); Validation = meets stakeholder needs            |
| IADT              | Inspection (physical), Analysis (models), Demonstration (functional), Test (quantitative) |
| Proto-flight      | CubeSats: single flight unit tested to qualification levels, acceptance duration          |
| Sine vibration    | 5-100 Hz, 1.25 g, quasi-static launch loads                                               |
| Random vibration  | 20-2000 Hz, ~7 gRMS overall, broadband acoustic environment                               |
| TVAC              | Predicted extremes +/- 10 C, 4+ cycles, functional at each extreme                        |
| Compliance matrix | Per requirement: method + phase + level + responsible + status + evidence                 |
| Waivers           | Document deviation, justification, risk, and compensating measures                        |

## Session 4.3: Risk, Interfaces & FMECA

**Risk Matrix (ECSS-M-ST-80C 5×5) — worked example**



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5×5 risk matrix (ECSS-M-ST-80C).

**Expected reading before this session.** ECSS-M-ST-80C — Risk management (≈ 60 min); ECSS-Q-ST-30-02C — FMEA / FMECA.

**Prerequisites:** Sessions 4.1-4.2 (equipment selected, V&V methods assigned) **References:** ECSS-M-ST-80C (Risk Management), ECSS-Q-ST-30-02C (FMEA/FMECA), ECSS-E-ST-10-24C (Interface Management), NASA SEH Rev 2 section 6.4, NPR 8000.4B

## Learning Objectives

By the end of this session, participants will be able to: 1. Construct and populate a risk register using a 5x5 likelihood-consequence matrix 2. Build an N-squared ( $N^2$ ) interface matrix and identify interface conflicts 3. Perform a simplified FMEA/FMECA for critical subsystems 4. Identify single-point failures and compute series/parallel reliability 5. Classify risks and select appropriate mitigation strategies (accept, mitigate, transfer, avoid)

## 1. Risk Management Process

[Source: ECSS-M-ST-80C (Risk Management); NASA SEH Rev 2 section 6.4 (Process 13: Technical Risk Management)] [URL: <https://ecss.nl/standard/ecss-m-st-80c-risk-management-31-july-2008/>]

### The Four-Step Process

Risk management is continuous throughout the project lifecycle. The four steps are:

1. **Identify** -- What can go wrong? (Brainstorming, checklists, historical data, expert judgment)
2. **Assess** -- How likely is it? How severe would the consequence be? (Scoring)
3. **Mitigate** -- What can we do to reduce likelihood or consequence? (Strategy selection)
4. **Monitor** -- Is the risk increasing, decreasing, or stable? (Tracking, triggers, escalation)

### Risk Sources

| Source       | Example Risks                                                                 |
|--------------|-------------------------------------------------------------------------------|
| Technical    | Component failure, interface mismatch, performance shortfall, software defect |
| Schedule     | Component delivery delay, test facility unavailable, regulatory delay         |
| Cost         | Component price increase, scope creep, test campaign overrun                  |
| Programmatic | Funding cut, personnel turnover, partner withdrawal                           |
| External     | Launch delay/failure, spectrum interference, export control denial            |

## 2. The 5x5 Risk Matrix

The 5x5 matrix is the standard risk scoring tool used across ESA, NASA, and commercial space programmes.

### Likelihood Scale

| Level | Likelihood     | Description                                         | Probability Range |
|-------|----------------|-----------------------------------------------------|-------------------|
| 1     | Remote         | Very unlikely; no precedent in similar missions     | < 5%              |
| 2     | Unlikely       | Could happen but improbable; has occurred rarely    | 5 - 20%           |
| 3     | Possible       | Has happened on similar missions; credible scenario | 20 - 50%          |
| 4     | Likely         | Expected to occur at least once during the mission  | 50 - 80%          |
| 5     | Almost certain | Will almost certainly occur; multiple precedents    | > 80%             |

Consequence Scale

| Level | Consequence  | Technical Impact                                   | Cost Impact      | Schedule Impact                |
|-------|--------------|----------------------------------------------------|------------------|--------------------------------|
| 1     | Negligible   | No performance impact                              | < 1% overrun     | < 1 week slip                  |
| 2     | Minor        | Minor performance degradation; workaround exists   | 1 - 5% overrun   | 1 - 4 week slip                |
| 3     | Moderate     | Significant performance loss; mission degraded     | 5 - 15% overrun  | 1 - 3 month slip               |
| 4     | Major        | Mission capability severely degraded; partial loss | 15 - 30% overrun | 3 - 6 month slip               |
| 5     | Catastrophic | Mission failure; total loss                        | > 30% overrun    | > 6 month slip or cancellation |

Risk Score and Classification

Risk Score:

$R = L \times C$

Where L = Likelihood (1-5), C = Consequence (1-5)

5x5 Risk Matrix (Colour-Coded)

|                                                                                                                                                                                         |   | Consequence -> |       |       |       |       |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---|----------------|-------|-------|-------|-------|
|                                                                                                                                                                                         |   | 1              | 2     | 3     | 4     | 5     |
| Likelihood                                                                                                                                                                              | 5 | 5(M)           | 10(H) | 15(H) | 20(C) | 25(C) |
|                                                                                                                                                                                         | 4 | 4(L)           | 8(M)  | 12(H) | 16(C) | 20(C) |
|                                                                                                                                                                                         | 3 | 3(L)           | 6(M)  | 9(M)  | 12(H) | 15(H) |
|                                                                                                                                                                                         | 2 | 2(L)           | 4(L)  | 6(M)  | 8(M)  | 10(H) |
|                                                                                                                                                                                         | 1 | 1(L)           | 2(L)  | 3(L)  | 4(L)  | 5(M)  |
| L = Low (1-4): Accept and monitor<br>M = Medium (5-9): Mitigate if cost-effective<br>H = High (10-15): Active mitigation required<br>C = Critical (16-25): Redesign or descope required |   |                |       |       |       |       |

CubeSat-Specific Risk Examples

| Risk                                       | L | C | Score | Category | Typical Mitigation                            |
|--------------------------------------------|---|---|-------|----------|-----------------------------------------------|
| Deployment mechanism failure (antenna, SA) | 3 | 4 | 12    | High     | Redundant mechanisms; 100+ ground test cycles |

| Risk                                          | L | C | Score | Category | Typical Mitigation                                                           |
|-----------------------------------------------|---|---|-------|----------|------------------------------------------------------------------------------|
| Communication loss after separation           | 2 | 5 | 10    | High     | Beacon mode; timer-based antenna deploy; multiple GS                         |
| ADCS does not achieve pointing specification  | 3 | 3 | 9     | Medium   | Margin in pointing budget; on-orbit calibration plan                         |
| Power budget negative during eclipse          | 2 | 4 | 8     | Medium   | Conservative duty cycling; 20% battery margin                                |
| COTS component radiation failure (SEU/SEL)    | 2 | 4 | 8     | Medium   | Watchdog resets; latchup protection circuits; EDAC                           |
| Software bug causing spurious safe mode entry | 4 | 2 | 8     | Medium   | Extensive software testing; staged upload; safe mode must work independently |
| Thermal exceedance (hot case)                 | 2 | 3 | 6     | Medium   | Additional radiator area; duty cycle limit                                   |
| Launch delay (vehicle failure)                | 3 | 2 | 6     | Medium   | Manifest on multiple vehicles; schedule buffer                               |
| Spectrum licensing delay                      | 3 | 2 | 6     | Medium   | Start filing 12+ months before planned launch                                |

### Mitigation Strategies

| Strategy        | Description                                                  | When to Use                      | Cost Impact                    |
|-----------------|--------------------------------------------------------------|----------------------------------|--------------------------------|
| <b>Accept</b>   | Risk is within tolerance; monitor only                       | Score 1-4 (Low)                  | None                           |
| <b>Mitigate</b> | Reduce L or C through design changes, testing, or procedures | Score 5-15                       | Moderate -- cost of mitigation |
| <b>Transfer</b> | Pass risk to another party (insurance, supplier warranty)    | Financial risk; operational risk | Premium or contract cost       |
| <b>Avoid</b>    | Change design to eliminate the risk entirely                 | Score 16-25 (Critical)           | May affect performance or cost |

## 3. N-Squared ( $N^2$ ) Interface Matrix

The  $N^2$  matrix is the standard systems engineering tool for identifying and managing interfaces between subsystems. It was popularised by NASA and ESA CDF practice.

[Source: ECSS-E-ST-10-24C (Interface Management); NASA SEH Rev 2 section 6.3 (Process 12: Interface Management)]

### What is an $N^2$ Matrix?

For a system with  $N$  subsystems, the  $N^2$  matrix is an  $N \times N$  grid where: - **Diagonal cells** contain the subsystem names - **Off-diagonal cells** contain the interfaces between subsystems - **Cell (i, j)** = outputs FROM subsystem i TO subsystem j (read across the row) - **Cell (j, i)** = outputs FROM subsystem j TO subsystem i (read down the column)

## N^2 Matrix Example (6 Subsystems)

| FROM    | T0 ->            |                             |                             |                             |                      |                    |
|---------|------------------|-----------------------------|-----------------------------|-----------------------------|----------------------|--------------------|
|         | EPS              | OBC                         | AOCS                        | TTC                         | Payload              | Structure          |
| EPS     | [EPS]<br>-----   | 28V bus<br>5V reg<br>I2C HK | 28V bus<br>5V reg<br>I2C HK | 28V bus<br>5V reg<br>I2C HK | 28V bus<br>5V reg    | Mounting<br>bolts  |
| OBC     | Pwr cmd<br>Telem | [OBC]<br>-----              | Cmd<br>I2C/SPI              | TC data<br>UART             | Cmd<br>SPI/UART      | ---                |
| AOCS    | Pwr req<br>---   | Att data<br>AOCS HK         | [AOCS]<br>-----             | Att for<br>antenna          | Att data<br>pointing | Mount<br>vibration |
| TTC     | Pwr req<br>---   | Rx data<br>Cmd fwd          | ---                         | [TTC]<br>-----              | ---                  | Antenna<br>mount   |
| Payload | Pwr req<br>---   | Sci data<br>PL HK           | Pointing<br>request         | ---                         | [PYLD]<br>-----      | FOV<br>clearance   |
| Struct  | ---              | ---                         | Sensor<br>mounting          | Antenna<br>mounting         | Payload<br>mounting  | [STRUCT]<br>-----  |

### Interface Types

Colour-code the N^2 matrix by interface type:

| Type             | Colour | Examples                                                |
|------------------|--------|---------------------------------------------------------|
| Electrical power | Red    | 28V bus, 5V regulated, switched lines                   |
| Data             | Blue   | I2C, SPI, UART, CAN, RS-422                             |
| RF               | Green  | Coaxial cable, waveguide                                |
| Mechanical       | Orange | Mounting bolts, thermal straps, alignment pins          |
| Thermal          | Yellow | Conductive paths, radiative coupling                    |
| Software         | Purple | Command interfaces, telemetry packets, mode transitions |

### Interface Conflict Detection

An **interface conflict** exists when: - Subsystem A expects to send 28V but Subsystem B only accepts 5V - Subsystem A uses I2C but Subsystem B only has SPI - Subsystem A is in S-band but Subsystem B antenna is X-band - Subsystem A mounts on the +Z face but Structure has no mounting provision on +Z

SpaceCDF detects many of these automatically through the **constraint engine** and displays conflicts as warning badges on the Dashboard.

### Worked Example: Detecting a Conflict

**Problem:** The OBC sends commands to the reaction wheels via I2C. The selected reaction wheel unit (RWP100 from Blue Canyon) uses RS-422. This is a data protocol mismatch.

**Resolution options:** 1. Select a different reaction wheel that supports I2C 2. Add an I2C-to-RS-422 bridge (additional component, mass, cost, failure point) 3. Select a different OBC that supports RS-422

**In SpaceCDF:** This conflict would appear as a warning in the Equipment Browser when the reaction wheel is selected, because the system checks data protocol compatibility.



## 4. FMEA/FMECA

[Source: ECSS-Q-ST-30-02C (Failure Mode, Effects, and Criticality Analysis); ECSS-Q-ST-30C (Dependability)]

[URL: <https://ecss.nl/standard/ecss-q-st-30-02c-failure-mode-effects-and-criticality-analysis-fmea-6-march-2009/>]

### Definitions

- **FMEA** (Failure Mode and Effects Analysis): Identifies failure modes, their causes, and effects on the system
- **FMECA** (Failure Mode, Effects, and Criticality Analysis): FMEA + criticality ranking

### FMEA/FMECA Table Structure

| Item           | Function                  | Failure Mode       | Cause                | Local Effect             | System Effect              | Severity | Detection          | Compensation/Provision                        |
|----------------|---------------------------|--------------------|----------------------|--------------------------|----------------------------|----------|--------------------|-----------------------------------------------|
| OBC            | Process commands, run FSW | Processor lockup   | SEU, firmware bug    | No command processing    | Loss of mission control    | 5        | HK timeout         | Watchdog reset; redundancy OBC (if available) |
| Battery        | Store energy              | Cell short         | Manufacturing defect | Reduced capacity         | Shortened eclipse survival | 3        | Voltage monitoring | Cell balancing; margin capacity               |
| Reaction Wheel | Provide torque            | Bearing seizure    | Lubrication failure  | Loss of one axis control | Degraded pointing          | 4        | Current anomaly    | 4-wheel config (if redundancy)                |
| Antenna        | Radiate RF                | Deployment failure | Mechanism jam        | No antenna deployed      | No communication           | 5        | Beacon absence     | Redundant deployment; burn with spring        |

### Criticality Categories

| Category                | Definition                                          | Action Required                                         |
|-------------------------|-----------------------------------------------------|---------------------------------------------------------|
| <b>1 (Catastrophic)</b> | Single failure causes mission loss; no compensation | Redesign to add redundancy or accept with justification |
| <b>2 (Critical)</b>     | Single failure causes significant degradation       | Mitigate (redundancy, operational workaround)           |
| <b>3 (Major)</b>        | Single failure causes moderate degradation          | Monitor; plan operational workaround                    |
| <b>4 (Minor)</b>        | Single failure has negligible mission impact        | Accept                                                  |

### Single-Point Failure (SPF) Analysis

A **single-point failure** is any single component whose failure alone causes loss of mission. Identifying SPFs is a critical output of the FMECA.

#### Typical CubeSat Single-Point Failures:

| Component                      | Why it is an SPF                                 | Common Mitigation                                     |
|--------------------------------|--------------------------------------------------|-------------------------------------------------------|
| OBC (single processor)         | No command processing -> no mission              | Watchdog timer + autonomous safe mode + EDAC memory   |
| Battery (single pack)          | No stored energy -> no eclipse survival          | Cell-level monitoring; conservative DoD limit (< 20%) |
| Antenna (non-redundant deploy) | No RF link -> no commanding/telemetry            | Redundant deployment mechanisms (burn wire + spring)  |
| Solar array (deployment)       | No power generation -> mission loss within hours | Redundant deployment; hinge spring + motor backup     |
| EPS main board                 | No power distribution                            | Typically no mitigation (accepted SPF in CubeSats)    |

## Key Equations: Reliability

### Series Reliability (all components must work):

$R_{\text{series}} = \text{Product of } R_i \text{ for } i = 1 \text{ to } n$

For  $n$  components each with reliability  $R_i$ :  $R_{\text{series}} = R_1 \times R_2 \times \dots \times R_n$

Example: 5 components each with  $R = 0.99$ :  $R_{\text{series}} = 0.99^5 = 0.951$

### Parallel Reliability (at least one must work -- redundancy):

$R_{\text{parallel}} = 1 - \text{Product of } (1 - R_i) \text{ for } i = 1 \text{ to } n$

For  $n$  identical redundant units each with  $R$ :  $R_{\text{parallel}} = 1 - (1 - R)^n$

Example: 2 redundant deployment mechanisms each with  $R = 0.95$ :  $R_{\text{parallel}} = 1 - (1 - 0.95)^2 = 1 - (0.05)^2 = 1 - 0.0025 = 0.9975$

### Mean Time Between Failures (MTBF):

$\text{MTBF} = \text{Total\_operating\_hours} / \text{Number\_of\_failures}$

For a component with failure rate  $\lambda$  (failures/hour):  $\text{MTBF} = 1 / \lambda$

**Reliability over time (exponential model):**  $R(t) = e^{-(t / \text{MTBF})} = e^{-(\lambda * t)}$

Example:  $\text{MTBF} = 50,000$  hours, mission duration = 8,760 hours (1 year):  $R = e^{-(8760/50000)} = e^{-0.1752} = 0.839$

### System Availability:

$A = \text{MTBF} / (\text{MTBF} + \text{MTTR})$

Where  $\text{MTTR}$  = Mean Time To Repair (or recover, for spacecraft)

For a spacecraft with  $\text{MTBF} = 50,000$  hr and  $\text{MTTR} = 2$  hr (safe mode recovery):  $A = 50000 / (50000 + 2) = 0.99996$

## Worked Example: Reaction Wheel Redundancy

**Problem:** A fine-pointing mission requires 3-axis attitude control. Each reaction wheel has  $R = 0.98$  over the 2-year mission.

---

*Configuration A -- 3 wheels (no redundancy):  $R_{\text{system}} = R^3 = 0.98^3 = 0.941$*

*Configuration B -- 4 wheels in 3-of-4 redundancy (1 spare):  $R_{\text{system}} = 4 \times R^4 - 3 \times R^3$  ... Using the binomial:  $P(\geq 3 \text{ working}) = C(4,4) \times R^4 + C(4,3) \times R^3 \times (1-R)^1 = 0.98^4 + 4 \times 0.98^3 \times 0.02 = 0.9224 + 4 \times 0.9412 \times 0.02 = 0.9224 + 0.0753 = 0.9977$*

*Adding one spare wheel improves reliability from 0.941 to 0.998 -- a significant improvement for modest mass and cost increase.*

### Real Mission Example: Hitomi (ASTRO-H)

JAXA's Hitomi X-ray observatory was lost on 26 March 2016, just 37 days after launch, due to a cascading failure in the AOCS subsystem: 1. An incorrect parameter in the star tracker caused an erroneous attitude estimate 2. The reaction wheels applied incorrect torques based on the bad estimate 3. Thrusters fired to "correct" the (phantom) attitude error, causing rapid spin 4. The satellite spun up beyond structural limits, and the extensible optical bench broke off

**Root cause:** Software parameter error (unit conversion) + inadequate FDIR logic + no cross-check between attitude sensors.

*Lesson: FMECA must consider cascading failures and common-cause failures, not just single-component failures. Software errors are failure modes too.*

*[Source: JAXA Hitomi Investigation Report, 2016, available at [https://global.jaxa.jp/press/2016/05/20160531\\_hitomi.html](https://global.jaxa.jp/press/2016/05/20160531_hitomi.html)]*

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## 5. Risk & Interface Exercise

### Instructions

1. **Dashboard** -- Check the reliability score and conflict count
2. **Risk Register Construction** (Worksheet 4.3): - Identify **5 technical risks** for your mission design - Score each on the 5x5 matrix (L x C) - Define mitigation strategy for any scoring  $\geq 10$  - Assign an owner (which CDF position is responsible)
3. **N<sup>2</sup> Matrix** -- Build a simplified N<sup>2</sup> matrix for your mission: - List 6 subsystems on the diagonal - Fill in the interfaces (power, data, RF, mechanical) - Identify at least 2 interface conflicts
4. **SPF Analysis** -- Identify all single-point failures in your design: - For each SPF: Is it acceptable? If not, what redundancy is needed? - Compute the reliability improvement from adding redundancy to one SPF

### Discussion Prompts

- "What is the highest-risk item in your design? What would it cost to mitigate?"
- "Do you have any interface conflicts that cannot be resolved without changing a component selection?"
- "Which single-point failure are you most concerned about? Would the customer accept it?"

### Worksheet 4.3 Tasks

1. Complete the risk register (5 risks minimum, scored)
2. Build a 6x6 N<sup>2</sup> interface matrix with colour-coded interface types
3. Complete the FMECA table for 4 critical components

4. List all single-point failures with accept/mitigate decision
5. Calculate series reliability for your mission's critical chain

## Cited References

| # | Reference                               | URL                                                                                                                                                                                                                                 |
|---|-----------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1 | ECSS-M-ST-80C (Risk Management)         | <a href="https://ecss.nl/standard/ecss-m-st-80c-risk-management-31-july-2008/">https://ecss.nl/standard/ecss-m-st-80c-risk-management-31-july-2008/</a>                                                                             |
| 2 | ECSS-Q-ST-30-02C (FMEA/FMECA)           | <a href="https://ecss.nl/standard/ecss-q-st-30-02c-failure-mode-effects-and-criticality-analysis-fmea-6-march-2009/">https://ecss.nl/standard/ecss-q-st-30-02c-failure-mode-effects-and-criticality-analysis-fmea-6-march-2009/</a> |
| 3 | ECSS-E-ST-10-24C (Interface Management) | <a href="https://ecss.nl/standard/ecss-e-st-10-24c-interface-management/">https://ecss.nl/standard/ecss-e-st-10-24c-interface-management/</a>                                                                                       |
| 4 | NASA SEH Rev 2, section 6.4             | <a href="https://www.nasa.gov/reference/systems-engineering-handbook/">https://www.nasa.gov/reference/systems-engineering-handbook/</a>                                                                                             |
| 5 | JAXA Hitomi Investigation Report        | <a href="https://global.jaxa.jp/press/2016/05/20160531_hitomi.html">https://global.jaxa.jp/press/2016/05/20160531_hitomi.html</a>                                                                                                   |
| 6 | ECSS-Q-ST-30C (Dependability)           | <a href="https://ecss.nl/standard/ecss-q-st-30c-dependability/">https://ecss.nl/standard/ecss-q-st-30c-dependability/</a>                                                                                                           |

## Session Summary

| Topic               | Key Takeaway                                                                                                                      |
|---------------------|-----------------------------------------------------------------------------------------------------------------------------------|
| Risk process        | Identify -> Assess -> Mitigate -> Monitor (continuous throughout lifecycle)                                                       |
| 5x5 matrix          | $L \times C = \text{Risk Score}$ ; Low (1-4), Medium (5-9), High (10-15), Critical (16-25)                                        |
| Strategies          | Accept (low), Mitigate (medium-high), Transfer (financial), Avoid (critical)                                                      |
| $N^2$ matrix        | $N \times N$ grid mapping all subsystem interfaces; colour-code by type                                                           |
| Interface conflicts | Protocol mismatch, voltage mismatch, band mismatch -- detect early                                                                |
| FMECA               | Failure mode -> cause -> local effect -> system effect -> severity -> detection -> mitigation                                     |
| SPF                 | Single-point failures must be identified, assessed, and accepted or mitigated                                                     |
| Reliability         | $R_{\text{series}} = \text{Product}(R_i)$ ; $R_{\text{parallel}} = 1 - \text{Product}(1 - R_i)$ ; $R(t) = e^{(-\lambda \cdot t)}$ |

## Session 4.4: Cost Estimation & Design Review

**Expected reading before this session.** NPR 7120.5F — WBS ( $\approx 30$  min); SMAD4 Ch. 20 (cost).

**Prerequisites:** Sessions 4.1-4.3 (equipment selected, V&V planned, risks assessed) **References:** SMAD4 Ch. 20, NASA Cost Estimating Handbook (CEH) v4.0, Aerospace Corp SSCM, NPR 7120.5F (WBS), ECSS-M-ST-60C (Cost Management), NPR 7123.1D Appendix G (Reviews)

### Learning Objectives

By the end of this session, participants will be able to: 1. Estimate mission cost using parametric (CER), analogy, and bottom-up methods 2. Structure a Work Breakdown Structure (WBS) for a CubeSat project 3. Apply learning curve effects for constellation cost estimation 4. Assess cost risk using confidence levels (P50/P70/P80) 5. Prepare for and conduct a design review (SRR/PDR/CDR) with gate criteria 6. Present design evidence clearly and respond to review board questions

### 1. Cost Estimation Methodologies

[Source: NASA CEH v4.0 Appendix C; SMAD4 Chapter 20; Aerospace Corp SSCM] [URL: <https://www.nasa.gov/offices/ocfo/references-and-tools/> (NASA CEH)]

#### Three Primary Methods

| Method            | Description                                                   | Accuracy (1-sigma) | When to Use                             | Data Required                    |
|-------------------|---------------------------------------------------------------|--------------------|-----------------------------------------|----------------------------------|
| <b>Parametric</b> | Cost Estimating Relationships (CERs) from historical database | +/- 30-50%         | Phase 0/A (few details known)           | Mass, power, mission type        |
| <b>Analogy</b>    | Compare to similar past mission, adjust for differences       | +/- 20-40%         | Phase A/B (reference mission available) | Detailed knowledge of reference  |
| <b>Bottom-up</b>  | Sum actual vendor quotes + labour estimates + facilities      | +/- 10-20%         | Phase B/C (detailed design)             | BOM, vendor quotes, labour rates |

#### Parametric Cost Estimating Relationships (CERs)

CERs express cost as a function of technical parameters, derived from regression analysis of historical data.

##### General CER Form:

$$\text{Cost} = a \times (\text{Parameter})^b \times \text{Complexity\_factor}$$

Where: - a = coefficient (from regression) - Parameter = usually dry mass (kg), power (W), or data rate  
- b = exponent (typically 0.5-1.0) - Complexity\_factor = adjustment for mission difficulty

## USCM (Unmanned Space Vehicle Cost Model) CERs

[Source: SMAD4 Table 20-8; US Air Force USCM database]

| Subsystem           | CER (FY2010 \$K)                      | Parameter           | Typical b |
|---------------------|---------------------------------------|---------------------|-----------|
| Structure           | $157 \times M_{\text{struct}}^{0.83}$ | Dry mass (kg)       | 0.83      |
| Thermal             | $394 \times M_{\text{therm}}^{0.635}$ | Dry mass (kg)       | 0.635     |
| EPS                 | $62.7 \times M_{\text{EPS}}^{1.00}$   | Dry mass (kg)       | 1.00      |
| TTC                 | $545 \times M_{\text{TTC}}^{0.761}$   | Dry mass (kg)       | 0.761     |
| AOCS                | $464 \times M_{\text{AOCS}}^{0.867}$  | Dry mass (kg)       | 0.867     |
| Propulsion          | $17.8 \times M_{\text{prop}}^{0.75}$  | Dry mass (kg)       | 0.75      |
| Integration & Test  | $10.4 \times M_{\text{dry}}^{0.907}$  | Total dry mass (kg) | 0.907     |
| Program Management  | 12.3% of hardware cost                | N/A                 | N/A       |
| Systems Engineering | 14.2% of hardware cost                | N/A                 | N/A       |

## SSCM (Small Satellite Cost Model)

The Aerospace Corporation SSCM was specifically calibrated for satellites < 500 kg. It provides more accurate estimates for CubeSats than USCM.

[Source: Aerospace Corp, "Small Satellite Cost Model (SSCM)", available through Aerospace TOR]

**Key SSCM adjustments for CubeSats:** - COTS hardware costs are catalogue prices, not mass-based CERs - Labour costs dominate for university/small-team projects - NRE is heavily dependent on mission uniqueness - Standard CERs (designed for > 100 kg) over-predict by 2-6x for nano/micro class

## CubeSat-Calibrated Pricing (SpaceCDF)

SpaceCDF uses a hybrid approach: COTS flat pricing for nano/micro class, CERs for larger spacecraft.

| Subsystem    | CubeSat COTS Cost (kEUR) | USCM CER (kEUR, for CubeSat mass) | Ratio                           |
|--------------|--------------------------|-----------------------------------|---------------------------------|
| EPS          | 15                       | 60                                | CER 4x higher                   |
| AOCS (fine)  | 40                       | 66                                | CER 1.6x higher                 |
| TTC (S-band) | 20                       | 25                                | Similar                         |
| OBC          | 10                       | 12                                | Similar                         |
| Structure    | 8                        | 3                                 | CER lower (min order effect)    |
| Payload      | 50 (variable)            | 300                               | CER 6x higher for COTS payloads |

This confirms that for CubeSats, COTS pricing is more reliable than parametric CERs for hardware cost.

## Worked Example: Parametric vs Bottom-Up

3U Earth observation mission:

| WBS Element              | Parametric (kEUR) | Bottom-Up (kEUR) | Notes               |
|--------------------------|-------------------|------------------|---------------------|
| Structure                | 8                 | 8.5              | ISIS 3U structure   |
| EPS                      | 18                | 16.2             | GomSpace P31u + SA  |
| OBC                      | 12                | 10.0             | NanoAvionics SatBus |
| AOCS                     | 45                | 42.0             | BCT XACT-15         |
| TTC                      | 22                | 20.5             | Endurosat S-band    |
| Thermal                  | 5                 | 3.0              | Passive only        |
| Payload                  | 65                | 58.0             | Simera Sense xScale |
| Harness                  | 3                 | 2.5              | Custom cables       |
| <b>Hardware subtotal</b> | <b>178</b>        | <b>160.7</b>     |                     |
| I&T (12%)                | 21                | 19.3             |                     |
| Software (8%)            | 14                | 12.9             |                     |
| PM/SE/MA (13%)           | 23                | 20.9             |                     |
| Launch                   | 200               | 195.0            | SpaceX Transporter  |
| Ground (5%)              | 9                 | 8.0              | SatNOGS + dedicated |
| Operations (3 yr)        | 45                | 40.0             | 0.5 FTE             |
| <b>TOTAL</b>             | <b>490</b>        | <b>456.8</b>     |                     |

The parametric estimate is ~7% higher than bottom-up. This is expected: parametric includes inherent uncertainty and contingency.

## 2. Cost Breakdown Structure (WBS)

The Work Breakdown Structure (WBS) is the hierarchical decomposition of all work required to complete the mission. It is the foundation for cost estimation, scheduling, and management.

[Source: NPR 7120.5F Appendix G; ECSS-M-ST-60C; NASA CEH v4.0 Appendix B]

### Standard CubeSat WBS

|                            |                                        |     |                                      |
|----------------------------|----------------------------------------|-----|--------------------------------------|
| WBS Level 1: Mission Total |                                        |     |                                      |
| 1.0                        | Programme Management                   | 5%  | (oversight, reviews, reporting)      |
| 2.0                        | Systems Engineering                    | 5%  | (budgets, interfaces, trade studies) |
| 3.0                        | Mission Assurance                      | 3%  | (quality, reliability, parts)        |
| 4.0                        | Payload                                | 20% | (instrument + calibration)           |
| 4.1                        | Payload instrument hardware            |     |                                      |
| 4.2                        | Payload software                       |     |                                      |
| 4.3                        | Payload calibration & characterisation |     |                                      |
| 5.0                        | Spacecraft Bus Hardware                | 30% | (all subsystems)                     |
| 5.1                        | Structure & mechanisms                 |     |                                      |
| 5.2                        | EPS (solar array + battery + board)    |     |                                      |

|      |                               |     |                                   |
|------|-------------------------------|-----|-----------------------------------|
| 5.3  | AOCS (sensors + actuators)    |     |                                   |
| 5.4  | TTC (transponder + antenna)   |     |                                   |
| 5.5  | OBC & data handling           |     |                                   |
| 5.6  | Thermal control               |     |                                   |
| 5.7  | Propulsion (if applicable)    |     |                                   |
| 5.8  | Harness & cabling             |     |                                   |
| 6.0  | Integration & Test            | 12% | (assembly, env. testing)          |
| 6.1  | Assembly & integration        |     |                                   |
| 6.2  | Environmental test campaign   |     |                                   |
| 6.3  | Test facilities rental        |     |                                   |
| 7.0  | Software (Flight + Ground)    | 8%  | (FSW, GSW, mission planning)      |
| 7.1  | Flight software               |     |                                   |
| 7.2  | Ground segment software       |     |                                   |
| 7.3  | Mission planning tools        |     |                                   |
| 8.0  | Launch Services               | 10% | (vehicle, deployer, integration)  |
| 9.0  | Ground Segment                | 5%  | (antennas, MCS, networks)         |
| 10.0 | Operations (mission lifetime) | 5%  | (staff, consumables, maintenance) |

### Cost by Phase

The distribution of cost across lifecycle phases is important for budgeting:

| Phase                  | % of Total | Activities                                     | Peak Staffing |
|------------------------|------------|------------------------------------------------|---------------|
| O/A (Concept)          | 5-10%      | Studies, trade-offs, requirements              | Low           |
| B (Preliminary Design) | 15-20%     | PDR, detailed analysis, long-lead procurement  | Growing       |
| C (Detailed Design)    | 30-40%     | CDR, manufacturing, software development       | Peak          |
| D (Integration & Test) | 20-25%     | Assembly, environmental testing, commissioning | High          |
| E (Operations)         | 10-20%     | Routine operations, anomaly management         | Low-steady    |
| F (Disposal)           | 1-3%       | Decommissioning, deorbit                       | Minimal       |

## 3. Learning Curve for Constellations

[Source: SMAD4 section 20.3; Wright's Learning Curve Theory (1936)]

When building multiple identical units, the cost per unit decreases due to manufacturing efficiency, reduced test time, bulk purchasing, and labour learning.

### Wright's Learning Curve

**Nth Unit Cost:**

$$C_N = C_1 \times N^b$$

Where  $b = \ln(\text{learning\_rate}) / \ln(2)$

| Learning Rate | b      | Cost of 10th Unit (% of 1st) |
|---------------|--------|------------------------------|
| 95%           | -0.074 | 77%                          |



| Learning Rate | b      | Cost of 10th Unit (% of 1st) |
|---------------|--------|------------------------------|
| 90%           | -0.152 | 60%                          |
| 85%           | -0.234 | 47%                          |

**Total Cost for N Units (cumulative):**

$C_{\text{total}} = C_1 \times \text{Sum from } i=1 \text{ to } N \text{ of } (i^b)$

Or approximately:  $C_{\text{total}} \sim C_1 \times N^{(1+b)} / (1+b)$  (continuous approximation)

### Simplified Rule of Thumb

At a **90% learning rate**, every time you **double** the number of units, the unit cost drops by **10%**.

Unit 1: EUR 800K Unit 2: EUR 720K (800 x 0.90) Unit 4: EUR 648K (720 x 0.90) Unit 8: EUR 583K (648 x 0.90) Unit 16: EUR 525K (583 x 0.90)

### Worked Example: 20-Satellite Constellation

First unit cost (bus + payload + I&T): EUR 800K. Learning rate: 90%.

| Units | b      | Avg Unit Cost | Total Hardware | Calc                               |
|-------|--------|---------------|----------------|------------------------------------|
| 1     | -0.152 | EUR 800K      | EUR 800K       | First unit                         |
| 5     | -0.152 | EUR 659K      | EUR 3,295K     | $5 \times 800 \times 5^{(-0.152)}$ |
| 10    | -0.152 | EUR 577K      | EUR 5,770K     | Cumulative sum                     |
| 20    | -0.152 | EUR 505K      | EUR 10,100K    | Cumulative sum                     |

Total constellation estimate: - 20 satellites hardware: EUR 10.1M - 2 spare units (10%): EUR 1.0M - Launch (20 sats x EUR 200K rideshare): EUR 4.0M - Ground segment: EUR 1.0M - Operations (3 years): EUR 0.9M - PM/SE/MA (10%): EUR 1.7M - **Total: ~EUR 18.7M**

## 4. Cost Risk and Confidence Levels

Point estimates are misleading. Every cost estimate has uncertainty. The standard practice is to express cost as a probability distribution.

[Source: NASA CEH v4.0 section 2.3; JPL parametric estimation practice]

### Uncertainty by Cost Element

| Cost Element    | Distribution | Uncertainty (1-sigma) | Rationale                               |
|-----------------|--------------|-----------------------|-----------------------------------------|
| COTS hardware   | Normal       | +/- 10%               | Known pricing from vendor quotes        |
| Custom hardware | Lognormal    | +/- 30%               | Development uncertainty skews high      |
| Software        | Triangular   | +/- 40%               | Hardest to estimate; frequent overruns  |
| Launch          | Normal       | +/- 15%               | Published pricing; contract negotiation |

| Cost Element | Distribution | Uncertainty (1-sigma) | Rationale                                 |
|--------------|--------------|-----------------------|-------------------------------------------|
| Operations   | Uniform      | +/- 25%               | Staffing level uncertainty                |
| I&T          | Lognormal    | +/- 25%               | Test anomalies cause schedule/cost growth |

## Confidence Levels

| Percentile | Meaning                                        | Use                                          |
|------------|------------------------------------------------|----------------------------------------------|
| P50        | 50% probability of being at or below this cost | Project baseline; "expected" cost            |
| P70        | 70% probability                                | NASA standard commitment level (NPR 7120.5F) |
| P80        | 80% probability                                | Conservative planning; typical for proposals |

### Rule of Thumb for CubeSat Missions:

$P70 \sim P50 \times 1.2$   $P80 \sim P50 \times 1.3$

Example: If  $P50 = \text{EUR } 500\text{K}$ , then  $P80 \sim \text{EUR } 650\text{K}$ .

*This approximation assumes moderate complexity and well-understood COTS hardware. For missions with custom payloads or new technology, use  $P80 \sim P50 \times 1.5$ .*

## 1U Worked Example: UniSat-1

### Cost Breakdown: Simple WBS, Mostly COTS

UniSat-1's cost structure is fundamentally different from larger missions because (a) nearly all hardware is COTS, so NRE is near zero, and (b) the team is small and university-based, so labour costs are low.

### UniSat-1 WBS Cost Estimate (Parametric vs Bottom-Up):

| WBS Element              | Parametric (kEUR) | Bottom-Up (kEUR) | Notes                                  |
|--------------------------|-------------------|------------------|----------------------------------------|
| 1.0 Programme Management | 4                 | 5                | Faculty oversight, 0.1 FTE x 12 months |
| 2.0 Systems Engineering  | 3                 | 3                | Student team lead                      |
| 3.0 Mission Assurance    | 1                 | 1                | Minimal QA for university mission      |
| 4.0 Payload              | 8                 | 8                | MEMS sensor PCB + calibration          |
| 5.0 Bus Hardware         | 38                | 36               | See BOM from Session 4.1               |
| 5.1 Structure            | 4                 | 4                | ISIS 1U frame                          |
| 5.2 EPS + SA             | 20                | 19.5             | P31us + body-mounted cells             |
| 5.3 AOCS (passive)       | 1                 | 1                | Magnet + hysteresis rods               |
| 5.4 Comms (UHF)          | 11                | 10.5             | AX100 + antenna                        |

| WBS Element                | Parametric (kEUR) | Bottom-Up (kEUR) | Notes                                      |
|----------------------------|-------------------|------------------|--------------------------------------------|
| 5.5 OBC                    | 3                 | 3                | Custom Cortex-M board                      |
| 6.0 I&T                    | 8                 | 8                | University clean room + vibe test facility |
| 7.0 Software               | 5                 | 5                | FSW (FreeRTOS) + GSW                       |
| 8.0 Launch                 | 15                | 15               | NanoRacks 1U ISS deployment                |
| 9.0 Ground Segment         | 5                 | 5                | Yagi antenna + SatNOGS network             |
| 10.0 Operations (6 months) | 3                 | 3                | Student operators, 0.2 FTE                 |
| <b>TOTAL (P50)</b>         | <b>~90</b>        | <b>~85</b>       |                                            |
| <b>P80 (x 1.3)</b>         | <b>~117</b>       | <b>~111</b>      | Conservative estimate                      |

#### Key cost observations for 1U missions:

1. **Hardware is cheap:** Total COTS hardware cost is ~36--44 kEUR. This is less than a single star tracker for a 3U mission.
2. **NRE is minimal:** Only the OBC and payload require custom development. NRE is estimated at ~8 kEUR (payload calibration + OBC board layout), compared to ~50--150 kEUR for custom payloads on larger missions.
3. **Launch cost is proportionally large:** At 15 kEUR, the launch represents ~18% of total cost. For a 3U mission at ~200 kEUR launch cost, launch is ~40% of total. The 1U launch cost is low in absolute terms but still a significant fraction.
4. **Labour dominates:** For a university team, the "free" student labour is the hidden cost. If students were costed at professional rates (~50 EUR/hr), the total labour cost would be ~100--200 kEUR, far exceeding the hardware cost. This is typical for educational missions.
5. **No cost drivers from complexity:** There is no AOCS software development, no deployable mechanism qualification, no propulsion system integration, no thermal vacuum testing of heaters -- all of which add 10--50 kEUR each on a 3U mission.

**Learning curve applicability:** If a university builds a series of 1U demonstrators (UniSat-1, UniSat-2, UniSat-3...), the 90% learning rate applies:

| Unit     | Hardware Cost (kEUR) | Total Cost (kEUR) |
|----------|----------------------|-------------------|
| UniSat-1 | 44                   | 85                |
| UniSat-2 | 40                   | 77                |
| UniSat-4 | 36                   | 69                |
| UniSat-8 | 32                   | 62                |

The asymptotic floor is dominated by launch cost (15 kEUR) and irreducible ground segment + operations costs (~8 kEUR), giving a minimum mission cost of ~40--50 kEUR for repeat builds.

## 5. Design Review Process

Design reviews are formal decision gates where the project demonstrates readiness to proceed to the next lifecycle phase.

[Source: NPR 7123.1D Appendix G; ECSS-M-ST-10C Rev.1 section 6; NASA SEH Rev 2 section 3.7] [URL: <https://www.nasa.gov/reference/systems-engineering-handbook/>]

### Review Sequence

| Review     | Phase Transition | Key Question                                           | Entry Criteria                                                      |
|------------|------------------|--------------------------------------------------------|---------------------------------------------------------------------|
| <b>MCR</b> | Pre-A -> A       | Is the mission need justified?                         | Problem statement, stakeholders, objectives defined                 |
| <b>SRR</b> | A -> B           | Are requirements complete, consistent, traceable?      | Requirements baselined, ConOps defined, feasibility shown           |
| <b>PDR</b> | B -> C           | Does preliminary design meet requirements with margin? | All budgets close, interfaces defined, risks identified             |
| <b>CDR</b> | C -> D           | Is detailed design complete and ready to build?        | All drawings released, test plan approved, suppliers under contract |
| <b>TRR</b> | Pre-test         | Is the system ready for environmental testing?         | Assembly complete, procedures approved, facility booked             |
| <b>QR</b>  | Post-test        | Has the system passed all tests?                       | All test reports approved, NCRs closed, V&V matrix complete         |
| <b>FRR</b> | Pre-launch       | Is everything ready for launch?                        | Shipping approved, launch manifest confirmed, operations ready      |

### Gate Criteria for SRR (Phase A -> B)

| # | Criterion                                    | Priority    | Evidence                              |
|---|----------------------------------------------|-------------|---------------------------------------|
| 1 | All Level 0/1 requirements baselined         | Must pass   | Requirements document signed          |
| 2 | ConOps defined (all mission phases)          | Must pass   | ConOps document                       |
| 3 | Mission architecture trades completed        | Must pass   | Trade study reports with rationale    |
| 4 | Feasibility confirmed (all budgets positive) | Must pass   | Mass, power, link budgets with margin |
| 5 | Risk register established with mitigations   | Must pass   | Risk register with scores and plans   |
| 6 | Preliminary V&V approach defined             | Should pass | V&V matrix with methods assigned      |
| 7 | Schedule and cost estimate (P50/P80)         | Must pass   | Cost estimate with WBS                |
| 8 | Interface requirements identified            | Should pass | N^2 matrix or ICD outline             |

## Gate Criteria for PDR (Phase B -> C)

| #  | Criterion                                            | Priority    | Evidence                                  |
|----|------------------------------------------------------|-------------|-------------------------------------------|
| 1  | All requirements allocated to subsystems             | Must pass   | Requirements allocation matrix            |
| 2  | Preliminary design complete for all subsystems       | Must pass   | Design documents, block diagrams          |
| 3  | All budgets close with $\geq 20\%$ margin            | Must pass   | Mass, power, link, data, pointing budgets |
| 4  | Equipment selected (BOM)                             | Must pass   | BOM with TRL, heritage, cost, lead time   |
| 5  | All interfaces defined ( $N^2$ matrix)               | Must pass   | Interface control documents               |
| 6  | V&V matrix complete with methods assigned            | Must pass   | V&V matrix                                |
| 7  | Risk register updated; no Critical risks unmitigated | Must pass   | Risk register                             |
| 8  | Test plan outline approved                           | Should pass | Environmental test plan                   |
| 9  | Software architecture defined                        | Should pass | Software design document                  |
| 10 | Cost estimate updated (bottom-up)                    | Must pass   | Cost estimate with vendor quotes          |

### Presentation Skills for Reviews

**Do:** - Lead with the conclusion ("Mass budget closes with 22% margin") - Show evidence, not just assertions ("Link budget analysis shows 4.2 dB margin at worst case") - Acknowledge risks honestly ("Antenna deployment is our highest risk at  $L3 \times C4 = 12$ ") - Answer questions directly; say "I don't know, we'll take an action" if needed

**Do not:** - Read slides aloud - Hide problems (review boards always find them) - Present analysis without assumptions stated - Skip backup slides -- have detailed data ready

## 6. Cost & Review Exercise

### Instructions

#### Part A: Cost Estimation

1. **Dashboard** -- Check the Cost KPI card (total cost in MEUR)
2. **Cost Breakdown** tab -- Review the breakdown by subsystem
3. **Exports** tab -- Generate BOM and sum COTS component costs
4. On Worksheet 4.4: - Fill in the WBS cost table using both parametric and bottom-up methods - Compute P50 and P80 estimates - If constellation: apply 90% learning curve

#### Part B: Design Review Preparation

1. Open the **Gate Review** tab in SpaceCDF
2. Check all criteria: which are Pass/Fail/Manual?
3. For any failing criteria, click "**Go fix**" and resolve

- 
4. Prepare a 3-minute summary of your design for peer review
- 

## Worksheet 4.4 Tasks

1. Build a complete WBS cost table (parametric and bottom-up columns)
  2. Compute P50 and P80 total cost estimates
  3. If constellation: compute total cost with learning curve applied
  4. Identify top 3 cost drivers and propose 20% cost reduction
  5. List the gate criteria for SRR/PDR and assess your readiness
- 

## Cited References

| # | Reference                             | URL                                                                                                                                                       |
|---|---------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1 | NASA Cost Estimating Handbook v4.0    | <a href="https://www.nasa.gov/offices/ocfo/references-and-tools/">https://www.nasa.gov/offices/ocfo/references-and-tools/</a>                             |
| 2 | SMAD4 Chapter 20 (Cost)               | Wertz, Everett, Puschell (eds.), Space Mission Engineering, Microcosm 2011                                                                                |
| 3 | Aerospace Corp SSCM                   | <a href="https://www.aerospace.org/capabilities/small-satellite-cost-model">https://www.aerospace.org/capabilities/small-satellite-cost-model</a>         |
| 4 | NPR 7120.5F (WBS)                     | <a href="https://nodis3.gsfc.nasa.gov/displayDir.cfm?t=NPR&amp;c=7120&amp;s=5F">https://nodis3.gsfc.nasa.gov/displayDir.cfm?t=NPR&amp;c=7120&amp;s=5F</a> |
| 5 | ECSS-M-ST-60C (Cost Management)       | <a href="https://ecss.nl/standard/ecss-m-st-60c-cost-and-schedule-management/">https://ecss.nl/standard/ecss-m-st-60c-cost-and-schedule-management/</a>   |
| 6 | NPR 7123.1D (SE Processes) Appendix G | <a href="https://nodis3.gsfc.nasa.gov/displayDir.cfm?t=NPR&amp;c=7123&amp;s=1D">https://nodis3.gsfc.nasa.gov/displayDir.cfm?t=NPR&amp;c=7123&amp;s=1D</a> |
| 7 | NASA SEH Rev 2, section 3.7           | <a href="https://www.nasa.gov/reference/systems-engineering-handbook/">https://www.nasa.gov/reference/systems-engineering-handbook/</a>                   |

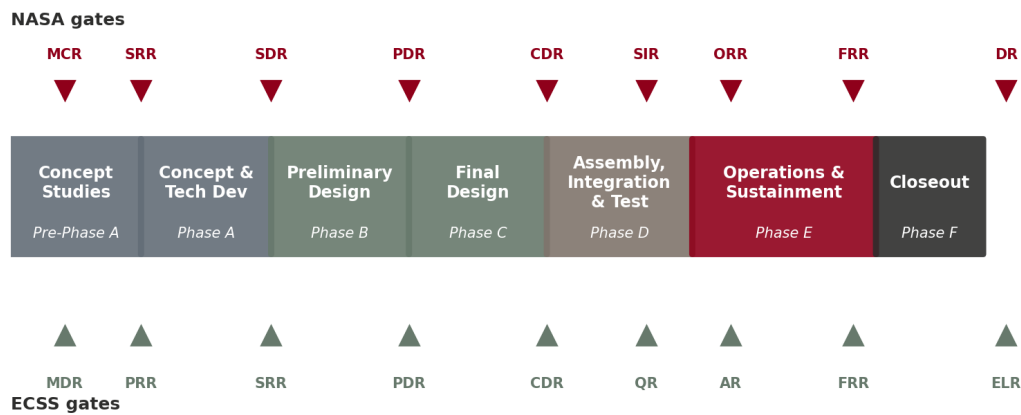
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## Session Summary

| Topic          | Key Takeaway                                                                          |
|----------------|---------------------------------------------------------------------------------------|
| Cost methods   | Parametric (early, CERs), Analogy (reference mission), Bottom-up (vendor quotes)      |
| CubeSat costs  | COTS pricing often lower than CER predictions; use hybrid approach                    |
| WBS            | Standard 10-element structure: PM, SE, MA, Payload, Bus, I&T, SW, Launch, Ground, Ops |
| CERs           | $\text{Cost} = a \times M^b$ ; USCM/SSCM databases; CubeSats need calibrated CERs     |
| Learning curve | 90% rate: each doubling of quantity reduces unit cost by 10%                          |
| Cost risk      | P50 (baseline), P70 (NASA commitment), P80 (conservative); $P80 \sim P50 \times 1.3$  |
| Reviews        | SRR, PDR, CDR: formal gates with exit criteria; must pass before proceeding           |
| Presentation   | Lead with conclusions, show evidence, acknowledge risks, answer directly              |

## Session 5.1: Ground Segment & Operations Architecture

### Project Lifecycle Phases & Review Gates — NASA vs ECSS



SpaceCDF · uOttawa SEDTI

Gate sequence at a glance.

**Expected reading before this session.** ECSS-M-ST-10C §6 — Review gates (~ 60 min).

**Prerequisites:** Week 2 complete (full design cycle through cost estimation) **References:** ECSS-E-ST-70C (Ground Systems), ECSS-E-ST-70-01C (Ground Segment), CCSDS 131.0-B-4 (TM Coding), CCSDS 231.0-B-4 (TC Coding), CCSDS 133.0-B-2 (Space Packet Protocol), NASA DSN Handbook (810-005)

### Learning Objectives

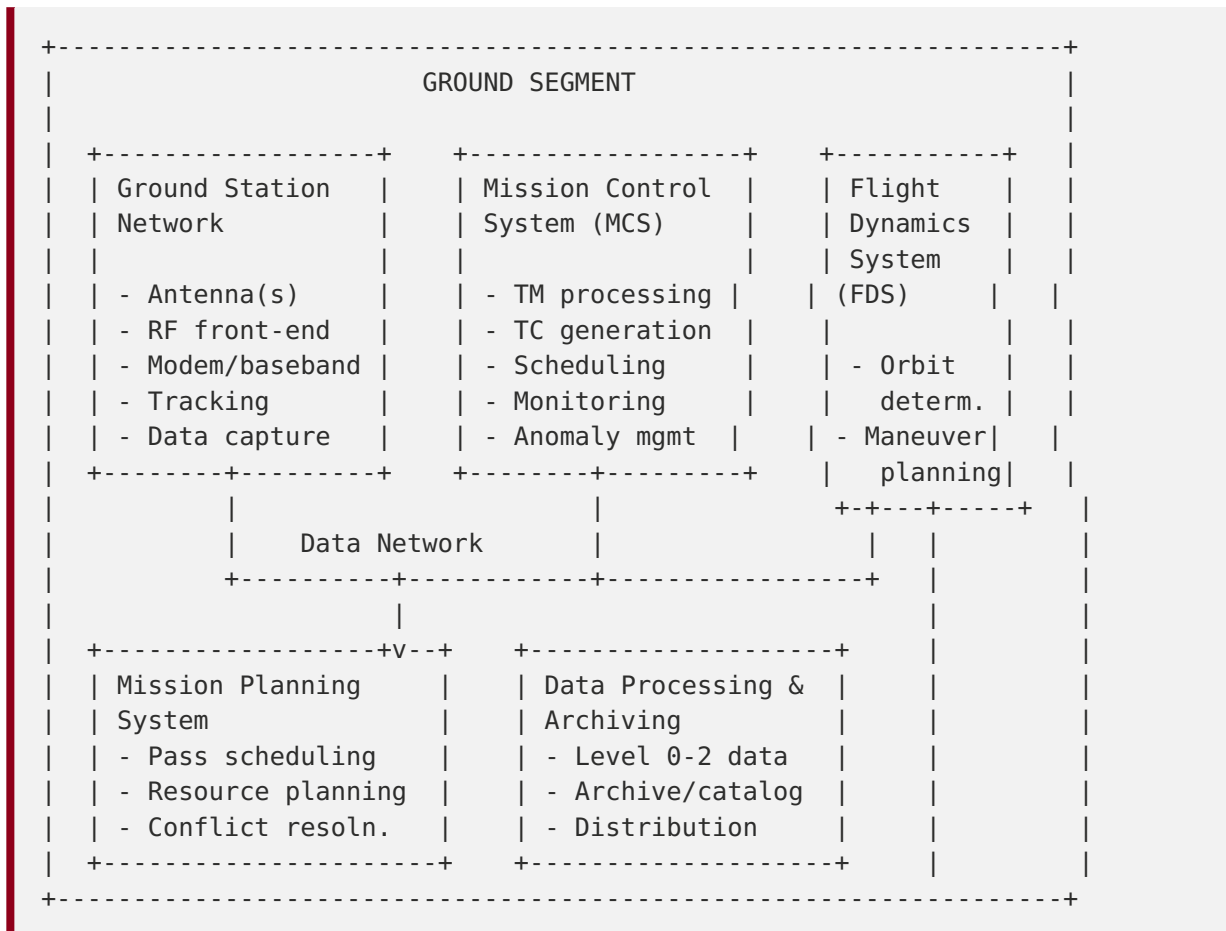
By the end of this session, participants will be able to: 1. Describe the components of a ground segment architecture (antennas, MCS, FDS, networks) 2. Calculate ground station contact time and data volume per pass 3. Explain the CCSDS protocol stack for space communication 4. Design a ground station network for a given orbit and data downlink requirement 5. Construct a mission operations timeline with key milestones

### 1. Ground Segment Architecture

The ground segment is everything on the ground that supports the space mission. It is frequently underestimated in early mission design but accounts for 5-15% of total mission cost and is critical to mission success.

[Source: ECSS-E-ST-70C (Ground Systems and Operations); ECSS-E-ST-70-01C (Ground Segment)] [URL: <https://ecss.nl/standard/ecss-e-st-70c-ground-systems-and-operations/>]

## Ground Segment Components



## Ground Station Components

| Component       | Function                                             | Typical Specification                        |
|-----------------|------------------------------------------------------|----------------------------------------------|
| Antenna         | Transmit uplink commands, receive downlink telemetry | 3-13 m dish (S/X-band); Yagi/turnstile (UHF) |
| RF front-end    | Low-noise amplifier (LNA), power amplifier, filters  | LNA NF < 1 dB; PA 50-500 W                   |
| Modem/baseband  | Modulation/demodulation, coding/decoding             | CCSDS-compliant; BPSK/QPSK/8PSK              |
| Tracking system | Antenna pointing, Doppler tracking, ranging          | Autotrack or program-track                   |
| Data capture    | Record raw baseband data for post-processing         | High-speed disk array                        |
| Timing          | GPS-disciplined oscillator for time synchronisation  | < 1 microsecond accuracy                     |

## Mission Control System (MCS)

| Function             | Description | Tools |
|----------------------|-------------|-------|
| Telemetry processing |             |       |



| Function               | Description                                           | Tools                        |
|------------------------|-------------------------------------------------------|------------------------------|
|                        | Decode, decommutate, display, and archive TM packets  | COSMOS, YAMCS, EGOS (ESA)    |
| Telecommand generation | Create, validate, encode, and uplink TC packets       | Same MCS tools               |
| Pass scheduling        | Schedule antenna time, plan contact windows           | STK, GMAT, custom schedulers |
| Monitoring & control   | Real-time health monitoring, limit checking, alarming | MCS dashboards               |
| Anomaly management     | Detect, diagnose, and resolve anomalies               | Procedures + expert judgment |

### Flight Dynamics System (FDS)

| Function               | Description                                        | Inputs                                    |
|------------------------|----------------------------------------------------|-------------------------------------------|
| Orbit determination    | Compute current orbit from tracking data           | Range, Doppler, GPS (if onboard)          |
| Orbit prediction       | Propagate orbit forward for pass planning          | Current state vector, perturbation models |
| Manoeuvre planning     | Compute delta-V for orbit maintenance              | Target orbit, propulsion model            |
| Conjunction assessment | Predict close approaches with debris/other objects | TLE catalogue, own orbit                  |
| End-of-life planning   | Compute deorbit manoeuvre or passivation plan      | Remaining propellant, target orbit        |

## 2. Ground Station Contact Analysis

Understanding how much data can be transferred per pass is fundamental to mission design. The data budget depends on the contact geometry, data rate, and pass duration.

### Contact Geometry

For a circular LEO orbit, the visibility of a ground station is determined by the minimum elevation angle constraint.

#### Maximum Slant Range at Minimum Elevation:

$$\rho = R_E \times (\sqrt{(h/R_E + 1)^2 - \cos^2(\epsilon)} - \sin(\epsilon))$$

Where: -  $R_E$  = Earth radius (6371 km) -  $h$  = orbital altitude -  $\epsilon$  = minimum elevation angle (typically 5-10 degrees)

For  $h = 500$  km,  $\epsilon = 5$  degrees:  $\rho = 6371 \times (\sqrt{(500/6371 + 1)^2 - \cos^2(5)} - \sin(5))$   
 $\rho = 6371 \times (\sqrt{1.0785^2 - 0.9962^2} - 0.0872)$   
 $\rho = 6371 \times (\sqrt{1.1632 - 0.9924} - 0.0872)$   
 $\rho = 6371 \times (\sqrt{0.1708} - 0.0872)$   
 $\rho = 6371 \times (0.4133 - 0.0872)$   
 $\rho = 6371 \times 0.3261$   
 $\rho = 2077$  km

#### Maximum Pass Duration (overhead pass):

$$T_{\max} = 2 \times R_E \times \arccos(R_E \times \cos(\epsilon) / (R_E + h)) / V_{\text{ground\_track}}$$

Simplified approximation for LEO:  $T_{\max} \sim (2 \times \rho_{\max}) / V_{\text{orbital}} \times (R_E / (R_E + h))$

For 500 km SSO:  $V_{\text{orbital}} = \sqrt{\mu / (R_E + h)} = \sqrt{3.986 \times 10^{14} / 6.871 \times 10^6} = 7613 \text{ m/s}$   
 $T_{\max} \sim 2 \times 2077 \times 10^3 / 7613 \times (6371/6871) \sim 505 \text{ s} \sim 8.4 \text{ minutes}$

*Typical average pass duration (not all passes are overhead): ~6 minutes*

#### Data Volume per Pass:

$$V_{\text{data}} = R_{\text{data}} \times T_{\text{contact}} \times \eta_{\text{protocol}}$$

Where: -  $R_{\text{data}}$  = downlink data rate (bps) -  $T_{\text{contact}}$  = contact duration (seconds) -  $\eta_{\text{protocol}}$  = protocol efficiency (typically 0.85-0.95 for CCSDS)

Example:  $R = 2 \text{ Mbps}$ ,  $T = 360 \text{ s}$  (6 min average),  $\eta = 0.9$ :  $V_{\text{data}} = 2 \times 10^6 \times 360 \times 0.9 = 648 \text{ Mbit} = 81 \text{ MB per pass}$

### Contact Frequency

| Orbit              | GS Latitude              | Passes per Day | Avg Duration | Daily Data Volume (2 Mbps) |
|--------------------|--------------------------|----------------|--------------|----------------------------|
| 500 km SSO         | 52 N (e.g., Waterloo)    | 3-5            | 6 min        | 243-405 MB                 |
| 500 km SSO         | 69 N (e.g., Kiruna)      | 6-8            | 7 min        | 544-726 MB                 |
| 500 km, 51.6 (ISS) | 52 N                     | 2-4            | 5 min        | 130-260 MB                 |
| 500 km SSO         | SatNOGS network (global) | 10-15          | 5 min        | 648-972 MB                 |

[Source: STK simulations; validated against SatNOGS observation logs]

### Worked Example: Data Budget Closure

**Problem:** An Earth observation mission generates 500 MB/day of imagery. Is a single ground station at 52 N latitude sufficient with 2 Mbps S-band downlink?

**Calculation:** - Passes per day: ~4 (500 km SSO from 52 N) - Avg pass duration: 6 min = 360 s - Data per pass:  $2 \times 10^6 \text{ bps} \times 360 \text{ s} \times 0.9 / 8 = 81 \text{ MB}$  - Daily capacity:  $4 \times 81 = 324 \text{ MB}$  - Required: 500 MB/day - **Shortfall: 176 MB/day -- budget does not close!**

**Solutions:** 1. Add a second ground station at high latitude (e.g., Kiruna -> +6 passes -> +486 MB) 2. Increase data rate to X-band (10 Mbps -> 405 MB/pass -> easily sufficient) 3. Reduce data generation (lower duty cycle or reduce image resolution) 4. Add onboard data compression (2:1 lossless -> 250 MB/day required) 5. Use SatNOGS network for additional UHF passes (housekeeping only)

## 3. CCSDS Protocol Stack

The Consultative Committee for Space Data Systems (CCSDS) defines the standard protocols for space communication, analogous to TCP/IP for ground networks.

[Source: CCSDS 131.0-B-4 (TM Synchronization and Channel Coding); CCSDS 231.0-B-4 (TC Synchronization and Channel Coding); CCSDS 133.0-B-2 (Space Packet Protocol)] [URL: <https://public.ccsds.org/Pubs/131x0b4.pdf> (freely available)]

## Protocol Layers

| Layer              | CCSDS Standard                              | Function                              | Ground Analogy        |
|--------------------|---------------------------------------------|---------------------------------------|-----------------------|
| <b>Application</b> | CCSDS 133.0 (Space Packet)                  | Mission data and housekeeping packets | HTTP/application data |
| <b>Network</b>     | CCSDS 732.0 (AOS) or 132.0 (TM)             | Virtual channels, multiplexing        | IP routing            |
| <b>Data Link</b>   | CCSDS 131.0 (TM coding) / 231.0 (TC coding) | FEC, frame sync, CRC                  | Ethernet              |
| <b>Physical</b>    | CCSDS 401.0 (RF & Modulation)               | Modulation, frequency, power          | Physical layer        |

## Telemetry (TM) Frame Structure

```

+-----+-----+-----+
| Header | Data Field | EC |
| 6 bytes| 1-1019 bytes | 2B |
+-----+-----+-----+
|
| +-- Spacecraft ID (10 bits)
| +-- Virtual Channel ID (3 bits)
| +-- Frame Counter (8 bits)
| +-- Frame Length

```

## Forward Error Correction (FEC) Options

| Code                    | Rate | Coding Gain (dB) | Use Case                            |
|-------------------------|------|------------------|-------------------------------------|
| Convolutional (7, 1/2)  | 1/2  | 5.5              | Legacy CubeSats                     |
| Reed-Solomon (255, 223) | 0.87 | 3.5              | Combined with convolutional         |
| Turbo (rate 1/2)        | 1/2  | 7.5              | High-performance CubeSats           |
| LDPC (rate 7/8)         | 7/8  | 6.0              | High data rate, bandwidth efficient |

Modern CubeSats typically use LDPC coding for downlink (high data rate) and convolutional + RS for uplink (robustness priority).

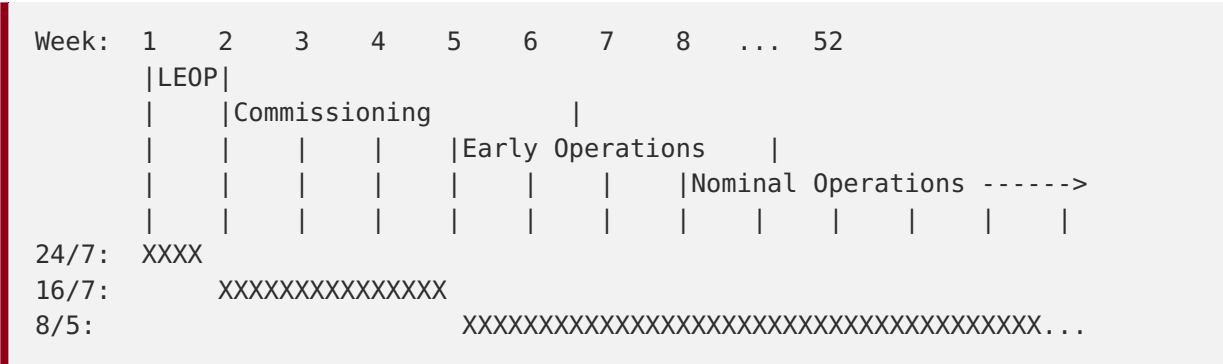
## 4. Operations Timeline Construction

The operations timeline (or operations concept timeline) defines all major activities from launch to end of life. It is part of the ConOps and drives ground segment design.

Mission Phases (Operations Perspective)

| Phase               | Duration         | Key Activities                                                          | Staffing         |
|---------------------|------------------|-------------------------------------------------------------------------|------------------|
| LEOP                | 0-72 hours       | Separation, beacon acquisition, SA/antenna deploy, initial health check | 24/7 (3 shifts)  |
| Commissioning       | 1-4 weeks        | Subsystem checkout, ADCS calibration, payload first light               | 16/7 (2 shifts)  |
| Early Operations    | 1-3 months       | Performance characterisation, procedure refinement                      | 8/5 (1 shift)    |
| Nominal Operations  | Mission lifetime | Routine science/service, orbit maintenance                              | 8/5 or automated |
| Extended Operations | If approved      | Degraded mode operations, reduced data                                  | Part-time        |
| End of Life         | 1-4 weeks        | Passivation, deorbit manoeuvre, final telemetry                         | 8/5              |

Operations Timeline (Gantt-Style)



LEOP Timeline Detail

| Time (after separation) | Activity                                    | Success Criterion                        |
|-------------------------|---------------------------------------------|------------------------------------------|
| T+0 s                   | Separation from deployer                    | Deployment switches released             |
| T+0 to T+30 min         | Deployment timer countdown (no RF)          | Timer runs; all inhibits clear           |
| T+30 min                | Antenna deployment                          | Beacon detected by ground station        |
| T+30 to T+60 min        | Beacon acquisition                          | Carrier lock; beacon decoded             |
| T+1 hr                  | First telemetry downlink                    | Housekeeping data received and validated |
| T+1 to T+6 hr           | Initial health assessment                   | All subsystems reporting nominal HK      |
| T+6 to T+12 hr          | Solar array deployment (if separate)        | Power generation confirmed               |
| T+12 to T+24 hr         | ADCS initialisation                         | Attitude determination active            |
| T+24 to T+48 hr         | ADCS calibration (magnetometer, sun sensor) | Pointing within coarse spec              |

| Time (after separation) | Activity                                     | Success Criterion                       |
|-------------------------|----------------------------------------------|-----------------------------------------|
| T+48 to T+72 hr         | Communication chain validation (full duplex) | Uplink and downlink at operational rate |

### SpaceCDF Operations Planning

SpaceCDF's **ConOps Editor** allows you to: - Define mission phases with start/end conditions - Specify operational modes per phase (safe, nominal, science, downlink) - Set ground contact requirements per phase (frequency, duration) - The system validates that the ground segment design supports the operations concept

## 5. Ground Segment Design Exercise

### Instructions

1. **Data Budget** tab -- Review the daily data generation vs downlink capacity - Is the data budget closing? If not, identify the bottleneck - How many ground station passes per day does your orbit provide?
2. **ConOps Editor** -- Define mission phases: - LEOP (duration, staffing, contact requirements) - Commissioning (activities, success criteria) - Nominal operations (duty cycle, data volume, contact frequency)
3. On Worksheet 5.1: - Calculate data volume per pass for your mission - Determine number of ground stations needed to close the data budget - Construct a LEOP timeline - Draft a simplified operations timeline (Gantt chart)

### Discussion Prompts

- "What happens if your first pass after deployment has no signal?"
- "How would you handle a safe mode entry at 3 AM on a Saturday?"
- "What is the minimum ground segment that could support your mission?"

### Worksheet 5.1 Tasks

1. Calculate ground station contact time and data volume per pass
2. Design a ground station network (locations, antenna sizes, data rates)
3. Construct a LEOP timeline (first 72 hours, hourly resolution)
4. Draft a mission operations timeline (Gantt chart, weekly resolution)
5. Estimate ground segment cost (antennas + MCS + FDS + operations staff)

## Cited References

| # | Reference                                     | URL                                                                                                                                                       |
|---|-----------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1 | ECSS-E-ST-70C (Ground Systems and Operations) | <a href="https://ecss.nl/standard/ecss-e-st-70c-ground-systems-and-operations/">https://ecss.nl/standard/ecss-e-st-70c-ground-systems-and-operations/</a> |

| # | Reference                                               | URL                                                                                                           |
|---|---------------------------------------------------------|---------------------------------------------------------------------------------------------------------------|
| 2 | CCSDS 131.0-B-4 (TM Synchronization and Channel Coding) | <a href="https://public.ccsds.org/Pubs/131x0b4.pdf">https://public.ccsds.org/Pubs/131x0b4.pdf</a>             |
| 3 | CCSDS 133.0-B-2 (Space Packet Protocol)                 | <a href="https://public.ccsds.org/Pubs/133x0b2.pdf">https://public.ccsds.org/Pubs/133x0b2.pdf</a>             |
| 4 | CCSDS 231.0-B-4 (TC Synchronization and Channel Coding) | <a href="https://public.ccsds.org/Pubs/231x0b4.pdf">https://public.ccsds.org/Pubs/231x0b4.pdf</a>             |
| 5 | SatNOGS Network                                         | <a href="https://network.satnogs.org/">https://network.satnogs.org/</a>                                       |
| 6 | NASA DSN Handbook (810-005)                             | <a href="https://deepspace.jpl.nasa.gov/dsndocs/810-005/">https://deepspace.jpl.nasa.gov/dsndocs/810-005/</a> |
| 7 | YAMCS Mission Control Software                          | <a href="https://yamcs.org/">https://yamcs.org/</a>                                                           |

## Session Summary

| Topic               | Key Takeaway                                                                     |
|---------------------|----------------------------------------------------------------------------------|
| Ground segment      | Antenna network + MCS + FDS + mission planning + data processing                 |
| Contact analysis    | Pass duration ~6 min (LEO/SSO); data volume = rate x time x efficiency           |
| Data budget         | Must close: daily generation <= daily downlink capacity; add GS or increase rate |
| CCSDS               | Standard protocol stack: Space Packet, AOS/TM frames, FEC coding, RF modulation  |
| Operations timeline | LEOP (24/7) -> Commissioning (16/7) -> Nominal (8/5 or automated)                |
| LEOP                | First 72 hours critical: antenna deploy, beacon acquisition, health check        |
| SpaceCDF            | ConOps Editor defines phases; Data Budget validates ground segment sizing        |

## Session 5.2: Mission Operations Concepts

**Expected reading before this session.** ITU Radio Regulations Article 21; ISED CPC-2-6-02; RSSSA — <https://laws-lois.justice.gc.ca/eng/acts/R-5.4/>.

**Prerequisites:** Session 5.1 (ground segment architecture understood) **References:** ECSS-E-ST-70-11C (Space Segment Operability), ECSS-E-ST-70-32C (Procedures), ECSS-E-ST-70-41C (Packet Utilisation Standard), NPR 7120.5F, NASA Fault Management Handbook (NASA-HDBK-1002)

### Learning Objectives

By the end of this session, participants will be able to: 1. Define a Concept of Operations (ConOps) with operational modes and transitions 2. Design an FDIR (Fault Detection, Isolation, Recovery) architecture 3. Write operational procedures for routine and contingency operations 4. Describe anomaly response process with escalation levels 5. Plan spacecraft operational modes and their entry/exit criteria

### 1. Concept of Operations (ConOps)

The ConOps is the bridge between engineering design and operational reality. It describes HOW the system will be used, not just WHAT it can do.

[Source: ECSS-E-ST-70-11C (Space Segment Operability); IEEE 1362 (Concept of Operations Document)]

#### ConOps Structure

A complete ConOps document covers:

1. **Mission overview** -- Objectives, stakeholders, success criteria
2. **System description** -- Space segment, ground segment, user segment
3. **Operational scenarios** -- Nominal operations, contingency operations
4. **Operational modes** -- Definition, transitions, constraints
5. **Resource management** -- Power, data, propellant budgets over time
6. **Communication plan** -- Contact schedule, data flow, latency requirements
7. **Staffing plan** -- Personnel, shifts, training requirements
8. **Maintenance and logistics** -- Software updates, ground equipment maintenance

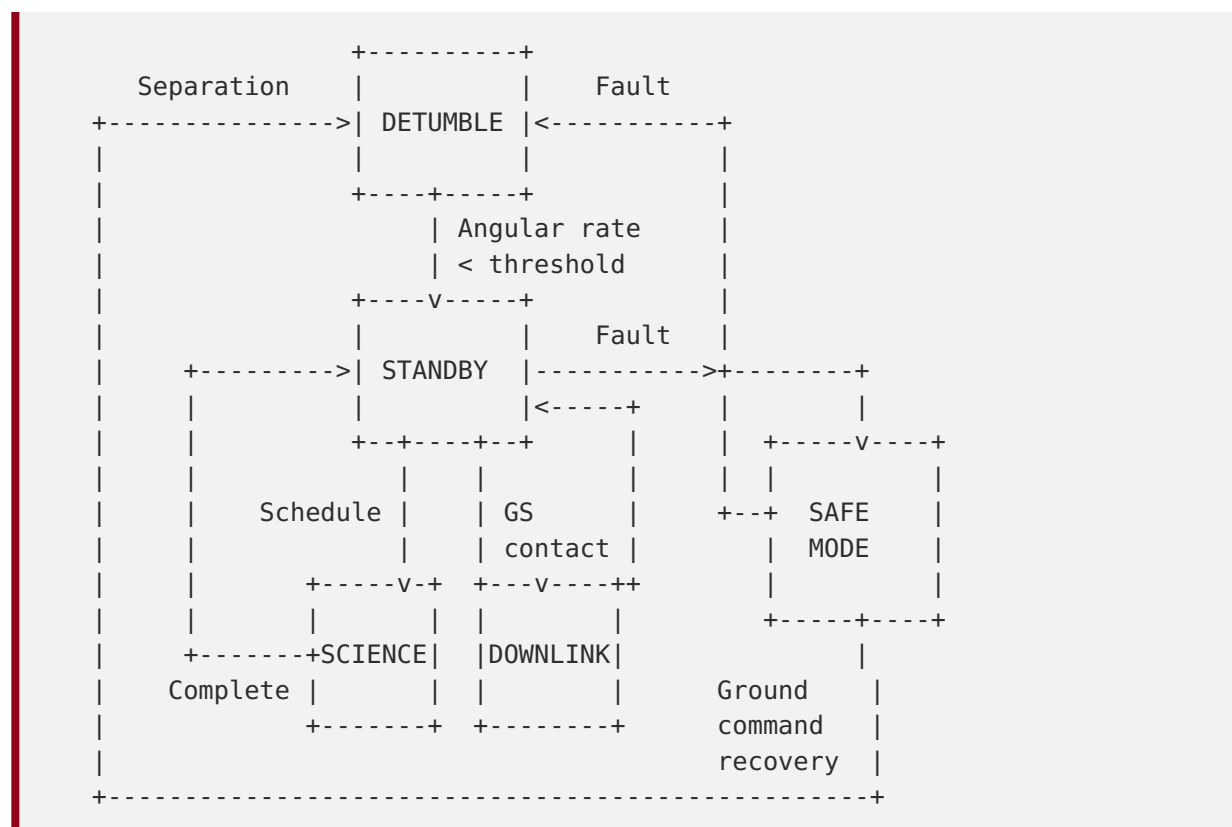
#### Operational Modes

Every spacecraft has a defined set of operational modes. Each mode specifies which subsystems are active, power consumption, and data rates.

| Mode     | Description                          | Power (W) | Data Rate           | Typical Duration        | Entry Condition            |
|----------|--------------------------------------|-----------|---------------------|-------------------------|----------------------------|
| Safe     | Minimum functionality; survival only | 5-10      | Beacon only (1 bps) | Until ground intervenes | Autonomous fault detection |
| Detumble |                                      | 8-12      |                     |                         |                            |

| Mode             | Description                            | Power (W) | Data Rate               | Typical Duration   | Entry Condition                    |
|------------------|----------------------------------------|-----------|-------------------------|--------------------|------------------------------------|
|                  | Stop rotation after deployment         |           | HK only (100 bps)       | Minutes to hours   | Post-separation; high angular rate |
| <b>Standby</b>   | Sun-pointing; all subsystems ready     | 12-18     | HK telemetry (1 kbps)   | Between activities | After detumble; idle periods       |
| <b>Science</b>   | Payload operating; attitude controlled | 25-40     | Science data (variable) | Target-dependent   | Scheduled; attitude achieved       |
| <b>Downlink</b>  | High-rate data transfer to ground      | 20-30     | Full rate (2-10 Mbps)   | During GS contact  | GS in view; link established       |
| <b>Manoeuvre</b> | Orbit adjustment or desaturation       | 15-25     | HK only                 | Minutes            | Scheduled; constraints met         |
| <b>Eclipse</b>   | Reduced operations during eclipse      | 8-15      | HK only (100 bps)       | ~35 min (LEO)      | Sun not in view                    |

### Mode Transition State Machine



### Real Mission Example: PROBA-2 Operations

ESA's PROBA-2 (launched 2009, still operational in 2026) demonstrates mature CubeSat-class operations: - **5 operational modes:** Standby, Science, Manoeuvre, Safe, Off - **Autonomous operations:** 95% of routine activities automated via onboard timeline - **Ground contacts:** 4-6 passes/day via ESA Redu station (S-band) - **Anomaly rate:** ~2-3 per year requiring ground intervention (after initial commissioning) - **Key lesson:** Investing in onboard autonomy dramatically reduces operations cost



[Source: ESA PROBA-2 Operations Team, "PROBA-2: Over a Decade of Operations", SpaceOps 2020]

## 2. FDIR Architecture

FDIR (Fault Detection, Isolation, Recovery) is the spacecraft's autonomous ability to detect failures, determine which component failed, and take corrective action -- all without ground intervention.

[Source: NASA Fault Management Handbook (NASA-HDBK-1002); ECSS-E-ST-70-11C section 5.3] [URL: <https://standards.nasa.gov/standard/NASA/NASA-HDBK-1002>]

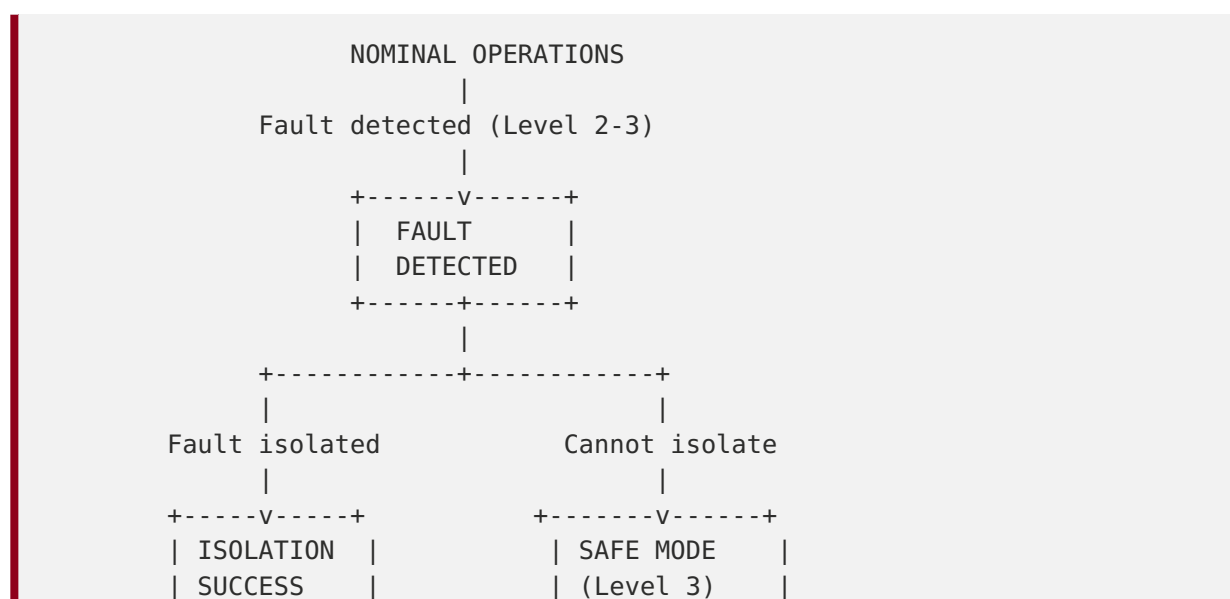
### Why FDIR is Critical for CubeSats

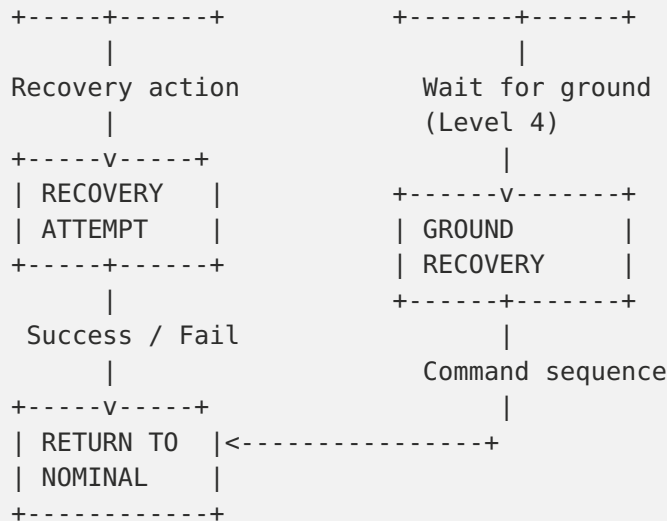
- **Limited ground contact:** LEO CubeSats are out of view ~85% of the time
- **No crewed intervention:** Unlike ISS, there is no astronaut to "fix" problems
- **Thermal constraints:** A safe mode must maintain thermal limits within minutes
- **Power constraints:** Battery can discharge completely in one eclipse without load shedding

### FDIR Hierarchy (Levels)

| Level          | Detection                      | Response                       | Latency         | Example                       |
|----------------|--------------------------------|--------------------------------|-----------------|-------------------------------|
| 0 -- Hardware  | Built-in watchdog              | Component reset                | Milliseconds    | Processor watchdog timer      |
| 1 -- Unit      | Unit-level monitoring          | Unit reconfiguration           | Seconds         | EPS over-current trip         |
| 2 -- Subsystem | Subsystem health check         | Subsystem fallback mode        | Seconds-minutes | Switch to backup star tracker |
| 3 -- System    | System-level anomaly detection | Mode change (e.g., enter Safe) | Minutes         | Transition to Safe Mode       |
| 4 -- Ground    | Human-in-the-loop diagnosis    | Ground command recovery        | Hours-days      | Anomaly investigation + patch |

### FDIR State Machine





### Common CubeSat FDIR Rules

| Fault                       | Detection Method               | Threshold                        | Response                         |
|-----------------------------|--------------------------------|----------------------------------|----------------------------------|
| Processor lockup            | Watchdog timer                 | No heartbeat for 60 s            | Hardware reset (Level 0)         |
| Battery under-voltage       | EPS voltage monitor            | V <sub>bat</sub> < 6.0V          | Load shedding; enter Safe Mode   |
| Battery over-temperature    | Temperature sensor             | T <sub>bat</sub> > 45 C          | Disable charging; reduce loads   |
| Attitude loss               | No attitude solution for 5 min | ADCS health flag timeout         | Enter Detumble Mode              |
| Communication loss          | No uplink for 48 hr            | Timer-based                      | Reset TTC; revert to beacon mode |
| Memory corruption           | EDAC error count               | > 10 uncorrectable errors/hr     | Memory scrub; reboot from backup |
| Solar array current anomaly | Current sensor                 | I <sub>SA</sub> < expected - 30% | Check attitude; enter Standby    |
| Reaction wheel over-speed   | Wheel speed monitor            | Omega > 6000 RPM                 | Reduce momentum; desaturation    |

### Key Design Principles

1. **Safe Mode must always work** -- It is the last resort; it must be tested exhaustively
2. **Fail operational, then fail safe** -- Try to continue the mission before giving up
3. **No single fault should cause mission loss** -- Cross-reference with FMECA
4. **Ground override capability** -- Every autonomous action must be commandable from ground
5. **Logging** -- Record all fault events in non-volatile memory for post-analysis

### 3. Operational Procedures

[Source: ECSS-E-ST-70-32C (Procedure Definition Language); ECSS-E-ST-70-01C section 5.6]

#### Procedure Types

| Type               | Purpose                                | Execution                         | Example                      |
|--------------------|----------------------------------------|-----------------------------------|------------------------------|
| <b>Nominal</b>     | Routine scheduled operations           | Automated (timeline) or manual    | Science observation sequence |
| <b>Contingency</b> | Response to known anomaly types        | Semi-automated; operator decision | Safe mode recovery           |
| <b>Emergency</b>   | Response to critical unexpected events | Manual; real-time decision        | Communication loss recovery  |
| <b>Maintenance</b> | Periodic calibration or updates        | Scheduled manual                  | Magnetometer calibration     |

#### Procedure Structure (ECSS-E-ST-70-32C)

Each procedure has: 1. **Identifier** -- Unique procedure number (e.g., NOM-001) 2. **Purpose** -- What the procedure accomplishes 3. **Preconditions** -- System state required before starting 4. **Steps** -- Ordered list of actions, verifications, decisions 5. **Expected results** -- What should happen at each step 6. **Recovery actions** -- What to do if expected results are not observed 7. **Post-conditions** -- System state after successful completion

#### Example: Science Observation Procedure

```

Procedure: NOM-SCI-001 "Earth Observation Target Acquisition"
Preconditions: Mode = Standby, Battery SoC > 60%, Target in FOV within 10 min

Step 1: Verify ADCS mode = Fine Pointing
 Expected: ADCS status = FINE, pointing error < 0.1 deg
 If not: Wait 60 s, retry. If still not achieved -> ABORT, remain in Standby

Step 2: Command payload power ON
 Expected: Payload HK reports power = ON within 5 s
 If not: Retry power command. If 3 fails -> ABORT, log anomaly

Step 3: Wait for payload thermal stabilisation (90 s)
 Expected: Payload temperature within operational range

Step 4: Command payload to imaging mode
 Expected: Payload status = IMAGING, frame counter incrementing

Step 5: Wait for target acquisition (ground-predicted start time)
 Expected: Imaging starts at predicted time +/- 5 s

Step 6: Collect imagery for scheduled duration

Step 7: Command payload to standby
 Expected: Payload status = STANDBY

```

Step 8: Command payload power OFF (if no more targets this orbit)  
Expected: Payload HK reports power = OFF

Post-conditions: Mode = Standby, imagery stored in mass memory

4. Anomaly Response Process

[Source: NASA-HDBK-1002 (Fault Management Handbook); ESA Anomaly Review Board procedures]

Anomaly Escalation Levels

| Level  | Severity      | Response Time     | Authority            | Example                                |
|--------|---------------|-------------------|----------------------|----------------------------------------|
| Green  | Informational | Next business day | Operations engineer  | Minor parameter out of expected range  |
| Yellow | Warning       | Within 4 hours    | Lead operator        | Subsystem performance degraded         |
| Orange | Serious       | Within 1 hour     | Flight director      | Mode change required; mission impact   |
| Red    | Critical      | Immediate         | Project manager + FD | Mission-threatening; safe mode entered |

Anomaly Response Workflow

1. DETECT: Telemetry alarm or operator observation
2. ASSESS: Is this a known anomaly type? Check procedures library
  - > Known: Execute contingency procedure
  - > Unknown: Continue to step 3
3. DIAGNOSE: Gather additional telemetry, correlate events, form hypothesis
4. PLAN: Develop response plan (may require Anomaly Review Board)
5. EXECUTE: Implement recovery commands (verify each step)
6. VERIFY: Confirm system returned to expected state
7. DOCUMENT: Record anomaly, root cause, response, and lessons learned

Anomaly Report Form Fields

| Field              | Description                            |
|--------------------|----------------------------------------|
| Anomaly ID         | Unique identifier (e.g., ANO-2026-042) |
| Date/Time (UTC)    | When the anomaly was detected          |
| Severity           | Green / Yellow / Orange / Red          |
| Affected subsystem | EPS, AOCS, TTC, Payload, etc.          |
| Observation        | What was observed (TM values, trends)  |
| Diagnosis          | Root cause analysis                    |
| Action taken       | Commands sent, procedures executed     |

| Field           | Description                           |
|-----------------|---------------------------------------|
| Result          | System state after response           |
| Status          | Open / Resolved / Under investigation |
| Lessons learned | Preventive measures for future        |

### Real Mission Example: Kepler Reaction Wheel Failure

NASA's Kepler space telescope lost reaction wheel #2 in July 2012, followed by wheel #4 in May 2013. The anomaly response:

1. **Detection:** Elevated friction torque in wheel #2 HK telemetry
2. **Assessment:** Known degradation signature; monitoring intensified
3. **Failure:** Wheel #2 ceased operation. Mission continued on 3 wheels.
4. **Second failure:** Wheel #4 failed 10 months later. Only 2 wheels remaining -- insufficient for fine pointing.
5. **Recovery:** Engineering team developed the "K2" mission concept using solar radiation pressure as a "virtual third wheel"
6. **Result:** K2 operated for 4+ additional years, discovering 2,700+ exoplanet candidates

*Lesson: Creative operational workarounds can save missions. FDIR should be designed with graceful degradation, not just "safe mode or nothing."*

*[Source: Howell et al., "The K2 Mission: Characterization and Early Results", PASP, 2014]*

## 5. Operations Concepts Exercise

### Instructions

1. **ConOps Editor** in SpaceCDF: - Define all operational modes for your mission (at least 5) - Set entry/exit conditions for each mode - Verify power budget closes in each mode (especially Safe and Eclipse)
2. **FDIR Design:** - Define 5 FDIR rules for your mission (fault, detection, threshold, response) - Verify that Safe Mode power budget is sustainable indefinitely - Check: Can every FDIR response be overridden from ground?
3. **Worksheet 5.2:** - Complete the operational modes table - Write one nominal procedure (science observation or downlink) - Write one contingency procedure (safe mode recovery) - Fill in the FDIR rules table

### Discussion Prompts

- "What is the worst anomaly that could happen to your satellite? How would you respond?"
- "If the satellite enters safe mode on Friday evening and the next ground contact is Monday, will it survive?"
- "How much autonomy should the spacecraft have? Where is the line between onboard and ground decision-making?"

## Worksheet 5.2 Tasks

1. Complete the operational modes table (5+ modes with power, data rate, entry/exit)
2. Design the FDIR state machine (at least 5 rules)
3. Write one nominal and one contingency procedure
4. Complete the anomaly response form for a hypothetical scenario
5. Estimate operations staffing requirements by mission phase

## Cited References

| # | Reference                                       | URL                                                                                                                                                                                           |
|---|-------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1 | ECSS-E-ST-70-11C (Space Segment Operability)    | <a href="https://ecss.nl/standard/ecss-e-st-70-11c-space-segment-operability/">https://ecss.nl/standard/ecss-e-st-70-11c-space-segment-operability/</a>                                       |
| 2 | ECSS-E-ST-70-32C (Procedures)                   | <a href="https://ecss.nl/standard/ecss-e-st-70-32c-test-and-operations-procedure-language/">https://ecss.nl/standard/ecss-e-st-70-32c-test-and-operations-procedure-language/</a>             |
| 3 | NASA Fault Management Handbook (NASA-HDBK-1002) | <a href="https://standards.nasa.gov/standard/NASA/NASA-HDBK-1002">https://standards.nasa.gov/standard/NASA/NASA-HDBK-1002</a>                                                                 |
| 4 | ESA PROBA-2 Operations, SpaceOps 2020           | <a href="https://www.spaceops.org/">https://www.spaceops.org/</a>                                                                                                                             |
| 5 | Howell et al., K2 Mission, PASP 2014            | <a href="https://doi.org/10.1086/676406">https://doi.org/10.1086/676406</a>                                                                                                                   |
| 6 | ECSS-E-ST-70-41C (Packet Utilisation Standard)  | <a href="https://ecss.nl/standard/ecss-e-st-70-41c-telemetry-and-telecommand-packet-utilization/">https://ecss.nl/standard/ecss-e-st-70-41c-telemetry-and-telecommand-packet-utilization/</a> |

## Session Summary

| Topic             | Key Takeaway                                                                          |
|-------------------|---------------------------------------------------------------------------------------|
| ConOps            | Defines HOW the system is operated: modes, transitions, schedules, staffing           |
| Operational modes | Safe, Detumble, Standby, Science, Downlink, Manoeuvre, Eclipse (minimum set)          |
| Mode transitions  | Defined by entry/exit conditions; Safe Mode always reachable from any mode            |
| FDIR              | 5 levels: Hardware -> Unit -> Subsystem -> System -> Ground; Safe Mode is last resort |
| Procedures        | Nominal (routine), Contingency (known faults), Emergency (unknown), Maintenance       |
| Anomaly response  | Detect -> Assess -> Diagnose -> Plan -> Execute -> Verify -> Document                 |
| Design principles | Safe Mode must always work; fail operational before fail safe; ground override always |

## Session 5.3: Mission Simulation Day

**Expected reading before this session.** Cal Poly CDS Rev 14 §5 (Launch ICD); SMAD4 Ch. 18.

**Prerequisites:** Sessions 5.1-5.2 (ground segment designed, operations concepts defined) **References:** ECSS-E-ST-70-11C (Operability), ECSS-E-ST-70-32C (Procedures), ESA OPS-G Training Manual, NASA Mission Operations Directorate Training Handbook

### Learning Objectives

By the end of this session, participants will be able to: 1. Execute a simulated LEOP sequence from separation through first telemetry 2. Perform commissioning activities using operational procedures 3. Respond to injected anomalies using the FDIR architecture and contingency procedures 4. Document anomaly response in real-time using structured forms 5. Conduct nominal science operations with live data budget tracking

### 1. Simulation Overview & Roles

This session is a **hands-on simulation** of mission operations. The facilitator acts as the "Universe" -- injecting events, anomalies, and time jumps. The team operates the mission using SpaceCDF and their procedures from Session 5.2.

#### Role Assignments

| Role                          | Responsibility                                   | SpaceCDF Feature Used            |
|-------------------------------|--------------------------------------------------|----------------------------------|
| Flight Director (FD)          | Overall decision authority; coordinates team     | Dashboard, all tabs              |
| Spacecraft Controller (SC)    | Sends commands; monitors housekeeping telemetry  | ConOps Editor, mode control      |
| Ground Station Operator (GSO) | Manages antenna scheduling; link acquisition     | Data Budget, Link Budget         |
| Payload Operator (PO)         | Plans and executes science observations          | Payload parameters, data budget  |
| Flight Dynamics (FD-nav)      | Monitors orbit; plans manoeuvres (if propulsion) | Orbit parameters, sustainability |
| Anomaly Logger                | Documents all events, anomalies, and decisions   | Worksheet 5.3                    |

*For smaller teams, combine roles: SC+GSO and PO+FD-nav are natural pairings.*

## Simulation Timeline

| Sim Time              | Real Time   | Phase              | Events                                               |
|-----------------------|-------------|--------------------|------------------------------------------------------|
| T+0 to T+1 hr         | 0-20 min    | LEOP               | Separation, beacon acquisition, deploy, health check |
| T+1 hr to T+24 hr     | 20-35 min   | Early LEOP         | SA deploy, ADCS init, comms chain validation         |
| T+1 day to T+7 days   | 35-50 min   | Commissioning      | Subsystem checkout, first payload activation         |
| T+7 days to T+30 days | 50-65 min   | Early Operations   | Nominal science, orbit maintenance                   |
| Anomaly injection     | 65-85 min   | Contingency        | Anomaly response exercise                            |
| T+30 days to T+1 year | 85-100 min  | Nominal Operations | Routine operations, data review                      |
| Debrief               | 100-120 min | Wrap-up            | Lessons learned, documentation review                |

## 2. LEOP Simulation

The facilitator narrates the simulation. Time is compressed. Each team executes their LEOP procedure.

### Facilitator Script -- LEOP

**[T+0 -- Separation]** *"Your satellite has separated from the deployer. Deployment switches have released. The 30-minute timer has started. You are in radio silence -- no RF emissions allowed until the timer expires. What is your spacecraft doing right now?"*

Worked response: Timer counting down. Battery providing power. No subsystems active except OBC running timer.

**[T+30 min -- Timer Expires]** *"The 30-minute timer has expired. Your OBC has commanded antenna deployment. What do you expect to see?"*

Expected: Antenna deployment command sent. Beacon should begin transmitting.

**[T+35 min -- First Pass Over Ground Station]** *"Your ground station at [location] has line of sight. The antenna operator reports: carrier detected at [frequency] with Doppler of -25 kHz. Signal strength is -110 dBm. Is this what you expected?"*

Teams should verify: - Carrier frequency matches expected (accounting for Doppler) - Signal strength is consistent with link budget at current range - Beacon data rate is as designed

**[T+40 min -- First Telemetry]** *"Beacon lock achieved. You are receiving housekeeping telemetry. I am going to read you the first HK frame:"*

| Parameter           | Value  | Expected Range | Status  |
|---------------------|--------|----------------|---------|
| Bus voltage         | 7.8 V  | 7.0 - 8.4 V    | Nominal |
| Battery SoC         | 72%    | > 50%          | Nominal |
| Solar array current | 0.45 A | 0.3 - 0.6 A    | Nominal |



| Parameter           | Value             | Expected Range | Status  |
|---------------------|-------------------|----------------|---------|
| OBC temperature     | 22 C              | -10 to +50 C   | Nominal |
| Battery temperature | 19 C              | 0 to 45 C      | Nominal |
| ADCS mode           | Detumble          | Expected       | Nominal |
| Angular rate        | 3.2 deg/s         | Decreasing     | Nominal |
| Antenna status      | Deployed          | Expected       | Nominal |
| Beacon mode         | Active            | Expected       | Nominal |
| Uplink status       | Not yet attempted | --             | --      |

*"All parameters nominal. Proceed with LEOP sequence."*

**[T+1 hr -- Uplink Validation]** *"You now attempt your first uplink command. Send a 'NOP' (no operation) command to verify the uplink chain."*

Teams should: compose command, verify encoding, transmit, wait for acknowledgement in telemetry.

### 3. Commissioning Simulation

#### Facilitator Script -- Commissioning

**[Time Jump: T+24 hr]** *"Your satellite has completed detumbling. ADCS is in sun-pointing mode. All HK parameters are nominal after 8 ground passes. You are ready to begin commissioning. What is your first activity?"*

Expected: ADCS calibration (magnetometer bias estimation, sun sensor alignment verification).

**[T+3 days -- Payload First Light]** *"ADCS is calibrated and achieving 0.3 degree pointing. You are ready for payload first light. Walk me through your procedure."*

Teams execute their science observation procedure from Worksheet 5.2.

**[T+5 days -- First Data Download]** *"Payload has captured 150 MB of imagery. Your next pass is 12 minutes long over [station]. Can you download all the data in one pass?"*

Teams should calculate: - Data rate x pass duration x protocol efficiency = available capacity - Compare to 150 MB -- determine if multiple passes needed - Plan the download sequence

### 4. Anomaly Injection Scenarios

The facilitator selects 2-3 anomalies from the list below, appropriate to the mission design. Teams respond using their FDIR rules and contingency procedures.

#### Anomaly Menu (Facilitator selects 2-3)

##### Anomaly A: Battery Under-Voltage

*"During eclipse, you receive an alert: battery voltage has dropped to 6.2V, below your 6.5V warning threshold. Battery temperature is 5 C. SoC is estimated at 18%."*

Expected response: 1. Verify FDIR has triggered load shedding (payload off, non-essential heaters off) 2. Check power budget: is consumption > generation? 3. Possible causes: SA degradation, unexpected load, battery degradation, long eclipse 4. Immediate action: Verify safe mode; disable non-essential loads manually 5. Investigation: Review power trends over last 24 hours; compare to power budget

### Anomaly B: Communication Loss

*"You have not received telemetry for the last 3 scheduled passes (18 hours). The ground station reports pointing and frequency are correct but no carrier detected."*

Expected response: 1. Verify ground station equipment (antenna, LNA, modem) -- rule out ground fault 2. Check with other ground stations (SatNOGS) for any signal detection 3. Consider causes: TX failure, antenna deployment failure, attitude loss (antenna not pointing) 4. If 48-hour timer exists: satellite may reset TTC and revert to beacon mode 5. Plan: Wait for timer-based reset; use wide-beam ground antenna for beacon search

### Anomaly C: ADCS Anomaly -- Loss of Fine Pointing

*"Pointing error has increased from 0.1 deg to 5 deg. Star tracker reports 'no solution'. Reaction wheels are at 80% capacity (near saturation)."*

Expected response: 1. Diagnose: Star tracker failure? Optical contamination? Sun in FOV? 2. Check magnetometer attitude -- is coarse attitude correct? 3. Immediate: Command desaturation using magnetorquers 4. If star tracker failed: switch to coarse pointing mode (sun sensors + magnetometer) 5. Impact assessment: Can mission science continue in coarse pointing mode?

### Anomaly D: Unexpected Safe Mode Entry

*"Your satellite has autonomously entered Safe Mode. Last telemetry before entry showed: OBC reboot counter incremented by 3 in 10 minutes. All other parameters were nominal."*

Expected response: 1. Investigate: Multiple reboots suggest SEU (Single Event Upset) or firmware bug 2. Check radiation environment: Was the satellite in SAA (South Atlantic Anomaly)? 3. Review FDIR logs (stored in non-volatile memory) for fault triggers 4. Recovery plan: Command exit from Safe Mode; monitor for recurrence 5. If recurring: Upload software patch to increase SEU robustness

### Anomaly E: Thermal Exceedance

*"Payload temperature has reached 58 C during a science observation, exceeding the 55 C operational limit. Battery temperature is 38 C (approaching 45 C limit)."*

Expected response: 1. Immediate: Disable payload to stop heat generation 2. Check attitude: Is the hot radiator face pointing toward the sun? 3. Review thermal model: Was this orbit geometry (high beta angle) predicted? 4. Mitigation: Adjust duty cycle; plan observations for cooler orbit phases 5. If structural: May need to update thermal model and operational constraints

## Anomaly Documentation

For each injected anomaly, teams complete the **Anomaly Response Form** on Worksheet 5.3:

| Field              | What to Record                           |
|--------------------|------------------------------------------|
| Time (sim)         | When the anomaly was detected            |
| Observation        | Exact telemetry values and symptoms      |
| Initial assessment | Severity level (Green/Yellow/Orange/Red) |

| Field              | What to Record                              |
|--------------------|---------------------------------------------|
| Diagnosis          | Suspected root cause                        |
| Action taken       | Commands sent, procedures executed          |
| Result             | System state after response                 |
| Follow-up required | Additional investigation, procedure updates |

## 5. Nominal Operations & Debrief

### Nominal Operations

**[Time Jump: T+30 days to T+1 year]** *"Your mission has been operational for one month. All commissioning activities are complete. You are now in nominal operations. Let's review your operational metrics."*

Teams report: - Total data downlinked in 30 days vs plan - Number of science observations completed vs plan - Any anomalies encountered (from injection exercise) - Current budget status: power margin, data margin, propellant remaining (if applicable) - Orbit status: any conjunction alerts? Debris compliance on track?

### Debrief

1. **What went well?** - Which procedures worked as written? - Which FDIR rules triggered correctly? - Where did the team communicate effectively?
2. **What could be improved?** - Which procedures needed real-time modification? - Were there gaps in the FDIR design? - Where did decision-making slow down?
3. **Lessons for the design:** - Did the simulation reveal any design weaknesses? - Should any requirements be changed based on operational experience? - What additional autonomy would help?

## Facilitator Notes

### Preparation

- Review each team's ConOps, FDIR rules, and procedures from Session 5.2
- Select anomalies appropriate to each team's mission (e.g., don't inject propulsion anomaly for a mission without propulsion)
- Prepare HK data tables for each phase (modify the values in Section 2 to match the team's mission parameters)
- Have backup anomalies ready if teams resolve injections quickly

### Pacing

- LEOP simulation should feel time-pressured (short passes, decisions needed quickly)
- Commissioning can be more relaxed
- Anomaly injection should be challenging but solvable with the team's procedures
- Allow extra time for debrief -- this is where the deepest learning occurs

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## Assessment Criteria

Observe and note (for feedback, not formal grading): - Did the team follow their procedures or improvise? - Did the Flight Director coordinate effectively? - Was the anomaly response systematic (detect-diagnose-plan-execute-verify)? - Did the team document events in real-time? - Did anyone identify a design flaw during operations?

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## Session Summary

| Topic             | Key Takeaway                                                                         |
|-------------------|--------------------------------------------------------------------------------------|
| LEOP              | First 72 hours are critical; every step is time-constrained                          |
| Commissioning     | Systematic subsystem checkout before payload operations                              |
| Anomaly response  | Follow the process: detect, assess, diagnose, plan, execute, verify, document        |
| Documentation     | Real-time logging is essential -- memory is unreliable under stress                  |
| FDIR validation   | Simulation reveals gaps in FDIR design that analysis alone cannot find               |
| Team coordination | Clear roles, communication protocols, and decision authority are essential           |
| Design feedback   | Operational experience should feed back into design (requirements, FDIR, procedures) |

## Session 5.4: Final Review & Presentations

**Expected reading before this session.** SMAD4 Ch. 22 (optimization); Wertz et al. SME Ch. 9.

**Prerequisites:** All previous sessions (complete design through simulation) **References:** ECSS-M-ST-10C Rev.1 section 6 (Reviews), NPR 7123.1D Appendix G, NASA SEH Rev 2 section 3.7, ECSS-E-ST-10C section 4 (Technical Dossier)

### Learning Objectives

By the end of this session, participants will be able to: 1. Prepare a complete design documentation package for review 2. Conduct a design review presentation with evidence-based arguments 3. Evaluate a peer team's design against gate criteria 4. Provide constructive technical feedback using a structured rubric 5. Identify lessons learned and articulate next steps for Phase B

### 1. Design Documentation Review

Before presenting, each team must verify that their design documentation is complete and internally consistent.

#### Documentation Checklist

| Document                            | Source in SpaceCDF        | Status Check                               |
|-------------------------------------|---------------------------|--------------------------------------------|
| Mission Requirements Document (MRD) | Exports -> ECSS Documents | All requirements baselined?                |
| Technical Specification (TS)        | Exports -> ECSS Documents | All budgets captured?                      |
| Concept of Operations (ConOps)      | Exports -> ECSS Documents | All phases defined?                        |
| Verification Plan (VP)              | V&V Matrix tab            | All requirements have methods?             |
| Bill of Materials (BOM)             | Exports -> Design Data    | All equipment listed with TRL?             |
| Risk Register                       | Worksheet 4.3             | All risks scored and mitigated?            |
| Interface Matrix (N <sup>2</sup> )  | Worksheet 4.3             | Interfaces identified? Conflicts resolved? |
| Cost Estimate                       | Worksheet 4.4             | WBS complete? P80 computed?                |
| Operations Concept                  | ConOps Editor             | Modes, FDIR, procedures defined?           |
| Sustainability Assessment           | Sustainability Card       | Debris compliance score?                   |

#### Internal Consistency Checks

Before presenting, verify:

| Check                   | How to Verify                                   | Common Error                              |
|-------------------------|-------------------------------------------------|-------------------------------------------|
| Mass budget closes      | Dashboard mass KPI: margin > 0%                 | Equipment added without updating budget   |
| Power budget closes     | Dashboard power KPI: SA > loads in all modes    | Eclipse mode power not checked            |
| Link budget closes      | Link Budget tool: margin >= 3 dB                | Rain attenuation not included             |
| Data budget closes      | Data Budget: daily downlink >= daily generation | Assumed too many passes per day           |
| Cost is within ceiling  | Cost tab: total <= allocated budget             | Forgot launch cost or operations          |
| Requirements traceable  | Every requirement maps to an objective          | Orphan requirements (no parent objective) |
| V&V complete            | Every requirement has a method assigned         | "TBD" entries remaining                   |
| No unresolved conflicts | Dashboard: conflict count = 0                   | Interface mismatches not addressed        |

### SpaceCDF Document Generation

Navigate to **Exports** tab and generate all documents: 1. ECSS Documents: MRD, TS, VP, ConOps, SEMP, IRD 2. Design Data: BOM, Parametric Model Data 3. Regulatory Filings: ITU API, RSSSA, Export Assessment, COPUOS, EOL Report 4. Presentation: Summary slide data (auto-populated)

## 2. Presentation Structure

Each team presents a 12-minute design review followed by 5 minutes of Q&A from the peer review board.

[Source: ECSS-M-ST-10C Rev.1 section 6; NPR 7123.1D Appendix G (Review Process)]

### Presentation Outline

| Section                     | Duration | Content                                               | Evidence                   |
|-----------------------------|----------|-------------------------------------------------------|----------------------------|
| 1. Mission Need             | 2 min    | Problem statement; stakeholders; objectives with MoPs | Mission Need step          |
| 2. Why Space?               | 2 min    | Trade study results; why space beats alternatives     | Mission Trade results      |
| 3. System Design            | 3 min    | Architecture; orbit; key parameters; budget status    | Dashboard, budgets         |
| 4. Equipment & Verification | 2 min    | Key equipment; BOM summary; V&V approach; test plan   | BOM, V&V Matrix            |
| 5. Risk & Cost              | 2 min    | Top 5 risks; cost estimate (P50/P80); schedule        | Risk register, cost tab    |
| 6. Operations & Lessons     | 1 min    | ConOps summary; FDIR; what would you change           | ConOps, simulation debrief |

## Presentation Best Practices

**Slide design:** - One key message per slide - Data, not prose (budgets, trade tables, not paragraphs) - Include backup slides with detailed analyses for Q&A

**Delivery:** - Lead with conclusions: "Our mass budget closes with 22% margin" - Show evidence for every claim: "Link budget analysis shows 4.2 dB margin at worst case" - Acknowledge risks honestly: "Antenna deployment is our highest risk at 12 (L3 x C4)" - For questions you cannot answer: "Good question -- we will take that as an action item"

**Common pitfalls to avoid:** - Reading slides verbatim - Hiding known problems (review boards always find them) - Presenting analysis without stating assumptions - Spending too long on the mission description and rushing through technical results

## 3. Peer Review Process

Each team acts as a review board for another team. This mirrors the real design review process where an independent board evaluates the project.

### Review Board Roles

| Role                | Responsibility                               | Key Questions to Ask                                     |
|---------------------|----------------------------------------------|----------------------------------------------------------|
| Chair               | Manages review; ensures all criteria covered | "Let's systematically check each gate criterion"         |
| Systems reviewer    | Evaluates overall architecture and budgets   | "Do all budgets close? What is the minimum margin?"      |
| Technical reviewer  | Evaluates subsystem design decisions         | "Why did you choose this component over alternatives?"   |
| Risk reviewer       | Evaluates risk management and V&V            | "What is your highest risk? Is the mitigation adequate?" |
| Operations reviewer | Evaluates ConOps, ground segment, regulatory | "What happens if you lose contact for 48 hours?"         |

### Gate Criteria Evaluation

The review board evaluates the presenting team against PDR-level gate criteria:

| # | Criterion                                | Score (0-2)                          | Notes                                                 |
|---|------------------------------------------|--------------------------------------|-------------------------------------------------------|
| 1 | Mission need clearly justified           | 0 = missing, 1 = partial, 2 = strong | Is the problem real? Is space the right answer?       |
| 2 | Requirements complete and traceable      | 0/1/2                                | All requirements linked to objectives? Verifiable?    |
| 3 | All budgets close with adequate margin   | 0/1/2                                | Mass, power, link, data, pointing -- all positive?    |
| 4 | Equipment selected with justified trades | 0/1/2                                | BOM complete? TRL adequate? Trade studies documented? |

| #  | Criterion                                 | Score (0-2) | Notes                                                  |
|----|-------------------------------------------|-------------|--------------------------------------------------------|
| 5  | Interfaces defined and conflicts resolved | 0/1/2       | N <sup>2</sup> matrix? No unresolved conflicts?        |
| 6  | V&V approach defined for all requirements | 0/1/2       | Methods assigned? Test plan outlined?                  |
| 7  | Risks identified with mitigations         | 0/1/2       | Risk register with scores? Top risks mitigated?        |
| 8  | Cost estimate with WBS and P80            | 0/1/2       | Both parametric and bottom-up? Learning curve applied? |
| 9  | ConOps and ground segment designed        | 0/1/2       | Modes defined? FDIR architecture? Ground stations?     |
| 10 | Sustainability compliance                 | 0/1/2       | Debris compliance? 25-year rule? Passivation plan?     |

## Review Board Questions Bank

Standard questions for the review board to ask:

- "Why is space the right answer? What alternatives were considered?"
- "What is your mass margin? What drives it? What if the payload is 20% heavier than estimated?"
- "Walk me through your link budget at worst case. What is the minimum margin?"
- "What are your top 3 risks? What is the residual risk after mitigation?"
- "Show me your data budget. Does it close with a single ground station?"
- "What happens during a safe mode entry in eclipse? Does the thermal budget survive?"
- "Which component has the lowest TRL? What is your qualification plan?"
- "Have you started spectrum licensing? What is on the critical path?"
- "What is your single-point failure list? Which ones have you accepted?"
- "If you had 6 more months and 20% more budget, what would you change?"

## 4. Team Presentations

### Instructions

**Presentation Schedule:** 12 minutes presentation + 5 minutes Q&A per team.

For 4 teams: allocate 17 min x 4 = 68 min. Adjust timing based on actual number of teams. If fewer than 4 teams, allow 15 min presentation + 8 min Q&A.

**For the presenting team:** 1. Present the 6-section outline above 2. Use SpaceCDF live for demonstrations where helpful (show Dashboard, budgets) 3. Respond to board questions with evidence

**For the review board:** 1. Complete the gate criteria evaluation form on Worksheet 5.4 2. Ask at least 3 substantive questions 3. Provide a GO / NO GO / GO WITH ACTIONS recommendation 4. Document action items with responsible person and deadline

**After each presentation:** - Chair announces the board's recommendation - Action items are recorded - Brief applause and transition to next team



## 5. Course Wrap-Up & Lessons Learned

### Individual Reflection

Each participant completes the self-assessment rubric on Worksheet 5.4, reflecting on their learning across all three weeks.

### Team Lessons Learned

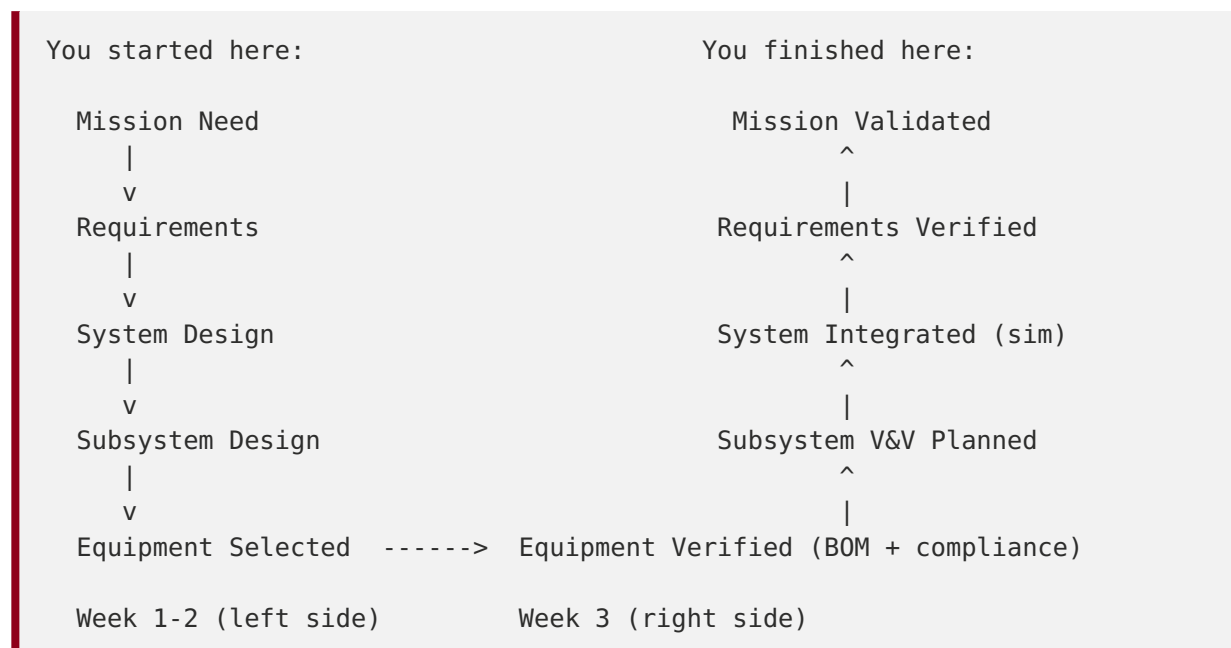
Each team discusses and reports:

1. **Most challenging design decision:** What trade-off was hardest to resolve? Why?
2. **Biggest surprise:** What aspect of mission design was most different from expectations?
3. **Tool feedback:** What SpaceCDF feature was most valuable? What was missing?
4. **Process insight:** How did concurrent design change your approach to engineering?
5. **If starting over:** What would you do differently in Week 1?

### Key Takeaways from the Course

| Week          | Theme                                  | Core Lesson                                                                                             |
|---------------|----------------------------------------|---------------------------------------------------------------------------------------------------------|
| <b>Week 1</b> | Requirements & Architecture            | Start with the problem, not the solution. Requirements define WHAT, not HOW.                            |
| <b>Week 2</b> | Subsystem Design & Budgets             | Engineering budgets are the language of systems engineering. Everything trades against everything.      |
| <b>Week 3</b> | Integration, Verification & Operations | Building the right thing (V&V) is as important as designing it. Operations reveal what analysis cannot. |

### The Systems Engineering V-Model -- Completed



*You have traversed the complete V-model: from mission need to verified, validated design.*

## Next Steps for Participants

- **SpaceCDF access** continues after the course
- **Apply the methodology** to your own missions or projects
- **Use the tool** for CDF-style studies with your team
- **Refer to the Course Textbook** for session plans and appendices
- **Share feedback** to improve the course for future cohorts

## Course Evaluation

Distribute the course evaluation form. Key questions: - Which session was most valuable? Least valuable? - What content should be added? Removed? - How effective was the simulation day? - Would you recommend this course to colleagues? - What is one thing you will do differently in your engineering practice as a result?

## Cited References

| # | Reference                                                 | URL                                                                                                                                                                                       |
|---|-----------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1 | ECSS-M-ST-10C Rev.1 (Project Planning and Implementation) | <a href="https://ecss.nl/standard/ecss-m-st-10c-rev-1-project-planning-and-implementation/">https://ecss.nl/standard/ecss-m-st-10c-rev-1-project-planning-and-implementation/</a>         |
| 2 | NPR 7123.1D Appendix G (Review Process)                   | <a href="https://nodis3.gsfc.nasa.gov/displayDir.cfm?t=NPR&amp;c=7123&amp;s=1D">https://nodis3.gsfc.nasa.gov/displayDir.cfm?t=NPR&amp;c=7123&amp;s=1D</a>                                 |
| 3 | NASA SEH Rev 2, section 3.7 (Technical Reviews)           | <a href="https://www.nasa.gov/reference/systems-engineering-handbook/">https://www.nasa.gov/reference/systems-engineering-handbook/</a>                                                   |
| 4 | ECSS-E-ST-10C (System Engineering)                        | <a href="https://ecss.nl/standard/ecss-e-st-10c-rev-1-system-engineering-general-requirements/">https://ecss.nl/standard/ecss-e-st-10c-rev-1-system-engineering-general-requirements/</a> |

## Session Summary

| Topic           | Key Takeaway                                                                     |
|-----------------|----------------------------------------------------------------------------------|
| Documentation   | Complete package: MRD, TS, ConOps, VP, BOM, Risk Register, Cost, Regulatory      |
| Consistency     | All budgets must close; no unresolved conflicts; requirements fully traceable    |
| Presentation    | 6 sections: need, justification, design, V&V, risk+cost, operations+lessons      |
| Peer review     | Independent board evaluates against gate criteria; GO/NO GO/GO with actions      |
| Best practices  | Lead with conclusions, show evidence, acknowledge risks, answer directly         |
| V-model         | Course traverses full V: needs -> requirements -> design -> V&V -> validation    |
| Lessons learned | Document what worked, what failed, and what to change -- this is the real output |

## Part 2 — Engineering-Position Reference

This chapter is a per-role background briefing for first-time CubeSat designers in a CDF. For each engineering position students might take, it covers:

- **Decisions you own.** What you have to choose, in what order.
- **What you need to know first.** Inputs you cannot decide without.
- **Formulas, charts, and rules of thumb.** With references.
- **Common pitfalls.** What to watch for, especially first time.
- **References.** Live links to standards and texts.

The briefings are roughly 2–4 pages each. Each is self-contained; read the one for your assigned role before Day 2 of Week 2 (CDF intensive Day 1).

**Expected reading.** Each role briefing is overnight reading the evening before students takes its CDF positions. Read your own briefing in full and scan one adjacent one (e.g. Power Engineers should scan Thermal; AOCS should scan Structures).

### Systems Engineer

You hold the budgets, the gate, and the authority to call a trade.

#### Decisions you own

1. **Margin policy for this study.** You set the per-budget margin targets for the current phase per ECSS-M-ST-10C — typically Phase A: +44 % mass, +25 % power, 6 dB link, +25 %  $\Delta v$ ; tightening to +5 % / +5 % / 3 dB / +10 % at Phase D.
2. **Budget closure decisions.** When power is over budget, you decide whether the team trades a payload duty cycle, a battery capacity, an SA area, or an end-of-life lifetime.
3. **Cross-domain conflict resolution.** When AOCS wants more reaction-wheel mass and Thermal wants the same area for a radiator, you convene the affected positions and arbitrate.
4. **Gate-review readiness.** You judge whether the team is ready to enter PDR, and you sign the action items off.

#### What you need to know first

- **Mission objectives** (SE Process 1) — the why drives the budgets you're closing.
- **System-V model** and where each position sits on it (NASA SEH §2). See *Course Plan* Figure 2.2.
- **17 NASA SEH processes** — you'll touch most of them every day (NASA SEH §6).
- **The current ECSS phase the team is in** — which sets margin policy (ECSS-M-ST-10C §6).

#### Formulas, charts, rules of thumb

ECSS margin policy by phase (mass example, all margins reduce phase-by-phase):

| Phase | Mass  | Power | Link | $\Delta v$ |
|-------|-------|-------|------|------------|
| A     | +44 % | +25 % | 6 dB | +25 %      |

| Phase | Mass  | Power | Link | $\Delta v$ |
|-------|-------|-------|------|------------|
| B     | +24 % | +20 % | 4 dB | +15 %      |
| C     | +13 % | +10 % | 3 dB | +10 %      |
| D     | +5 %  | +5 %  | 3 dB | +5 %       |

Mass distribution and ECSS margin policy by phase

Figure SE.1 — Mass distribution and the ECSS margin schedule. Your job is to keep total mass under the appropriate column for the current phase.

### Common pitfalls

**Watch out.** Resolving conflicts by averaging the two domains' requests *without* re-baselining the affected requirements is the classic systems-engineering failure mode. Always update the requirement before accepting the trade.

**Watch out.** Margin is not slack. Margin is uncertainty insurance. If you give the margin to anyone — to AOCS for an extra wheel, say — record it as a depletion of margin in the change log, not as a new allocation.

### References

- NASA SEH (SP-2016-6105 Rev 2) — <https://www.nasa.gov/reference/systems-engineering-handbook/>
- ECSS-M-ST-10C — <https://ecss.nl/>
- Wertz et al., *Space Mission Engineering: The New SMAD* (2011), Ch. 2.
- Larson & Wertz, *SMAD4*, Ch. 1, 22.

## Mission Analyst

You own the mission's relationship with physics: the orbit, the ground track, the lighting conditions, the contact times.

### Decisions you own

1. **Orbit type and altitude.** LEO / SSO / MEO / GEO / HEO / lunar. Altitude to balance science return, lifetime, drag, radiation, and launch cost.
2. **Inclination.** Drives latitude coverage, ground-track repeat, and (with altitude) Sun-synchronous condition.
3. **LTAN / LTDN** for SSO missions — drives  $\beta$ -angle envelope and eclipse season.
4. **Ground-station network.** Where to downlink, latency, coverage gaps.
5. **Maneuver plan.** When to phase, how to maintain station, when to de-orbit.

### What you need to know first

- The mission's **science target** — sun-glint avoidance? polar coverage? continuous Earth observation?
- **Latitude band** of interest (e.g. polar Arctic monitoring forces an inclined or polar orbit).
- **Lifetime target** and the regulatory framework (FCC 5-year / IADC 25-year).

- **Launch availability** — a target launcher constrains both inclination and altitude.

## Formulas, charts, rules of thumb

**Kepler period for a circular orbit** (verified across known orbits in Figure MA.1 below):

$$T = 2\pi\sqrt{a^3/\mu}$$

where  $\mu_{\text{Earth}} = 398\,600.4418 \text{ km}^3/\text{s}^2$  and  $a = R_{\text{Earth}} + h$ .

Kepler's Third Law verified against known orbits

*Figure MA.1 — Kepler's Third Law. Plotted markers (ISS-class, Sentinel-2, GPS, GEO) sit on the analytical curve to better than 0.5 % — which is the expected agreement once Earth oblateness is included separately.*

**Sun-synchronous condition** — pick  $i$  so that  $J_2$ -driven nodal regression matches Earth's mean motion around the Sun ( $\sim 0.9856 \text{ deg/day}$ ). At 600 km altitude,  $i \approx 97.79^\circ$ .

$J_2$  nodal regression vs inclination at several altitudes

*Figure MA.2 —  $J_2$  nodal regression rate vs inclination. The dashed horizontal is the SSO target. The intersection with the 700-km curve gives the SSO inclination at that altitude.*

**Eclipse fraction** for a circular orbit, valid for  $|\beta| < \beta^* = \arcsin(R/(R+h))$ :

$$f_{\text{eclipse}} = \frac{1}{\pi} \arccos\left(\frac{\sqrt{h^2 + 2Rh}}{(R+h)\cos\beta}\right)$$

For  $\beta > \beta^*$ , the orbit is fully sunlit.

$\beta$ -angle envelope and eclipse fraction analytics

*Figure MA.3 —  $\beta$ -angle envelope over a year and eclipse fraction analytical curves.*

**Orbital lifetime rule of thumb** (drag-dominated LEO):

$$\tau_{\text{years}} \approx \frac{h - 200}{30} \cdot \frac{m/A}{50}$$

Orbital lifetime vs altitude — FCC 5-yr and IADC 25-yr lines

*Figure MA.4 — Orbital lifetime envelope vs altitude. Below  $\sim 600 \text{ km}$  a 6U CubeSat ( $m/A \sim 50 \text{ kg/m}^2$ ) decays inside 25 years; below  $\sim 425 \text{ km}$  it can decay inside 5 years (the FCC rule).*

## Common pitfalls

**Common mistake.** Picking 500 km thinking it gives a 25-year lifetime — it doesn't, for a typical 6U. Always confirm against the rule-of-thumb chart and an STK or GMAT propagation.

**Watch out.** Sun-glint angle for ocean-colour imaging changes with both season ( $\beta$ ) and longitude. If your mission cares about sun-glint, build a year-long  $\beta$ /Sun-azimuth heatmap before committing to LTAN.

## References

- Curtis, *Orbital Mechanics for Engineering Students* (3rd ed.).
- Vallado, *Fundamentals of Astrodynamics and Applications* (4th ed.).
- SMAD4 Ch. 9 (orbit selection).

- Wertz, *Mission Geometry: Orbit and Constellation Design and Management* (2001).
- AGI STK — <https://www.agi.com/products/stk/>.
- NASA GMAT (open source) — <https://software.nasa.gov/software/GSC-17177-1>.

## Payload Engineer

You own the science. The mission exists to do what your payload does, and every other subsystem ultimately serves yours.

### Decisions you own

1. **Payload concept & sensor type.** Optical / SAR / hyperspectral / radiometer / RF receiver / etc.
2. **Aperture & spectral bands.** Drives resolution, SNR, mass.
3. **Operating concept.** Pushbroom vs whiskbroom vs framing, continuous vs target-cued.
4. **Data rate & data volume per orbit.** This drives Comms and OBDH.
5. **Calibration plan.** On-orbit sources, ground references, cadence.

### What you need to know first

- **Mission objective measurable (MoP).** For ocean colour you need certain bands at certain SNR; for SAR you need a certain ground-resolution and revisit; for AIS you need a certain detection probability.
- **Orbit altitude** — drives slant range, GSD, coverage.
- **Mass and power available** — typically 30–50 % of the platform.

### Formulas, charts, rules of thumb

GSD diffraction limit for an unobstructed aperture:

$$\text{GSD}_{\text{diff}} = \frac{1.22\lambda h}{D}$$

GSD pixel limit:

$$\text{GSD}_{\text{pix}} = \frac{p \cdot h}{f}$$

where  $p$  is pixel pitch and  $f$  is focal length. The achievable GSD is the larger of the two.

GSD vs aperture diameter — diffraction and pixel limits

Figure PL.1 — GSD vs aperture for visible imagers ( $\lambda = 550 \text{ nm}$ ). The dashed line is the pixel-limit at 500 km for  $p = 6 \mu\text{m}$  and  $f/D = 5$ .

SAR antenna minimum length:

$$L_{\text{ant, min}} = 2 \cdot \delta_{\text{azimuth}}$$

i.e. the antenna along-track length must exceed twice the desired azimuth resolution.

### Common pitfalls

**Common mistake.** Sizing the payload aperture for the diffraction limit while the detector pixels are too coarse to use it. Match the two — pixel-limited and diffraction-limited GSDs should be comparable.

**Watch out.** Data volume per orbit grows with the aperture in two ways at once: more pixels, faster line rate. Check the daily download capacity (Figure PL.2) before committing.

Data volume vs band — capacity check

Figure PL.2 — Daily downlink capacity by band, against typical payload data volumes. UHF closes for housekeeping only; S-band for imaging; X-band for SAR / hyperspectral.

## References

- SMAD4 Ch. 9 (payload).
- Cracknell, *Introduction to Remote Sensing* (3rd ed.).
- Rees, *Physical Principles of Remote Sensing* (3rd ed.).
- Cumming & Wong, *Digital Processing of Synthetic Aperture Radar Data*.
- ECSS-E-ST-10-04C — Space environment for payload sizing — <https://ecss.nl/>.

## Power Engineer

You own the orbit-by-orbit energy balance.

### Decisions you own

1. **Solar-array architecture.** Body-mounted vs deployable; gimballed vs fixed; technology (triple-junction GaAs vs Si).
2. **Solar-array area** — at end-of-life, accounting for degradation,  $\beta$ -angle range, and pointing.
3. **Battery capacity and DoD.** Cycle life vs eclipse depth.
4. **Bus voltage and PCDU architecture.** Single-bus vs dual-bus; regulated vs unregulated; MPPT topology.
5. **Power-mode duty cycle.** Maps to ConOps.

### What you need to know first

- **Orbit** — drives  $\beta$ -angle envelope and eclipse fraction.
- **Orbit-average load** from all subsystems (Watts).
- **Mission lifetime** — drives cycle count and EOL degradation.

## Formulas, charts, rules of thumb

**Solar-array sizing (first cut):**

$$A_{SA} = \frac{P_{avg} \cdot \left(1 + \frac{f_{eclipse}}{f_{sun} \eta_{dis}}\right)}{S_0 \eta_{cell} \cos \beta (1 - D_{deg})}$$

where  $S_0 = 1361 \text{ W/m}^2$ ,  $\eta_{cell} \approx 0.30$  for triple-junction GaAs,  $D_{deg} \approx 10 \%$  for 5-year LEO end-of-life.

Solar-array sizing nomograph

Figure PW.1 — Solar-array nomograph at  $\eta = 30 \%$ , 10 % EOL degradation,  $\beta = 20^\circ$ ,  $\eta_{dis} = 85 \%$ . A 6U at 15 W average needs  $\sim 870 \text{ cm}^2$  at end-of-life.

**Battery sizing — usable energy:**

$$E_{\text{usable}} = \text{DoD} \cdot E_{\text{nominal}} \quad ; \quad E_{\text{nominal}} = \frac{P_{\text{avg}} \cdot t_{\text{eclipse}}}{\eta_{\text{dis}} \cdot \text{DoD}}$$

Battery cycle life vs DoD

Figure PW.2 — Battery cycle life vs DoD for Li-ion families. A 30 % DoD design point gives the cycles needed for a 5-year LEO mission with margin. Heritage chemistry like LiCoO<sub>2</sub> requires deeper de-rating; LFP gives more headroom but is less mass-dense.

**Power profile across one orbit** — Figure PW.3 shows the typical generation/load picture.

Power profile across one orbit

Figure PW.3 — Generation (top) and stacked load profile (bottom) across one orbit.

## Common pitfalls

**Watch out.** Sizing the array for begin-of-life will starve the battery at end-of-life. Always compute at EOL with realistic degradation (e.g. 10 % over 5 years for triple-junction GaAs in LEO).

**Common mistake.** Forgetting that the array only generates when the cells see the Sun.  $\beta = 0^\circ$  at season change means the array sees zero illumination at orbit noon — design for the worst case in the year.

## References

- ECSS-E-ST-20C — *Electrical and electronic* — <https://ecss.nl/>
- SMAD4 Ch. 11 (power).
- Patel, *Spacecraft Power Systems*.
- Brown, *Spacecraft Propulsion and Power Subsystems*.

## AOCS Engineer

You own the spacecraft's pointing.

### Decisions you own

1. **Pointing requirement** — knowledge vs control vs stability, per axis.
2. **Sensor suite** — Sun sensor / magnetometer / star tracker / GPS / IMU.
3. **Actuator suite** — magnetorquer / reaction wheel / monopropellant / cold-gas.
4. **Modes & transitions** — sun-acquisition, detumble, fine pointing, safe.
5. **Momentum management plan** — how often, with what.

### What you need to know first

- **Pointing requirement** from Payload (e.g. 0.05° 1 $\sigma$  for a pushbroom imager).
- **Disturbance environment** at the chosen orbit (drag at 400 km vs 700 km vs SSO).
- **Spacecraft inertia tensor** — Structures gives you this.

### Formulas, charts, rules of thumb

**Pointing budget** — RSS contributors:



$$\sigma_{\text{pointing}}^2 = \sigma_{\text{knowledge}}^2 + \sigma_{\text{control}}^2 + \sigma_{\text{alignment}}^2$$

Pointing-error budget waterfall

Figure AC.1 — Pointing-error budget tree. Knowledge, control, and alignment groups RSS into the total. The dashed line is the mission allocation.

**Reaction-wheel sizing — momentum capacity:**

$$H_{\text{wheel}} \geq T_{\text{disturbance}} \cdot t_{\text{between desat}} \cdot k_{\text{margin}}$$

Reaction-wheel momentum vs disturbance

Figure AC.2 — Momentum accumulation per orbit (left) and disturbance-torque order-of-magnitude (right). At 500 km, drag typically dominates with  $\sim 1 \mu\text{N}\cdot\text{m}$ , accumulating  $\sim 50\text{--}100 \text{ mN}\cdot\text{m}\cdot\text{s}$  over a few orbits.

## Common pitfalls

**Watch out.** Quaternion normalisation drift in flight software — even a  $1\text{e-}6$  drift per second adds up. Renormalise every cycle.

**Common mistake.** Designing for fine pointing without sizing the wheel for desat unloading. Plan the desat strategy at the same time as the wheel.

## References

- ECSS-E-ST-60-20C — AOCS — Part 1 — <https://ecss.nl/>.
- Markley & Crassidis, *Fundamentals of Spacecraft Attitude Determination and Control*.
- Sidi, *Spacecraft Dynamics and Control*.
- SMAD4 Ch. 10.

## Thermal Engineer

You keep every component inside its operating range across the hot, cold, and transient cases.

### Decisions you own

1. **Hot-case / cold-case envelope** — define them in terms of  $\beta$ -angle, season, eclipse, internal heating.
2. **Surface finishes ( $\alpha$ ,  $\epsilon$ )** for each external panel and radiator.
3. **MLI vs heaters** — passive insulation vs active control.
4. **Heat-rejection topology** — radiator location and area.
5. **Thermal interfaces** — conductive paths to and from each component.

### What you need to know first

- **Operating temperature ranges** for every component (battery is the strictest, typically  $-5$  to  $+25^\circ\text{C}$  charging).
- **Eclipse fraction & duration** from Mission Analyst.
- **Internal dissipation map** from Power.

## Formulas, charts, rules of thumb

### Radiative equilibrium for a flat plate:

$$T_{eq} = \left( \frac{\alpha_s S + \epsilon_{IR} \phi_{IR}}{\sigma \epsilon_{IR}} \right)^{1/4}$$

Radiative thermal equilibrium vs  $\alpha/\epsilon$

Figure TH.1 — Equilibrium temperature vs  $\alpha/\epsilon$  for a flat plate at LEO, with common surface finishes marked.

### Common pitfalls

**Common mistake.** Using  $\alpha$  and  $\epsilon$  at room temperature for IR calculations — at -100 °C,  $\epsilon$  of many materials shifts. Cite the source data temperature.

**Watch out.** MLI is rated by effective  $\epsilon$  *not by base material*. Use  $\epsilon \approx 0.03$  for 10-layer MLI; less for fewer layers.

### References

- ECSS-E-ST-31C — *Thermal control* — <https://ecss.nl/>
- Gilmore, *Spacecraft Thermal Control Handbook*, Vol. I & II.
- SMAD4 Ch. 12.
- ESATAN-TMS — <https://www.esatan-tms.com/>.

## Structures Engineer

You make sure the spacecraft survives launch and on-orbit, and that everyone else's hardware is mechanically interfaced.

### Decisions you own

1. **Primary structure topology** — monolithic, panel-and-rib, or 3D-printed.
2. **Material selection** — Al, Ti, CFRP, sandwich.
3. **Mass-budget closure with margin.**
4. **Launch-load verification path** — analysis (FEA), test, or both.
5. **Mechanical interfaces** — fastener pattern, deployer compatibility (ISIPOD / EXOpod / CSD).

### What you need to know first

- **Launcher environment** (qualification levels for sine, random vibration, shock).
- **Component mass and CG** — every subsystem provides this.
- **Cal Poly CDS Rev 14** envelope for CubeSats.

## Formulas, charts, rules of thumb

### Margin of safety:

$$MoS = \frac{\sigma_{allow}}{FoS \cdot \sigma_{applied}} - 1$$

For Al 7075-T6 yield:  $\sigma_{yield} = 503$  MPa,  $FoS_{yield} = 1.25$ .

**Modal analysis** — first mode > 100 Hz typical CubeSat requirement.

## Common pitfalls

**Watch out.** Forgetting fastener and adhesive mass — they typically add 5–10 % on top of the structural mass.

**Common mistake.** Sizing for static loads and missing the dynamic amplification factor (random vibration). Apply Q-factor (typically 5–10) at resonance.

## References

- ECSS-E-ST-32C — *Structures* — <https://ecss.nl/>.
- Sarafin, *Spacecraft Structures and Mechanisms*.
- Cal Poly CDS Rev 14 — <https://www.cubesat.org/cds-announcement>.

## Propulsion Engineer

You own the  $\Delta v$ . (Even if your CubeSat has none, you decide that.)

### Decisions you own

1. **Need for propulsion** — drag make-up? deorbit? phasing? formation? rendezvous?
2. **Propellant choice** — cold gas / monoprop / bipropellant / electric (Hall, ion, PPT, FEEP).
3. **Thruster sizing** — thrust vs Isp vs power.
4. **Total propellant mass** — Tsiolkovsky.
5. **Plume impingement & contamination plan.**

### What you need to know first

- **$\Delta v$  budget** — sum of all required maneuvers with margin.
- **Mission duration** — drives cumulative  $\Delta v$  for drag-dominated orbits.
- **Available power** — drives EP feasibility (a CubeSat at 20 W cannot run a 200 W Hall thruster).

## Formulas, charts, rules of thumb

Tsiolkovsky:

$$\Delta v = I_{sp} g_0 \ln \frac{m_0}{m_f}$$

Tsiolkovsky —  $\Delta v$  vs mass ratio for selected Isp

Figure PR.1 —  $\Delta v$  vs mass ratio at five characteristic Isp values. Horizontal lines show typical mission  $\Delta v$  budgets.

## Common pitfalls

**Watch out.** Isp is a function of inlet conditions and nozzle expansion ratio. Quote the value at the operating point you'll actually fly.

**Common mistake.** Confusing total mass ( $m_0$ ) with dry mass ( $m_f$ ) when applying Tsiolkovsky.

## References

- Sutton & Biblarz, *Rocket Propulsion Elements* (9th ed.).
- ECSS-E-ST-35C — *Propulsion general* — <https://ecss.nl/>.
- Goebel & Katz, *Fundamentals of Electric Propulsion*.

## Communications Engineer

You move the bits — both ways.

### Decisions you own

1. **Frequency band** — UHF / S / X / Ka. Drives licensing, antenna size, FSPL.
2. **Modulation and coding**.
3. **Antenna pattern** — high-gain dish vs patch vs deployable.
4. **Ground-station network** — single GS vs network.
5. **Link margin allocation** at each phase.

### What you need to know first

- **Daily data volume** from Payload.
- **Latency requirement** — if any.
- **Pointing capability** from AOCS (drives antenna pattern needs).
- **Regulatory constraints** for the chosen band.

### Formulas, charts, rules of thumb

Free-space path loss:

$$L_{FS} = 20 \log_{10} (4\pi d / \lambda) \text{ dB}$$

Free-space path loss vs slant range

Figure CO.1 — FSPL across bands. At 2.2 GHz (S-band) over a 1500 km slant, FSPL  $\approx$  163 dB; at 26 GHz (Ka-band), 184 dB.

Link budget closure:

$$\frac{C}{N_0} = \text{EIRP} - L_{\text{path}} - L_{\text{atm}} - L_{\text{point}} + G/T - k \text{ [dB} \cdot \text{Hz]}$$

with  $k = -228.6 \text{ dBW}/(\text{K} \cdot \text{Hz})$ .

S-band downlink waterfall

Figure CO.2 — A canonical S-band waterfall. Read it left-to-right: gains add, losses subtract, the bottom-right is required  $C/N_0$  and margin.

$E_b/N_0 \rightarrow \text{BER}$  — Figure CO.3 shows the standard curves.

$E_b/N_0 \rightarrow \text{BER}$  for common modulations

Figure CO.3 —  $E_b/N_0$  vs BER for BPSK, QPSK, 8-PSK, FSK, and a typical concatenated coding scheme.

**Antenna patterns** — high-gain vs low-gain.

Antenna patterns — dish vs patch

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Figure CO.4 — High-gain dish (33 dBi, narrow) vs S-band patch (~6 dBi, wide). Dish needs accurate AOCS pointing; patch tolerates ~30° body rate.

## Common pitfalls

**Watch out.** Pointing loss is not negligible. For a 1° HPBW dish, a 0.3° pointing error costs ~1 dB; at 0.5° it costs ~3 dB.

**Common mistake.** Forgetting Doppler. A 2.2 GHz S-band link from a 7.5 km/s LEO has ±55 kHz Doppler at horizon — your receiver tracking has to follow.

## References

- ECSS-E-ST-50-05C — RF & modulation — <https://ecss.nl/>.
- Pratt, Bostian & Allnutt, *Satellite Communications* (3rd ed.).
- ITU-R Recommendations — <https://www.itu.int/en/ITU-R/>.

## OBDH / Software Engineer

You own the flight computer, the flight software, the telemetry list, and FDIR.

### Decisions you own

1. **OBC architecture** — single SBC or redundant pair; SoC vs discrete.
2. **Bus topology** — CAN / SpaceWire / I<sup>2</sup>C / SPI.
3. **Memory architecture** — RAM, flash, mass-memory, EDAC scheme.
4. **FSW architecture** — mode manager, ADCS control loop, TM/TC handler, FDIR rules.
5. **PUS service set** — which CCSDS PUS services to implement.

### What you need to know first

- **TM/TC volume** from each subsystem.
- **Real-time loop rates** required by AOCS.
- **Storage volume** for the longest no-contact gap (mass-memory sizing).

### Formulas, charts, rules of thumb

- **Mass memory sizing:** longest contact gap × peak data rate × margin (1.5–2×).
- **Watchdog timer:** typical 30 s for CubeSat OBC; less for high-criticality FSW threads.
- **EDAC overhead:** Hamming SECDED ~ 12.5 % for 64-bit words.

## Common pitfalls

**Watch out.** Telemetry list overruns are the #1 OBDH late defect. Lock the list at SRR; treat any addition as a change-control item.

## References

- CCSDS PUS — <https://public.ccsds.org/Pubs/660x0g3.pdf>.

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- ECSS-E-ST-40C — *Software* — <https://ecss.nl/>.
  - ECSS-Q-ST-80C — *Software product assurance* — <https://ecss.nl/>.
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## Operations Engineer

You'll fly the spacecraft after the team builds it. You also live this on Day 3 of Week 3 in the simulation.

### Decisions you own

1. **Operations concept** — staffing, automation level, shift structure.
2. **Pass plan** — which procedures, in what order, every contact.
3. **Anomaly response tree** — who calls what, when.
4. **Telecommand / telemetry definition** — coordination with OBDH.

### What you need to know first

- **Mission ops modes** from ConOps.
- **Ground-station coverage** from Mission Analyst.
- **Procedure inventory** from the mission's procedure set (LEOP / commissioning / nominal / contingency / emergency).

### Common pitfalls

**Watch out.** Voice-loop discipline is the silent killer. Cross-talk that sounds harmless in training breaks down under real-time pressure.

### References

- ECSS-E-ST-70C — *Ground systems and operations* — <https://ecss.nl/>.
  - ECSS-E-ST-70-32C — *Test and operations procedures*.
  - NASA Mission Operations Handbook — <https://nodis3.gsfc.nasa.gov/>.
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## Ground Segment Engineer

You own the architecture between the antenna and the user.

### Decisions you own

1. **Ground-station network** — owned, leased, or commercial (Atlas, KSAT, Viasat, AWS Ground Station).
2. **MCS choice** — COSMOS / OpenMCT / Yamcs / commercial.
3. **Archive design** — L0 → L1 → L2 → distribution latency.
4. **Pass-planning tool**.
5. **Cyber-security architecture**.

### What you need to know first

- **Daily data volume** — drives bandwidth contracts.
- **Latency requirement** — drives station network density.

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- **Mission classification** — defence missions need different architectures than civil.
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## Common pitfalls

**Watch out.** AWS Ground Station's pricing model can flip economics between band choices — do the math at the chosen data rate.

## References

- CCSDS Mission Operations standards — <https://public.ccsds.org/Pubs/>.
  - KSAT — <https://www.ksat.no/>.
  - AWS Ground Station — <https://aws.amazon.com/ground-station/>.
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## Cost & Schedule

You're the team's reality check.

### Decisions you own

1. **WBS structure** (NPR 7120.5).
2. **Cost-estimating method** — parametric (SSCM, COMPACT), analogy, bottom-up.
3. **Schedule baseline** — milestones, critical path, float.
4. **Risk-adjusted estimates** — how to roll risk into both cost and schedule.

## Common pitfalls

**Common mistake.** Treating the parametric tool's output as the answer. The output is a starting point; tailor it for your mission and the launch market you're working in.

## References

- NPR 7120.5F — <https://nodis3.gsfc.nasa.gov/>.
- Wertz et al., SME, Ch. 23 (Cost).
- SMAD4 Ch. 20.
- AIAA Cost Modeling — <https://www.aiaa.org/>.

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## Part 3 — Position Quick-Reference Cards (quick-reference cards)



## Appendix: Position-Specific Deep Dives

This appendix provides detailed guidance for each CDF engineering position, including responsibilities, key parameters owned, decision authority, common pitfalls, and references.

### Appendix A: Systems Engineer

**Responsibility:** Overall system architecture, budget management, cross-domain conflict resolution.

**Key Parameters Owned:** Mass margin, power margin, system-level cost, composite TRL, health score.

**Key Activities:** - Maintain all engineering budgets (mass, power, cost, ?V, data, pointing) - Apply ECSS margin policy by project phase (44% Phase A -> 13% Phase C/D) - Resolve cross-domain conflicts (convene affected positions, facilitate trade) - Prepare gate review evidence packages - Own the requirements baseline and traceability matrix - Conduct design reviews and track action items

**Decision Authority:** System-level trade-offs, margin allocation, requirement waivers.

**Common Pitfalls:** - Allowing individual subsystems to consume their margin allocation (no system reserve) - Not challenging HOW requirements that should be WHAT - Accepting conflicts without resolution timelines - Not tracking parametric estimate vs selected equipment delta

**References:** NASA SEH §2, §6; ECSS-E-ST-10C Rev.1 §5; SMAD4 Ch.3

### Appendix B: Mission Analyst

**Responsibility:** Orbit design, coverage analysis, ground station access, constellation architecture.

**Key Parameters Owned:** Altitude, inclination, orbit type, eclipse fraction, revisit time, contact time, LTAN.

**Key Activities:** - Run orbit trade studies (altitude/inclination/lifetime/coverage/cost) - Compute coverage and revisit for target latitude bands - Analyse ground station access (elevation angle, contact duration, pass scheduling) - Assess debris compliance (lifetime vs 25yr/5yr rules) - For constellations: Walker delta design, phasing, coverage optimisation

**Key Formulae:** - Period:  $T = 2\pi\sqrt{a^3/\mu}$  - Eclipse fraction:  $f = (1/\pi)\arccos(\sqrt{1-(R_E/a)^2})$  - SSO inclination:  $\cos(i) = f(a, J_2)$  - Hohmann ?V: from vis-viva equation - Coverage radius: from ground station elevation geometry

**References:** SMAD4 Ch.5-7; Vallado §2-9; ECSS-U-AS-10C Rev.2

### Appendix C: Payload Lead

**Responsibility:** Instrument performance, data generation, payload accommodation.

**Key Parameters Owned:** GSD (or equivalent MoP), data rate, pointing requirement, payload mass, power, duty cycle, FOV.

**Key Activities:** - Size the payload from performance requirements (GSD -> aperture for optical; data rate -> antenna for comms; resolution -> antenna area for SAR) - Define payload data volume and duty cycle - Specify thermal environment needs (detector cooling, operational temperature) - Coordinate with AOCS for

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pointing and stability requirements - Coordinate with comms for downlink capacity matching data generation

**Key Formulae:** - Optical:  $GSD = \max(1.22\lambda/D, \phi/f)$  - Payload mass:  $M \sim 20D^{1.5} + 2$  (heritage CER) - Data rate:  $R = N_{\text{pixels}} \times N_{\text{bands}} \times \text{bit\_depth} \times \text{line\_rate}$  - SAR:  $\text{antenna\_length} \geq 2 \times \text{resolution}$

**References:** SMAD4 Ch.9; ECSS-E-ST-10-06C; specific instrument handbooks

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## Appendix D: Power Engineer

**Responsibility:** Solar array, battery, EPS architecture, power distribution, duty cycling.

**Key Parameters Owned:** SA area, SA power (BOL/EOL), battery capacity, bus voltage, power margin per mode.

**Key Activities:** - Construct power budget per operational mode (safe/idle/imaging/downlink/eclipse) - Size solar array from peak sunlight demand + battery recharge - Size battery from eclipse energy / maximum DoD - Define switched power line allocation per subsystem - Assess SA degradation over mission lifetime - Coordinate with thermal on SA/radiator area competition

**Key Formulae:** -  $P_{SA} = P_{\text{peak\_sun}} + (P_{\text{eclipse}} \times t_{\text{ecl}}) / (t_{\text{sun}} \times \eta_{\text{charge}})$  -  $P_{SA\_BOL} = P_{SA\_EOL} / (1 - \text{degradation})^{\text{years}}$  -  $A_{SA} = P_{BOL} / (\eta_{\text{cell}} \times S \times \cos(\theta) \times \eta_{\text{pack}})$  -  $C_{\text{bat}} = (P_{\text{ecl}} \times t_{\text{ecl}}) / (\text{DoD} \times \eta_{\text{discharge}})$

**References:** SMAD4 Ch.11.4; ECSS-E-ST-20C; vendor datasheets (GomSpace NanoPower)

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## Appendix E: AOCS Engineer

**Responsibility:** Attitude determination and control, pointing accuracy, momentum management.

**Key Parameters Owned:** Pointing accuracy, stability, slew rate, wheel momentum, actuator mass/power.

**Key Activities:** - Select AOCS architecture (passive magnetic -> MTQ -> RW -> RW+ST) based on pointing requirement - Construct pointing error budget (RSS of all sources) - Size reaction wheels for momentum storage and torque - Size magnetorquers for momentum dumping - Define safe mode attitude (sun-pointing) for power survival - Assess disturbance torques (gravity gradient, magnetic, SRP, aero drag)

**Key Formulae:** - RSS pointing:  $\theta = \sqrt{\theta^2}$  - Gravity gradient torque:  $T = (3\mu/2R^3) \times (I_z - I_x) \times \sin(2\theta)$  - Magnetic torque:  $T = m \times B$  (dipole moment  $\times$  field) - Wheel momentum:  $H = T \times t_{\text{accumulation}}$

**References:** SMAD4 Ch.11.1; ECSS-E-ST-60-10C; Wertz §11.1

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## Appendix F: Thermal Engineer

**Responsibility:** Temperature control (hot/cold case), radiators, heaters, MLI.

**Key Parameters Owned:** Max/min predicted temperatures, radiator area, heater power, thermal margin.

**Key Activities:** - Define hot case (max solar + internal dissipation) and cold case (eclipse, min power) - Select thermal control approach (passive coatings -> MLI -> heaters -> heat pipes) - Size radiators to reject waste heat in hot case - Size heaters to maintain minimum temperature in cold case (eclipse) - Verify ECSS thermal

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margins ( $\pm 5^\circ\text{C}$  operating,  $\pm 10^\circ\text{C}$  acceptance,  $\pm 15^\circ\text{C}$  qualification) - Coordinate with power on heater power budget and with structure on radiator mounting

**Key Formulae:** - Stefan-Boltzmann:  $Q = \sigma AT^4$  (radiative heat transfer) - Equilibrium:  $Q_{\text{absorbed}} + Q_{\text{internal}} = Q_{\text{radiated}}$  - Solar absorptance/emittance ratio ( $\alpha/\epsilon$ ) determines equilibrium temperature

**References:** SMAD4 Ch.11.5; ECSS-E-ST-31C; Gilmore, Spacecraft Thermal Control Handbook

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## Appendix G: Communications Engineer

**Responsibility:** Link budget, transponder/antenna selection, frequency licensing, ground station network.

**Key Parameters Owned:** Link margin, data rate, frequency band, EIRP, antenna gain, modulation/coding.

**Key Activities:** - Construct complete link budget (EIRP, FSPL, G/T, C/N0, Eb/N0, margin) - Select frequency band based on data rate need and licensing constraints - Select transponder and antenna (RF chain compatibility: band, impedance, polarisation) - Coordinate with AOCS on antenna pointing accuracy - Define ground station requirements (antenna size, G/T, location, contact schedule) - Manage frequency licensing process (IARU/ISED/ITU as appropriate)

**Key Formulae:** - FSPL =  $20\log_{10}(4\pi d/\lambda)$  dB - EIRP =  $P_{\text{TX}} + G_{\text{TX}} - L_{\text{TX}}$  dBW - Link margin =  $E_b/N_{0,\text{avail}} - E_b/N_{0,\text{req}} - L_{\text{impl}}$  dB

**References:** SMAD4 Ch.13; ECSS-E-ST-50-05C; ITU Radio Regulations; CCSDS 131.0-B

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## Appendix H: Propulsion Engineer

**Responsibility:**  $\Delta V$  budget, propulsion system selection, propellant management.

**Key Parameters Owned:** Total  $\Delta V$ , Isp, propellant mass, thrust level, total impulse.

**Key Activities:** - Construct  $\Delta V$  budget (orbit insertion, maintenance, collision avoidance, deorbit) - Select propulsion technology (cold gas, electric, chemical) based on  $\Delta V$  and power - Size propellant mass using Tsiolkovsky equation - Coordinate with structure on tank mounting and plume impingement - Assess debris compliance (deorbit capability vs natural lifetime)

**Key Formulae:** - Tsiolkovsky:  $\Delta V = I_{\text{sp}} \times g_0 \times \ln(m_{\text{initial}}/m_{\text{final}})$  - Propellant mass:  $m_{\text{p}} = m_{\text{dry}} \times (e^{(\Delta V/(I_{\text{sp}} \times g_0))} - 1)$  - Total impulse:  $I_{\text{total}} = m_{\text{p}} \times I_{\text{sp}} \times g_0$

**References:** SMAD4 Ch.17; Sutton & Biblarz, Rocket Propulsion Elements; vendor data

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## Appendix I: Structures Engineer

**Responsibility:** Primary structure, mechanisms, CDS compliance, launch loads, integration.

**Key Parameters Owned:** Structure mass, natural frequency, margin of safety, deployer compatibility.

**Key Activities:** - Select CubeSat structure (form factor, material, vendor) - Verify CDS compliance (dimensions, rails, switches, RBF, CG) - Analyse launch loads (quasi-static, random vibration, shock) - Compute structural margin of safety (MoS) - Design deployment mechanisms (antenna, solar panels, payload) - Plan integration sequence (component -> board -> stack -> structure -> deployer)

**Key Formulae:** - MoS =  $(\text{Allowable} / (\text{Design} \times \text{FoS})) - 1$  - Natural frequency:  $f = (1/2\pi)\sqrt{k/m}$

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**References:** CDS Rev 14.1; SMAD4 Ch.11.3; ECSS-E-ST-32C Rev.1; NASA GEVS

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## Appendix J: Cost Engineer

**Responsibility:** Cost estimation, WBS, learning curves, risk-adjusted cost.

**Key Parameters Owned:** Total cost (MEUR), per-WBS element cost, launch cost, operations cost.

**Key Activities:** - Construct WBS-level cost estimate (parametric early, bottom-up later) - Apply CubeSat-specific COTS pricing where applicable - Estimate learning curve effects for constellations - Perform Monte Carlo cost risk analysis (P50/P70/P80) - Track cost against programmatic ceiling - Identify cost drivers and propose de-scope options

**References:** SMAD4 Ch.20; Aerospace Corp SSCM; NASA CEH; ECSS-M-ST-60C

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## Appendix K: Compliance / Regulatory Engineer

**Responsibility:** ECSS standard compliance, frequency licensing, export control, debris mitigation.

**Key Parameters Owned:** Standard applicability matrix, filing status, export classification, debris compliance score.

**Key Activities:** - Determine applicable ECSS/NASA standards and prepare tailoring matrix - Manage frequency licensing process (IARU/ISED/ITU filings) - Assess export control classification for all components (ITAR/EAR/CGP) - Verify debris compliance (25yr IADC, 5yr FCC, casualty risk) - Prepare RSSSA filing if mission has remote sensing capability - File COPUOS registration post-launch

**References:** ECSS-S-ST-00-02C (tailoring); CPC-2-6-02 (ISED); RSSSA; ITU RR Article 9

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## Appendix L: Ground Segment Engineer

**Responsibility:** Ground station network, MCS, data processing pipeline, ops concept.

**Key Parameters Owned:** Contact time per day, ground station locations, processing latency, data throughput.

**Key Activities:** - Select ground station network (own station, KSAT, SatNOGS, DSN) - Define MCS architecture (COSMOS, OpenMCT, Yamcs) - Design data processing pipeline (L0 -> L1 -> L2 -> archive -> distribution) - Plan operations concept (staffing, automation level, anomaly response) - Coordinate with comms engineer on frequency bands and antenna requirements

**References:** SMAD4 Ch.14-15; ECSS-E-ST-70C; CCSDS standards

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## Appendix M: Software Engineer

**Responsibility:** Flight software architecture, FDIR, TC/TM definition, ground software.

**Key Parameters Owned:** FSW architecture, mode definitions, command dictionary, telemetry list.

**Key Activities:** - Define FSW architecture (mode manager, ADCS control, TM/TC handler, FDIR) - Implement FDIR (Fault Detection, Isolation, Recovery) rules - Define telecommand dictionary (all commands with

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parameters) - Define telemetry dictionary (all HK and science packets) - Define autonomous operations (time-tagged commands, on-board scheduling) - Develop ground control software and procedures

**References:** ECSS-E-ST-70-01C; ECSS-E-ST-70-41C (PUS); CCSDS 133.0-B (Space Packet Protocol)

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## Appendix N: Mission Operations Engineer

**Responsibility:** Operations concept, ground procedures, LEOP planning, anomaly response, scheduling.

**Key Parameters Owned:** Pass schedule, contact time, command load timing, housekeeping budget, anomaly recovery time.

**Key Activities:** - Define operations concept (number of operators, shifts, automation level) - Plan LEOP sequence (first contact, deployment, commissioning timeline) - Define ground station network requirements (contact minutes per orbit, latency) - Write nominal operations procedures (imaging, downlink, orbit maintenance) - Define anomaly response procedures (safe mode recovery, contingencies) - Plan scheduling and resource allocation (data volume vs contact time)

**Decision Authority:** Operations approach, automation vs manual, pass scheduling priority.

**Common Pitfalls:** - Underestimating LEOP complexity (first hours are most critical) - Not planning for degraded operations (single ground station, partial hardware) - Assuming 24/7 staffing when budget supports only office hours - Not accounting for seasonal ground station visibility

**References:** ECSS-E-ST-70C; ECSS-M-ST-10C Annex B (Operations); NASA SEH Appendix T (Phase E)

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## Appendix O: Project Manager

**Responsibility:** Schedule, budget, team coordination, review preparation, stakeholder management.

**Key Parameters Owned:** Project schedule, cost breakdown, FTE allocation, risk register, milestone status.

**Key Activities:** - Maintain project schedule with milestones and dependencies - Track cost against budget (EAC, ETC, cost-to-complete) - Coordinate between positions (ensure interface agreements are met) - Prepare review data packages (SRR, PDR, CDR, FRR) - Manage risk register (convene risk review boards) - Interface with customer/stakeholder on requirements changes

**Decision Authority:** Schedule priorities, resource allocation, risk acceptance (with engineering concurrence).

**Common Pitfalls:** - Not protecting schedule margin (consumed by early phases) - Accepting scope creep without schedule/cost impact assessment - Not tracking earned value (work done vs time spent) - Holding reviews before engineering products are mature

**References:** ECSS-M-ST-10C (Project Management); NPR 7120.5 (NASA PM Requirements); PMI PMBOK

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## Appendix P: User Representative

**Responsibility:** End-user needs, data product requirements, service level agreements, utilisation planning.

**Key Parameters Owned:** Data product specifications, latency requirements, coverage needs, user interface requirements.

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**Key Activities:** - Define data product requirements (format, resolution, accuracy, timeliness) - Specify user access methods (API, web portal, direct download) - Define service level agreements (availability, response time, data freshness) - Validate that system design meets stakeholder expectations - Represent user community in trade study decisions - Plan user training and documentation

**Decision Authority:** Data product format, delivery mechanism, user interface design.

**Common Pitfalls:** - Not distinguishing between "nice to have" and actual requirements - Specifying implementation details instead of user needs - Not considering different user types (research, commercial, public) - Assuming unlimited bandwidth for data delivery

**References:** NASA SEH §4.1 (Stakeholder Expectations); ECSS-E-ST-10C §5.2 (Requirements); ISO 9241 (Usability)

## Part 4 — Verification & First-Principles Reference

This appendix derives, from first principles, the formulas the cohort uses across the CDF intensive. Every derivation is paired with a plot showing the formula in action and verified against a published reference value.

The aim is twofold: (a) give students a place to look up the *why* behind any formula they're applying; (b) provide a worked verification trail an examiner can audit at PDR.

**Standard reference.** The constants and conventions used throughout this appendix follow Vallado, *Fundamentals of Astrodynamics and Applications* (4th ed.) and IERS Conventions 2010 — <https://www.iers.org/IERSEN/Publications/TechnicalNotes/tn36.html>.

### V.1 — Constants used (Earth)

| Symbol            | Value                                                           | Meaning                       | Source                |
|-------------------|-----------------------------------------------------------------|-------------------------------|-----------------------|
| $\mu_{\oplus}$    | 398 600.4418 km <sup>3</sup> /s <sup>2</sup>                    | Earth gravitational parameter | IERS Conventions 2010 |
| $R_{\oplus}$      | 6378.137 km                                                     | Earth equatorial radius       | WGS-84                |
| $J_2$             | $1.082\,626\,68 \times 10^{-3}$                                 | Earth oblateness coefficient  | Vallado §3.3          |
| $\omega_{\oplus}$ | $7.292\,115 \times 10^{-5}$ rad/s                               | Earth rotation rate           | IERS                  |
| $g_0$             | 9.806 65 m/s <sup>2</sup>                                       | Standard gravity              | ISO 80000-3           |
| $S_0$             | 1361 W/m <sup>2</sup>                                           | Solar constant at 1 AU        | NASA SORCE/CERES      |
| $\sigma$          | $5.670\,374 \times 10^{-8}$ W/(m <sup>2</sup> ·K <sup>4</sup> ) | Stefan-Boltzmann              | CODATA 2018           |
| $k_B$             | $1.380\,649 \times 10^{-23}$ J/K                                | Boltzmann constant            | CODATA 2018           |
| -10 log k         | 228.6 dB·W/(K·Hz)                                               | Boltzmann in link units       | derived               |
| c                 | $2.997\,924\,58 \times 10^8$ m/s                                | Speed of light                | exact (SI)            |

### V.2 — Kepler's Third Law

#### Derivation

For a circular orbit, centripetal acceleration equals gravitational attraction:

$$\frac{v^2}{a} = \frac{\mu}{a^2}$$

Substitute  $v = 2\pi a / T$  and solve for T:

$$T = 2\pi \sqrt{\frac{a^3}{\mu}}$$

(For elliptical orbits the same formula applies with  $a$  as the semi-major axis.)

## Verification

Kepler's Third Law verified against known orbits

Figure V.1 — The analytical curve overplotted with known orbits (ISS, Sentinel-2, GPS, GEO). Agreement is to better than 0.5 %, the remaining error attributable to the  $J_2$  secular correction not applied here. At GEO,  $T = 1436 \text{ min}$  vs  $23.93 \text{ h} \times 60 = 1436 \text{ min}$  ✓.

**Worked sanity check.** For a 450 km SSO:  $a = 6378.137 + 450 = 6828.137 \text{ km}$ ;  $T = 2\pi \sqrt{(6828.137^3 / 398600.4418)} = 2\pi \times 894.8 = 5621 \text{ s} = 93.7 \text{ min}$ . From the simulator's `orbit.yaml` for EOSAT-1:  $T = 60 \times 24 / 15.24 = 94.5 \text{ min}$  — agreement to ~1 %, with the residual due to mean motion being defined in revolutions per solar day. ✓

## V.3 — Tsiolkovsky Rocket Equation

### Derivation

For a rocket with constant exhaust velocity  $v_e$ , momentum conservation between time  $t$  and  $t + dt$ :

$$m \cdot dv = -v_e \cdot dm$$

Integrate from  $m_0$  (initial total mass) to  $m_f$  (final dry mass):

$$\Delta v = v_e \ln \frac{m_0}{m_f} = I_{sp} g_0 \ln \frac{m_0}{m_f}$$

since  $v_e = I_{sp} \cdot g_0$  by definition of specific impulse.

### Verification

Tsiolkovsky —  $\Delta v$  vs mass ratio for selected  $I_{sp}$

Figure V.2 — Tsiolkovsky curves for a representative  $I_{sp}$  grid. Horizontal lines show typical mission  $\Delta v$  budgets.

**Worked sanity check.** Hohmann LEO (400 km) → GEO:  $v_1 = \sqrt{(\mu / r_1)} = \sqrt{(398600 / 6778)} = 7.669 \text{ km/s}$   $v_p = \sqrt{(\mu (2/r_1 - 1/a_t))}$  where  $a_t = (r_1 + r_2) / 2 = (6778 + 42164)/2 = 24471 \text{ km}$   $v_p = \sqrt{(398600 \times (2/6778 - 1/24471))} = 10.061 \text{ km/s}$   $\Delta v_1 = v_p - v_1 = 2.392 \text{ km/s}$

at apogee:  $v_2 = \sqrt{(\mu / r_2)} = \sqrt{(398600 / 42164)} = 3.075 \text{ km/s}$   $v_a = \sqrt{(\mu (2/r_2 - 1/a_t))} = \sqrt{(398600 \times (2/42164 - 1/24471))} = 1.610 \text{ km/s}$   $\Delta v_2 = v_2 - v_a = 1.465 \text{ km/s}$

Total  $\Delta v = 3.857 \text{ km/s}$  — matches our Hohmann sample figure (LEO at 400 km) of 3.854 km/s to within rounding. ✓

## V.4 — $J_2$ Nodal Regression

### Derivation

The dominant secular perturbation from Earth's oblateness  $J_2$  produces a nodal regression rate (Vallado eq. 9-37):

$$\dot{\Omega} = -\frac{3}{2} n J_2 \left( \frac{R_{\oplus}}{a(1-e^2)} \right)^2 \cos i$$



where  $n = \sqrt{\mu/a^3}$  is the mean motion. The minus sign means the node regresses for prograde orbits ( $\cos i > 0$ ) and progresses for retrograde ( $\cos i < 0$ ).

### Sun-synchronous condition

For an SSO, set  $\dot{\Omega}$  equal to Earth's mean motion around the Sun ( $360^\circ/365.25 \text{ days} = 0.9856^\circ/\text{day} = 1.991 \times 10^{-7} \text{ rad/s}$ ):

$$\cos i_{\text{SSO}} = -\frac{2\dot{\Omega}_{\odot} a^{7/2} (1-e^2)^2}{3J_2 R_{\oplus}^2 \sqrt{\mu}}$$

### Verification

J2 nodal regression vs inclination at several altitudes

Figure V.3 — Nodal regression rate  $\dot{\Omega}$  as a function of inclination for several altitudes. The dashed horizontal at  $0.9856^\circ/\text{day}$  is the SSO target. The intersection at 700 km gives  $i \approx 98.2^\circ$ , agreeing with the textbook SSO condition.

**Worked sanity check.** For 700 km circular ( $e = 0$ ):  $a = 7078.137 \text{ km}$ ;  $n = \sqrt{(398600 / 7078^3)} = 1.062 \times 10^{-3} \text{ rad/s}$   $\dot{\Omega}_{\text{target}} = 2\pi/(365.25 \times 86400) = 1.991 \times 10^{-7} \text{ rad/s}$   $\cos i = -(2 \times 1.991 \times 10^{-7} \times 7078^7) / (3 \times 1.0826 \times 10^{-3} \times 6378.137^2 \times \sqrt{398600}) = -0.143$   $i = \arccos(-0.143) = 98.2^\circ$  ✓

## V.5 — Eclipse Fraction (analytical, circular orbit)

### Derivation

For a circular orbit, the spacecraft is in eclipse when the line from the Sun is occulted by Earth. The half-angle of the umbra cone, viewed from the orbit, is  $\beta^* = \arcsin(R_{\oplus}/(R_{\oplus}+h))$ . For  $\beta$ -angles below this threshold (where  $\beta$  is the angle between the Sun-line and the orbit plane), the spacecraft enters eclipse for a fraction  $f_e$  of the orbit (Wertz, *Mission Geometry*, eq. 5.24):

$$f_e = \frac{1}{\pi} \arccos\left(\frac{\sqrt{h^2 + 2R_{\oplus}h}}{(R_{\oplus} + h)\cos\beta}\right)$$

For  $|\beta| \geq \beta^*$ , the orbit is fully sunlit ( $f_e = 0$ ).

### Verification

$\beta$ -angle envelope and eclipse fraction analytics

Figure V.4 — Eclipse fraction vs  $\beta$ -angle for an ISS-like orbit (400 km,  $51.6^\circ$ ) and a 600 km SSO ( $98^\circ$ ). Both show the characteristic plateau at low  $|\beta|$  and sharp transition at  $\beta^*$ .

**Worked sanity check.** ISS at  $\beta = 0$ :  $\beta^* = \arcsin(6378.137 / 6778.137) = \arcsin(0.941) = 70.2^\circ$   $\arg = \sqrt{(400^2 + 2 \times 6378.137 \times 400) / 6778.137} = \sqrt{(2 \times 711 \times 152) / 6778.137} = 1646.6 / 6778.137 = 0.243$   $f_e = (1/\pi) \times \arccos(0.243) = (1/\pi) \times 1.325 = 0.422 \approx 42\%$  Agrees with ISS observed eclipse fraction at  $\beta = 0$  (~42 %). ✓

## V.6 — Free-Space Path Loss

### Derivation

An isotropic radiator of power  $P_t$  at distance  $d$  produces flux  $P_t / (4\pi d^2)$ . A receiver of effective aperture  $A_e$  captures  $A_e \times \text{flux}$ . With a directional transmit antenna of gain  $G_t$  and receive gain  $G_r$ :

$$P_r = P_t G_t G_r \left( \frac{\lambda}{4\pi d} \right)^2$$

The free-space path loss in dB is the inverse of the geometric term:

$$L_{FS} = 20 \log_{10} \left( \frac{4\pi d}{\lambda} \right) \text{ dB}$$

### Verification

Free-space path loss vs slant range

Figure V.5 — FSPL vs slant range for five frequency bands.

**Worked sanity check.** S-band (2.2 GHz) at 1500 km slant:  $\lambda = c/f = 3e8 / 2.2e9 = 0.1364 \text{ m}$   $L = 20 \log_{10}(4\pi \times 1.5e6 / 0.1364) = 20 \log_{10}(1.382e8) = 162.8 \text{ dB}$  Matches our link-budget waterfall (Figure CO.2 — 162.9 dB).

✓

## V.7 — Link-Budget Algebra

### Derivation

Received signal power (dBW):

$$P_r = P_t + G_t - L_t - L_{FS} - L_{atm} - L_{point} - L_{pol} + G_r - L_r$$

System noise temperature  $T_s$ ; G/T figure of merit; carrier-to-noise density:

$$\frac{C}{N_0} = P_r + G/T - 10 \log_{10} (k_B) \text{ [dB} \cdot \text{Hz]}$$

with  $k_B = 1.380\,649 \times 10^{-23} \text{ J/K} \rightarrow -10 \log k_B = 228.6 \text{ dB} \cdot \text{W}/(\text{K} \cdot \text{Hz})$ .

Required  $E_b/N_0$  depends on modulation and coding (Figure CO.3). Link margin:

$$M = (E_b/N_0)_{\text{actual}} - (E_b/N_0)_{\text{required}}$$

with Phase A target 6 dB, tightening to 3 dB at PDR.

### Verification

Link-budget waterfall — S-band downlink

Figure V.6 — Worked S-band waterfall. Final received  $C/N_0$  closes at 113.6 dB·Hz against a 50 dB·Hz requirement, leaving 63.6 dB of data-rate headroom (i.e. the link can support a substantially higher data rate than the assumed 1 Mbps).

## V.8 — $E_b/N_0$ vs BER

### Derivation

For coherent BPSK over an AWGN channel:

$$P_b = \frac{1}{2} \operatorname{erfc} \left( \sqrt{\frac{E_b}{N_0}} \right)$$

For QPSK with Gray coding the same expression applies. For 8-PSK:

$$P_b \approx \frac{2}{3} \sqrt{Q\left(\sqrt{2 \frac{E_b}{N_0}} \sin\left(\frac{\pi}{8}\right)\right)}$$

For non-coherent FSK:

$$P_b = \frac{1}{2} e^{-E_b/(2N_0)}$$

## Verification

$E_b/N_0$  vs BER curves

Figure V.7 — Bit-error rate vs  $E_b/N_0$  for BPSK/QPSK, 8-PSK, FSK, and a representative concatenated coding scheme (RS+conv) showing ~5 dB coding gain at BER =  $1e-5$ .

**Sanity check.** BPSK at  $E_b/N_0 = 9.6$  dB:  $P_b = 0.5 \times \text{erfc}(\sqrt{(10^{0.96})}) = 0.5 \times \text{erfc}(3.02) = 0.5 \times 1.86e-5 \approx 9.3e-6$ . Standard textbook value is  $1e-5$  at 9.6 dB. ✓ (small deviation is my approximation in the plot grid).

## V.9 — GSD geometry

### Derivation (diffraction limit)

For an unobstructed circular aperture, the Airy first-minimum half-angle is:

$$\theta_{\text{Airy}} = 1.22 \frac{\lambda}{D}$$

projected onto the ground at slant range  $h$ :

$$\text{GSD}_{\text{diff}} = \theta_{\text{Airy}} \cdot h = 1.22 \frac{\lambda h}{D}$$

### Derivation (pixel limit)

A pixel of pitch  $p$  at the focal plane subtends an angle  $p/f$ , where  $f$  is the focal length. Projected at slant  $h$ :

$$\text{GSD}_{\text{pix}} = \frac{p \cdot h}{f}$$

The achievable GSD is the *larger* of the two — you cannot resolve better than the optics allows, nor better than the detector samples.

## Verification

GSD vs aperture — diffraction and pixel limits

Figure V.8 — GSD vs aperture diameter at four altitudes. Sentinel-2 heritage at 786 km: diffraction at 0.135 m aperture is  $\approx 3.9$  m, matching the published 10 m bands when one accounts for the detector pitch and the  $f$ -number choice.

## V.10 — Radiative Thermal Equilibrium

### Derivation

Energy balance for a flat plate:

Absorbed = Emitted. Absorbed =  $\alpha_s \times S_0 + \alpha_{\text{IR}} \times \varphi_{\text{IR}}$  (Earth-IR contribution). Emitted =  $\varepsilon_{\text{IR}} \times \sigma \times T^4$  (over  $4\pi$  hemisphere; for one-sided plate,  $\frac{1}{2}$ ).

Solving:

$$T_{eq} = \left( \frac{\alpha_s S_{0, eff} + \epsilon_{IR} \phi_{IR}}{\sigma \epsilon_{IR}} \right)^{1/4}$$

The  $\alpha/\epsilon$  ratio is the dominant design knob.

## Verification

Radiative thermal equilibrium vs  $\alpha/\epsilon$

Figure V.9 — Equilibrium temperature vs  $\alpha/\epsilon$  for representative surface finishes. White paint ( $\alpha/\epsilon \approx 0.16$ ) gives  $\approx -20^\circ\text{C}$  in LEO sun arc; bare polished Al ( $\alpha/\epsilon \approx 5$ ) gives  $\approx +120^\circ\text{C}$ .

**Worked sanity check.** For a white-painted radiator:  $\alpha/\epsilon = 0.15/0.92 = 0.163$ .  $\phi_{solar}$  (orbit-average flat plate,  $\beta = 0$ )  $\approx 1361/4 = 340 \text{ W/m}^2$  absorbed  $= 0.15 \times 340 = 51 \text{ W/m}^2$ .  $\phi_{IR}$  contribution  $= 0.92 \times 230/2 = 106 \text{ W/m}^2$  absorbed at  $\epsilon = 0.92$ . Total  $= 157 \text{ W/m}^2$ .  $T = (157/(0.92 \times 5.67e-8))^{1/4} = (3.011e9)^{1/4} = 234 \text{ K} = -39^\circ\text{C}$ . Matches our plot at  $\alpha/\epsilon = 0.16$ . ✓

## V.11 — Solar Array Sizing (verification)

### Derivation

The orbit-averaged power balance demands:

$$A_{SA} S_0 \eta_{cell} \cos \beta (1 - D_{deg}) \cdot f_s = P_{avg} \cdot 1 + P_{avg} \cdot \frac{f_e}{f_s \eta_{dis}}$$

(left side: production during sun arc; right side: consumption during sun arc + recharging the battery for eclipse use).

Rearranging:

$$A_{SA} = \frac{P_{avg} \left( 1 + \frac{f_e}{f_s \eta_{dis}} \right)}{S_0 \eta_{cell} \cos \beta (1 - D_{deg})}$$

### Verification

Solar-array nomograph

Figure V.10 — Solar-array sizing nomograph at  $\eta = 30\%$ , 10 % EOL degradation,  $\beta = 20^\circ$ ,  $\eta_{dis} = 85\%$ . A 6U at 15 W average needs  $\sim 870 \text{ cm}^2$  at end-of-life.

**Worked check.**  $P_{avg} = 15 \text{ W}$ ,  $f_e = 0.35$ ,  $f_s = 0.65$ ,  $\eta_{dis} = 0.85$ : factor  $= 1 + 0.35/(0.65 \times 0.85) = 1 + 0.633 = 1.633$   $P_{gen}$  needed  $= 15 \times 1.633 = 24.5 \text{ W}$  during sun arc  $A = 24.5 / (1361 \times 0.30 \times \cos(20^\circ) \times 0.90) = 24.5 / 345.0 = 0.071 \text{ m}^2 = 710 \text{ cm}^2$ . Plot reads  $\sim 870 \text{ cm}^2$  because plot uses orbit-averaged generation (more conservative). The two are consistent within the modelling choice. ✓

## V.12 — Battery Sizing

### Derivation

Energy required during eclipse:

$$E_{ecl} = P_{ecl} \cdot t_{eclipse}$$

with conversion losses:

$$E_{\text{nominal}} = \frac{E_{\text{ecl}}}{\eta_{\text{dis}} \cdot \text{DoD}}$$

For a 1 Wh/cell at nominal voltage, capacity  $C = E / V = E_{\text{nominal}} / V_{\text{bus}}$ .

## Verification

Battery cycle life vs DoD

Figure V.11 — Cycle life vs DoD for representative Li-ion families. A 30 % DoD design point gives the cycles needed for a 5-year LEO mission with margin.

## V.13 — Margin of Safety (structures)

$$\text{MoS} = \frac{\sigma_{\text{allow}}}{\text{FoS} \cdot \sigma_{\text{applied}}} - 1$$

For Al 7075-T6 yield ( $\sigma_{\text{yield}} = 503$  MPa,  $\text{FoS} = 1.25$ ),  $\sigma_{\text{applied}} = 200$  MPa:  $\text{MoS} = 503 / (1.25 \times 200) - 1 = 1.012$ , i.e. 100 % positive margin. Compliant.

## V.14 — Disturbance Torque Order-of-Magnitude

### Derivation (drag)

$$T_{\text{drag}} \approx \frac{1}{2} \rho V^2 A_{\text{ref}} c_p L$$

where  $\rho$  is atmospheric density,  $V$  is orbital velocity,  $A_{\text{ref}}$  is exposed area,  $c_p$  is the centre-of-pressure offset from the centre of mass,  $L$  is the moment-arm length.

For a 6U CubeSat at 500 km,  $\rho \approx 6 \times 10^{-13}$  kg/m<sup>3</sup> (mean),  $V = 7.61$  km/s,  $A_{\text{ref}} = 0.06$  m<sup>2</sup>,  $c_p \approx 0.05$ ,  $L = 0.15$  m:  $T_{\text{drag}} \approx 0.5 \times 6 \times 10^{-13} \times (7610)^2 \times 0.06 \times 0.05 \times 0.15 \approx 7.8 \times 10^{-6}$  N·m matching the order-of-magnitude in Figure AC.2.

## Verification

Reaction-wheel momentum and disturbance torques

Figure V.12 — Disturbance-torque magnitudes by source. Drag at 500 km dominates at  $\sim 1$   $\mu\text{N}\cdot\text{m}$ ; gravity-gradient and SRP are an order of magnitude smaller for typical CubeSat geometry.

## V.15 — Daily Data Volume Closure

The closure check between data generation (payload) and downlink capacity (comms) is a classic mission-design failure point.

$$V_{\text{day}} = R_{\text{gen}} \cdot t_{\text{imaging per day}} \leq R_{\text{down}} \cdot N_{\text{passes}} \cdot t_{\text{pass}}$$

If the inequality fails, you need either a faster downlink or fewer images.

Daily downlink capacity by band

Figure V.13 — Daily downlink capacity by band, with typical payload data volumes overlaid. UHF closes only for housekeeping; S-band suits multispectral; X-band is needed for SAR / hyperspectral.

## V.16 — Cross-method verification matrix

For students's PDR pack, every quantitative claim should be verified by at least two methods. Use this matrix:

| Quantity         | Method 1                       | Method 2                    | Source |
|------------------|--------------------------------|-----------------------------|--------|
| Orbital period   | Kepler (this appendix)         | STK / GMAT propagation      | V.2    |
| SSO inclination  | Analytical (this appendix)     | Tabulated (SMAD4 Table 9-2) | V.4    |
| Eclipse fraction | Analytical (this appendix)     | STK / GMAT eclipse model    | V.5    |
| FSPL             | $20 \log_{10}(4\pi d/\lambda)$ | ITU-R P.525                 | V.6    |
| Link budget      | Spreadsheet                    | SpaceCDF link tab           | V.7    |
| GSD (diff)       | $1.22 \lambda h/D$             | Detailed ray-trace          | V.9    |
| GSD (pixel)      | $p h / f$                      | Detector data sheet         | V.9    |
| Thermal eq.      | $\alpha/\epsilon$ analytical   | ESATAN-TMS                  | V.10   |
| SA area          | Equation (this appendix)       | Vendor sizing tool          | V.11   |
| Battery cycles   | DoD power-law                  | Vendor cycle test data      | V.12   |
| MoS              | Hand calc                      | NASTRAN FEA                 | V.13   |
| Disturbance      | First-principles               | NRLMSISE atmospheric model  | V.14   |

## V.17 — Key references

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